Nonparametric Statistics

Lecture Notes following Jung, *Nonparametric Statistics with R* (Ch. 1–12, Appendix A)

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Chapter 1. Introduction to Nonparametric Statistics

1.1 What is nonparametric statistics?

Nonparametric statistics (nonparametric statistical methodology) is the theory and practice of statistical inference (estimation and hypothesis testing) under *few assumptions about the population.* Three guiding questions:

1. To compare the means of two samples we would use a t-test if both were normal. Which test should we use if normality is badly violated?
2. The coefficient of variation (standard deviation divided by the mean) is a common index of inequality. From the scholarship amounts $x_1, \ldots, x_n$ of $n = 100$ randomly chosen students we compute $s/\bar{x}_n$. How accurate is this estimate?
3. Poll results for a politician's approval are observed over time (Figure 1.1 of the book). The relation between time and approval is clearly not linear. How can we estimate the change of approval over time accurately?

The first question (testing without normality) is answered by classical nonparametric, *distribution-free* methods: tests based on signs and ranks, and permutation tests. The second concerns the uncertainty of an estimator; the mathematical-statistical calculation is hard in general, and the *bootstrap* replaces it by computation. The third concerns estimating a curve; *nonparametric function estimation* (smoothing) uses the observations near each point to estimate the curve there.

1.1.1 Nonparametric models

Assumptions about a population are usually made through a parametric model. If $X_1, \ldots, X_n$ are drawn from a population, one often assumes that its density f belongs to the **normal family**

$$\begin{gathered} \mathcal{F}_G = \{\phi_{\mu,\sigma} : \mu \in (-\infty, \infty),\ \sigma > 0\}, \\ \phi_{\mu,\sigma}(x) = \phi(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{(x-\mu)^2}{2\sigma^2} \Big), \quad -\infty < x < \infty. \end{gathered} \tag{1.1}$$

This is the normality assumption, a *parametric model*: the population is determined by the two parameters μ, σ. (A "parameter" is mathematically an undetermined constant, and in computing a variable that determines a property of a system.) Nonparametric statistics avoids such a model and offers methods valid whatever family f belongs to. Examples of families used instead:

- $\mathcal{F}_1 = \{f : \int x^2 f(x)\, dx < \infty\}$, all distributions with finite mean and variance;
- $\mathcal{F}_2 = \{f : f(\mu - x) = f(\mu + x),\ x \in (-\infty, \infty) \text{ for some } \mu\}$, distributions whose density is symmetric about some μ (the mean need not exist).

These families contain uncountably many distributions, and parametrizing them needs infinitely many parameters, whereas $\mathcal{F}_G$ is indexed by a two-dimensional parameter space. Note $\mathcal{F}_G \subset \mathcal{F}_1$ and $\mathcal{F}_G \subset \mathcal{F}_2$. Most parametric models (normal, gamma, ...) lie in $\mathcal{F}_1$. A strong assumption on the population means a small family; assuming only $f \in \mathcal{F}_1$ is a very weak assumption.

1.1.2 Nonparametric statistics and parameters

In nonparametric statistics a **parameter** is a characteristic of an element of the family. For $f \in \mathcal{F}_1$, the mean $\mu_f := \int x f(x)\, dx$ and variance $\sigma_f^2 = \int (x - \mu_f)^2 f(x)\, dx$ are parameters; but knowing them

does not determine f. For $f \in \mathcal{F}_1$ or $\mathcal{F}_2$ the median is also a parameter. Nonparametric inference is estimation and testing of parameters defined on a very large family; when the assumptions are extremely weak (e.g. no normality, only $f \in \mathcal{F}_2$), the methods are called **distribution-free**.

A parameter may be simple (mean, median) or anything that characterizes the population. Under normality all quantiles are determined by the mean and variance; under $\mathcal{F}_1$ they are not, and a quantile (percentile) is itself a parameter to estimate. The density f or the distribution function $F(x) = \int_{-\infty}^{x} f(t)\,dt$ may also be the target. Accordingly:

1. **Location and scale parameters**: mean, median, variance. Classical distribution-free methods transform the data into signs, ranks or scores; e.g. the Wilcoxon signed-rank test, the Kruskal–Wallis test, the Friedman test.
2. **The density as a parameter**: the field of *density estimation*.
3. **Parameters in complex models**: extensions to ANOVA, regression, multivariate analysis.

For instance, the usual simple linear regression model assumes

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

With the X_i fixed, the Y_i are independent normal with means $\beta_0 + \beta_1 X_i$ and variance σ^2, so the family

$$f(y_1, \ldots, y_n) \in \Big\{ \prod_{i=1}^{n} \phi(y_i; \beta_0 + \beta_1 X_i, \sigma^2) : \beta_0, \beta_1 \in \mathbb{R},\ \sigma^2 > 0 \Big\}$$

is a parametric model with three parameters (ϕ as in (1.1)). The *distribution-free* linear regression model is

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} f, \quad f \in \mathcal{F}_1,$$

and a *nonparametric* regression model is, e.g.,

$$Y_i = m(X_i) + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2),$$

where $m(\cdot)$ is any curve in a function space $\mathcal{H}$ (e.g. smooth or differentiable functions). Both $\mathcal{F}_1$ and $\mathcal{H}$ are infinite-dimensional, so these families cannot be described by finitely many parameters. In distribution-free regression the interest is in β_0, β_1, approached much like the median in a one-sample model; in nonparametric regression the function-valued parameter $m(\cdot)$ is estimated.

1.1.3 Three tools of nonparametric statistics

Nonparametric methods are needed because:

- they are developed under very weak assumptions (or very large model spaces), so errors from violating assumptions are small;
- they also apply under parametric models, and mathematical calculation can be replaced by computation;
- the model space is so large that they perform very well in estimating complex systems.

Correspondingly there are three kinds of methodology:

1. **Permutation tests.** Proposed by R. A. Fisher; widely used since the 1990s as computation became cheap. They follow Fisher's idea of *conditional inference*: inference given the observed data, i.e. treating the observed values as fixed.

2. **The bootstrap** (Bradley Efron). Obtains the sampling distribution of a statistic from the given sample, without a parametric model. Like permutation tests it relies on computation and repetition; both are *resampling* methods.
3. **Smoothing.** Estimating a density $f(\cdot)$ or regression function $m(\cdot)$ without a specific form is hard because the function space is infinite-dimensional. Smoothing estimates $m(x)$ or $f(x)$ at a point x from the observations near x; as x varies, the observations used change smoothly, giving a smooth estimate $\hat{m}(\cdot)$.

Classical sign and rank methods were mostly developed before the 1970s and can be understood as special permutation tests. From the 1980s, smoothing-based nonparametric function estimation (density estimation, nonparametric regression) became the mainstream, while permutation tests and the bootstrap proved highly effective for complex statistics of complex populations, under both parametric and nonparametric models.

1.2 Parametric models and statistical inference

Statistical inference splits into estimation and hypothesis testing. To explain point and interval estimation, standard errors and testing, we first review standard parametric models.

1.2.1 Standard parametric models

Normal distribution. If $X_1, \ldots, X_n$ come from a normal population, the family is $\mathcal{F}_G$; we write $X \sim N(\mu, \sigma^2)$. The density (1.1) is determined by the location parameter $\mu \in (-\infty, \infty)$ and the scale parameter $\sigma^2 > 0$, with $E(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$. $N(0,1)$ is the *standard normal*, and any $X \sim N(\mu, \sigma^2)$ standardizes to $Z = (X - \mu)/\sigma \sim N(0,1)$. With $\Phi(z) = F_Z(z) = P(Z \le z)$,

$$F_X(x) = P(X \le x) = P\{Z = (X-\mu)/\sigma \le (x-\mu)/\sigma\} = \Phi\{(x-\mu)/\sigma\}.$$

Definition (Quantile)

For a distribution with distribution function F and $p \in (0,1)$, the p**-quantile** is

$$Q_p = \min\{z : F(z) \ge p\},$$

the smallest point at which the cumulative probability reaches p. As a function of p it is the **quantile function** $Q(p) = Q_p$.

If F is invertible, $Q_p = F^{-1}(p)$; for the normal, the quantile function is $\Phi^{-1} : (0,1) \to (-\infty, \infty)$. For a significance level $\alpha \in (0,1)$ the z**-value** is $z_\alpha = \Phi^{-1}(1-\alpha)$; e.g. $z_{0.05} = \Phi^{-1}(0.95) \approx 1.645$, $z_{0.025} = \Phi^{-1}(0.975) \approx 1.96$.

Binomial distribution. A Bernoulli trial has two outcomes (success/failure, positive/negative, ...), coded 1 and 0. If independent Bernoulli trials have common success probability $p = P(\text{“1”}) \in (0,1)$, the number X of successes in n trials follows the binomial distribution $X \sim B(n,p)$; the parameter is p (n is fixed). Its probability mass function and distribution function are

$$f(x) = P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n,$$

$$F_B(x) = P(X \le x) = \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k}.$$

F_B is a step function, so it is not invertible, but $Q_p = \min\{x : F_B(x) \ge p\}$ is still defined, and:

- for an integer $x \in \{0, \dots, n\}$, if $p \in (F(x-1), F(x)]$ then $Q_p = x$;
- for every $p \in (0,1)$, $F(Q_p - 1) < p \le F(Q_p)$, hence

$$P(X \le Q_p) \ge p, \qquad P(X \ge Q_p) > 1 - p.$$

Proof (supplied). F is nondecreasing, right-continuous and constant on each $[x, x+1)$, so $\{z : F(z) \ge p\} = [x_0, \infty)$ for the smallest integer x_0 with $F(x_0) \ge p$, and $Q_p = x_0$. Minimality gives $F(x_0 - 1) < p \le F(x_0)$. If $p \in (F(x-1), F(x)]$ then x is that smallest integer, so $Q_p = x$. Finally $P(X \le Q_p) = F(Q_p) \ge p$ and $P(X \ge Q_p) = 1 - F(Q_p - 1) > 1 - p$. ■

The mean and variance are $E(X) = \sum_{x=0}^{n} x f(x) = np$ and $\mathrm{Var}(X) = np(1-p)$ (X is a sum of n independent Bernoulli(p) variables, each with mean p and variance $p(1-p)$).

1.2.2 Parameter estimation and hypothesis testing

Example 1.1

Eight randomly chosen men in their 50s joined a weight-loss program. After 10 weeks their weight losses (kg) were $3, 2, 0, 8, 12, 11, 5, -2$ (the last participant gained weight). How much weight loss can a man in his 50s expect from the program, and is the program effective?

The first question is estimation, the second hypothesis testing.

Estimation. The population in Example 1.1 is the distribution of weight losses, and the "expected loss" is a location parameter (mean, median, mode). Under normality ($f \in \mathcal{F}_G$) mean, median and mode all equal μ, and $\bar{X}_n = \frac{1}{n}\sum X_i$ is known to be the best estimator. In particular:

1. $E(\bar{X}_n) = \mu$: it is **unbiased**, with **standard error** $\mathrm{se} = (\mathrm{Var}(\bar{X}_n))^{1/2} = \sigma/\sqrt{n}$, the smallest among unbiased estimators.
2. $\bar{X}_n \sim N(\mu, \sigma^2/n)$ exactly; if σ^2 is known an exact confidence interval follows: the 95% interval $\mu \in \bar{X}_n \pm 1.96\sigma/\sqrt{n}$ has exact coverage because $Z = (\bar{X}_n - \mu)/\mathrm{se} \sim N(0,1)$.
3. If σ^2 is unknown, studentize with $s^2 = \sum_{i=1}^{n}(X_i - \bar{X}_n)^2/(n-1)$:

$$T = \frac{\bar{X}_n - \mu}{\widehat{\mathrm{se}}}, \qquad \widehat{\mathrm{se}} = \frac{s}{\sqrt{n}}.$$

   Under normality $T \sim t_{n-1}$, and the 95% interval is $\mu \in \bar{X}_n \pm t_{0.025,n-1}s/\sqrt{n}$, where $t_{\alpha,n-1}$ is the $1-\alpha$ quantile of t_{n-1}.

Unbiasedness and small standard error are desirable. Under a parametric model (e.g. normal) one can find an optimal estimator that is unbiased with the smallest standard error; mathematical statistics is largely about finding optimal estimators under various parametric assumptions. But if the population is far from normal, the standard error of the estimator (the measure of accuracy) can be very large or very small compared with $\widehat{\mathrm{se}}$, and the standard error of $\bar{X}_n$, "optimal" under normality, can be much larger than that of other estimators. The nonparametric methods to come guarantee a certain performance for any population in a very large family, i.e. under very weak assumptions.

Hypothesis testing. The second question of Example 1.1 is a test: choose between two hypotheses about a parameter, the null H_0 and the alternative H_1, using the data. "Is the program effective?" becomes $H_0 : \mu = 0$ (no effect) versus $H_1 : \mu > 0$ (effect). A **test statistic** takes typically different values under the two hypotheses; e.g. $T = \bar{X}_n/\widehat{\mathrm{se}}$ tends to be large under H_1 and near 0 under H_0 (where $T \sim t_{n-1}$). The decision rule usually has the form

- if $T > c$, reject H_0 (accept H_1);
- if $T < c$, fail to reject H_0 (insufficient evidence against H_0).

The **rejection region** determined by c is chosen so that the probability of a type I error (rejecting a true H_0) is at most the significance level α (usually 0.05). The closer the type I error probability is to α, the higher the **power** (probability of accepting H_1 when it is true), so if possible c is set so that the type I error probability is exactly α. For the t-test the **critical value** is $c = t_{\alpha,n-1}$: reject when $T > t_{\alpha,n-1}$. Under normality the t-test is the most powerful among tests of the location parameter with type I error controlled at α. If normality fails badly, the t-test no longer guarantees control of the type I error at the given level; permutation tests, in particular, control it exactly under very weak assumptions.

- Tests are classified by the direction of H_1. For $H_0 : \mu = \mu_0$ and $T = (\bar{X}_n - \mu_0)/\widehat{\text{se}}$:
  - lower-tailed: $H_1 : \mu < \mu_0$, rejection region $\{T : T < -t_{\alpha,n-1}\}$;
  - upper-tailed: $H_1 : \mu > \mu_0$, rejection region $\{T : T > t_{\alpha,n-1}\}$;
  - two-sided: $H_1 : \mu \neq \mu_0$, rejection region $\{T : |T| > t_{\alpha/2,n-1}\}$.
- The **p-value** measures how inconsistent the data are with H_0: the probability, under H_0, of a test statistic at least as extreme as the observed T_{obs}, with $T \sim t_{n-1}$:
  - lower-tailed: p-value $= P(T < T_{\text{obs}})$;
  - upper-tailed: p-value $= P(T > T_{\text{obs}})$;
  - two-sided: p-value $= P(T < -|T_{\text{obs}}|$ or $T > |T_{\text{obs}}|)$.

  The smaller the p-value, the harder it is to believe H_0. Rejecting H_0 when the p-value is below α is the same rule as the rejection region above.

The statistician's job in testing: set appropriate assumptions on the population; express the hypotheses mathematically in terms of parameters; choose a test statistic distinguishing H_0 from H_1; find the rejection region. In nonparametric statistics, where the assumptions are very weak, the key is a **distribution-free test statistic**, one that does not depend on the population distribution. Finding the rejection region then means finding the distribution of the statistic under H_0, the **null distribution** (t_{n-1} for the t-statistic under normality).

Example 1.2

Assume the data of Example 1.1 come from a normal population: $X_1, \ldots, X_8 \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$. Answer the questions at $\alpha = 0.05$. The sample mean and standard deviation are $\hat{\mu} = \bar{x}_n = 4.875$ and $\hat{\sigma} = s = 5.1$, so the expected loss is estimated as 4.875 kg, and the 95% confidence interval $\mu \in \bar{x}_n \pm t_{0.025,7} s/\sqrt{8}$ is computed in R:

```
> x = c(3, 2, 0, 8, 12, 11, 5, -2)
> n = length(x)
> (ci = mean(x) + c(-1,1) * qt(1-0.025,n-1) * sd(x) / sqrt(n))
[1] 0.6253086 9.1246914
```

To test effectiveness, take $H_0 : \mu = 0$ versus $H_1 : \mu > 0$ and run the t-test:

```
> t.test(x, alternative = "greater")$p.value
[1] 0.01504372
```

The p-value is below 5%, so the program has a significant effect at $\alpha = 5\%$.

Duality of tests and confidence intervals. Suppose inference on μ uses a point estimate (or its standardized form T) and its sampling distribution. When testing $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$: if μ_0 lies in the 95% confidence interval (L, U), the level-5% test does not reject H_0; if $\mu_0 \notin (L, U)$ it rejects. Conversely, if the level-α test of $H_0 : \mu = \mu'$ fails to reject, μ' lies in the interval. Thus tests can be obtained from confidence intervals and vice versa.

Example 1.3 (Tests and confidence intervals under normality)

For $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, the 95% interval for μ is $CI(\mu) = \bar{X}_n \pm t_{0.025,n-1} s/\sqrt{n}$. The level-5% t-test of $H_0 : \mu = \mu_0$ versus $H_1 : \mu \neq \mu_0$ rejects when $|\bar{X}_n - \mu_0| > t_{0.025,n-1} s/\sqrt{n}$. If the test fails to reject,

$$|\bar{X}_n - \mu_0| \leq t_{0.025,n-1} s/\sqrt{n} \iff \mu_0 \in \bar{X}_n \pm t_{0.025,n-1} s/\sqrt{n}, \tag{1.2}$$

so $\mu_0 \in CI(\mu)$; if it rejects, $\mu_0 \notin CI(\mu)$. Conversely, the test that rejects H_0 exactly when μ_0 is not in the 95% interval is the t-test.

Normality and the central limit theorem. For real data, normality almost never holds exactly. A population with a symmetric, bell-shaped density is approximately normal, and the "normality assumption" is generally accepted in that approximate sense; the parametric methods above (intervals, t-tests) are then approximately correct. Data often differ greatly from normal, e.g. discrete data (from a binomial population), ordinal data (survey answers: very satisfied, satisfied, neutral, dissatisfied, very dissatisfied), or continuous data from a heavily skewed distribution. Still, t-based inference is often used for them, because by the *central limit theorem* the large-sample distribution of the statistic agrees with its null distribution under normality.

Theorem 1.1 (Central limit theorem)

Let $f \in \mathcal{F}_1$ and $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f$. The sample mean $\bar{X}_n$ has mean $E(\bar{X}_n) = E(X_1) = \mu_f$ and variance $\text{Var}(\bar{X}_n) = \frac{1}{n}\text{Var}(X_1) = \frac{\sigma_f^2}{n}$. When n is large, the standardized sample mean is approximately standard normal:

$$\frac{\bar{X}_n - \mu_f}{\sqrt{\sigma_f^2/n}} \xrightarrow{d} N(0,1),$$

or $\bar{X}_n \dot{\sim} N(\mu_f, \sigma_f^2/n)$.

Proof (supplied). The mean and variance follow from linearity and independence: $E\bar{X}_n = \frac{1}{n}\sum EX_i = \mu_f$ and $\text{Var}\,\bar{X}_n = \frac{1}{n^2}\sum \text{Var}\,X_i = \sigma_f^2/n$. Assume $\sigma_f^2 > 0$ and let $Y_i = (X_i - \mu_f)/\sigma_f$, so $EY_i = 0$, $EY_i^2 = 1$, and $Z_n := \frac{\bar{X}_n - \mu_f}{\sigma_f/\sqrt{n}} = \frac{1}{\sqrt{n}}\sum_{i=1}^n Y_i$. Let $\varphi(t) = Ee^{itY_1}$. Since $E|Y_1|^2 < \infty$, φ is twice differentiable with $\varphi(0) = 1$, $\varphi'(0) = iEY_1 = 0$, $\varphi''(0) = -EY_1^2 = -1$, so $\varphi(t) = 1 - \frac{t^2}{2} + o(t^2)$ as $t \to 0$. By independence,

$$Ee^{itZ_n} = \varphi(t/\sqrt{n})^n = \left(1 - \frac{t^2}{2n} + o(\tfrac{1}{n})\right)^n \longrightarrow e^{-t^2/2}$$

for every fixed t, using $(1 + a_n/n)^n \to e^a$ when $a_n \to a$ (valid for complex a_n). The limit is the characteristic function of $N(0,1)$, so by Lévy's continuity theorem $Z_n \xrightarrow{d} N(0,1)$. ∎

In Theorem 1.1, $X_n \xrightarrow{d} f$ means that the distribution of X_n converges to f as $n \to \infty$ (*convergence in distribution*); $\bar{X}_n \dot{\sim} f_n$ means that for large n the distribution of $\bar{X}_n$ is approximately f_n. By the CLT, statistics of the form of a mean or sum can be approximated by a normal distribution. Thus

the usefulness of the normal model extends beyond normal-like populations to every population for which the large-sample CLT approximation applies to the estimator or test statistic.

1.3 Why nonparametric methods are needed

Methods developed under parametric (especially normal) models remain useful when the population is not exactly normal. Still, nonparametric methods are needed for three reasons:

- They are developed under very weak assumptions (or very large model spaces), so errors from violated assumptions are very small. In particular, in small samples the normal approximation via the CLT cannot be used.
- They also apply under parametric models, and mathematical calculation is easily replaced by computation. Even with large samples, when the CLT does not apply, the sampling distribution of an estimator or the null distribution of a test statistic can be approximated nonparametrically.
- The nonparametric model space is so large that they excel at estimating complex systems. Just as the normal family approximately contains many similar distributions, nonparametric statistics approximates an infinite-dimensional (nonparametric) family by finite-dimensional families of very high dimension.

1.4 Distribution-free and robust methods

A **robust** statistical method performs well even when the data contain **outliers**.

Example 1.4

The copper content (parts per million) of 24 wholemeal flour products, in increasing order (Analytical Methods Committee, 1989):

```
2.20 2.20 2.40 2.40 2.50 2.70 2.80 2.90
3.03 3.03 3.10 3.37 3.40 3.40 3.40 3.50
3.60 3.70 3.70 3.70 3.70 3.77 5.28 28.50
```

The last value 28.50 looks like an outlier. Whether a value is an outlier should not be decided by the statistician alone but with the judgment of experts of the field: 28.50 may be a typo for 2.85, a measurement-device error, or a genuine value. The mean and standard deviation change drastically with and without it:

- mean: 4.26 (with), 3.21 (without);
- standard deviation: 5.2 (with), 0.68 (without).

If 28.50 really is an outlier, the sample mean and standard deviation should not be used to summarize the data. By contrast, the robust summaries, the sample **median** and the **median absolute deviation** (MAD)[1], give

- median: 3.385 (with), 3.37 (without);
- MAD: 0.53 (with), 0.50 (without),

[1]The MAD measures spread: it is the median of the absolute median-centered deviations $|x_i - \text{median}\{x_i\}|$, $\text{MAD}(x) = \text{median}\{|x_i - \text{median}\{x_i\}|\}$. Compare the standard deviation, the root mean square of the mean-centered deviations, $\text{sd}(x) = \left(\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2\right)^{1/2}$.

and change little. Distribution-free nonparametric methods keep their properties whatever the population. Data with an outlier can be viewed as a sample from a very heavy-tailed population in which very large values occur; so distribution-free methods are robust. (Not every robust method is nonparametric.) Several nonparametric methods in this book are better known as robust methods, and are motivated that way when they appear.

Chapter 2. Distribution-Free Inference for One Population

Overview

Using a simple random sample from a population with distribution function $F(x)$, we test hypotheses about the population median $\theta = Q(0.5)$, e.g. the one-sided problem

$$H_0 : \theta = \theta_0, \qquad H_1 : \theta > \theta_0.$$

Example 2.1 (IQ test)

Ten randomly chosen 6th graders of an elementary school received a special one-week training to raise IQ scores, then took an IQ test designed for upper-grade pupils:

$$(98, 121, 110, 89, 109, 108, 102, 92, 131, 114).$$

IQ scores are standardized to mean 100 and standard deviation 10. Is the special training effective?

Example 2.2 (Sodium intake)

The recommended sodium intake for healthy adult men is 1500 mg. The daily intakes of 10 randomly chosen dieting adults were $1200, 2300, 5800, 1100, 1200, 1700, 1400, 1900, 3800, 1300$ (mg). Is the daily intake of dieting adults larger than recommended?

In Example 2.1 the population can be taken normal: with no effect the mean is $\theta = 100$, with an effect $\theta > 100$, so the family is $\{N(\theta, 10) : \theta \in [50, 150]\}$ and the hypotheses are $H_0 : \theta = 100$, $H_1 : \theta > 100$. With known $\sigma = 10$,

$$Z = \frac{\bar{X}_n - \theta_0}{\sigma/\sqrt{n}} = \frac{\bar{X}_n - 100}{10/\sqrt{10}}$$

is standard normal under H_0 (its null distribution). The observed $Z_{\text{obs}} = 7.4$ gives the one-sided p-value $P(Z > 7.4) \approx 6 \times 10^{-14}$, so H_1 (the training is effective) is accepted. In Example 2.2 normality is implausible: the 10 values are right-skewed (long heavy right tail), so even symmetry is hard to assume. When neither normality nor another parametric model is reasonable, distribution-free tests are used.

2.1 The sign test

In Example 2.2 we may use the simplest distribution-free test, the **sign test**. Assume $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ with $F \in \mathcal{F} = \{F\}$, no restriction on the population. Every population has a median $\theta = Q_F(0.5)$, and we test

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_1 : \theta > \theta_0. \tag{2.1}$$

A distribution-free test needs a test statistic whose null distribution does not depend on the actual population. Under H_0 the population can be any distribution with median θ_0; but by definition of the median, half of the population lies below θ_0 and half above. So transform

$$S_i = S_i(X_i, \theta_0) = \begin{cases} +, & X_i > \theta_0, \\ -, & X_i < \theta_0. \end{cases} \tag{2.2}$$

Then $P(S_i = +) = 1/2$ for every population with median θ_0.[2] The S_i are independent, so the number of "+" signs is $B(n, 1/2)$ under H_0. In summary:

- test statistic of the sign test: $S = \sum_{i=1}^{n} I(S_i = +) = \sum_{i=1}^{n} I(X_i > \theta_0)$;
- null distribution: under H_0, $S \sim B(n, 1/2)$.

Under H_0, $P(X_i > \theta_0) = 1/2$; if in fact $\theta > \theta_0$ (H_1 true), the probability that $S_i = +$ exceeds $1/2$, so S tends to be larger (Figure 2.2 of the book). The distribution-free sign test rejects H_0 when S is large. Since the null distribution is binomial, it is also called the **binomial test**.

2.1.1 The p-value and rejection region of the sign test

The p-value of the sign test is the probability under H_0 of observing $S \sim B(n, 1/2)$ more extreme (in the direction of H_1) than the observed S_{obs}. For the upper-tailed test (2.1), $p = P(S \geq S_{\text{obs}})$.

The upper-tailed test rejects H_0 when S is at least a critical value c_{val}, $S \geq c_{\text{val}}$. To make the type I error at most, and as close as possible to, α, take the smallest c_{val} with

$$P(S \geq c_{\text{val}} \mid H_0) = \sum_{k=c_{\text{val}}}^{n} \binom{n}{k} \left(\frac{1}{2}\right)^n \leq \alpha.$$

In terms of quantiles,

$$c_{\text{val}} = Q_{1-\alpha} + 1, \tag{2.3}$$

where Q_p is the p-quantile of $B(n, 1/2)$.

Proof of (2.3) (supplied). Let F be the distribution function of $B(n, 1/2)$. For an integer c, $P(S \geq c) \leq \alpha \iff 1 - F(c-1) \leq \alpha \iff F(c-1) \geq 1 - \alpha$. By definition $Q_{1-\alpha}$ is the smallest integer x with $F(x) \geq 1 - \alpha$, so the smallest admissible c satisfies $c - 1 = Q_{1-\alpha}$. ■

Example 2.3

In Example 2.1 we test $H_0 : \theta = 100$ against $H_1 : \theta > 100$ (the training is effective) with the sign test. Giving "+" to the observations above $\theta_0 = 100$, $S_{\text{obs}} = 7$ of the $n = 10$ values are positive.

```
> iq = c(98, 121, 110, 89, 109, 108, 102, 92, 131, 114)
> ifelse(iq > 100, "+", "-")
 [1] "-" "+" "+" "-" "+" "+" "+" "-" "+" "+"
```

The null distribution is $S \sim B(10, 1/2)$ and large S_{obs} favors H_1, so the p-value is $P(S \geq S_{\text{obs}}) = \sum_{x=S_{\text{obs}}}^{n} f_B(x; n, \frac{1}{2})$:

```
> (pvalue = 1-pbinom(Sobs-1,n,1/2))
[1] 0.171875
```

(or `sum(dbinom(Sobs:n,n,1/2))`). The p-value is large, so there is not enough evidence that the training is effective.

Example 2.4

With $n = 10$ observations as in Example 2.1, what is the rejection region of the sign test at $\alpha = 0.05$? The distribution of S is discrete, so the type I error cannot be controlled at exactly 0.05. With rejection region $S \geq 9$ the type I error is

```
> S0 = 9
```

[2] More precisely, $P(S_i = +) = P(X_i > \theta_0) = P(X_i \geq \theta_0) = 1/2$ requires $P(X = \theta_0) = 0$ at the median θ_0.

```
> 1 - pbinom(S0-1, n, 1/2) # P(S >= S0) = 1 - P(S <= S0 -1)
[1] 0.01074219
```

With $S \geq 8$ it exceeds α:

```
> S0 = 8; 1 - pbinom(S0-1, n, 1/2)
[1] 0.0546875
```

Only with $S \geq 9$ or $S \geq 10$ is the type I error controlled below α, so the rejection region is $S \geq 9$.

Large-sample approximation. The binomial $B(n, p)$ is well approximated by the normal for large n (roughly $np \geq 10$) by the central limit theorem. The null distribution of the sign statistic is $B(n, \frac{1}{2})$, so for large n, under H_0,

$$S \dot{\sim} N\Big(\frac{n}{2}, \frac{n}{2}(1-\tfrac{1}{2})\Big)$$

($X_n \dot{\sim} F$ means the distribution of X_n is approximated by F asymptotically), and the p-value of S_{obs} is approximately

$$p = P(S \geq S_{\text{obs}}) = P\Big(\frac{S - 0.5n}{\sqrt{0.25n}} \geq \frac{S_{\text{obs}} - 0.5n}{\sqrt{0.25n}}\Big) \approx P\Big(Z \geq \frac{S_{\text{obs}} - 0.5n}{\sqrt{0.25n}}\Big),$$

with $Z \sim N(0,1)$; i.e. $p \approx 1 - \Phi\big((S_{\text{obs}} - 0.5n)/\sqrt{0.25n}\big)$. The approximate rejection region at level α is

$$S \geq 0.5n + z_\alpha \sqrt{0.25n}.$$

Example 2.5

The approximate p-value of the sign test for Example 2.1 using the large-sample approximation:

```
> Sz = (Sobs - n/2) / sqrt(0.25*n)
> 1-pnorm(Sz)
[1] 0.1029516
```

With only $n = 10$ observations, however, the large-sample approximation should not be used.

Ties. If X_i is discrete rather than continuous, observations equal to the hypothesized median θ_0 may occur, i.e. a "tie" $X_i = \theta_0$ in (2.2). The simplest treatment deletes tied observations from the test. If m observations equal θ_0, the signs S_i are defined only for the remaining $n' = n - m$, and the null distribution of $S = \sum_{i=1}^{n'} S_i$ is still $B(n', 1/2)$.[3]

Lower-tailed and two-sided tests. Everything above was for the upper-tailed test. For the lower-tailed test $H_0 : \theta = \theta_0$, $H_1 : \theta < \theta_0$, or the two-sided test $H_0 : \theta = \theta_0$, $H_1 : \theta \neq \theta_0$, only the meaning of "extreme" changes. In the lower-tailed test small S_{obs} favors H_1, so $p = P(S \leq S_{\text{obs}}) = \sum_{x=0}^{S_{\text{obs}}} f_B(x; n, \frac{1}{2})$; in the two-sided test, since the null distribution is symmetric, the p-value is twice the smaller one-sided p-value, $p = 2\min\{P(S \geq S_{\text{obs}}), P(S \leq S_{\text{obs}})\}$.

[3] If X_i is discrete and $P(X_i = \theta_0) = \epsilon > 0$ at the median θ_0, then $P(S_i = +) = P(X_i > \theta_0 \mid X_i \neq \theta_0) = \frac{P(X_i \geq \theta_0) - \epsilon}{1-\epsilon} \geq \frac{.5-\epsilon}{1-\epsilon}$ and $P(S_i = -) = P(X_i < \theta_0 \mid X_i \neq \theta_0) = \frac{P(X_i \leq \theta_0) - \epsilon}{1-\epsilon} \geq \frac{.5-\epsilon}{1-\epsilon}$, so $P(S_i = +) \approx P(S_i = -) \approx 1/2$ and the null distribution of the tie-deleted sign test is approximately $B(n, 1/2)$. If $P(X_i > \theta_0) = P(X_i < \theta_0)$, then $P(S_i = +) = P(S_i = -) = 1/2$ and the null distribution is exactly $B(n, 1/2)$.

2.1.2 Advantages of the sign test

The sign test is exact under very few assumptions, i.e. over a very wide family of populations. Outliers affect it only through "+" or "−", so it is robust. It needs very simple data: it applies even if the data are only signs.

What is its advantage over the usual t-test (or Z-test)? To answer we must know in which populations the sign test is more powerful. Let $X_1, \ldots, X_n$ come from a distribution with median θ, and test $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$ with rejection region $S \geq c_{\text{val}}$. If the true median is $\theta_1 (> \theta_0)$, what is the power? We need the distribution of S under $H_1 : \theta = \theta_1$. Each $I(X_i > \theta_0)$ is Bernoulli with success probability $P(X_i > \theta_0)$, which the value θ_1 alone does not determine; it depends on the distribution of X_i under H_1. E.g. under $\theta = \theta_1$ it differs between $X_i \sim N(\theta_1, 1)$ and $X_i \sim \text{Uniform}(\theta_1 - 1, \theta_1 + 1)$, and so does the power. If the exact alternative distribution is known with $p_1 = P(X_i > \theta_0)$, the power is

$$P(S \geq c_{\text{val}}) = P(S \geq c_{\text{val}} \mid S \sim B(n, p_1)) = 1 - F_B(c_{\text{val}} - 1; n, p_1) \approx 1 - \Phi\Big(\frac{c_{\text{val}} - np_1}{\sqrt{np_1(1-p_1)}}\Big),$$

the last by the large-sample approximation.

Example 2.6

For the data of Example 2.1 assume $X_i \sim N(\theta, 10^2)$, and compare the power of two tests of $H_0 : \theta = 100$ versus $H_1 : \theta > 100$:

(a) Z-test with rejection region $\bar{X}_n > 100 + z_{0.05}10/\sqrt{10}$ (type I error 5%);

(b) sign test with rejection region $S \geq 8$ (type I error 5.5%).

If in fact $\theta = 105$, what is the probability that each test accepts H_1 (the power)? For the Z-test, under $\theta = 105$, $Z_n = (\bar{X}_n - 105)/\sqrt{10} \sim N(0, 1)$, so

$$P(\bar{X}_n > 100 + z_{0.05}10/\sqrt{10}) = P\Big(Z_n > \frac{-5 + z_{0.05}\sqrt{10}}{\sqrt{10}}\Big) \approx 0.47, \tag{2.4}$$

a power of 47%. For the sign test, under $\theta = 105$, $X_i \sim N(105, 10^2)$ and each sign is + with probability $p_1 = P(X_i > 100) = 1 - \Phi(-1/2) \approx 0.69$. So $S \sim B(n, 0.69)$ and the power of the sign test is $P(S \geq 8) \approx 0.36$. When the population really is normal, the Z-test is much more powerful.

The sign test can be more powerful than the Z-test when the population is not normal: for the heavy-tailed Laplace distribution, the sign test beats the Z-test.

Example 2.7 (Example 2.6 continued)

Suppose the alternative population is Laplace$(105, \sqrt{50})$ (the scale $\sqrt{50}$ makes the standard deviation 10). The Z-statistic $\bar{X}_n$ is no longer exactly normal, but the Laplace is symmetric, so by the CLT $\bar{X}_n$ is well approximated by a normal, and the power of the Z-test is approximately 0.47 (from (2.4)). For the sign test, $p_1 = P(X_i > 100 \mid X_i \sim \text{Laplace}(105, \sqrt{50})) \approx 0.75$, giving

$$P(S \geq 8) \approx 0.54,$$

higher than the Z-test.

In summary, the sign test is more powerful, hence more efficient, for heavy-tailed populations.

2.1.3 Estimating the median

Using the sample median $\hat{\theta} = \text{median}\{X_1, \ldots, X_n\}$ and order statistics, we obtain an estimator and a distribution-free confidence interval for the population median θ. Because the sign test is distribution-free, the estimator associated with it is the sample median (the *Hodges–Lehmann estimator* of the sign test, see below), and the order-statistic interval keeps its confidence level whatever the population.

Confidence interval. Let $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ be the order statistics. They relate simply to the sign statistic $S_\theta = \sum_{i=1}^{n} I(X_i > \theta)$: the number of observations larger than θ is S_θ. For a confidence level $1-\alpha$ we look for order statistics $(X_{(l)}, X_{(u)})$ such that the median θ lies in

$$X_{(l)} \le \theta < X_{(u)} \tag{2.5}$$

with probability at least $1-\alpha$. $X_{(l)} \le \theta$ means at most $n-l$ observations exceed θ, and $\theta < X_{(u)}$ means at least $n+1-u$ observations exceed θ.[4] Hence (2.5) is equivalent to

$$n - u + 1 \le \sum_{i=1}^{n} I(X_i > \theta) \le n - l,$$

and since $S_\theta = \sum I(X_i > \theta) \sim B(n, 1/2)$ we seek (l, u) with

$$P(n-u+1 \le S_\theta \le n-l) = \sum_{x=n-u+1}^{n-l} \binom{n}{x} (0.5)^n = \sum_{x=l}^{u-1} \binom{n}{x} (0.5)^n \ge 1-\alpha \tag{2.6}$$

of smallest length $(u-1)-l$. For intervals with both endpoints, it is customary to make the probabilities on both sides outside the interval equal: the probability $P(\theta \ge X_{(u)})$ that θ exceeds the upper limit and $P(\theta < X_{(l)})$ that it is below the lower limit are forced to be equal,

$$P(\theta \ge X_{(u)}) = P(\theta < X_{(l)}) \le \alpha/2. \tag{2.7}$$

Then $P(X_{(l)} \le \theta < X_{(u)}) = 1 - P(\theta \ge X_{(u)}) - P(\theta < X_{(l)}) \ge 1-\alpha$. The events in (2.7) are $\{S_\theta \le n-u\}$ and $\{S_\theta \ge n-l+1\}$, and by the symmetry of $B(n, \frac{1}{2})$, $P(S_\theta \le n-u) = P(S_\theta \ge u)$ and $P(S_\theta \ge n-l+1) = P(S_\theta \le l-1)$. So (2.7) reads

$$P(S_\theta \ge u) = P(S_\theta \le l-1) \le \alpha/2,$$

which forces $u = n-l+1$. Therefore take the largest l with $P(S_\theta \le l-1) = P(S_\theta < l) \le \alpha/2$,

$$l := Q_B(\alpha/2).$$

With this,[5] the $1-\alpha$ confidence interval is $X_{(Q_B(\alpha/2))} \le \theta < X_{(n-Q_B(\alpha/2)+1)}$.[6]

Example 2.8

The 95% confidence interval for the median sodium intake of Example 2.2:

```
> x = sort(c(1200, 2300, 5800, 1100, 1200, 1700, 1400, 1900,
    3800, 1300))
> n = length(x)
```

[4] The inequalities in (2.5) were chosen so that the interval is equivalent to a condition on S_θ. If the population is continuous, $P(X_{(l)} = \theta) = 0$, so open and closed intervals have the same confidence level.

[5] More precisely, if $P(S_\theta \le Q_B(\alpha/2)) > \alpha/2$ then $l = Q_B(\alpha/2)$; if $P(S_\theta \le Q_B(\alpha/2)) = \alpha/2$ then $l = Q_B(\alpha/2)+1$. For the usual levels ($\alpha = 0.05$ or 0.01) the second case essentially never occurs, so $l = Q_B(\alpha/2)$ is fine.

[6] Intervals like (2.5) can also be obtained from the duality between tests and confidence intervals; see Song et al. (2015), Section 2.4.

```
> alpha = 0.05
> l = qbinom(alpha/2, size = n, prob = 1/2)
> u = n + 1 - l
> (ci = x[c(l,u)])
[1] 1200 3800
```

The coverage probability of $(1200, 3800)$ exceeds 95%:

```
> (coverage = sum(dbinom(l:(u-1), n, prob = 1/2)) )
[1] 0.9785156
```

For large n an approximate interval uses the large-sample approximation $S_\theta \dot{\sim} N(0.5n, 0.25n)$ in (2.6): find integers l^*, u^* between 1 and n with

$$\frac{l - 0.5n}{\sqrt{0.25n}} \approx -z_{\alpha/2}, \qquad \frac{u - 1 - 0.5n}{\sqrt{0.25n}} \approx z_{\alpha/2}.$$

The normal-approximation interval $(X_{(l^*)}, X_{(u^*)})$ is also distribution-free and asymptotically valid.

Inference for quantiles. The sign test for the median extends naturally to quantiles. E.g. test the 10% quantile $\theta_{0.1} = Q(0.1)$. Under $H_0 : \theta_{0.1} = \theta_0$, the sign of $X_i - \theta_0$ is $-$ with probability $1/10$ and $+$ with probability $9/10$. So the null distribution of $S_{\theta_0} = \sum_{i=1}^n I(X_i > \theta_0)$ is $B(n, 0.9)$, and the lower-tailed test $(H_1 : \theta_{0.1} < \theta_0)$ has p-value $p = P(S_{\theta_0} \le S_{\text{obs}}) = F_B(S_{\text{obs}}; n, 0.9)$.

A distribution-free confidence interval for the p-quantile $Q(p)$ also uses order statistics. Let $N_\theta = \#\{i : X_i \le \theta\} \sim B(n, p)$ at $\theta = Q(p)$ (continuous population). As before, $X_{(l)} \le Q(p) < X_{(u)} \iff l \le N_\theta \le u - 1$, so we seek (l, u) with

$$P(l \le N_\theta \le u - 1) = \sum_{x=l}^{u-1} \binom{n}{x} p^x (1-p)^{n-x} \ge 1 - \alpha.$$

(For $p = 1/2$ this is (2.6).)

2.2 The Wilcoxon signed-rank test

If the population f can be assumed symmetric about the median θ, there is a more powerful test than the sign test. The population then belongs to

$$f \in \mathcal{F}_2 = \{f : f(\theta - x) = f(\theta + x),\ x \in (-\infty, \infty) \text{ for some } \theta \in (-\infty, \infty)\},$$

and if the mean exists it equals the median. Under this assumption consider the two-sided test: with $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} f \in \mathcal{F}_2$, for the median θ,

$$H_0 : \theta = \theta_0, \qquad H_1 : \theta \ne \theta_0. \tag{2.8}$$

Wilcoxon (1945) extracted more information from the data than the signs, expecting a more efficient (more powerful) test. The **signed-rank test statistic** uses both signs and ranks relative to θ_0. The sign is $I(X_i - \theta_0)$ (i.e. 1 if $X_i > \theta_0$, 0 if $X_i < \theta_0$; observations with $X_i = \theta_0$ are deleted), and the rank R_i is the position of the absolute deviation $D_i = |X_i - \theta_0|$ in increasing order: if the ith absolute deviation is the i'th smallest $(D_i = D_{(i')})$, then $R_i = i' \in \{1, \ldots, n\}$.

Example 2.9

For observations $11, 20, 16, 5, 3, 17$ and $\theta_0 = 15$, Wilcoxon's signs and ranks are:

X_i	11	20	16	5	3	17
deviation $X_i - \theta_0$	-4	$+5$	$+1$	-10	-12	$+2$
sign S_i	$-$	$+$	$+$	$-$	$-$	$+$
sign $I(X_i > \theta_0)$	0	1	1	0	0	1
absolute deviation D_i	4	5	1	10	12	2
rank R_i	3	**4**	**1**	5	6	**2**

Wilcoxon's signed-rank statistic is the sum of the ranks with positive sign,

$$SR_+ = \sum_{i=1}^{n} I(X_i > \theta_0) \cdot R_i.$$

If the null hypothesis "symmetric about θ_0" is true, then given that the absolute deviation $|X_i - \theta_0|$ equals d, the deviation is positive or negative with exactly equal probability:

$$P(X_i > \theta_0 \mid |X_i - \theta_0| = d) = \frac{1}{2} = P(X_i < \theta_0 \mid |X_i - \theta_0| = d). \tag{2.9}$$

So under H_0, $I(X_i > \theta_0)R_i$ and $I(X_i < \theta_0)R_i$ have the same expectation, and SR_+ should be neither too large nor too small. If $\theta > \theta_0$, deviations of larger rank tend to be positive, so SR_+ is large; if $\theta < \theta_0$, SR_+ is small. For the two-sided alternative we reject H_0 when SR_+ is too large or too small.

In Example 2.9 the ranks with positive sign are $1, 2, 4$, so $SR_+ = 1 + 2 + 4 = 7$. This is smaller than the negative ranks $3, 5, 6$, but is it small enough to reject H_0? The answer comes from the null distribution of the statistic, which also gives the p-value and the rejection region.

Example 2.10

In R the signed-rank statistic for the data of Example 2.9:

```
> x = c(11,20,16,5,3,17); theta0 = 15
> D = x - theta0 # deviations
> sign(D) # signs
[1] -1  1  1 -1 -1  1
> rank(abs(D)) # ranks of absolute deviations
[1] 3 4 1 5 6 2
> (SR = sum( ( sign(D)>0 ) * rank(abs(D)) ) )
[1] 7
```

2.2.1 Null distribution of the Wilcoxon signed-rank statistic and the p-value

By (2.9), under H_0 each rank R_i has a positive or negative sign with probability $1/2$. The key observation is that the ranks $1, 2, \ldots, 6$ are regarded as fixed (whatever continuous population the observations come from, the ranks are always $1, 2, \ldots, 6$). Under H_0 each rank independently gets a positive sign with probability $1/2$, i.e. enters SR_+ with probability $1/2$. So the signed ranks S_iR_i take the

$$(\pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6) \tag{2.10}$$

$2^6 = 64$ possible values, each sign arising with probability $1/2$, so each configuration in (2.10) has probability $(1/2)^6 = 1/64$. All cases under H_0:

(signed ranks)	ranks in SR_+	SR_+	probability
$(-1,-2,-3,-4,-5,-6)$	none	0	1/64
$(+1,-2,-3,-4,-5,-6)$	1	1	1/64
$(-1,+2,-3,-4,-5,-6)$	2	2	1/64
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(+1,+2,-3,+4,-5,-6)$	1,2,4	7	1/64
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$(+1,+2,+3,+4,+5,+6)$	1,2,3,4,5,6	21	1/64

The last two columns give the null distribution of SR_+; tabulating gives its probability mass function (Figure 2.3 of the book: $n = 6$, no ties, symmetric about 10.5).

For Example 2.9, the probability of a value at most the observed $SR_+ = 7$ is, from the null distribution, $P(SR_+ \le 7) = 18/64 \approx 0.28$. This is the p-value of the lower-tailed test; since the null distribution is symmetric, the p-value of the two-sided test (2.8) is 0.56, and $H_0 : \theta = \theta_0$ is not rejected.

To derive the null distribution of SR_+ we fixed the ranks R_i and resampled only the signs $S_i = +, -$; this is generally called the **re-randomization method**. The null distribution of SR_+ depends only on the sample size n and is not a well-known parametric distribution (whereas the sign statistic is binomial). Since it is used often, R's `dsignrank()` gives its probability mass function; it is called **Wilcoxon's signed-rank distribution**.

To get the rough shape of the null distribution, compute its mean and variance. Under H_0 the $I(X_i > \theta_0)$ are independent, equal to 1 or 0 with probability 1/2 each; with $V_i \equiv I(X_i > \theta_0)$, $V_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(1/2)$ and

$$SR_+ = \sum_{i=1}^{n} V_i R_i \overset{d}{=} \sum_{i=1}^{n} i V_i. \tag{2.11}$$

Justification of (2.11) (supplied). Under H_0, by (2.9) the sign vector $(V_1, \ldots, V_n)$ is independent of the absolute deviations $(D_1, \ldots, D_n)$, hence of the ranks, and consists of i.i.d. Bernoulli(1/2) variables. Given the ranks, $(R_1, \ldots, R_n)$ is a permutation π of $(1, \ldots, n)$, and $\sum_i V_i R_i = \sum_k k V_{\pi^{-1}(k)}$, where $(V_{\pi^{-1}(k)})_k$ is again i.i.d. Bernoulli(1/2). So the conditional distribution given the ranks is that of $\sum_k k V_k$, and so is the unconditional one. ■

Hence

$$E(SR_+) = \sum_{i=1}^{n} i E(V_i) = \frac{1}{2}\sum_{i=1}^{n} i = \frac{n(n+1)}{4},$$

$$\text{Var}(SR_+) = \sum_{i=1}^{n} i^2 \,\text{Var}(V_i) = \frac{1}{4}\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{24},$$

and the distribution is symmetric about the mean $n(n+1)/4$ (replacing V_i by $1 - V_i$, which has the same law, maps SR_+ to $\frac{n(n+1)}{2} - SR_+$). With observed value SR_{obs} from a sample of size n, the p-values are:

- upper alternative $H_1 : \theta > \theta_0$: $p_r = P(SR_+ \ge SR_{\text{obs}})$;
- lower alternative $H_1 : \theta < \theta_0$: $p_l = P(SR_+ \le SR_{\text{obs}}) = F_W(SR_{\text{obs}}; n)$;
- two-sided $H_1 : \theta \ne \theta_0$: by symmetry of the signed-rank distribution, $p = 2\min\{p_r, p_l\}$.

2.2.2 Large-sample approximation

For large n, computing SR_+ for all 2^n signed-rank configurations is infeasible. Then the mean and variance give the standardized statistic, approximately standard normal for large n:

$$Z := \frac{SR_+ - E(SR_+)}{\sqrt{\mathrm{Var}(SR_+)}} = \frac{SR_+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}} \dot{\sim} N(0,1). \tag{2.12}$$

Proof of (2.12). The normal approximation is the central limit theorem, as for the sign statistic S. By (2.11), $SR_+ = \sum_i iV_i = \sum_i U_i$, where under H_0 the $U_i = iV_i$ are independent with

$$U_i = \begin{cases} 0 & \text{with probability } 1/2, \\ i & \text{with probability } 1/2. \end{cases}$$

The U_i are not identically distributed, so we use Lyapunov's central limit theorem.[7] (Supplied verification.) Take $\delta = 1$: $|U_i - i/2| = i/2$ always, so $\sum_i E|U_i - \mu_i|^3 = \sum_i i^3/8 = O(n^4)$, while $s_n^2 = n(n+1)(2n+1)/24$ gives $s_n^3 \asymp n^{9/2}$. The ratio is $O(n^{-1/2}) \to 0$, so Lyapunov's condition holds and $Z \xrightarrow{d} N(0,1)$. ■

The approximate p-value of the upper-tailed test is

$$P(SR_+ \geq SR_{\text{obs}}) = P\Big(Z \geq \frac{SR_+ - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}\Big) \approx 1 - \Phi\Big(\frac{SR_{\text{obs}} - n(n+1)/4}{\sqrt{n(n+1)(2n+1)/24}}\Big). \tag{2.13}$$

Example 2.11

To see whether counseling changes depression scores of 12 patients, scores before counseling were (in patient-ID order) $(1.83, 0.5, 1.62, 2.48, 1.68, 1.88, 1.55, 3.06, 1.30, 0.39, 0.94, 1.34)$ and after counseling $(0.88, 0.65, 0.59, 2.05, 1.06, 1.29, 1.06, 3.14, 1.29, 0.43, 0.57, 0.92)$. To see whether counseling lowers the score, run the signed-rank test with the large-sample approximation. For paired tests, record the differences of the pairs as a new vector `x`; large values mean the depression score decreased. This is an upper-tailed test, so use (2.13):

```
> x.before = c(1.83, 0.5, 1.62, 2.48, 1.68, 1.88, 1.55, 3.06,
    1.30, 0.39, 0.94, 1.34)
> x.after = c(0.88, 0.65, 0.59, 2.05, 1.06, 1.29, 1.06, 3.14,
    1.29, 0.43, 0.57, 0.92)
> x = x.before - x.after
> D = x - 0
> (SR = sum( ( sign(D)>0 ) * rank(abs(D)) ) )
[1] 69
> n = length(x)
> Z_SR = (SR - n*(n+1)/4)/ sqrt(n*(n+1)*(2*n+1)/24)
> (p = 1-pnorm(Z_SR))
[1] 0.009301465
```

The p-value is small: there is substantial evidence that counseling lowers depression scores.

[7] Lyapunov's CLT: let $X_1, \ldots, X_n$ be independent with means μ_i and variances σ_i^2, and $s_n^2 = \sum_{i=1}^n \sigma_i^2$. If for some $\delta > 0$ Lyapunov's condition $\lim_{n\to\infty} \frac{1}{s_n^{2+\delta}} \sum_{i=1}^n E\big[|X_i - \mu_i|^{2+\delta}\big] = 0$ holds, then $s_n^{-1} \sum_i (X_i - \mu_i) \xrightarrow{d} N(0,1)$.

2.2.3 Ties

Example 2.12

For 10 healthy volunteers we want to know whether blood pressure rises after exercise compared with before. After an hour of high-intensity exercise the differences in blood pressure were $-3, -2, -2, 0, 1, 2, 4, 8, 11, 11$. Assuming the differences are symmetric about the median θ, we test whether the median difference exceeds 0.

These data have two kinds of ties:

- observations equal to the hypothesized median θ_0 (here 0);
- observations with equal absolute deviations: here $-2, -2, 2$ all have $D_i = |X_i - \theta_0| = 2$, and $11, 11$ must get the same rank.

The first kind is handled as in the sign test: delete the observation and take the sample size as 9. For the second kind one usually uses **mid-ranks**. Exactly one absolute deviation is smaller than 2, so the three tied values would occupy ranks $2, 3, 4$; since they cannot be ordered, each gets the average $3 = \frac{2+3+4}{3}$. The signed-rank statistic with ties is then computed from:

X_i	-3	-2	-2	0	1	2	4	8	11	11
deviation $X_i - \theta_0$	-3	-2	-2	0	1	2	4	8	11	11
sign S_i	$-$	$-$	$-$		$+$	$+$	$+$	$+$	$+$	$+$
sign $I(X_i > \theta_0)$	0	0	0		1	1	1	1	1	1
absolute deviation D_i	3	2	2		1	2	4	8	11	11
rank R_i	5	3	3		**1**	**3**	**6**	**7**	**8.5**	**8.5**

The signed-rank statistic is the sum of positive ranks, $SR_+ = 1 + 3 + 6 + 7 + 8.5 + 8.5 = 34$. (The book prints 39, an arithmetic slip; the R output below confirms the ranks.) Its null distribution differs from the tie-free case: the R_i are not the integers $1, 2, \ldots, n$ but $1, 3, 3, 3, 5, 6, 7, 8.5, 8.5$, so the last equality of (2.11) fails. The re-randomization principle still applies: under H_0, fix the ranks R_i and let the signs $V_i = I(S_i = +)$ be independent with probability 1 or 0 each, $V_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(1/2)$; with all 2^n cases, re-randomization gives the null distribution.

Example 2.13

R code deriving the null distribution by re-randomization with ties, for testing whether the median of the data below is 0:

```
> toy.data = c(-3,-2,-2,0,1,2,4,8,11,11); theta0 = 0
> dev = toy.data - theta0
> reduced.dev = dev[sign(dev) != 0 ]
> (ranks = rank(abs(reduced.dev)))
[1] 5.0 3.0 3.0 1.0 3.0 6.0 7.0 8.5 8.5
```

All sign configurations are generated by `permutations()` of the `gtools` package: the rows of `allsignflips` are all length-n vectors with entries in `v = c(1,-1)` (with repetition, `repeats.allowed = TRUE`); there are 2^n rows.

```
> n = length(ranks)
> allsignflips = gtools::permutations(2, n, v = c(1,-1),
+     repeats.allowed = TRUE)
> nperm = nrow(allsignflips) # equals 2^n
```

Compute the rank sum for each of the 2^n sign flips:

```
> SRvec = vector(length = nperm) # rank sum in each case
> for (i in 1:nperm){
+   SRvec[i] = sum(ranks[allsignflips[i,] > 0])
+ }
```

The number of sign flips giving a rank sum is its frequency; dividing by 2^n gives the probability mass function of the null distribution.

```
> N = n*(n+1)/2
> dsignrank.tie = vector(length = N+1)
> for (sr in 0:N){
+   dsignrank.tie[sr+1] = sum(SRvec == sr) / nperm
+ }
> data.frame(SR = 0:N, SRdensity = dsignrank.tie) %>% head(3)
  SR   SRdensity
1  0 0.001953125
2  1 0.001953125
3  2 0.000000000
```

Because of the ties no rank equals 2, so $SR_+ = 2$ has probability 0. By contrast `dsignrank()` cannot produce the exact null distribution with ties:

```
> dsignrank(0:2,n)
[1] 0.001953125 0.001953125 0.001953125
```

2.2.4 Estimation and confidence interval for θ via the signed-rank test

For observations $X_1, \ldots, X_n$, the averages of all pairs

$$W_{ij} := \frac{X_i + X_j}{2}, \qquad 1 \le i \le j \le n,$$

are the **Walsh averages**. For $i = j$, $W_{ii} = X_i$; there are $N = n(n+1)/2$ Walsh averages. The number of Walsh averages larger than a given value θ equals the signed-rank statistic for hypothesized median θ:

$$SR_+(\theta) = \#\{W_{ij} > \theta\}. \tag{2.14}$$

To see this roughly, suppose the observations are ordered and distinct ($X_1 < X_2 < \cdots < X_n$, i.e. $X_i = X_{(i)}$). For $i < j$, $W_{ij} < W_{i,j+1}$ and $W_{ij} < W_{i+1,j}$; and $W_{ij} > \theta$ means X_i is closer to θ than X_j. The largest Walsh average is $W_{nn} = X_n$, the second is $W_{n-1,n}$, and the third is $W_{n-1,n-1} = X_{n-1}$ or $W_{n-2,n}$. Arrange the Walsh averages from W_{11} to W_{nn}. If $\theta > W_{nn}$, clearly $SR_+ = 0 = \#\{W_{ij} > \theta\}$. Next, if $\theta \in (W_{n-1,n}, W_{nn})$ then $\#\{W_{ij} > \theta\} = 1$, and the only observation above θ is X_n with $|X_n - \theta| < |X_{n-1} - \theta|$, so $SR_+ = 1$. By the same logic, each time θ passes a Walsh average W_{ij}, SR_+ increases by 1.

Proof of (2.14) (supplied). Assume no ties among the $|X_k - \theta|$ and no W_{ij} equal to θ, and let $d_k = X_k - \theta$. Then $W_{ij} > \theta \iff d_i + d_j > 0$. Assign each pair $\{i, j\}$ $(i \le j)$ to the index $k \in \{i, j\}$ with the larger $|d_k|$. The sign of $d_i + d_j$ is the sign of d_k, since the other term has smaller absolute value (or is d_k itself if $i = j$). The pairs assigned to k are $\{k, m\}$ with $|d_m| \le |d_k|$, i.e. exactly R_k pairs, where R_k is the rank of $|d_k|$. Hence

$$\#\{W_{ij} > \theta\} = \sum_{k:d_k>0} R_k = SR_+(\theta).$$

■

By (2.14), the signed-rank statistic is the sign statistic of the Walsh averages (# of W_{ij} larger than θ). Just as the sign test (number of observations above θ) gave intervals $X_{(l)} \le \theta < X_{(u)}$, the

signed-rank test gives intervals from the order statistics $W_{(1)} < \cdots < W_{(n(n+1)/2)}$ of the Walsh averages.[8]

For a confidence level $1-\alpha$ we take order statistics $W_{(l)}, W_{(u)}$ such that the median θ lies in

$$W_{(l)} \le \theta < W_{(u)} \tag{2.15}$$

with probability at least $1-\alpha$. As for the sign test, choose l, u so that the two outside probabilities are equal; each must be at most $\alpha/2$ for coverage $1-\alpha$:

$$P(\theta \ge W_{(u)}) = P(\theta < W_{(l)}) \le \frac{\alpha}{2}. \tag{2.16}$$

The distribution of the Walsh averages depends on the population, so instead use the following relation, which allows the Wilcoxon signed-rank distribution of SR_+ with the unknown population: with $N = n(n+1)/2$ and $a \in \{1, \ldots, N\}$,

$$\begin{aligned} SR_+ > N - a &\iff \theta < W_{(a)}, \\ SR_+ \le N - a &\iff \theta \ge W_{(a)}. \end{aligned} \tag{2.17}$$

(By (2.14), $SR_+(\theta) > N-a$ means more than $N-a$ Walsh averages exceed θ, i.e. $\theta < W_{(a)}$.) The distribution of SR_+ is symmetric about $n(n+1)/4$, and by (2.17), (2.16) becomes

$$P(SR_+ \ge u) = P(SR_+ \le l-1) \le \frac{\alpha}{2}.$$

Here $u = n(n+1)/2 - l + 1$. So with the largest l satisfying $P(SR_+ \le l-1) = P(SR_+ < l) \le \alpha/2$,

$$l := Q_{SR}(\alpha/2),$$

the $1-\alpha$ interval is $W_{(Q_{SR}(\alpha/2))} \le \theta < W_{(n(n+1)/2 - Q_{SR}(\alpha/2)+1)}$. As the confidence level shrinks, what remains is the median of the Walsh averages,

$$\hat{\theta} = \text{median}\{W_{(ij)}\},$$

an estimator of the population median θ: the **Hodges–Lehmann estimator**.[9]

Example 2.14

For Example 2.11, as an answer to how much counseling lowers depression scores, compute the nonparametric estimate of the median θ of the score differences and a 95% confidence interval, assuming the population is symmetric about its median θ. The Walsh averages come from `Rfit::walsh()`.

```
> w = sort(Rfit::walsh(x)) # sorted walsh averages
> median(w)
[1] 0.4225
> alpha = 0.05
> u = n*(n+1)/2 + 1 - qsignrank(alpha/2,n)
> l = qsignrank(alpha/2,n)
> c(l,u)
[1] 14 65
> (ci = c(w[l], w[u]))
```

[8] This interval can be justified by inverting the test. Whether $H_0 : \theta = \theta_0$ is rejected depends on $SR_+(\theta_0)$, i.e. on the relative position of θ_0 among the Walsh averages; the values θ_0 not rejected at level α form the $1-\alpha$ interval. Both methods give the same interval.

[9] Hodges and Lehmann (1963) proposed obtaining estimators of a location parameter from tests; these are called Hodges–Lehmann estimators. The Hodges–Lehmann estimator from the signed-rank test is the median of the Walsh averages; from the sign test it is the sample median. It was also proposed independently by Sen (1963), and is sometimes called the Hodges–Lehmann–Sen estimator.

```
[1] 0.145 0.685
```

The median estimate is 0.4225 and the 95% interval is $(0.145, 0.685)$.

2.2.5 The R function `wilcox.test()`

The signed-rank test is implemented in R's `wilcox.test()`. For the counseling data of Example 2.11, with the differences in vector `x`: `alternative = "greater"` gives the upper-tailed test, `exact = F` the large-sample approximation, and `correct = F` turns off the continuity correction used in normal approximations of discrete distributions.

```
> wilcox.test(x, alternative = "greater", exact = F, correct =F)

        Wilcoxon signed rank test

data:  x
V = 69, p-value = 0.009301
alternative hypothesis: true location is greater than 0
```

This matches Example 2.11. For the exact test using the exact null distribution of the signed-rank statistic, set `exact = TRUE` (exact only without ties); no continuity correction is needed.

```
> wilcox.test(x, alternative = "greater", exact = T)

        Wilcoxon signed rank exact test

data:  x
V = 69, p-value = 0.008057
alternative hypothesis: true location is greater than 0
```

The two-sided test and the accompanying Walsh-average interval:

```
> wilcox.test(x, conf.int = T)

        Wilcoxon signed rank exact test

data:  x
V = 69, p-value = 0.01611
alternative hypothesis: true location is not equal to 0
95 percent confidence interval:
 0.145 0.685
sample estimates:
(pseudo)median
        0.4225
```

This matches Example 2.14.

2.3 When should nonparametric methods be used?

If the population is approximately normal, or the sample mean is approximately normal by the CLT, the Z- or t-test gives inference on the mean. In comparison, the sign test is exact and distribution-free for any population, and the signed-rank test is exact when the population is symmetric. Should we then always prefer the signed-rank test over the sign test for symmetric populations? Relatedly, may we use the signed-rank test when the population is normal?

This section compares the three tests of the location parameter (t-test, sign test, signed-rank test). Their estimators and intervals for θ also differ: the t-test gives $\bar{X}_n$ with interval $\bar{X}_n \pm z_{\alpha/2}\hat{\sigma}/\sqrt{n}$; the signed-rank test gives median$\{W_{(ij)}\}$ with interval $(W_{(l)}, W_{(u)})$; the sign test gives median$\{X_i\}$ with interval $(X_{(l)}, X_{(u)})$. Which is best?

Under a parametric model (e.g. normal) there is an optimal estimator ($\bar{X}_n$ under normality) and an optimal test (the t-test). Under nonparametric models the family is so large that no method is optimal in every case: the t-test is best for the normal but may not be for the uniform. So we compare efficiencies in several representative populations.

2.3.1 Comparing inference methods

Comparing estimators. Which of $\bar{X}_n$, median$\{W_{(ij)}\}$, median$\{X_i\}$ is the best estimator of the population mean θ (or median θ)? The usual criterion is the mean squared error (MSE); the smaller the MSE, the better the estimator. The MSE of an estimator $\hat{\theta}$ of θ is

$$\mathrm{MSE}(\hat{\theta}, \theta) = E(\hat{\theta} - \theta)^2.$$

If $\hat{\theta}$ is unbiased, $\mathrm{MSE}(\hat{\theta}, \theta) = \mathrm{Var}(\hat{\theta})$, so we compare variances or standard deviations (standard errors). The expectation $E(\cdot)$ in the MSE is with respect to the sampling distribution of $\hat{\theta}$, which is determined by the population (the distribution of X_i) and the sample size n.

Relative efficiency of two estimators. If the target is always θ, write $\mathrm{MSE}(\hat{\theta}, \theta) = \mathrm{MSE}(\hat{\theta})$. Smaller MSE is better, so define the **efficiency**

$$e(\hat{\theta}) = \frac{c}{\mathrm{MSE}(\hat{\theta})},$$

where the constant c is the minimal $\mathrm{MSE}(\hat{\theta})$ over the family, so that $e(\hat{\theta}) \in [0, 1]$; the closer to 1, the more efficient. To compare $\hat{\theta}_1$ and $\hat{\theta}_2$ use the **relative efficiency**

$$RE(\hat{\theta}_1, \hat{\theta}_2) = \frac{e(\hat{\theta}_1)}{e(\hat{\theta}_2)} = \frac{\mathrm{MSE}(\hat{\theta}_2)}{\mathrm{MSE}(\hat{\theta}_1)}. \tag{2.18}$$

If $RE(\hat{\theta}_1, \hat{\theta}_2) > 1$, $\hat{\theta}_1$ is more efficient; if < 1, $\hat{\theta}_2$ is. Like the MSE, the relative efficiency depends on the population and the sample size n.

Example 2.15

Estimating the mean of a normal population: the relative efficiency of $\bar{X}_{10}$ (from 10 observations) and $\bar{X}_{40}$ (from 40). If the population variance is σ^2 (unknown), $\mathrm{MSE}(\bar{X}_n) = \sigma^2/n$, so

$$RE(\bar{X}_{40}, \bar{X}_{10}) = \frac{\mathrm{MSE}(\bar{X}_{10})}{\mathrm{MSE}(\bar{X}_{40})} = \frac{\sigma^2/10}{\sigma^2/40} = 4 :$$

quadrupling the sample size quadruples the efficiency.

Example 2.16

Estimating the mean μ of $N(\mu, \sigma^2)$: relative efficiency of the median $m_n \equiv$ median$\{X_1, \ldots, X_n\}$ and the mean $\bar{X}_n$. The exact MSE of m_n depends on the distribution of order statistics and has no neat form, but for large n, $m_n \dot{\sim} N(\mu, \sigma^2\pi/(2n))$ is known,

so $\text{MSE}(m_n) \approx \text{Var}(m_n) \approx \sigma^2\pi/(2n)$ and

$$RE(\bar{X}_n, m_n) = \frac{\text{MSE}(m_n)}{\text{MSE}(\bar{X}_n)} \approx \frac{\sigma^2\pi/(2n)}{\sigma^2/n} = \pi/2 = 1.57 : \tag{2.19}$$

for a normal population the sample mean is about 1.57 times as efficient as the sample median.

When the exact MSE cannot be computed for fixed n, as in Example 2.16, or when the relative efficiency depends on n, we compare efficiencies as $n \to \infty$, the **asymptotic relative efficiency** (ARE):

$$ARE(\hat{\theta}_1, \hat{\theta}_2) = \lim_{n\to\infty} RE(\hat{\theta}_1, \hat{\theta}_2). \tag{2.20}$$

For a normal population, by (2.19), $ARE(\bar{X}_n, m_n) = 1.57$: the sample mean is more efficient than the sample median.

Relative efficiency of two confidence intervals. Consider two intervals $\theta \in (L_1, U_1)$ and $\theta \in (L_2, U_2)$ with the same level $1-\alpha$. If their coverage probabilities (the probability the interval contains the parameter) are exact, $P(\theta \in (L_i, U_i)) = 1-\alpha$ for $i = 1, 2$, the shorter $\ell_i = U_i - L_i$, the more efficient. So define the efficiency of (L_i, U_i) as $e((L_i, U_i)) = 1/\ell_i^2$.[10]

If the point estimator $\hat{\theta}$ used to build an interval is asymptotically $N(\theta, \sigma_{\hat{\theta}}^2/n)$, the asymptotic interval is $C_{\hat{\theta}} \equiv \hat{\theta} \pm z_{\alpha/2}\sigma_{\hat{\theta}}/\sqrt{n}$. Also $\text{MSE}(\hat{\theta}) \approx \sigma_{\hat{\theta}}^2/n$, so the length of $C_{\hat{\theta}}$ is approximately $2z_{\alpha/2}\sigma_{\hat{\theta}}/\sqrt{n} \approx 2z_{\alpha/2}\sqrt{\text{MSE}(\hat{\theta})}$ and the efficiency is $e(C_{\hat{\theta}}) \approx \frac{1}{4z_{\alpha/2}^2\,\text{MSE}(\hat{\theta})}$. For two estimators $\hat{\theta}_1, \hat{\theta}_2$ with asymptotic intervals $C_{\hat{\theta}_1}, C_{\hat{\theta}_2}$, by (2.18) and (2.20),

$$ARE(C_{\hat{\theta}_1}, C_{\hat{\theta}_2}) = \lim_{n\to\infty} \frac{e(C_{\hat{\theta}_1})}{e(C_{\hat{\theta}_2})} = \lim_{n\to\infty} \frac{\text{MSE}(\hat{\theta}_2)}{\text{MSE}(\hat{\theta}_1)} = ARE(\hat{\theta}_1, \hat{\theta}_2) :$$

the ARE of two asymptotic intervals equals the ARE of the corresponding estimators.

Comparing tests. From a random sample $X_1, \ldots, X_n$ test $H_0 : \theta = \theta_0$ against $H_1 : \theta < \theta_0$ for the median θ. If the population is normal, the median is the mean and the t-test controls the type I error at exactly the given level: $P(\sqrt{n}(\bar{X}_n - \theta_0)/s_n < -t_{\alpha,n-1}) = \alpha$. If the population is not normal, the type I error can exceed the level. Some examples of type I error and size of parametric tests in various populations:

Example 2.17

Type I error of the t-test of $H_0 : \theta = 0$ when the population with median $\theta = 0$ is Uniform$(-1, 1)$. By the CLT, for large n the statistic $T_n = \sqrt{n}(\bar{X}_n - \theta_0)/s_n$ is approximately standard normal, so $P(T_n < -t_{\alpha,n-1}) \approx \alpha$ and the type I error is approximately controlled. For very small n the type I error can be approximated by a simple simulation: with $n = 5$ at $\alpha = 0.05$, the type I error is $P(T_5 < -t_{0.05,4}) = 0.056$, slightly above the level. With $n = 10$ the distribution of T_n is almost standard normal and the type I error is close to 0.05.

[10] Defining the efficiency as c/ℓ_i^2 for some $c > 0$ instead of $1/\ell_i^2$ does not change relative efficiencies. Interval efficiency is not standard; for convenience we define "efficiency" first and then look at relative efficiency. Using the reciprocal of the squared length makes it comparable to the relative efficiency of estimators.

Example 2.18

Type I error of the t-test of $H_0 : \theta = 0$ when the population with median $\theta = 0$ is the slash distribution, whose tails are much longer than normal. ($Z \sim N(0,1)$ and $U \sim \text{Uniform}(0,1)$ independent: $X = Z/U$ is *slash*-distributed, and $E(X)$ does not exist.) For small n, e.g. $n = 10$ at $\alpha = 0.05$, the type I error of the t-test is about 0.029 (99% CI $(0.028, 0.030)$), a test more conservative than the level (rejecting H_0 less often than necessary). The CLT does not apply, so there is no guarantee that the type I error converges to 0.05 as n grows: for $n = 100$ it is still about 0.031, well below the level.

Example 2.19

Now an asymmetric population. The median and mean differ, so the t-test cannot be used for hypotheses about the median. We consider (a) using the t-test anyway and (b) setting new hypotheses about the population mean and running the t-test at $\alpha = 0.05$; as a simple asymmetric population take the exponential.

(a) The population is exponential with median $\theta = \log(2)$; test $H_0 : \theta = \log(2)$ against $H_1 : \theta < \log(2)$ with the t-test. The exponential with mean $\mu = 1$ has median $\theta = \mu \log(2) = \log(2) < 1$, so the statistic $t_n = \sqrt{n}(\bar{X}_n - \log(2))/s_n$ diverges to infinity instead of converging to the standard normal. The type I error $P(t_n < -t_{\alpha,n-1}) \approx 0$.

(b) The population is exponential with mean $\mu = 1$; test $H_0 : \mu = 1$ against $H_1 : \mu < 1$ with the t-test. By the CLT t_n is approximately standard normal, but because of the skewness of the exponential the approximation is good only for very large n (convergence to normality is slow). At $n = 100$, t_n is not close enough to normal, and the type I error is $P(t_{100} < -t_{\alpha,n-1}) = 0.074 > \alpha$, not controlled. If instead of $-t_{\alpha,n-1} \approx -z_\alpha \approx -1.66$ we use $c = -1.925$, wrong rejections become less frequent and the type I error $P(t_{100} < -1.92) \approx 0.049 < \alpha$ is controlled.

Unlike these examples, nonparametric tests control the type I error exactly (or conservatively) for every population. Two tests are compared by the principles:

1. both must control the type I error exactly or conservatively;
2. among tests satisfying the first principle, the one more likely to reject H_0 when H_1 is true (higher power) is better.

A test whose type I error exceeds the level cannot in principle be compared. As in Example 2.19(b), we may adjust the rejection region so that the type I error is controlled and then compare powers. Comparing powers requires the distribution of the test statistic when H_1 is true; but unlike H_0, H_1 comprises many cases.[11] E.g. the distribution of t_n differs between $\theta = \theta_0 - 1$ and $\theta = \theta_0 - 2$ even under normality. Comparing tests A and B, A may be more powerful in some alternatives and B in others; the power also depends on n and α. The best approach is to assume the most plausible alternatives in the given situation and compare the two powers there.

Example 2.20

Using a random sample of size n, test the median θ; earlier studies show the population is symmetric. Compare the t-test, the sign test and the signed-rank test in the cases:

(a) $n = 10$ (very small) and $n = 100$ (large enough);

[11] $H_0 : \theta = \theta_0$ also comprises many cases. If the family is $\{N(\theta, 1) : \theta \in (-\infty, \infty)\}$ there is only one case under H_0, $N(\theta_0, 1)$; but for $\{N(\theta, \sigma^2) : \theta \in (-\infty, \infty), \sigma^2 > 0\}$ there are many, depending on σ^2. Under this parametric model the null distribution of t_n is the same whatever σ^2. For a nonparametric model $\{f : \text{distribution with median } \theta,\ \theta \in (-\infty, \infty)\}$, the null distribution of a test statistic may depend on f even under H_0.

(b) the population is normal $X \sim N(\theta, 1)$ (thin tails), Laplace with median θ, $X \sim$ Laplace$(\theta, 1)$ (heavy tails), or slash with median θ, $X = \theta + Z/U$ (very heavy tails).

Powers at $\alpha = 0.05$ for the six combinations (six alternative distributions), by simulation:

population	sample size	t-test	sign test	signed-rank test
Normal, $\theta = 1/4$	$n = 10$	17.8%	4.5%	15.3%
Normal, $\theta = 1/4$	$n = 100$	80.0%	61.2%	78.2%
Laplace, $\theta = 1/4$	$n = 10$	14.7%	5.3%	14.3%
Laplace, $\theta = 1/4$	$n = 100$	55.1%	70.3%	68.2%
Slash, $\theta = 1/2$	$n = 10$	8.7%	4.3%	12.7%
Slash, $\theta = 1/2$	$n = 100$	10.8%	60.3%	58.9%

For a normal population the t-test is the most powerful at every sample size (as is also known theoretically). With $n = 100$ the power of the signed-rank test is almost that of the t-test. This conclusion also holds for very thin-tailed populations (e.g. uniform centered at θ).
For heavy-tailed populations the conclusion depends on n. With small n the sign test's power drops sharply and the signed-rank and t-tests are more powerful. With large n the sign and signed-rank tests are much more powerful than the t-test. For the extremely heavy-tailed slash distribution (where the CLT fails) the t-test's power is very small, while for the Laplace the power differs little from nonparametric methods. Especially for very small samples ($n = 10$) the t-test's power is almost that of nonparametric methods, so it is acceptable there.

Asymptotic relative efficiency of two tests. As above, the comparison of two tests can change with n, and for a fixed alternative the power of a test increases with n. So another way of comparing two tests: at a given alternative $\theta = \Delta$, compare the sample sizes needed to reach the same power, e.g. $\beta = 80\%$. For a given population (e.g. slash), let N_t be the sample size for which the level-α t-test "δ_t" has power β at $\theta = \Delta$, and N_{SR} the sample size for which the signed-rank test "δ_{SR}" has power β in the same situation. The relative efficiency of the t-test to the signed-rank test is defined as[12]

$$RE(\delta_t, \delta_{SR}) = \frac{N_{SR}}{N_t}.$$

In Example 2.20, for the slash distribution the t-test needed $N_t = 100$ for power $\beta = 10\%$, whereas the signed-rank test reaches $\beta = 10\%$ already with $N_{SR} = 10$, so the relative efficiency of the t-test to the signed-rank test is $N_{SR}/N_t = 10/100 = 0.1$ (the signed-rank test is 10 times as efficient). Remember that the relative efficiency depends on the population under the alternative.

The asymptotic relative efficiency is the relative efficiency as the alternative value Δ approaches θ_0. When Δ and θ_0 are very close, alternative and null are hard to distinguish and very large samples are needed. As $\Delta \to \theta_0$, $N_{SR} \to \infty$ and $N_t \to \infty$, and the limit of their ratio is the ARE. The ARE of the t-test to the signed-rank test is

$$ARE(\delta_t, \delta_{SR}) = \lim_{\Delta \to \theta_0} \frac{N_{SR}}{N_t}.$$

2.3.2 Asymptotic relative efficiency of the t-test, sign test and signed-rank test

We compare the AREs of the three tests in several representative families. If the population is not symmetric, only the sign test controls the type I error, so we consider symmetric populations only, where hypotheses about the median θ are hypotheses about the mean. We compare the uniform

[12] This is called Pitman's relative efficiency.

(almost no tails), the normal, the logistic (heavier tails but close to normal), the Laplace (heavy tails), and the slash (tails so heavy the mean does not exist) (Figure 2.4 of the book).

Each of the three tests corresponds to an estimator of θ: the t-test to $\bar{X}_n$, the signed-rank test to median$\{W_{(ij)}\}$, the sign test to median$\{X_i\}$. For large n the rejection regions are given by the asymptotic null distributions of the estimators, and when these are normal the ARE of two tests equals the ARE of the two estimators:

$$ARE(\delta_t, \delta_{SR}) = ARE(\bar{X}_n, \text{median}\{W_{(ij)}\}).$$

The table compares the AREs of the three tests (or estimators):

distribution	ARE(sign, t)	ARE(SR, t)	ARE(sign, SR)
Uniform	$1/3$	1	$1/3$
Normal	$2/\pi \approx 0.64$	$3/\pi \approx 0.954$	$2/3$
Logistic	$\pi^2/12 \approx 0.82$	$\pi^2/9 \approx 1.09$	$3/4$
Laplace	2	1.5	$4/3$
Slash	∞	∞[13]	1.09^*

The ARE of the CLT-based t-test relative to nonparametric methods worsens as the tails get heavier. For very heavy tails (Laplace and slash) the sign test has the highest ARE. The ARE of the signed-rank test relative to the t-test is at worst 0.954 (in fact at least 0.864 over all symmetric populations), so it is efficient for every population.

Where the table comes from (supplied)

Let the population have density f symmetric about θ, variance σ^2 (possibly infinite). As $n \to \infty$:

- $\bar{X}_n \dot{\sim} N(\theta, \sigma^2/n)$ (CLT);
- the sample median $m_n \dot{\sim} N(\theta, \frac{1}{4nf(\theta)^2})$;
- the Hodges–Lehmann estimator median$\{W_{(ij)}\} \dot{\sim} N(\theta, \frac{1}{12n(\int f^2)^2})$.

Taking ratios of these variances,

$$ARE(\text{sign}, t) = 4\sigma^2 f(\theta)^2, \qquad ARE(SR, t) = 12\sigma^2\Big(\int f^2\Big)^2, \qquad ARE(\text{sign}, SR) = \frac{f(\theta)^2}{3(\int f^2)^2}.$$

Check: for $N(0,1)$, $f(0)^2 = 1/(2\pi)$ and $\int \phi^2 = 1/(2\sqrt{\pi})$, giving $2/\pi$, $3/\pi$, $2/3$. For Uniform$(-1,1)$, $\sigma^2 = 1/3$, $f(0) = 1/2$, $\int f^2 = 1/2$, giving $1/3$, 1, $1/3$. For the Laplace density $\frac{1}{2}e^{-|x|}$, $\sigma^2 = 2$, $f(0) = 1/2$, $\int f^2 = 1/4$, giving 2, $3/2$, $4/3$. For the logistic, $\sigma^2 = \pi^2/3$, $f(0) = 1/4$, $\int f^2 = 1/6$, giving $\pi^2/12$, $\pi^2/9$, $3/4$.
Sketch for the median. For continuous f with $f(\theta) > 0$, $P(m_n \le \theta + t/\sqrt{n}) = P(\#\{X_i \le \theta + t/\sqrt{n}\} \ge \lceil n/2 \rceil)$. The count is $B(n, p_n)$ with $p_n = F(\theta + t/\sqrt{n}) = \frac{1}{2} + f(\theta)t/\sqrt{n} + o(n^{-1/2})$, and the normal approximation to the binomial gives

$$P(m_n \le \theta + t/\sqrt{n}) \to \Phi(2f(\theta)t).$$

So $\sqrt{n}(m_n - \theta) \xrightarrow{d} N(0, 1/(4f(\theta)^2))$.

[13] For the slash distribution, $\text{MSE}(\bar{X}_n) = \infty$, so the ARE relative to the t-test is ∞. (* For the slash, ARE(sign, SR) cannot be obtained as a ratio of the first two columns and is computed directly from the formulas below.)

Chapter 3. Permutation Tests

Overview

A **permutation test** estimates the null distribution of a test statistic by **re-randomization**; it was proposed by Fisher and Pitman in the 1930s. Re-randomization requires a great many repetitions, so the method only became prominent recently, with fast computers. The English word "permutation" in permutation test means mathematically an ordered arrangement, but also "to shuffle" (permute). Strictly speaking "shuffle test" would be more precise and less misleading, but "permutation test" is established and is used here.

3.1 Data-generating models

Permutation tests apply in two probability models: the **randomization model**, which chooses at random the subjects (people or other units) receiving a treatment of interest, and the **population model**, which samples subjects at random from populations. We focus on the two-sample situation.

Example 3.1 (Randomization model)

A new treatment for recovery after surgery is compared with the conventional one.[a] Of $N = 7$ patients scheduled for heart surgery at a Seoul hospital this March, $m = 4$ were chosen at random to receive the new treatment and the remaining $n = 3$ the conventional one. The new-treatment patients were discharged after full recovery on days $19, 22, 24, 26$, and the conventional patients on days $23, 33, 40$. Test H_0: "there is no difference between the new and conventional treatments" against H_1: "the new treatment shortens recovery time."

[a] Example adapted from Ernst (2004).

Example 3.2 (Population model)

We compare the antenna lengths of two subspecies of the tsetse fly, the forest-dwelling *Glossina fusca* and the river-dwelling *Glossina palpalis*. Nine forest flies and six river flies were caught and their antenna lengths measured; the flies can be regarded as random samples from each population:

forest flies: 1.24, 1.38, 1.36, 1.40, 1.38, 1.48, 1.38, 1.54, 1.56
river flies: 1.14, 1.20, 1.30, 1.26, 1.28, 1.18

Test H_0: "there is no difference in antenna length (distribution) between the two species" against H_1: "the antenna lengths differ."

3.2 The randomization model

In Example 3.1, let $X_1, \ldots, X_m$ be the recovery times of the m patients receiving the new treatment and $Y_1, \ldots, Y_n$ those of the n conventional patients. We want a test rejecting H_0 when the difference of mean recovery times $\bar{Y}_n - \bar{X}_m$ is large: with $D = \bar{Y}_n - \bar{X}_m$ as test statistic, we measure how extreme the observed D_{obs} is relative to the null distribution of D.

In the randomization model, the **random assignment** of treatments is the basis of inference. In Example 3.1 the N patients were fixed before the random assignment of treatments. If H_0 is true (the two treatments have the same effect), each patient's recovery time is the same whatever

treatment the patient receives. E.g. the first new-treatment patient (19 days) would also have needed 19 days under the conventional treatment. In the randomization model the patients are fixed in advance, so **the observations are not random; only the treatment they receive is random** (because of random assignment). The assignment that m specific patients out of N get the new treatment and the other n the conventional one is just one of $\binom{N}{m}$ equally likely random assignments.[14] Each assignment occurs with probability $1/\binom{N}{m}$, so the null distribution of D is the distribution of the differences D arising under the $\binom{N}{m}$ assignments.

In Example 3.1, the table below lists D for all $\binom{7}{4} = 35$ possible assignments. Row 5 is the observed difference $D_{\text{obs}} = 9.25$. Under H_0 the null distribution of D puts probability 1/35 on each of the 35 values (Figure 3.1 of the book shows its histogram and dot plot). The null distribution of a test statistic obtained in this way is called the **randomization distribution**.

	new				old			D		new				old			D
1	19	22	23	24	26	33	40	11.000	19	19	24	33	40	22	23	26	−5.333
2	19	22	23	26	24	33	40	9.833	20	19	26	33	40	22	23	24	−6.500
3	19	22	23	33	24	26	40	5.750	21	22	23	24	26	19	33	40	6.917
4	19	22	23	40	24	26	33	1.667	22	22	23	24	33	19	26	40	2.833
5	19	22	24	26	23	33	40	9.250	23	22	23	24	40	19	26	33	−1.250
6	19	22	24	33	23	26	40	5.167	24	22	23	26	33	19	24	40	1.667
7	19	22	24	40	23	26	33	1.083	25	22	23	26	40	19	24	33	−2.417
8	19	22	26	33	23	24	40	4.000	26	22	23	33	40	19	24	26	−6.500
9	19	22	26	40	23	24	33	−0.083	27	22	24	26	33	19	23	40	1.083
10	19	22	33	40	23	24	26	−4.167	28	22	24	26	40	19	23	33	−3.000
11	19	23	24	26	22	33	40	8.667	29	22	24	33	40	19	23	26	−7.083
12	19	23	24	33	22	26	40	4.583	30	22	26	33	40	19	23	24	−8.250
13	19	23	24	40	22	26	33	0.500	31	23	24	26	33	19	22	40	0.500
14	19	23	26	33	22	24	40	3.417	32	23	24	26	40	19	22	33	−3.583
15	19	23	26	40	22	24	33	−0.667	33	23	24	33	40	19	22	26	−7.667
16	19	23	33	40	22	24	26	−4.750	34	23	26	33	40	19	22	24	−8.833
17	19	24	26	33	22	23	40	2.833	35	24	26	33	40	19	22	23	−9.417
18	19	24	26	40	22	23	33	−1.250									

Table 3.2: all random assignments possible in Example 3.1.

The observed $D_{\text{obs}} = 9.25$ is relatively large (far right) in the null distribution. Under H_0 the probability that D is at least 9.25 is $3/35 \approx 0.08$, fairly small, so there is some evidence that the two treatments differ.

This method is called the **randomization test**. To obtain its null distribution, the randomization distribution, we shuffle the given treatment assignment; so it can also be called a shuffle test or permutation test.

p-value and type I error of the randomization test. In the randomization model the p-value of the observed statistic D_{obs} is the probability, under H_0, of a value of D at least as extreme as D_{obs}; "extreme" is determined by H_1. For H_1: "the new treatment shortens recovery," larger D is more extreme under H_0. (If H_1 were "the new treatment lengthens recovery," smaller, i.e. more negative, D would be more extreme.) In the one-sided test of Example 3.1, $D \geq D_{\text{obs}}$ is extreme, and under H_0 the null distribution of D is the discrete uniform distribution on the $\binom{N}{m}$ values d_i, so

$$p = P(D \geq D_{\text{obs}} \mid H_0) = \frac{\sum_{i=1}^{\binom{N}{m}} I(d_i \geq D_{\text{obs}})}{\binom{N}{m}}. \tag{3.1}$$

[14] The binomial coefficient $\binom{N}{m}$ ("N choose m") is the number of ways of choosing m of N objects without order: $\binom{N}{m} = \frac{N!}{m!(N-m)!}$.

From the table, 3 of the d_i are at least D_{obs}, so $p = 3/35$.

By (3.1), p-values are multiples of $1/\binom{N}{m}$. If the d_i have no ties, the null distribution of p is the discrete uniform distribution on $\{1/\binom{N}{m}, 2/\binom{N}{m}, \ldots, 1\}$, each value with probability $1/\binom{N}{m}$. At significance level $\alpha = k/\binom{N}{m}$ the type I error of the randomization test is

$$P(\text{reject } H_0 \mid H_0) = P(p \le \alpha \mid H_0) = P\left(\sum_{i=1}^{\binom{N}{m}} I(d_i \ge D_{\text{obs}}) \le k \;\middle|\; H_0 \right) = \frac{k}{\binom{N}{m}} = \alpha, \tag{3.2}$$

so the type I error is controlled exactly at level α. In this sense the randomization test is called an *exact inference*.

Proof of (3.2) (supplied). Under H_0 the multiset $\{d_1, \ldots, d_{\binom{N}{m}}\}$ does not depend on which assignment was realized, and the realized assignment is uniform over the $\binom{N}{m}$ assignments, so D_{obs} is uniform over the d_i. With no ties, let $d_{(1)} > d_{(2)} > \ldots$; if $D_{\text{obs}} = d_{(j)}$ then $\sum_i I(d_i \ge D_{\text{obs}}) = j$. This count is at most k exactly when D_{obs} is one of the k largest values, which has probability $k/\binom{N}{m}$. ■

Example 3.3

In Example 3.1, what is the p-value if H_1 is "the new treatment lengthens recovery time"? This is lower-tailed, so

$$p = P(D \le D_{\text{obs}} \mid H_0) = \frac{\sum_{i=1}^{\binom{N}{m}} I(d_i \le D_{\text{obs}})}{\binom{N}{m}} \tag{3.3}$$

is 33/35.

In two-sided tests the meaning of "extreme" needs care. Whether D is positive or negative, larger $|D|$ favors H_1, so if H_0 is rejected when $|D| = |\bar{Y}_n - \bar{X}_m|$ is large,

$$p = P(|D| \ge |D_{\text{obs}}|).$$

The randomization distribution is not symmetric about 0, so doubling a one-sided p-value is not exact: $P(|D| \ge |D_{\text{obs}}|) \ne 2\min\{P(D \ge D_{\text{obs}}), P(D \le D_{\text{obs}})\}$ in general.

Example 3.4

In Example 3.1, let H_1 be "the new treatment's recovery time differs from the conventional one." The upper-tailed p-value is 3/35 and the lower-tailed one 33/35; the two-sided p-value is 4/35.

Test statistics of the randomization test. So far we used $D = \bar{Y}_n - \bar{X}_m$. Why the difference of the two treatment means? Would another statistic, e.g. a randomization test based on the two-sample t-statistic, be more efficient? The efficiency of a randomization or permutation test depends on the choice of statistic and on the population from which the observations come. Other statistics usable for two-sample comparisons as in Example 3.1:

- the sum $\sum_{i=1}^{n} Y_i$ of the observations of one treatment;
- the two-sample t-statistic $T = (\bar{Y}_n - \bar{X}_m)/\widehat{\text{se}}$, where $\widehat{\text{se}} = \sqrt{\frac{1}{m} + \frac{1}{n}}\, s_p$ and $s_p^2 = \{\sum_{i=1}^{m}(X_i - \bar{X}_m)^2 + \sum_{i=1}^{n}(Y_i - \bar{Y}_n)^2\}/(m+n-2)$;
- the difference of medians $\text{median}\{Y_i\} - \text{median}\{X_i\}$, useful for asymmetric distributions;

- the difference of trimmed means. E.g. the 10% trimmed mean is the mean of the middle 90% of the data after deleting the largest 5% and the smallest 5%; useful for symmetric distributions with outliers.

In the randomization test the observations $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$ are fixed, so the total $\sum_i^m X_i + \sum_i^n Y_i$ is the same under every assignment. Hence the statistic first considered,

$$D = \frac{1}{n}\sum_{i=1}^{n} Y_i - \frac{1}{m}\sum_{i=1}^{m} X_i = \frac{m+n}{mn}\sum_{i=1}^{n} Y_i - \frac{1}{m}\Big(\sum_{i=1}^{m} X_i + \sum_{i}^{n} Y_i\Big),$$

is an increasing function of $\sum_{i=1}^n Y_i$, and the randomization test with $\sum_{i=1}^n Y_i$ coincides with the one using D. The two-sample t-statistic T is also an increasing function of D, so the randomization p-value with T equals that with D. In contrast, using the difference of medians $\text{median}\{Y_i\} - \text{median}\{X_i\}$ can give a different result from D.

Proof that T is increasing in D (supplied). Let $\bar{Z}$ and $SST = \sum_i (X_i - \bar{Z})^2 + \sum_j (Y_j - \bar{Z})^2$ be the grand mean and total sum of squares of the pooled data; both are the same for every assignment. The ANOVA decomposition gives

$$SST = (m+n-2)s_p^2 + \frac{mn}{m+n}D^2,$$

because the between-group sum of squares is $m(\bar{X}_m - \bar{Z})^2 + n(\bar{Y}_n - \bar{Z})^2 = \frac{mn}{m+n}(\bar{Y}_n - \bar{X}_m)^2$. Hence

$$T = \frac{D}{\sqrt{\frac{1}{m}+\frac{1}{n}}\sqrt{(SST - \frac{mn}{m+n}D^2)/(m+n-2)}} = g(D),$$

and $g(x) = cx/\sqrt{a - bx^2}$ ($a = SST$, $b = \frac{mn}{m+n}$, $c > 0$) satisfies $g'(x) = ca/(a - bx^2)^{3/2} > 0$ on $bx^2 < a$, where D always lies. So T is a strictly increasing function of D. ■

3.3 The population model

In Example 3.2, m flies were sampled at random from population 1 (forest tsetse flies) with antenna lengths $X_1, \ldots, X_m$, and n flies from population 2 (river tsetse flies) with $Y_1, \ldots, Y_n$. We want the two-sample t-statistic as test statistic. Since the observations were sampled at random from populations, the hypotheses can be stated through the distributions F of population 1 and G of population 2:

$$H_0 : F = G \quad \text{vs.} \quad H_1 : F \neq G. \tag{3.4}$$

In the population model the randomness of sampling from the population is the basis of inference. Random sampling yields $X_i \overset{\text{i.i.d.}}{\sim} F$, so the observations represent the population F well. Hence the results of tests and estimates in the population model can be claimed to hold for the whole population.[15] By contrast, inference in the randomization model is inference about the given data (the sample at hand).

If each population is normal with equal variances, $F \equiv N(\mu_1, \sigma^2)$ and $G \equiv N(\mu_2, \sigma^2)$, and the hypotheses become hypotheses about the location parameters of a parametric model:

$$H_0 : \mu_2 - \mu_1 = 0 \quad \text{vs.} \quad H_1 : \mu_2 - \mu_1 \neq 0. \tag{3.5}$$

[15] In complete random sampling the observations are independent and each follows its population. In observational studies the observations are often not truly independent, or it is impossible to sample so that they exactly follow the population. Here we assume the samples are approximately independent and approximately follow the population (i.e. roughly satisfy the i.i.d. assumption). In the randomization model of Section 3.2, random assignment to the given units can be implemented exactly.

Here the two-sample t-test is known to be the most efficient test, and the null distribution of the two-sample t-statistic is the t_{m+n-2} distribution. If normality cannot be assumed (i.e. for the hypotheses (3.4) rather than (3.5)), can we obtain the null distribution of t?

The answer is the **permutation test** in the population model. Its idea: if H_0 is true, the two populations F and G coincide, so the $N = m+n$ random variables all come from the same distribution, whichever population (F or G) they come from. So the observed sample $X_1, \ldots, X_m, Y_1, \ldots, Y_n$ is one of the $\binom{N}{m}$ possible partitions of these N values into population 1 (m values) and population 2 (n values). Given the values $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$, each of the $\binom{N}{m}$ partitions has the same probability $1/\binom{N}{m}$ of having occurred. The distribution of t over the $\binom{N}{m}$ partitions is the **permutation distribution** (shuffle distribution).[16] A test using it as the null distribution of t to compute p-values is called a **permutation test**.

To understand the permutation distribution, arrange the data as a table with the antenna length (variable `AL`) and a categorical variable (`label`) indicating forest or river fly. The different partitions (shuffles) in the permutation test amount to shuffling the `label` variable of the data (columns 1–2 are the original data). The third variable `permuted.label` in the R output is one of the $\binom{N}{m}$ values obtained by shuffling `label`, and under H_0 the original labels and `permuted.label` have the same probability.

```
   AL  label permuted.label
1  1.24 forest  river
2  1.38 forest  forest
3  1.36 forest  forest
4  1.40 forest  forest
5  1.38 forest  forest
6  1.48 forest  forest
7  1.38 forest  river
8  1.54 forest  forest
9  1.56 forest  forest
10 1.14 river   river
11 1.20 river   forest
12 1.30 river   forest
13 1.26 river   river
14 1.28 river   river
15 1.18 river   river
```

For the tsetse data of Example 3.2, let us obtain the null distribution (permutation distribution) of the test statistic. There are $\binom{15}{9} = 5005$ shuffles, so listing all of them by hand as in Example 3.1 is impractical. The `label` and `permuted.label` above are two of the 5005 cases; the two-sample t-statistic computed from each gives $t_{\text{obs}} = 4.08$ (from `label`) and $t_i = 2.81$ (from `permuted.label`). Doing this for all 5005 cases gives the permutation distribution (Figure 3.2 of the book; for R see Section 3.6.1). The two-sided p-value is $p = P(|T| \geq |t_{\text{obs}}| \mid H_0)$; in Example 3.2, 11 of the $|t_i|$ are at least $|t_{\text{obs}}| = 4.08$, so $p = 11/5005 \approx 0.002$ and the antenna lengths of the two tsetse species differ significantly.

3.3.1 Conditional inference

The permutation test in the population model is a **conditional inference** method. The permutation distribution is not the exact null distribution, but the null distribution under the condition that, when H_0 is true, the observed values (above, the values of `AL`) are exactly the values we observed. Moreover, whatever the population distribution, the permutation distribution given the

[16] The number of distinct values of the permutation distribution is $\binom{N}{m} = N!/(m!n!)$, the number of ways (combinations) of splitting N objects into m and n. This differs from the number of permutations of m out of N, ${}_NP_m = N!/m!$, and from the number $N!$ of orderings of all N objects. In permutation distributions and tests, "permutation" means shuffling, not ordered arrangement; the number of possible shuffles is $\binom{N}{m}$.

observed values is distribution-free, because with the data values fixed it is the distribution of the statistic over the $\binom{N}{m}$ shuffles, each with the same probability (a discrete uniform distribution).

Note that the permutation test in the population model is exactly the same as the randomization test in the randomization model. For this reason the permutation test, like the randomization test, is a **conditionally exact inference** (conditional because the data values are fixed). In formulas: if $\alpha = k/\binom{N}{m}$,

$$P(\text{reject } H_0 \mid H_0, \text{data}) = P(p \le \alpha \mid H_0, \text{data}) = \alpha.$$

That is, given the data the level is controlled exactly. Since it is controlled for every data set from the population,

$$P(\text{reject } H_0 \mid H_0) = E\{P(\text{reject } H_0 \mid H_0, \text{data})\} = E\{\alpha\} = \alpha,$$

so the level α is controlled unconditionally too.

3.3.2 The random permutation test

A permutation (randomization) test computes the statistic for all shuffles and obtains the exact null distribution. In Example 3.1 there were $\binom{7}{4} = 35$ shuffles, and in Example 3.2 $\binom{15}{9} = 5005$; the number of shuffles grows very fast with the sample size. With $(n, m) = (10, 10)$ it is about 6.7×10^{11}, too many for a personal computer.

A simple remedy is random sampling from the permutation distribution: recomputing the statistic for a few thousand randomly sampled shuffles takes little time on most computers. The p-value of this **random permutation test**, or Monte Carlo permutation test, is obtained as follows, for a test rejecting H_0 when $T_{\text{obs}} \ge c$ for a critical value c:

1. Choose the number M of random samples approximating the permutation distribution; $M = 999$ or $M = 1999$ is common.
2. Compute T_{obs} from the unshuffled data.
3. For $i = 1, \ldots, M$ generate a random shuffle splitting the N units into m and n, and compute T_i from the shuffled data.
4. The proportion of $\{T_{\text{obs}}, T_1, \ldots, T_M\}$ at least T_{obs} is the approximate p-value. With $T_0 = T_{\text{obs}}$,

$$\hat{p} = \frac{\sum_{i=0}^{M} I(T_i \ge T_{\text{obs}})}{M+1}.$$

In the third step we generate M independent shuffles splitting N into m and n, i.e. we sample M times from the $\binom{N}{m}$ shuffles with equal probability. To program this we either shuffle the category of each unit (in Example 3.2 the value of `label`) or shuffle the units' ids (in Example 3.2 the numbers `1:15` in the first column).

Shuffling the unit ids. Generate a random permutation of $\{1, \ldots, N\}$, assign the first m units to the first category and the remaining n to the second. E.g. for Example 3.1 with $N = 7$, $m = 4$, the following R code generates one random shuffle:

```
> N <- 7
> rand.permutation <- sample(1:N, N, replace = FALSE)
> rand.permutation
[1] 7 5 4 1 6 2 3
```

In this shuffle, units $7, 5, 4, 1$ are assigned to the first category and units $6, 2, 3$ to the second.

If the permutation $(5, 7, 4, 1, 6, 2, 3)$ happened to be generated, it differs from $(7, 5, 4, 1, 6, 2, 3)$ but has the same probability (each $1/N!$) and gives the same shuffle. A random permutation is one of

the $N!$ permutations chosen at random, and $m!n!$ of them are equivalent to $(7, 5, 4, 1, 6, 2, 3)$ (give the same shuffle). So each distinct shuffle has probability $(m!n!)/N! = 1/\binom{N}{m}$, and shuffling unit ids by a random permutation is an exact way of producing a random shuffle.

Shuffling category values. The second way is to shuffle the values of the categories. If the categories before shuffling are `New,New,New,New,Old,Old,Old`, applying the permutation $(7, 5, 4, 1, 6, 2, 3)$ from the code above gives `Old,Old,New,New,Old,New,New`. (The first unit takes the category of the original 7th value, the second unit that of the 5th, and so on.) In this shuffle units 3, 4, 6, 7 are in the first category and units 1, 2, 5 in the second. Compared with shuffling ids, the same permutation gives a different shuffle, but shuffling the categories also uses a random permutation and is an exact way of producing an arbitrary shuffle. The following R code shows both ways of shuffling the category values with a random permutation:

```
> N = 7
> label = c("New","New","New","New","Old","Old","Old")
> set.seed(0)
> perm.id = sample(1:N, N, replace = FALSE)
> perm.id
[1] 6 1 4 3 7 2 5
> label[perm.id]
[1] "Old" "New" "New" "New" "Old" "New" "Old"
> set.seed(0)
> permuted.label = sample(label, N, replace = FALSE)
> permuted.label
[1] "Old" "New" "New" "New" "Old" "New" "Old"
```

Accuracy of the random permutation p-value. The random permutation p-value $\hat{p}$ approximates the permutation p-value p; more precisely it estimates p from M random samples, where M is the number of random shuffles. How far is $\hat{p}$ from the exact p? $\hat{p}$ is essentially the mean of the independent Bernoulli variables $I(T_i \geq T_{\text{obs}})$ with $E\{I(T_i \geq T_{\text{obs}})\} = P(T_i \geq T_{\text{obs}}) = p$, so

$$E(\hat{p}) = p, \qquad \text{se}(\hat{p}) = \sqrt{\frac{p(1-p)}{M}} \approx \sqrt{\frac{\hat{p}(1-\hat{p})}{M}}.$$

With $M = 999$, $\hat{p} = 0.01$ has standard error 0.003 and $\hat{p} = 0.1$ has 0.009, so $\hat{p}$ is quite accurate.

3.4 How permutation tests work

Permutation tests are nonparametric tests needing very few assumptions about the population, and they are conditionally exact given the data. They can be schematized as follows:

1. State H_0 and H_1, and choose a test statistic D that tends to be large under H_1.
2. Check that the data are **exchangeable** under H_0, and determine the possible shuffles within the range of exchangeability.
3. If the number of shuffles is not large (e.g. $M = 2000$ or fewer), obtain for exact inference the distribution of the statistic over all shuffles (the re-randomized or permutation distribution),[17] i.e. recompute the statistic d_i for each shuffle.
4. The p-value is the proportion of the d_i at least the unshuffled value D_{obs}.

[17] If there are too many shuffles, estimate the null distribution by the random permutation test.

Whatever statistic is chosen in step 1, the permutation test controls the type I error exactly or conservatively. If the family of populations can be specified, choosing a statistic with higher power in that family is better. E.g. for a two-sample test with approximately normal populations, choose the two-sample t-statistic (or the difference of means). If the distribution looks skewed or has outliers, a difference of medians or trimmed means raises the power. Replacing observations by special scores such as ranks or signs and computing the statistic from them often reproduces classical nonparametric methods.

Permutation tests need very few assumptions about the population; the needed assumption is essentially the **exchangeability of the data under the null hypothesis** described in step 2. Exchangeability is the key assumption that allows the null distribution to be derived in a permutation test. Writing a reordering of n observations as a (one-to-one) permutation function $\pi : \{1, \ldots, n\} \to \{1, \ldots, n\}$, exchangeability is defined as follows.[18]

Definition 3.1 (Exchangeable random variables)

Let $F(x_1, \ldots, x_n)$ be the joint distribution of random variables $X_1, \ldots, X_n$. The random variables are **exchangeable** if for every permutation $\pi : \{1, \ldots, n\} \to \{1, \ldots, n\}$ of their order, the joint distribution of $X_{\pi(1)}, \ldots, X_{\pi(n)}$ is also $F(x_1, \ldots, x_n)$.

Exchangeability is a very weak assumption. E.g. independent identically distributed (i.i.d.) random variables are exchangeable whatever their distribution. Even identically distributed but dependent random variables can be exchangeable. So exchangeability is a very weak nonparametric assumption.

Why exchangeability makes the permutation test valid (supplied)

Let $Z = (Z_1, \ldots, Z_N)$ be exchangeable under H_0 and $D = D(Z)$ a statistic. Let $\{\pi_1, \ldots, \pi_K\}$ be the $K = N!$ permutations, and $p(Z) = \frac{1}{K} \sum_k I(D(Z_{\pi_k}) \geq D(Z))$ the permutation p-value. Then $P(p(Z) \leq \alpha \mid H_0) \leq \alpha$.

Proof. By exchangeability $Z_{\pi_k} \stackrel{d}{=} Z$ for each k, so for U uniform on $\{1, \ldots, K\}$ independent of Z, $Z_{\pi_U} \stackrel{d}{=} Z$. Given the multiset $\{D(Z_{\pi_k})\}_k$ (which is the same for Z and every Z_{π_U}, as the permutations form a group), $D(Z_{\pi_U})$ is a uniformly chosen element of it. The p-value is the proportion of elements at least the chosen one, and a uniformly chosen element is among the top j (counted with ties) with probability at most j/K. So $P(p \leq \alpha) \leq \alpha$, with equality when αK is an integer and there are no ties. Restricting to the $\binom{N}{m}$ two-sample splits (each realized by $m!n!$ permutations) gives the same statement.

Some examples of exchangeability of random variables (or data):

Example 3.5 (i.i.d. random variables)

In a two-population model, observations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are independent, each by simple random sampling. If $H_0 : F = G$ is true, all observations are independent with the same distribution ($F = G$), so they are exchangeable. Likewise, observations i.i.d. in a one-population model are exchangeable, whatever the population and whether or not H_0 holds.

[18] A permutation function π is a bijection of a set onto itself. E.g. on $\{1, 2, 3\}$, $\pi(1) = 2$, $\pi(2) = 1$, $\pi(3) = 3$ swaps the first two elements 1 and 2.

Example 3.6 (Compound symmetry)

To assess elementary pupils' achievement, first choose an elementary school in Seoul, then randomly sample n upper-grade pupils of that school and observe their common achievement test scores $X_1, \dots, X_n$. Assume the scores are normal and, since all pupils in the sample attend the same school, strongly positively correlated: for all $i \neq j$, $X_i \sim N(\mu, \sigma^2)$ and $\mathrm{Corr}(X_i, X_j) = \rho > 0$. Then the joint distribution of $(X_1, \dots, X_n)$ is multivariate normal $N_n(M, \Sigma)$ with mean and covariance

$$M = \begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{pmatrix}, \qquad \Sigma = \sigma^2 \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & & \ddots & \\ \rho & \rho & \cdots & 1 \end{pmatrix}.$$

Reordering the variables does not change the mean or the covariance, so $(X_1, \dots, X_n)$ is exchangeable but not independent. Covariance matrices of this form arise often in cluster samples; this is the *compound symmetry* structure.

The next example is a case where the data are not exchangeable even under the null hypothesis.

Example 3.7

In a two-population model, $X_1, \dots, X_m \overset{\text{i.i.d.}}{\sim} F$ and $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} G$ are independent, each a simple random sample, with $F = N(\mu_1, 1)$ and $G = N(\mu_2, 4)$; we test $H_0 : \mu_1 = \mu_2$. Even under H_0, F and G differ, so the joint distributions of $(X_1, X_2, \dots)$ and $(X_1, Y_1, \dots)$ differ: these observations are not exchangeable.

Applying a permutation test to data that are not exchangeable under H_0 may fail to control the type I error exactly, so in principle permutation tests cannot be applied.

3.5 Advantages and disadvantages of permutation tests

Most tests in Part I of this book can be understood as permutation tests. Their advantages:

1. They apply to any data satisfying exchangeability: the assumptions on the population can be very weak, so they are nonparametric methods applicable to very broad families.
2. There is no restriction on the choice of test statistic.
3. The null distribution of any statistic is very easy to obtain.

Disadvantages:

1. Computation takes longer than for parametric methods and traditional (rank- and sign-based) nonparametric procedures.
2. Because much computation is needed, they are hard to use for inference other than testing via p-values (e.g. for confidence intervals).
3. With shuffles that do not satisfy exchangeability, the type I error may not be controlled.

3.6 Various permutation tests

3.6.1 Two-sample permutation tests

To introduce permutation tests, Section 3.1 looked at randomization or permutation tests in the randomization model (two treatments) or the population model (two populations). Both use the same test procedure, so both are called the two-sample permutation test. With F the distribution of the first population and G that of the second, the null hypothesis is

$$H_0 : F = G, \tag{3.6}$$

and three alternatives can be considered (Figure 3.3 of the book):

- $H_1 : F(x) \leq G(x)$ for all x: "observations of population 1 tend to be larger than those of population 2."
- $H_1 : F(x) \geq G(x)$ for all x: "observations of population 1 tend to be smaller than those of population 2."
- $H_1 : F(x) \neq G(x)$: "the two distributions differ (in location, in variance, or in shape)."

If the two populations are equal under H_0 and differ only in the location parameter under H_1, the location difference Δ can be used as the parameter. Then

$$F(x) = G(x + \Delta), \tag{3.7}$$

or in terms of densities $f(x) = g(x + \Delta)$, so for $X \sim F$, $Y \sim G$,

$$P(X \leq x) = P(Y \leq x + \Delta), \tag{3.8}$$

i.e. $X + \Delta \stackrel{d}{=} Y$ or $X \stackrel{d}{=} Y - \Delta$: the second population is the first one shifted by Δ (to the right). Then the null hypothesis (3.6) is simply a statement about the parameter Δ, $H_0 : \Delta = 0$, and the three alternatives become lower-tailed, upper-tailed and two-sided:

- (lower-tailed) $H_1 : \Delta < 0$: "population 2 lies to the left of population 1";
- (upper-tailed) $H_1 : \Delta > 0$: "population 2 lies to the right of population 1";
- (two-sided) $H_1 : \Delta \neq 0$.

A simple estimator of Δ, or one of the following, as test statistic D tends to take different values under the null and the alternative:

- the sum of the observations of the new treatment $\sum_{i=1}^{n} Y_i$;
- the two-sample t-statistic $t = (\bar{Y}_n - \bar{X}_m)/\widehat{\text{se}}$;
- the difference of medians or of trimmed means.

With m observations from the first population, n from the second, and $N = m + n$, the null distribution of D is obtained from all $\binom{N}{m}$ shuffles or approximated by random permutations. If $D = \bar{X} - \bar{Y}$ is used and $d_1, \ldots, d_M$ are the values from random permutations, the p-values are as follows (write d_0 for the observed D_{obs}):

- lower-tailed: $p_l = \sum_{i=0}^{M} I(d_i \leq D_{\text{obs}})/(M + 1)$;
- upper-tailed: $p_u = \sum_{i=0}^{M} I(d_i \geq D_{\text{obs}})/(M + 1)$;
- two-sided: $p = \sum_{i=0}^{M} I(|d_i| \geq |D_{\text{obs}}|)/(M + 1)$.

Generating all shuffles in the two-sample test. To generate the null distribution of a statistic D in a two-sample test from all shuffles, all $\binom{N}{m}$ cases must be generated. The R code below is an example. (Note: for large N this approach is not recommended.)

```
> dat = read.csv("data/tsetse.csv"); N = 15; m = 9; n = 6
> head(dat,3)
    AL  label
1 1.24 forest
2 1.38 forest
3 1.36 forest
```

First fix the total sample size N and the size m of the first sample. To shuffle the unit ids we use `gtools::combinations()`.

```
> allperms <- gtools::combinations(N,m)
> head(allperms,3)
     [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9]
[1,]    1    2    3    4    5    6    7    8    9
[2,]    1    2    3    4    5    6    7    8   10
[3,]    1    2    3    4    5    6    7    8   11
> nrow(allperms)
[1] 5005
```

Each row of `allperms` holds the $m = 9$ ids of the first sample after shuffling, and there are $\binom{N}{m} = 5005$ distinct rows. With $D = \bar{Y}_n - \bar{X}_m$ the null distribution is generated by

```
> D.perm = vector(length = choose(N,m))
> for (i in 1:choose(N,m)){
+   permuted.x = dat$AL[allperms[i,]]
+   permuted.y = dat$AL[-allperms[i,]]
+   D.perm[i] = mean(permuted.y) - mean(permuted.x)
+ }
> head(D.perm,3)
[1] -0.18666667 -0.07000000 -0.08666667
```

Instead of the `for` loop one can use `apply` for speed:

```
> D.perm = apply(allperms,1,
+               function (perm.x.id){
+                 permuted.x = dat$AL[perm.x.id]
+                 permuted.y = dat$AL[-perm.x.id]
+                 mean(permuted.y) - mean(permuted.x)
+               } )
> head(D.perm,3)
[1] -0.18666667 -0.07000000 -0.08666667
```

In any case the "exact" two-sided p-value (null hypothesis (3.6)) is

```
> Dobs = with(dat,
+   mean(AL[label == "river"]) - mean(AL[label == "forest"])
+ )
> (p = mean(abs(D.perm) >= abs(Dobs)))
[1] 0.002197802
```

Random permutation test for two samples. A random permutation test needs no enumeration of cases and is simpler to implement. The R code below computes the lower-tailed p-value with the observed D_{obs} above by random permutations. There is no need to compute the statistic for each case of the shuffles; generate random permutations with `sample()` and repeat (`replicate()`) the computation M times.

```
> M = 1999
> D.perm = replicate(M, {
+   x.id = sample(1:N,m, replace =FALSE)
+   permuted.x = dat$AL[x.id]
+   permuted.y = dat$AL[-x.id]
+   mean(permuted.y) - mean(permuted.x)
+ })
> (p = mean(c(Dobs,D.perm) <= Dobs))
[1] 5e-04
```

3.6.2 One-sample permutation tests

For a test about the location parameter of one population from a sample, let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$. We compare $H_0 : \theta = \theta_0$ for the median $\theta = Q_F(0.5)$ of F with one of

$$H_1 : \theta > \theta_0, \qquad H_1 : \theta < \theta_0, \qquad H_1 : \theta \neq \theta_0.$$

The test statistic can be one of:

- the t-statistic $T = \frac{\bar{X}_n - \theta_0}{s/\sqrt{n}}$, with $s^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X}_n)^2$ the sample variance;
- the sign statistic $S = \sum_{i=1}^n I(S_i = +) = \sum_{i=1}^n I(X_i > \theta_0)$;
- Wilcoxon's signed-rank statistic $SR_+ = \sum_{i=1}^n I(X_i > \theta_0) \cdot R_i$.

With the statistic chosen, the first of the four steps of Section 3.4 is done. What, then, is the permutation distribution (null distribution under H_0) of the statistic?

First consider the sign statistic S. Under H_0, an observation X_i is above the median θ_0 with the same probability as below it ($P(X_i > \theta_0) = P(X_i < \theta_0) = 1/2$), so shuffling the signs of the n observations leaves the distribution unchanged: the joint distribution of $(S_1, \ldots, S_n)$ equals that of $(-S_1, S_2, \ldots, S_n)$ (the joint distribution is the same however many "$-$" signs are attached and wherever). So the permutation distribution here is obtained not by shuffling the order of observations but by flipping their signs.[19] There are $M = 2^n$ possible sign flips, through which the permutation distribution of S is obtained. It does not depend on the observed values (only on the signs S_i), and coincides with the null distribution $B(n, 1/2)$ of S derived when the sign test was introduced.

For the permutation distribution of Wilcoxon's SR_+, recall that each observation's contribution to SR_+ splits into a rank R_i and a sign S_i. If F is symmetric about θ_0, an observation X_i with rank R_i is equally likely to lie left or right of θ_0. That is, the joint distribution of $((S_1, R_1), \ldots, (S_n, R_n))$ given $(R_1, \ldots, R_n)$ equals that of $((-S_1, R_1), \ldots, (S_n, R_n))$, however many "$-$" signs are attached and wherever. When the population is symmetric about θ_0, the conditional null distribution (permutation distribution) of $X_1, \ldots, X_n$ given the ranks is obtained by flipping the signs of the observations, and this is Wilcoxon's signed-rank distribution.

Summarizing, the sign test and the Wilcoxon signed-rank test are permutation tests with statistics S and SR_+. Also, the permutation distribution of SR_+ cannot be obtained when the population does not satisfy the symmetry assumption: with the ranks R_i fixed, the distribution of S_i changes with the sign flips, violating exchangeability, so re-randomization cannot be performed.

Next consider the permutation distribution of the t-statistic. In a permutation test the t-statistic can be used even if normality is violated, but the population must be assumed symmetric. Compare each X_i with the hypothesized median θ_0: with $Z_i = X_i - \theta_0$ and sign $S_i = +I(X_i > \theta_0) - I(X_i < \theta_0)$,

$$X_i - \theta_0 = S_i |Z_i|.$$

[19] Again, in this case the shuffles differ from permutations, which is common. That is why "shuffle distribution" and "shuffle test" are more precise terms than permutation distribution and permutation test.

Given the observed $(|Z_1|, \ldots, |Z_n|)$, the conditional distributions of S_i and $-S_i$ coincide,

$$P(S_i = \text{“}+\text{”} \mid |Z_i|) = P(S_i = \text{“}-\text{”} \mid |Z_i|) = 1/2,$$

so, as for the permutation distribution of SR_+, the permutation distribution of T is obtained using the observations shuffled by the $M = 2^n$ distinct sign flips,

$$X_i^{\text{perm}} = \theta_0 \pm S_i|Z_i|.$$

[20]

3.6.3 Testing independence for categorical data

The data of the next example are categorical.

Example 3.8

Is excessive salt intake related to cardiovascular disease? The data below record whether 30 cardiovascular patients and 30 people of similar health without the disease consume too much salt:

	high salt	low salt	total
cardiovascular disease	5	25	30
no disease	2	28	30
total	7	53	60

We want to test with these data whether excessive salt intake and cardiovascular disease are related. One may compare the proportion of excessive salt takers between the populations “disease” and “no disease,” or the proportion of patients between the populations “high salt” and “low salt.” The populations and parameters considered differ with how the data were sampled, but the test procedure is the same. Concretely, let p_1 be the proportion of patients in the high-salt group and p_2 that in the low-salt group, and test

$$H_0 : p_1 = p_2 \quad \text{vs} \quad H_1 : p_1 > p_2.$$

(The alternative may also be two-sided.) In categorical data analysis this is called a **test of homogeneity**, or a **test of independence** if the data are a random sample, and the χ^2 test can be used; but it is accurate only for large samples and cannot be applied here. Instead, a permutation test with a simple statistic can be used. Let the statistic D be the number of patients who take too much salt and have the disease (under the alternative D tends to be large). The observed value is $D_{\text{obs}} = 5$.

Rearranged as a data table (rows for the $n = 60$ patients, two columns for the two variables), with `Cardio = Yes` for disease and `Salt = H` for excessive intake:

```
   Cardio Salt
1      No    H
2      No    H
3     Yes    H
4     Yes    H
5     Yes    H
6     Yes    H
7     Yes    H
8      No    L
```

[20] For reference, the one-sample t-test and the one-sample permutation test with the t-statistic have very similar efficiency (Helwig, 2019).

```
9       No    L
10      No    L
...
```

If $H_0 : p_1 = p_2$ is true, the probability of disease is the same whether or not one is in the high-salt group (the probability of `Cardio = Yes` is the same for `Salt = H` and `Salt = L`). We do not know this probability $p = p_1 = p_2$ exactly. Fisher's exact test for two categorical variables fixes that half of the `Cardio` observations (30 of 60) are `Yes` and that 7 values of `Salt` are `Yes`, and considers all shuffles possible under H_0. Such shuffles fix the values of one variable (e.g. `Cardio`) and shuffle the values of the other (`Salt`). Of the $N = 60$ observations, $k = 7$ have `Salt = H`, so there are $\binom{60}{7} = 386206920$ distinct shuffles, and all of them give the exact permutation distribution of D. (Using it as the null distribution gives the p-value of Fisher's exact test.)

The fixed variable `Cardio` has only two possible values, so the number of possible values of D (8 here) is much smaller than the number of shuffles (about 3.8×10^8). So it is not necessary to consider all shuffles to obtain the permutation distribution. Fisher's exact test could be used from the 1930s because this permutation distribution is a well-known distribution (the hypergeometric distribution[21]).

Example 3.9 (Example 3.8 continued)

Compare the observed $D_{\text{obs}} = 5$ with the null distribution of D to get the p-value. D can take the values $d = 0, 1, \ldots, 7$, and $P(D = d)$ can be obtained from the hypergeometric distribution (or as the proportion of all shuffles giving $D = d$). The R code below extracts the hypergeometric probabilities directly to get the null distribution of D:

```
> nulldist = dhyper(0:7, m = 30, n = 30, k = 7)
> round(nulldist,2)
[1] 0.01 0.05 0.16 0.29 0.29 0.16 0.05 0.01
```

The alternative is upper-tailed, so the p-value is

$$P(D \geq D_{\text{obs}}) = P(D \geq 5) = P(D = 5) + P(D = 6) + P(D = 7) = 0.211,$$

not enough evidence to reject H_0.

For reference, writing the two variables as C_i, S_i, under H_0 the joint distribution of $((C_1, S_1), (C_2, S_2))$ equals that of $((C_2, S_1), (C_1, S_2))$. So (C_i) is exchangeable and the condition of the permutation test is satisfied.

[21] When m of N patients have the disease and $N - m$ do not, and k of them are chosen at random and assigned to the high-salt group, the number X of patients with disease assigned to the high-salt group follows the hypergeometric distribution with probability mass function $f(x) = P(X = x) = \binom{m}{x}\binom{N-m}{k-x}\Big/\binom{N}{k}$. In Example 3.8, $N = 60$, $m = 30$, $k = 7$.

Chapter 4. Inference for Two Populations

Overview

Comparing two populations is one of the most basic and most frequent problems in statistics. This chapter uses two running examples.

Example 4.1

Acute leukemia splits into two types, acute myelogenous leukemia (AML) and acute lymphocytic leukemia (ALL). The expression level of a particular gene was measured for 72 acute leukemia patients: $m = 47$ have ALL and $n = 25$ have AML (Golub et al., 1999). The first three rows of the data are shown below; the distributions of the expression level (`ge`) by type are in Figure 4.1 of the book.

```
> leukemia4= read.csv(file = "data/leukemia4.csv")
> head(leukemia4,3)
   ge label
1 328   ALL
2 295   ALL
3 276   ALL
```

Is there a difference in gene expression between the two types ALL and AML?

Example 4.2

To test the hypothesis that the higher the residual food intake (RFI) of Korean cattle, the better the meat quality, the RFI and marbling score (Marb) of 7 cattle were recorded. RFI was classified as high or low.

RFI	Marb
High	740, 610, 720
Low	580, 680, 850, 510

4.1 Differences between two populations

For comparing two populations we assume data

$$X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F, \qquad Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G.$$

Of the two models where permutation tests apply (randomization and population models) this is the population model; the technical steps of the test are the same for data from a randomization model, so for simplicity we assume the population model. We are interested in inference about the difference between the unknown populations, which can be described on several levels:

1. **Same shape and dispersion, different location.** For a location shift parameter Δ, $F(x) = G(x + \Delta)$ or $f(x) = g(x + \Delta)$.[22] If the shape $f(x)$ is a well-known distribution (e.g. normal), inference is parametric; with no restriction on the shape it is nonparametric.
2. **Same shape, different dispersion and location.** With a shift Δ and scale $\gamma > 0$, relative to the density $f(x)$ of the first population the second has density $g(x) = \frac{1}{\gamma} f(\frac{x-\Delta}{\gamma})$. If $\gamma > 1$, G has the larger variance.

[22] If $\Delta > 0$, G lies to the right of F; with the medians θ_F or means μ_F as reference, $\theta_F + \Delta = \theta_G$ and $\mu_F + \Delta = \mu_G$.

3. **Same shape and location, different dispersion.** The model is $g(x) = \frac{1}{\gamma} f\left(\frac{x}{\gamma}\right)$ for a scale $\gamma > 0$.
4. **Different shapes, possibly with different location or dispersion.** The difference cannot be described by finitely many parameters, but generally the shift or scale parameter is taken as the parameter of interest and a distribution-free test is used. Even under a null hypothesis of equal location or dispersion the two populations may differ.
5. **Different shapes and no parameter of interest.** The hypotheses are not about a parameter: $H_0 : F = G$, $H_1 : F \neq G$.

With $H_0 : F = G$ the exchangeability condition of the permutation test is satisfied. In the location-shift model (case 1) and the scale model (case 3), the null hypotheses $H_0 : \Delta = 0$ and $H_0 : \gamma = 1$ make the two populations equal (case 5 too). In cases 2 and 4, however, $H_0 : \Delta = 0$ does not imply equal populations, and the permutation test cannot be applied.

4.2 Tests for location

4.2.1 The two-sample t-test and permutation tests

First the traditional case where normality can be assumed. The first population F is $N(\mu_F, \sigma_F^2)$ and the second G is $N(\mu_G, \sigma_G^2)$. In this parametric model the location parameter Δ is the difference of means $\Delta = \mu_G - \mu_F$ and the scale parameter γ the ratio of standard deviations $\gamma = \sigma_G/\sigma_F$.

In case 1 above we compare the means of two normal distributions with equal variances, and the two-sample t-test using

$$T = \left(\frac{1}{n} + \frac{1}{m}\right)^{-\frac{1}{2}} \frac{\bar{Y}_n - \bar{X}_m}{s_p}$$

is exact and the most powerful. Under H_0, T follows t_{m+n-2}, which gives the p-value.

In case 2 we compare the means of two normal distributions with different variances, and the two-sample (unequal variance) t-test with Welch's statistic

$$T = \frac{\bar{Y}_n - \bar{X}_n}{\sqrt{\frac{s_X^2}{m} + \frac{s_Y^2}{n}}}$$

is known to be approximately exact. Under H_0, T is approximately t_ν, with degrees of freedom approximated by

$$\nu \approx \left(\frac{s_X^2}{m} + \frac{s_Y^2}{n}\right)^2 \Big/ \left(\frac{s_X^4}{m^2(m-1)} + \frac{s_Y^4}{n^2(n-1)}\right).$$

Permutation tests for location. When normality is badly violated, or the sample is too small for the exactness of the two-sample t-test, a permutation test can be performed, with null hypothesis $H_0 : F = G$. As statistic one can use the sum $\sum_{i=1}^n Y_i$ of the second sample, the two-sample t-statistic T, the difference of medians, or the difference of trimmed means. The permutation test is exact whichever statistic is used, but depending on the shape of the population some statistics are preferred (more powerful); e.g. for asymmetric distributions the difference of medians is more powerful (see Chapter 3).

As in one-sample tests, instead of the observations themselves one can use their ranks, signs or scores and run a permutation test. Traditional nonparametric (distribution-free) methods for two-sample problems were developed on exactly such transformations.

4.2.2 The Wilcoxon rank-sum test

The **rank-sum test** (Wilcoxon, 1945) proposed by Wilcoxon is the most widely used nonparametric method for comparing the locations of two populations. It is often used when the two-sample t-test is not exact or lacks power, e.g. for heavy-tailed populations. It is a permutation test using the sum of ranks as statistic, instead of comparing means or medians of the two samples.

Wilcoxon's rank-sum statistic is defined as follows. Arrange $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$ in order and assign ranks from the smallest value. The statistic is the sum of the ranks of one of the two samples (e.g. the Y's). With S_i $(i = 1, \ldots, m)$ the ranks of the X observations and R_i $(i = 1, \ldots, n)$ those of the Y observations,

$$W = \sum_{i=1}^{n} R_i.$$

Example 4.3

The ranks of the data of Example 4.2:

```
> cowdat = read.csv("data/cow.csv")
> ranks = rank(cowdat$Marb)
> data.frame(cowdat, ranks)
   RFI Marb ranks
1 High  740     6
2 High  610     3
3 High  720     5
4  Low  580     2
5  Low  680     4
6  Low  850     7
7  Low  510     1
```

With ties among the observations, mid-ranks are used (e.g. if there were two smallest values 510, both would get rank 1.5). Taking the `RFI==Low` sample as Y, the observed rank-sum statistic is $W = 2 + 4 + 7 + 1 = 14$:

```
> (W.obs = sum( ranks[cowdat$RFI == "Low"] ))
[1] 14
```

If $H_0 : F = G$ is true, the observations $(X_1, \ldots, X_m, Y_1, \ldots, Y_n)$ are exchangeable, and so are the ranks $(S_1, \ldots, S_m, R_1, \ldots, R_n)$. The null distribution of Wilcoxon's rank-sum statistic W is its permutation distribution over all $\binom{m+n}{n}$ shuffles. For the marbling data of Example 4.2 there are $\binom{7}{4} = 35$ shuffles, and the null distribution of W over all of them is:

```
> allperms.id = gtools::combinations(7,4)
> W.perm = apply(allperms.id,1,
+                function(perm.y.id){ sum(ranks[perm.y.id]) })
> table(W.perm)
```

w	10	11	12	13	14	15	16	17	18	19	20	21	22
$P(W = w)$	$\frac{1}{35}$	$\frac{1}{35}$	$\frac{2}{35}$	$\frac{3}{35}$	$\frac{4}{35}$	$\frac{4}{35}$	$\frac{5}{35}$	$\frac{4}{35}$	$\frac{4}{35}$	$\frac{3}{35}$	$\frac{2}{35}$	$\frac{1}{35}$	$\frac{1}{35}$

(Figure 4.2 of the book plots it.) The value of W depends only on the ranks of the observations, not on their values, so the null distribution of W does not depend on the population. The distribution-free permutation test using this null distribution is Wilcoxon's rank-sum test.

Example 4.4

In Example 4.2 let $H_0 : F = G$ and $H_1 : F(x) \le G(x)$ (the population with `RFI == High`, "X", lies to the right of the population with `RFI == Low`, "Y"). The smaller the rank sum W of the `RFI == Low` sample, the more extreme under H_0. So the p-value of $W_{\text{obs}} = 14$ is $P(W \le W_{\text{obs}}) = P(W \le 14) = 11/35$: we cannot be sure that higher RFI improves meat quality.

For reference, using the rank sum $W' = \sum_{i=1}^{m} S_i$ of the other sample (the "X" observations) as statistic gives the same permutation test, since $W + W' = N(N+1)/2$ is constant.

The Mann–Whitney test. With observations $X_1, \dots, X_m$ from the first population and $Y_1, \dots, Y_n$ from the second, Mann and Whitney proposed as statistic for the location difference the number of pairs (X_i, Y_j) with Y_j larger than X_i:

$$U = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j > X_i),$$

the **Mann–Whitney U-statistic**. With ties in the data, $I(Y_j > X_i) + \frac{1}{2} I(Y_j = X_i)$ is used instead of $I(Y_j > X_i)$. The U-statistic is closely related to Wilcoxon's rank-sum statistic:

$$W = U + \frac{n(n+1)}{2}.$$

Proof (supplied). Assume no ties. The rank of Y_j in the pooled sample is one plus the number of observations smaller than Y_j:

$$R_j = 1 + \#\{k : Y_k < Y_j\} + \#\{i : X_i < Y_j\}.$$

Summing over j, $\sum_j \#\{k : Y_k < Y_j\} = 0 + 1 + \cdots + (n-1)$ (the jth smallest Y has $j-1$ smaller Y's), so $W = n + \frac{n(n-1)}{2} + U = U + \frac{n(n+1)}{2}$. ■

So the permutation test with U (Mann–Whitney test) gives always the same result as the permutation test with W (Wilcoxon rank-sum test); both are called the **Mann–Whitney–Wilcoxon test**.

The distribution of the Mann–Whitney U is that of W shifted left by $n(n+1)/2$; knowing one gives the other. When $H_0 : F = G$ is true and there are no ties, the distribution of W is called the **Wilcoxon rank-sum distribution**.[23] It is completely determined by m, n and does not depend on the data values, so the Mann–Whitney–Wilcoxon test is distribution-free.

When (m, n) is not large, the exact rank-sum distribution over all $\binom{m+n}{m}$ shuffles is easily computed; for large samples the number of shuffles is huge and exact p-values take very long. Then one may use a random permutation test, but the usual approach is the large-sample approximation of the rank-sum distribution introduced next.

4.2.3 Large-sample approximation

In the leukemia gene-expression data of Example 4.1, let the ALL expression levels be $X_1, \dots, X_m \sim F$ and the AML ones $Y_1, \dots, Y_n \sim G$, and apply the rank-sum test. Here $m = 47$, $n = 25$ are too large to consider all shuffles for the null distribution. The rank sum $W = \sum_{i=1}^{n} R_i$ is of the form of a "mean" of ranks, so the CLT can approximate its distribution by a normal for large m, n.

[23] The distribution of W is sometimes called the Wilcoxon rank-sum distribution and so is that of U; one determines the other. Here (no ties) we call the distribution of W the Wilcoxon rank-sum distribution. R's `dwilcox()`, `qwilcox()` etc. give the distribution of U for small m, n.

Recall the assumptions of the CLT. The most basic CLT says that for i.i.d. observations $Z_1, \ldots, Z_n$ the sample mean is approximately normal for large n (Theorem 1.1):

$$\bar{Z}_n = \frac{1}{n}\sum_{i=1}^{n} Z_i \dot{\sim} N\Big(\mu_Z, \frac{\sigma_Z^2}{n}\Big), \tag{4.1}$$

with $E(Z_1) = \mu_z$ and $\mathrm{Var}(Z_1) = \sigma_Z^2$. Let $N = m + n$. In the rank-sum statistic, under H_0 the ranks $R_1, \ldots, R_n$ all have the same distribution (discrete uniform on $\{1, \ldots, N\}$). But if $R_1 = 1$, R_2 cannot be 1, so the ranks are not independent and the basic CLT (4.1) does not apply. Instead, the ranks $R_1, \ldots, R_n$ can be seen as a sample drawn *without replacement* with equal probabilities from the finite population $\{1, \ldots, N\}$. The rank sum W is the sum (mean $\times n$) of a sample of size n drawn without replacement from a finite population, and a CLT for finite populations applies.

For reference, the basic CLT (4.1) applies to means of samples drawn *with replacement*, whether the population is infinite or finite.

Samples from a finite population. Consider a finite population $\mathcal{A}_N = \{A_1, \ldots, A_N\}$ of N units. Each A_i is a number and may take any fixed value; e.g. ranks of observations $A_1 = 1, \ldots, A_N = N$, or with ties several units with the same value, $A_1 = 1.5$, $A_2 = 1.5$, $A_3 = 3, \ldots, A_N = N$. The population mean and variance of this finite population are

$$\mu = \frac{1}{N}\sum_{i=1}^{N} A_i \equiv \bar{A}, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(A_i - \mu)^2 = \frac{\sum_{i=1}^{N} A_i^2}{N} - (\bar{A})^2 \equiv \overline{A^2} - (\bar{A})^2. \tag{4.2}$$

If X is drawn at random from the discrete uniform distribution on $\mathcal{A}_N$, then $\mu = E(X)$ and $\mathrm{Var}(X) = \sigma^2$.

Sampling with replacement. If $X_1, \ldots, X_n$ are drawn with replacement from $\mathcal{A}_N$,

$$X_i \overset{\text{i.i.d.}}{\sim} \mathrm{Uniform}(\mathcal{A}_N), \tag{4.3}$$

so $E(\bar{X}_n) = \mu$, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, and by the basic CLT (4.1), for large n, $\bar{X}_n \dot{\sim} N(\mu, \sigma^2/n)$, i.e. as $n \to \infty$

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

Sampling without replacement. If $X_1, \ldots, X_n$ are drawn without replacement from $\mathcal{A}_N$, the sample $\{X_1, \ldots, X_n\}$ is one of the $\binom{N}{n}$ ways of choosing n of the N units $A_1, \ldots, A_N$. The distribution of $\{X_1, \ldots, X_n\}$ is uniform over all size-n subsets of $\mathcal{A}_N$: for every subset $\{p_1, \ldots, p_n\} \subset \{1, \ldots, N\}$, the probability that the sample is $\{A_{p_1}, \ldots, A_{p_n}\}$ is

$$P(\{X_1, \ldots, X_n\} = \{A_{p_1}, A_{p_2}, \ldots, A_{p_n}\}) = \frac{1}{\binom{N}{n}}.$$

The sample mean $\bar{X}_n$ of a sample drawn without replacement is a good estimator of the population mean μ.

Theorem 4.1

Let $X_1, \ldots, X_n$ be drawn without replacement from a finite population $\mathcal{A}_N$ of N units with mean μ and variance σ^2, and let $\bar{X}_n$ be the sample mean. Then

$$E(\bar{X}_n) = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n}\frac{N-n}{N-1}.$$

Proof. The values of $X_1, \dots, X_n$ are n of $A_1, \dots, A_N$. Let $Z_i = 1$ if A_i is among the sample values and $Z_i = 0$ otherwise. Then $X_1 + \cdots + X_n \overset{d}{=} \sum_{i=1}^{N} Z_i A_i$. The indicators $(Z_1, \dots, Z_N)$ are not independent, but each Z_i has the same marginal distribution with $P(Z_i = 1) = n/N$. So

$$E(\bar{X}_n) = \frac{1}{n} \sum_{i=1}^{N} A_i P(Z_i = 1) = \frac{1}{n} \cdot \frac{n}{N} \sum_{i} A_i = \mu.$$

For the variance we need the joint distribution of (Z_i, Z_j). (Supplied.) For $i \neq j$, $P(Z_i = Z_j = 1) = \binom{N-2}{n-2} / \binom{N}{n} = \frac{n(n-1)}{N(N-1)}$, so

$$\operatorname{Var}(Z_i) = \frac{n}{N}\Big(1 - \frac{n}{N}\Big), \qquad \operatorname{Cov}(Z_i, Z_j) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = -\frac{n}{N} \cdot \frac{N-n}{N(N-1)}.$$

The variance of $\bar{X}_n$ is unchanged by shifting all A_i by a constant, so take $\mu = 0$; then $\sum_i A_i^2 = N\sigma^2$ and $\sum_{i \neq j} A_i A_j = (\sum_i A_i)^2 - \sum_i A_i^2 = -N\sigma^2$. Hence

$$\operatorname{Var}\Big(\sum_i Z_i A_i\Big) = N\sigma^2 \cdot \frac{n}{N}\frac{N-n}{N} + N\sigma^2 \cdot \frac{n}{N}\frac{N-n}{N(N-1)} = n\sigma^2 \frac{N-n}{N}\Big(1 + \frac{1}{N-1}\Big) = n\sigma^2 \frac{N-n}{N-1},$$

and dividing by n^2 gives the claim. ∎

Compared with the variance σ^2/n of the mean of a sample drawn with replacement, the variance of the mean of a sample drawn without replacement is smaller by the factor $\frac{N-n}{N-1}$, the **finite-population correction factor**. For sample size $n = 1$, $\frac{N-n}{N-1} = 1$, and it decreases as n grows. For $n = N$ (the whole population drawn) the variance of the sample mean must be 0, and indeed $\frac{N-n}{N-1} = 0$.

The sample size n is always smaller than N, so for a large-sample approximation of $\bar{X}_n$, N must grow along with n. The CLT for sample means drawn without replacement from a finite population:

Theorem 4.2 (Hájek (1960); Li and Ding (2017))

For every $N = 1, 2, 3, \dots$ let a finite population $\mathcal{A}_N = \{A_{N1}, \dots, A_{NN}\}$ be given, with mean μ_N and variance σ_N^2. Let $\bar{X}_n$ be the mean of a sample $X_1, \dots, X_n$ drawn without replacement from $\mathcal{A}_N$. If $\min\{n, N-n\} \to \infty$ and (Hájek's condition, supplied)

$$\frac{\max_i (A_{Ni} - \mu_N)^2}{\min\{n, N-n\}\, \sigma_N^2} \to 0,$$

then

$$\frac{\bar{X}_n - E(\bar{X}_n)}{\sqrt{\operatorname{Var}(\bar{X}_n)}} = \frac{\bar{X}_n - \mu_N}{\sigma_N/\sqrt{n}} \sqrt{\frac{N-1}{N-n}} \xrightarrow{d} N(0,1).$$

Proof (sketch, supplied). Without the displayed condition the statement fails (e.g. a population with one huge value), so some such condition is needed; for ranks $A_{Ni} = i$ it holds, since $\max_i (i - \frac{N+1}{2})^2 \le N^2/4$ while $\sigma_N^2 = (N^2-1)/12$. Write $S = \sum_i Z_i (A_{Ni} - \mu_N)$ with the inclusion indicators Z_i of Theorem 4.1. Hájek couples sampling without replacement with *Poisson sampling*: let $I_1, \dots, I_N$ be i.i.d. Bernoulli(n/N) and $S^* = \sum_i I_i (A_{Ni} - \mu_N)$. S^* is a sum of independent terms with variance $\frac{n}{N}(1 - \frac{n}{N}) N \sigma_N^2$, and the displayed condition is exactly Lindeberg's condition for it, so $S^* / \operatorname{sd}(S^*) \xrightarrow{d} N(0,1)$. Given $K = \sum_i I_i$, the selected set under Poisson sampling is a uniform K-subset. One adds or removes $|K - n|$ randomly chosen units to obtain a uniform n-subset, i.e. a realization of S. Since $|K - n| = O_p(\sqrt{n(N-n)/N})$, one shows $E(S - S^*)^2 / \operatorname{Var}(S) \to 0$. Hence $S / \operatorname{sd}(S)$ has the same normal limit, and $\operatorname{Var}(S) = n^2 \operatorname{Var}(\bar{X}_n)$ by Theorem 4.1. ∎

Large-sample approximation of the Wilcoxon rank-sum distribution. The rank sum $W = \sum_{i=1}^{n} R_i$ is the sum of n observations drawn without replacement from the finite population $\mathcal{A}_N = \{1, \ldots, N\}$. Under H_0 this is simple random sampling without replacement, and by (4.2) the population mean and variance of $\mathcal{A}_N$ are

$$\mu_N = \frac{N+1}{2}, \qquad \sigma_N^2 = \frac{(N+1)(2N+1)}{6} - \Big(\frac{N+1}{2}\Big)^2 = \frac{N^2-1}{12}.$$

The rank sum is related to the mean rank by $W = n\bar{R}$, so by Theorem 4.1

$$E(W) = n\mu = \frac{n(N+1)}{2}, \qquad \mathrm{Var}(W) = n^2 \frac{\sigma^2}{n} \frac{N-n}{N-1} = \frac{nm(N+1)}{12},$$

and by Theorem 4.2, for large n and m,

$$\frac{W - E(W)}{\sqrt{\mathrm{Var}(W)}} \dot{\sim} N(0,1). \tag{4.4}$$

So the null distribution of W can be approximated by the normal $N(E(W), \mathrm{Var}(W))$.

With ties among the observations the ranks considered are not $\{1, \ldots, N\}$ but, e.g.,

$$\mathcal{A}'_N = \{1.5, 1.5, 4, 4, 4, \ldots, N\}.$$

Then $\mathcal{A}'_N$ is taken as the population, $\mathrm{Var}(W)$ is recomputed, and the CLT is applied. The mean is still $E(W) = n(N+1)/2$ as without ties.

Example 4.5

In the leukemia data of Example 4.1, $m = \#ALL = 47$ and $n = \#AML = 25$, large enough. There are tied observations, and the rank-sum statistic is $W_{\mathrm{obs}} = 1113.5$. The approximate null distribution by (4.4) for the data with ties has $E(W) = 912.5$, $\mathrm{Var}(W) = 7147.917$:

$$W \dot{\sim} N(912.5, 7147.917). \tag{4.5}$$

(Figure 4.3 of the book shows this normal-approximated null distribution with the observed value 1113.5.) The two-sided p-value by the normal approximation is

$$p_G = P\Big(\frac{|W - E(W)|}{\sqrt{\mathrm{Var}(W)}} \geq \frac{|W_{\mathrm{obs}} - E(W)|}{\sqrt{\mathrm{Var}(W)}}\Big) \approx 2P(Z \geq 2.37) = 0.017.$$

For comparison, the random permutation test with the rank-sum statistic: from $M = 1999$ random permutations (approximate) null distribution, 14 of the shuffled statistics w_i were larger than W_{obs}, so

$$P(W \geq W_{\mathrm{obs}}) \approx \frac{1 + \sum_{i=1}^{M} I(w_i \geq W_{\mathrm{obs}})}{M+1} = \frac{15}{2000},$$

and by symmetry of the permutation distribution

$$p_{RP} = P(|W - E(W)| \geq |W_{\mathrm{obs}} - E(W)|) = 2P(W \geq W_{\mathrm{obs}}) \approx \frac{30}{2000} = 0.015,$$

almost the same as the normal approximation.

Large-sample approximation of permutation distributions. In two-sample permutation tests the statistic is often the mean of the observations sampled from one of the populations. The rank-sum statistic is n times the mean of the ranks of the "Y" sample; the statistic $\bar{D} =$

$\frac{D}{n} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (the mean of the second sample) used to introduce permutation tests is also the mean of n observations drawn without replacement. So the permutation distribution of $\bar{D}$ can be approximated by the CLT. The finite population is the pooled sample $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$, whose population mean is the pooled mean and whose population variance is the variance of the pooled sample without bias correction:

$$\mu = \frac{\sum_i^m X_i + \sum_i^n Y_i}{m+n}, \qquad \sigma^2 = \frac{\sum_{i=1}^m (X_i - \mu)^2 + \sum_{i=1}^n (Y_i - \mu)^2}{m+n}.$$

The permutation distribution of $\bar{D}$ has mean $E(\bar{D}) = \mu$ and variance $\mathrm{Var}(\bar{D}) = \frac{\sigma^2}{n}\frac{m}{m+n-1}$, so for large m, n

$$\frac{\bar{D} - \mu}{\sigma/\sqrt{n}}\sqrt{\frac{N-1}{N-n}} \xrightarrow{d} N(0,1). \tag{4.6}$$

Example 4.6

For the leukemia data of Example 4.1, a random permutation test with $\bar{D} = \frac{1}{n}\sum_{i=1}^{n} Y_i$ (the mean expression score of AML patients) gives the estimated permutation distribution shown as a histogram in Figure 4.4 of the book. Compared with the observed $\bar{D}_{\text{obs}} = 233.12$, 17 of the $B = 1999$ randomly generated $\bar{d}_i$ exceed $\bar{D}_{\text{obs}}$. Assuming the permutation distribution is symmetric, the p-value is

$$p \approx 2 \times (1 + 17)/2000 = 0.018 :$$

enough evidence that the expression scores of AML and ALL patients differ. The approximate null distribution of $\bar{D}$ from the large-sample approximation (4.6) closely approximates the permutation distribution.

The permutation distribution of $\bar{D}$ is well approximated by a normal because $\bar{D}$ is a mean. Is the permutation distribution of the difference of means $\bar{Y}_n - \bar{X}_m$ or of the t-statistic also well approximated by a normal? If so, what are the mean and variance of that permutation distribution?[24] Even when the exact asymptotic distribution is unknown, if a normal approximation is expected, one can approximate its two parameters μ_n, σ_n^2.

If the permutation distribution is approximately $N(\mu_n, \sigma_n^2)$, the sample mean and variance of the statistics $\{\bar{d}_1, \ldots, \bar{d}_B\}$ from random shuffles estimate μ_n, σ_n^2. The dotted density in Figure 4.4 is the normal approximated from the random permutation distribution,

$$N(\hat{\mu}_D, \hat{\sigma}_D^2), \qquad \hat{\mu}_D = \frac{1}{M}\sum_{i=1}^{M} \bar{d}_i, \quad \hat{\sigma}_D^2 = \frac{1}{M-1}\sum_{i=1}^{M} (\bar{d}_i - \hat{\mu}_D)^2,$$

very similar to the normal from the large-sample approximation.

In the same way the permutation distribution of the t-statistic can be approximated by a normal. Figure 4.5 of the book compares the permutation distribution of the t-statistic generated by random shuffles (histogram) with the approximating distribution (dotted density). If the mean and variance of the permutation distribution are μ_{RP}, σ_{RP}^2, the approximating distribution is $N(\mu_{RP}, \sigma_{RP}^2)$. The observed t-statistic is $T_{\text{obs}} = 2.51$; the random permutation p-value is $p_{RP} = \{1 + \sum_{i=1}^{M} I(|T_i| \geq |T_{\text{obs}}|)\}/(M+1) = 0.0105$, and the approximate p-value is $p_N = P(|Z| \geq |T_{\text{obs}} - \mu_{RP}|/\sigma_{RP}) \approx 0.014$. At the 5% level this gives the same conclusion as Example 4.6.

When the permutation test statistic is a mean, a difference of means, a standardized difference of means, or some function of them, its large-sample distribution is normal, so the permutation

[24] With random partitions of a finite population into "Y" and "X" samples, the difference of means $\bar{Y}_n - \bar{X}_m$ is not like sampling with replacement from an infinite population, where $\bar{Y}_n$ and $\bar{X}_m$ are independent. To get $\mathrm{Var}(\bar{Y}_n - \bar{X}_m)$ one must compute $\mathrm{Corr}(\bar{Y}_n, \bar{X}_m)$ (by Lehmann (1975), $\mathrm{Corr}(\bar{Y}_n, \bar{X}_m) = -1$: the total is fixed, so $m\bar{X}_m + n\bar{Y}_n$ is constant). The asymptotic distribution of the t-statistic under sampling from a finite population also requires care.

distribution can be approximated by a normal. If the large-sample distribution of the statistic is not normal, there is no guarantee. Also, keep in mind that large-sample approximations are for large samples.

4.2.4 Estimating the shift parameter and confidence intervals

In the location-shift model, where the two populations have the same shape and dispersion but different locations, assume $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F(x)$ and $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} F(x-\Delta)$ for a shift parameter Δ; there is no restriction on the shape of F. If F has a mean, the mean $E(Y_1)$ of the second population exceeds the mean $E(X_1)$ of the first by Δ. In Section 4.2.2 we saw that the Mann–Whitney–Wilcoxon test of $H_0 : \Delta = 0$ against $H_1 : \Delta < 0$ (or $\Delta > 0$, $\Delta \neq 0$) is distribution-free.

Here we use the Mann–Whitney–Wilcoxon test to estimate Δ (point and interval estimation). Of the two forms of the statistic (Wilcoxon's rank sum and Mann–Whitney's U), we use Mann and Whitney's

$$U = U(0) = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j - X_i > 0).$$

There are mn differences $D_{ij} = Y_j - X_i$; denote them $\{D_1, \ldots, D_{mn}\}$. The Mann–Whitney U is the number of the mn differences D_i that are positive (compared with 0). For a value Δ_0 of the shift,

$$U(\Delta_0) = \sum_{i=1}^{m} \sum_{j=1}^{n} I(Y_j - X_i > \Delta_0) = \sum_{i=1}^{mn} I(D_i > \Delta_0),$$

which tests the more general null hypothesis $H_0 : \Delta = \Delta_0$. As in Section 2.2.4, where the Wilcoxon signed-rank statistic for a single population's location was the sign statistic of the Walsh averages relative to θ_0 and gave a confidence interval, here we use the confidence interval given by the number of differences D_i greater than Δ_0.

With $D_{(k)}$ the kth smallest difference, choose two integers $1 \leq k_l < k_u \leq mn$ and set the interval for Δ as

$$D_{(k_l)} \leq \Delta < D_{(k_u)}, \tag{4.7}$$

choosing k_l, k_u so that $P(D_{(k_l)} \leq \Delta < D_{(k_u)}) \geq 1 - \alpha$ for the level $1 - \alpha$, where Δ is the true unknown shift. Using the relation between $U(\Delta)$ and the D_i, and the facts that $U(\Delta)$ always has the distribution of the Mann–Whitney U whatever the value of Δ and that this distribution is symmetric about $E(U) = mn/2$, one shows

$$P(D_{(k_l)} \leq \Delta < D_{(k_u)}) = P(k_l \leq U(\Delta) \leq k_u - 1) \geq 1 - \alpha. \tag{4.8}$$

Proof of (4.8) (supplied). $D_{(k_l)} \leq \Delta$ iff at least k_l differences are $\leq \Delta$, i.e. $U(\Delta) = \#\{D_i > \Delta\} \leq mn - k_l$; and $\Delta < D_{(k_u)}$ iff at most $k_u - 1$ differences are $\leq \Delta$, i.e. $U(\Delta) \geq mn - k_u + 1$. At the true Δ, $Y_j - \Delta \overset{d}{=} X_i$, so $U(\Delta)$ is the Mann–Whitney statistic of the samples X and $Y - \Delta$, which come from the same population: $U(\Delta)$ has the null distribution of U. That distribution is symmetric about $mn/2$, i.e. $mn - U \overset{d}{=} U$, so $P(mn - k_u + 1 \leq U(\Delta) \leq mn - k_l) = P(k_l \leq U(\Delta) \leq k_u - 1)$. ■

As for the signed-rank interval for location, with the quantile function Q_U of the null distribution of U, setting

$$k_l = Q_U(\alpha/2), \qquad k_u = mn - k_l + 1 = mn - Q_U(\alpha/2) + 1$$

makes the level of the interval (4.7) at least $1 - \alpha$. As the level of (4.7) is made smaller and smaller, what remains is the median of the differences D_i,

$$\hat{\Delta} := \text{median}\{D_1, \ldots, D_{mn}\} = \text{median}\{Y_j - X_i : i = 1, \ldots, m,\ j = 1, \ldots, n\},$$

the Hodges–Lehmann estimator of Δ.

Example 4.7

In Example 4.2, assume $X_i \overset{d}{=} Y_j - \Delta$; find the Hodges–Lehmann estimate of Δ and a two-sided interval at level 80%. First compute and sort the differences $D_{ij} = Y_j - X_i$:

```
> x = cowdat$Marb[cowdat$RFI == "High"]
> y = cowdat$Marb[cowdat$RFI == "Low"]
> m = length(x); n = length(y)
> D = sapply(y, function(y){y-x}) # Delta = E(Y) - E(X)
> (D.ord = sort(D))
 [1] -230 -210 -160 -140 -100  -60  -40  -30   70  110  130  240
```

All 12 differences: their median, the Hodges–Lehmann estimate, is $\hat{\Delta} = -50$. The interval (4.7) is computed at $\alpha = 0.2$; the quantile of U comes from `qwilcox()`:

```
> alpha = 0.2
> kl = qwilcox(alpha/2,m,n)
> ku = m*n - kl + 1
> c(kl,ku)
[1]  2 11
> (ci = D.ord[c(kl,ku)])
[1] -210  130
```

The 80% interval for Δ is $\Delta = E(Y) - E(X) \in [-210, 130)$. By (4.8) its coverage probability comfortably exceeds 80%:

```
> (coverage = sum(dwilcox(kl:(ku-1),m,n)))
[1] 0.8857143
```

With such small samples no inference can guarantee much accuracy.

Large-sample intervals. For large m, n the quantile function $Q_U(\cdot)$ of U is hard to compute exactly; since U is approximately normal, the normal quantile function approximates $Q_U(\cdot)$. Using $U = W - n(n+1)/2$,

$$\mu_U := E(U) = E(W) - \frac{n(n+1)}{2} = \frac{nm}{2}, \qquad \sigma_U^2 := \mathrm{Var}(U) = \mathrm{Var}(W) = \frac{nm(n+m+1)}{12},$$

so $U \dot{\sim} N(\mu_U, \sigma_U^2)$ and $Q_U(\alpha)$ for $\alpha \in (0,1)$ is approximated by the α-quantile of $N(\mu_U, \sigma_U^2)$:

$$\alpha \le P(U \le Q_U(\alpha)) = P\Big(\frac{U - \mu_U}{\sigma_U} \le \frac{Q_U(\alpha) - \mu_U}{\sigma_U}\Big) \approx P(Z \le \Phi^{-1}(\alpha)).$$

With $\Phi^{-1}(\alpha) = -z_\alpha$,

$$Q_U(\alpha) \approx \mu_U - \sigma_U z_\alpha = \frac{nm}{2} - z_\alpha \sqrt{\frac{nm(n+m+1)}{12}}.$$

With the large-sample orders $\tilde{k}_l = \lfloor \frac{nm}{2} - z_{\alpha/2}\sqrt{\frac{nm(n+m+1)}{12}} \rfloor$ and $\tilde{k}_u = \lfloor \frac{nm}{2} + z_{\alpha/2}\sqrt{\frac{nm(n+m+1)}{12}} + 1 \rfloor$, the interval $D_{(\tilde{k}_l)} \le \Delta < D_{(\tilde{k}_u)}$ is an asymptotically exact $1-\alpha$ interval for the shift Δ.

Example 4.8

In Example 4.1, assume $X_i \overset{d}{=} Y_j - \Delta$ and find a two-sided 95% interval for Δ:

```
> # confidence interval large sample
> x = leukemia4$ge[leukemia4$label == "ALL"]
> y = leukemia4$ge[leukemia4$label == "AML"]
```

```
> D = sapply(y, function(y){y-x})
> D.ord = sort(D)
```

There are 1175 differences, not printed. The orders $\tilde{k}_l, \tilde{k}_u$ are generally not integers, so the nearest integers are used:

```
> m = length(x); n = length(y)
> ew = n * (n+m + 1)/2 # mean of W
> vw = n*m*(n+m+1)/12  # variance of W
> eu = ew - n*(n+1)/2  # mean of U
> vu = vw              # variance of U
> (ci.limits = round( eu + 1.96 * sqrt(vu) *c(-1,1)))
[1] 422 753
> (ci = D.ord[ci.limits])
[1]  13 133
```

The 95% interval for the shift is $\Delta \in (13, 133)$: the gene expression of the AML population is significantly higher than that of the ALL population.

4.2.5 Summary: tests for location

Wilcoxon proposed the rank-sum test (Wilcoxon, 1945) together with the signed-rank test, and Mann and Whitney (1947) studied the properties of the rank-sum distribution and its normal approximation in the form of the U-statistic. Nearly 80 years later, the Mann–Whitney–Wilcoxon test is still the most representative nonparametric test for the locations of two samples. That it is a permutation test using a particular transformation of the data (to ranks) was already used by Wilcoxon, but the term "permutation test" was not used: it had been reserved for tests that shuffle the labels while keeping the observed values (i.e. permutation tests using $\bar{Y} - \bar{X}$ and the like). Today "permutation test" generally refers to methods using as null distribution the permutation distribution conditional on (with the original values fixed) the observed values.

Permutation tests are conditionally exact. In particular, the Mann–Whitney–Wilcoxon test considers only the relative positions of the data, not their values, so it gives the same result whatever population generated the data, as long as the order is the same: a distribution-free method. But the type I error is controlled exactly only if the two populations are completely identical under the null ($H_0 : F = G$). If the two populations have the same median but differ in dispersion or shape, the Wilcoxon test does not guarantee exactness.[25]

Under the null hypothesis, all observations X_i are an independent sample from F, so $U_i = F(X_i) \overset{\text{i.i.d.}}{\sim} \text{Uniform}(0, 1)$. Multiplying the transformed U_i by $N + 1$ and computing the expectations of their order statistics gives $1, 2, \ldots, N$, i.e. the ranks. Using this, the Wilcoxon test can be understood as a permutation test that first transforms the unknown distribution F into the uniform distribution and uses the expectations of the transformed values (ranks) instead of the observations X_i. In this process the actual population distribution F is used indirectly, which is why the null distribution is the same for every population: a distribution-free method.

[25] When two populations with the same shape and median but different dispersion are compared, Fligner and Policello (1981) showed that a test using a standardized form of the Wilcoxon rank-sum statistic is approximately exact (the Fligner–Policello test). More recently Chung and Romano (2016) dealt with this situation, showing that a method similar to the derivation of Welch's t-test is approximately exact even under different shapes and medians. Without rank transformation, Yuen's (1974) Yuen–Welch t-test, which uses trimmed means in Welch's t-test, is also known to be approximately exact when normality is badly violated.

4.2.6 The R function `wilcox.test()`

The Mann–Whitney–Wilcoxon test of the location difference of two populations is implemented in R's `wilcox.test()`, also used for the one-sample Wilcoxon signed-rank test (see Section 2.2.5). Some usages follow. With the `leukemia4` data frame of Example 4.1, let x_i be the values of `ge` with `label == "ALL"` and y_i those with `label == "AML"`; in the model $x_i = \mu_1 + \epsilon_i$, $y_i = \mu_2 + e_i$ we run the rank-sum test of

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_1 \neq \mu_2.$$

The argument `conf.int = TRUE` also computes a 95% confidence interval for the shift $\Delta = \mu_1 - \mu_2$[26]; with `exact = FALSE` the large-sample approximation is used.

```
> wilcox.test(ge ~ label, exact = FALSE, conf.int = TRUE, data =
    leukemia4)

        Wilcoxon rank sum test with continuity correction

data:  ge by label
W = 386.5, p-value = 0.01771
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -133.99994  -12.99998
sample estimates:
difference in location
             -68.00001
```

With no ties and small samples, the exact test and interval use the Wilcoxon rank-sum distribution as below. These data have ties, so the test is not exact and a warning appears.

```
wilcox.test(ge ~ label, exact = TRUE, data = leukemia4)
```

It can also be used as follows, with the observations of the two samples in vectors `x` and `y`. Adding `alternative = "greater"` performs the upper-tailed test ($H_1 : \mu_1 < \mu_2$); "greater" is interpreted as $y > x$ in the order of the function's arguments.

```
> wilcox.test(y,x, exact = FALSE, correct = FALSE, alternative = "
    greater")

        Wilcoxon rank sum test

data:  y and x
W = 788.5, p-value = 0.008715
alternative hypothesis: true location shift is greater than 0
```

4.3 Paired two-sample comparisons

Paired two-sample comparisons are best understood as tests about one population.

Example 4.9

A cosmetics company developed a new skin-protection cream. To compare the effect of the new cream (A) with the existing cream (B), 8 users were selected; cream A was applied to one arm and cream B to the other, the skin was exposed to sunlight for a fixed time, and the degree of burn was measured. Can cream A be said to protect the skin better than cream B? (Higher values mean a more severe burn.)

[26] In Sections 3.6.1, 4.1 and 4.2.4 the shift was defined the other way, $\Delta = \mu_2 - \mu_1$.

	1	2	3	4	5	6	7	8
A	72	109	74	33	54	62	41	18
B	34	72	165	14	92	75	58	85

In this situation we compare populations A and B, but each observation of A is paired with one of B. User 1 used both creams A and B, so the two observations 72, 34 cannot be regarded as independent. Comparing the location (mean or median) of population A with that of population B in this situation is a **paired comparison**.

With observations $X_1, \ldots, X_n, Y_1, \ldots, Y_n$, each (X_i, Y_i) comes from the ith user and so is not independent, but the pairs

$$\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$$

are independent. So assume that the differences $D_i = Y_i - X_i$ of the two effects for each user satisfy

$$D_i = \Delta + \epsilon_i, \qquad i = 1, \ldots, n,$$

with the ϵ_i i.i.d. The distribution of ϵ_i (median 0) may or may not be normal. If each population is normal, the distribution of D_i (and of ϵ_i) is normal too; but if the populations are not normal, or the two distributions differ, the distribution of ϵ_i cannot be taken to be normal.

In a paired comparison we test, for the location difference Δ of the two populations,

$$H_0 : \Delta = 0 \quad \text{vs} \quad H_1 : \Delta \neq 0.$$

(One-sided tests $H_1 : \Delta < 0$ or $H_1 : \Delta > 0$ are possible too.) Since the differences D_i are a sample from one population, a location test for one population is performed: the one-sample t-test under normality, otherwise the sign test or the Wilcoxon signed-rank test. See Chapter 2.

4.4 Tests for the scale parameter

So far we have looked at tests of the locations of two populations and distribution-free estimation of the shift Δ. Here we look at distribution-free tests of differences in dispersion. Inference about dispersion is less common, but when normality cannot be assumed, rank-based distribution-free tests can be considered.

4.4.1 When the two locations are equal

First assume the two populations have the same location and differ only in scale: $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} f$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} g$, with

$$X_i = \mu + \sigma_1 e_i, \qquad Y_i = \mu + \sigma_2 \epsilon_i,$$

where $e_1, \ldots, e_m, \epsilon_1, \ldots, \epsilon_n$ are independent with the same distribution with median 0. The two populations share the location μ and the ratio of dispersions is $\gamma = \sigma_2/\sigma_1$: the dispersion of the second population is γ times that of the first. Figure 4.6 of the book shows the two distributions when the distribution of ϵ and e is normal and $\gamma = 2$.

Example 4.10

A food factory got complaints that the package weights of curry vary from product to product, and wants to introduce a new process to keep the weights constant. Randomly sampled weights from the old process were $110, 115, 120, 105, 103, 117$ g, and from the new process $109, 118, 113, 112, 119$ g. Does the new process keep the weights more constant than the old

one?

Assume the mean weights of the old and new processes are equal, and consider rank-based distribution-free test statistics.

Suppose $H_0 : \gamma = 1$ is true, i.e. the two populations are equal. The observations $X_1, \ldots, X_m$ and $Y_1, \ldots, Y_n$ all come from the same distribution, so the null distribution of Wilcoxon's rank-sum statistic $W = \sum_{i=1}^{n} R_i$ of the ranks R_i of the "Y" observations in the pooled sample is the distribution-free Wilcoxon rank-sum distribution. But when the alternative is true, even if the second population is less dispersed than the first ($\gamma < 1$), the rank sum W shows no tendency to become smaller or larger, so it cannot be used to test dispersion.

The alternative is to transform the observations not to ranks but to **scores** reflecting the difference in dispersion. If the "Y" observations are less dispersed than the "X" observations, they will cluster in the middle; so we can devise scores that are small for observations in the middle and large otherwise (or, conversely, large in the middle). The table below records, for the data of Example 4.10 (the new process is the second population), the ranks and several scores reflecting dispersion:

process	Old	Old	New	Old	New	New	Old	Old	New	New	Old
observation	103	105	109	110	112	113	115	117	118	119	120
rank	1	2	3	4	5	6	7	8	9	10	11
AB score	1	2	3	4	5	6	5	4	3	2	1
ST score	1	4	5	8	9	11	10	7	6	3	2
Mood score	25	16	9	4	1	0	1	4	9	16	25

The graph of ranks against the three scores (Figure 4.7 of the book) shows the difference clearly.

The **AB score** is the Ansari–Bradley score: score 1 to the smallest and largest ranks, 2 to the second smallest and second largest, and so on up to the middle rank. The **Ansari–Bradley statistic** T_{AB} is the sum of the Ansari–Bradley scores of the second sample; it tends to be large when the second population is less dispersed.

The **Siegel–Tukey (ST) score** is very similar to the Ansari–Bradley score, but devised so that without ties all ST scores are distinct. The Siegel–Tukey statistic T_{ST} is the sum of the Siegel–Tukey scores of the second sample. It was specially devised so that under the null its distribution coincides with the Wilcoxon rank-sum distribution.

The **Mood score** is the squared difference between each rank R_i and the middle rank $(N+1)/2$, $a_M(R_i) = (R_i - (N+1)/2)^2$; it was devised to be more sensitive to observations in the tails, making differences in scale easier to detect. The Mood statistic is also the sum of the Mood scores of the second sample.

Under $H_0 : \gamma = 1$ or $H_0 : f = g$, the null distributions of the Ansari–Bradley, Siegel–Tukey and Mood statistics are obtained as the permutation distributions of the scores. Under the lower-tailed alternative $H_1 : \gamma < 1$ (the second population is less dispersed), larger Ansari–Bradley and Siegel–Tukey statistics and smaller Mood statistics are more extreme, and the p-values come from the permutation distributions of the statistics. These p-values depend on the relative order of the observations but not on the actual values, so they are distribution-free. These distribution-free tests are the Ansari–Bradley, Siegel–Tukey and Mood tests.

Example 4.11

For the data of Example 4.10, the Ansari–Bradley statistic is $T_{AB} = 3+5+6+3+2 = 19$, the Siegel–Tukey statistic $T_{ST} = 5+9+11+6+3 = 34$, and the Mood statistic $T_M = 9+1+0+9+16 = 35$. Generate the exact permutation distribution of each statistic and test $H_0 : \gamma = 1$ against $H_1 : \gamma < 1$. First prepare the data:

```
> x = c(110, 115, 120, 105, 103, 117)
> y = c(109, 118, 113, 112, 119)
> curry = data.frame(label = c(rep("Old",6),rep("New",5)),
+                    x = c(x,y),rank = rank(c(x,y)))
> curry2 = cbind(curry %>% arrange(x),
+                AB = c(1:6, 5:1),
+                ST = c(1,4,5,8,9,11,10,7,6,3,2)) %>%
+   mutate(Mood = (rank-6)^2)
> head(curry2,3)
  label   x rank AB ST Mood
1   Old 103    1  1  1   25
2   Old 105    2  2  4   16
3   New 109    3  3  5    9
```

The three statistics are the sums of the scores of the observations with `label == "New"`:

```
> stats.obs = curry2 %>% filter(label == "New") %>% summarise(
    sum(AB),sum(ST),sum(Mood)) %>% unlist()
> stats.obs
 sum(AB)  sum(ST) sum(Mood)
      19       34        35
```

There are $\binom{11}{5} = 462$ shuffles, so we perform the exact permutation test. Generate the null distribution of each statistic and compute the p-values, taking care with the meaning of "extreme":

```
> N = 11; n = 5
> allperms = gtools::combinations(N,n) # perm. ids for "y" group
> stats.perm = apply(allperms,1,
+       function (perm.y.id){
+         curry2[perm.y.id,] %>% summarise(sum(AB),sum(ST),sum(
    Mood)) %>% unlist()
+       } )
> p.AB = mean(stats.perm[1,] >= stats.obs[1])
> p.ST = mean(stats.perm[2,] >= stats.obs[2])
> p.Mood = mean(stats.perm[3,] <= stats.obs[3])  # smaller is more extreme
> (p.vals = c(p.AB = p.AB, p.ST = p.ST, p.Mood = p.Mood))
     p.AB      p.ST    p.Mood
0.2294372 0.2683983 0.1861472
```

The three permutation p-values are all large, so $H_0 : \gamma = 1$ is not rejected. (Figure 4.8 of the book shows the null distributions of the statistics with the observed values.)

The Siegel–Tukey statistic was devised so that its null distribution is the Wilcoxon rank-sum distribution, a great advantage in the 1960s, when generating permutation distributions took days, but hardly an advantage now that computers are fast. Moreover the Siegel–Tukey scores are not exactly symmetric, so scoring from the right changes the value of the statistic slightly, and the same data can give different test results; it is now rarely used. Many other scores have been proposed besides these three, but the most basic and intuitive Ansari–Bradley test is the most frequently used test of scale.

In R, `ansari.test()`, `mood.test()` and `SiegelTukeyTest()` of the DescTools package perform approximate tests.

Example 4.12

For the data of Example 4.10, run the scale tests with `ansari.test()`, `mood.test()` and `SiegelTukeyTest()`, testing $H_0 : \gamma = 1$ against $H_1 : \gamma < 1$ as in Example 4.11. `ansari.test()` gives exact p-values for small samples.

```
> ansari.test(y,x,alternative = "less")

        Ansari-Bradley test

data:  y and x
AB = 19, p-value = 0.2294
alternative hypothesis: true ratio of scales is less than 1
```

`DescTools::SiegelTukeyTest()` computes the Siegel–Tukey statistic leaving out the median (here the value with ST score "11"). With such small samples this changes the p-value considerably.

```
> DescTools::SiegelTukeyTest(y,x,alternative = "less",exact =
    TRUE)

        Siegel-Tukey-test for equal variability

data:  y and x
ST = 23, p-value = 0.4571
alternative hypothesis: true ratio of scales is less than 1
```

Finally `mood.test()` computes the p-value from the standardized statistic with the large-sample approximation; for small samples the permutation p-value is more accurate.

```
> mood.test(y,x,alternative = "less")

        Mood two-sample test of scale

data:  y and x
Z = -0.98058, p-value = 0.1634
alternative hypothesis: less
```

4.4.2 When the two locations differ

If the two populations may differ in both dispersion and location,

$$X_i = \mu_1 + \sigma_1 e_i, \qquad Y_i = \mu_2 + \sigma_2 \epsilon_i,$$

how can we test $\gamma = \sigma_2/\sigma_1$ when $\mu_1 = \mu_2$ cannot be assumed?

If the populations can be assumed close to normal, the ratio of the two sample variances, using the estimators $S_1^2 = \sum_{i=1}^{m}(X_i - \bar{X})^2/(m-1)$ and $S_2^2 = \sum_{i=1}^{n}(Y_i - \bar{Y})^2/(n-1)$ of the population variances,

$$F = \frac{S_2^2}{S_1^2}, \tag{4.9}$$

follows the $F_{n-1,m-1}$ distribution (asymptotically) under H_0 ($\sigma_1^2 = \sigma_2^2$), which gives a test. (The book writes $F_{m-1,n-1}$; with S_2^2, from the n observations of Y, in the numerator, the degrees of freedom are $(n-1, m-1)$.)

When the populations are very different from normal, even if $H_0 : \gamma = 1$ is true, the two populations have different locations, so the exchangeability of the Ansari–Bradley permutation test (or of permutation tests with other scores) fails and the type I error cannot be controlled. The location-adjusted

observations $\{X_i - \mu_1, Y_j - \mu_2 : i = 1, \ldots, m,\ j = 1, \ldots, n\}$ are exchangeable under H_0, but μ_1, μ_2 are unknown and cannot be used. Higgins (2004) proposed estimating the unknown locations and running an approximate permutation test with the location-adjusted observations $X_i - \hat{\mu}_1$, $Y_j - \hat{\mu}_2$.

First, some statistics usable when the populations are very different from normal but have the same shape apart from location and dispersion. The following three statistics tend to be large when the alternative $\gamma > 1$ is true:

1. the ratio F (4.9) of the two sample variances; with heavy tails or outliers its variability is large;
2. the ratio of median absolute deviations (MAD),
$$R = \frac{\text{median}\{|Y_i - \text{median}\{Y_i\}|\}}{\text{median}\{|X_i - \text{median}\{X_i\}|\}};$$
its variability is small even with heavy tails or outliers;
3. the Ansari–Bradley statistic of the median-adjusted observations: using that the median-adjusted observations $X'_i = X_i - \text{median}\{X_i\}$ and $Y'_i = Y_i - \text{median}\{Y_i\}$ have approximately the same location, compute the Ansari–Bradley statistic of $\{X'_1, \ldots, X'_m, Y'_1, \ldots, Y'_n\}$. (As location estimate one may use the median of the Walsh averages instead of median$\{X_i\}$, and the Siegel–Tukey or Mood statistic instead of Ansari–Bradley.) Since T_{AB} (the sum of the second sample's scores) is large when the second sample is *less* dispersed, here it is small when $\gamma > 1$.

A permutation test can be run with each statistic. Under H_0 the locations differ, so $\{X_i, Y_j\}$ are not exchangeable, but the location-adjusted deviations $X'_i = X_i - \text{median}\{X_i\}$, $Y'_j = Y_j - \text{median}\{Y_j\}$ are (approximately) exchangeable, and shuffling them gives an approximate permutation test. (With the F statistic (4.9) use the mean $\bar{X}$ instead of the median median$\{X_i\}$ to form X'_i.)

The procedure of the approximate permutation test with the Ansari–Bradley statistic: first compute T_{AB} from the adjusted observations $\mathbb{X}' = \{X'_1, \ldots, X'_m, Y'_1, \ldots, Y'_n\}$. Then, over all shuffles of the adjusted observations $\mathbb{X}'$ (or M random shuffles), compute the Ansari–Bradley statistic repeatedly, and use the distribution of $\{T^{*i}_{AB} : i = 1, \ldots, M\}$ as the null distribution. For $H_0 : \gamma = 1$ against $H_1 : \gamma > 1$, smaller T_{AB} is more extreme, so $p = M^{-1} \sum_{i=1}^{M} I(T^{*i}_{AB} \le T_{AB})$. (The book writes "$\ge$"; with T_{AB} the sum of the second sample's scores, $\gamma > 1$ makes it small.)

Example 4.13

For the acute leukemia data of Example 4.1, test whether the dispersions of ALL and AML differ. Vector `x` holds the ALL expression values and `y` the AML ones. First, the F-test with the ratio F (4.9) of the sample variances gives a fairly large p-value, not enough evidence that the variances differ. The samples are large enough to trust the F-test.

```
> var.test(x,y)$p.value
[1] 0.1986019
```

Next, median-adjusted statistics. The two locations cannot be assumed equal, so instead of the Ansari–Bradley test directly we must use the median-adjusted observations:

```
> ansari.test(x - median(x), y - median(y))

        Ansari-Bradley test

data:  x - median(x) and y - median(y)
AB = 933, p-value = 0.1347
alternative hypothesis: true ratio of scales is not equal to 1
```

Without median adjustment the differences in scale and location are confounded and cannot be distinguished. A wrong example:

```
> ansari.test(x,y)

        Ansari-Bradley test

data:  x and y
AB = 958.5, p-value = 0.03518
alternative hypothesis: true ratio of scales is not equal to 1
```

The Siegel–Tukey function `SiegelTukeyTest()` has an argument for median adjustment; the two calls below give the same result.

```
DescTools::SiegelTukeyTest(x,y,adjust.median = TRUE)
DescTools::SiegelTukeyTest(x - median(x), y - median(y))
```

4.5 The Kolmogorov–Smirnov test

So far we have looked at tests of location and dispersion of two populations. This section examines the **Kolmogorov–Smirnov** (KS) test, which tests a broad alternative of difference between two populations without fixing one parameter (location or dispersion).

With distribution functions F, G of the two populations and observations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$, we test

$$H_0 : F = G \quad \text{vs.} \quad H_1 : F \neq G.$$

The alternative is not written in terms of a particular parameter, so under it G and F may differ in shape even with equal means and variances (e.g. G more right-skewed than F). Then permutation tests using the Wilcoxon rank-sum or Ansari–Bradley statistic, sensitive to the shift Δ or scale γ, have very low power and may fail to reject H_0 even when the distributions are very different.

The Kolmogorov–Smirnov statistic directly compares the **empirical distribution functions** obtained from each sample. First, the empirical distribution function $\hat{F}_X$ of a sample $X_1, \ldots, X_m$. The (cumulative) distribution function of a random variable X (or a population) gives for each $x \in (-\infty, \infty)$ the probability $F(x) = P(X \leq x)$ that $X \leq x$. The true F is unknown, so a good estimator of $F(x)$ is the proportion of observations at most x,

$$\hat{F}_X(x) = \hat{P}(X \leq x) = \frac{1}{m} \sum_{i=1}^{m} I(X_i \leq x),$$

the empirical distribution function. E.g. Figure 4.9 of the book compares the empirical distribution function of $m = 40$ observations randomly drawn from the standard normal with the standard normal distribution function.

At each x, $\hat{F}_X(x)$ lies around the true $F_X(x)$. Indeed $E(\hat{F}_X(x)) = F_X(x)$ and $\text{Var}(\hat{F}_X(x)) = F_X(x)(1 - F_X(x))/n$ (here $n = m$, the sample size), so for large samples the empirical distribution function is very close to the true one.

To compare the distributions of two samples, compare their empirical distribution functions. Figure 4.10 of the book shows those of the ALL and AML samples of the acute leukemia patients of Example 4.1: AML values tend to be larger than ALL values, and the larger the difference of the two empirical distribution functions (the vertical distance between the lines), the more we believe the distributions differ.

The **Kolmogorov–Smirnov statistic** (KS statistic) K is the maximal difference of the two empirical distribution functions $\hat{F}_X(x)$ and $\hat{F}_Y(x)$, scaled using the sample sizes:

$$K := \sqrt{\frac{mn}{m+n}} \sup_x \{|\hat{F}_X(x) - \hat{F}_Y(x)|\}.$$

Under $H_0 : F = G$ both $\hat{F}_X(x)$ and $\hat{F}_Y(x)$ approach $F(x)$, so the statistic is multiplied by $\sqrt{\frac{mn}{m+n}} = \left(\frac{1}{m} + \frac{1}{n}\right)^{-1/2}$ to keep it from becoming too small. The two empirical distribution functions are step functions with at most $m+1$ and $n+1$ distinct values, so to compute K we need not examine the difference at every x, only at each observation X_i, Y_j. When the populations differ (the alternative is true), whatever the form of the difference, K tends to be large, so the KS statistic tests any form of alternative. But because the KS test is sensitive to every possible alternative, it is less powerful than tests sensitive only to specific alternatives (e.g. the Wilcoxon rank-sum test, sensitive only to location differences).

The KS test rejects H_0 when K is large. The null distribution of K is known to depend only on m, n, whatever the null distribution $F = G$: the KS test is distribution-free. The exact null distribution of K is complicated to describe, but if the null distribution $F = G$ is continuous, its large-sample distribution is the Kolmogorov distribution.[27] R's `ks.test()` uses the exact null distribution of K for small samples without ties, and otherwise the Kolmogorov distribution, the large-sample distribution of K. Alternatively a (random) permutation test with the KS statistic K can be run.

Example 4.14

For the two empirical distribution functions of the acute leukemia patients of Example 4.1, the maximal difference is $D = \sup_x\{|\hat{F}_X(x) - \hat{F}_Y(x)|\} = \max_{i,j}\{|\hat{F}_X(z) - \hat{F}_Y(z)| : z = X_i \text{ or } Y_j\} = 0.41617$, so the observed KS statistic is $K_{\text{obs}} = \sqrt{\frac{mn}{m+n}} D = \sqrt{\frac{47\times 25}{47+25}} \times 0.41617 \approx 1.68$. Compared with the Kolmogorov distribution, the large-sample distribution of K (Figure 4.11 of the book), $p = P(K \geq K_{\text{obs}}) \approx 0.007$: strong evidence that the gene expression of the two types of acute leukemia patients differs. R's `ks.test()` approximates the p-value simply; vector `x` holds the ALL values and `y` the AML values.

```
> ks.test(x,y)

        Two-sample Kolmogorov-Smirnov test

data:  x and y
D = 0.41617, p-value = 0.007014
alternative hypothesis: two-sided
```

4.6 Two-sample tests: which test to use?

Which test has the highest power while controlling the type I error when comparing the locations of two populations? In nonparametric statistics, with its very broad range of populations, the answer is not simple. The Wilcoxon–Mann–Whitney test, or permutation tests, require the two distributions to be equal under the null and can be used only when this assumption holds. In this section, for tests of the locations of two populations, we treat separately the case where the distributions are equal under the null and the case where under the null the locations are equal but the dispersions differ.

[27] If the population $F = G$ under the null is continuous, the asymptotic distribution of the KS statistic is $\lim_{m,n\to\infty} P(K \leq x) = 1 - 2\sum_{j=1}^{\infty}(-1)^{j-1}\exp(-2j^2x^2)$, the Kolmogorov distribution.

4.6.1 Comparison in the location-shift model

Assume observations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} G$ and a shift Δ with

$$X_i \overset{d}{=} Y_j - \Delta.$$

That is, the mean (or median) of G exceeds that of F by Δ, a location-shift model in which the difference of the two populations is described only by Δ. When $H_0 : \Delta = 0$ is true, the two distributions are equal.

We compare four tests: (1) the two-sample t-test, (2) the Wilcoxon–Mann–Whitney rank-sum test, (3) the permutation test with the two-sample t-statistic (t-permutation test) and (4) the permutation test with the difference of medians (m-permutation test).

Type I error. All four tests control the type I error approximately below α. The nonparametric rank-sum test and the permutation tests are theoretically exact, so they control the type I error for any population. The two-sample t-test controls it exactly only for normal populations; for populations with thinner tails than normal (e.g. uniform) its type I error is not much larger than α, and for large m, n it controls it approximately. (The same held for one-sample tests of location.) For heavy-tailed or asymmetric populations (e.g. Laplace or exponential) the two-sample t-test is rather conservative (type I error $< \alpha$).

Power comparison. Before comparing, recall that for large samples the two-sample t-test and the permutation test with the two-sample t-statistic are similar. The table below compares which test is most powerful in which situation (Keller-Mcnulty and Higgins, 1987; Higgins, 2004).

	small samples	large samples
thin tails (uniform)	t-perm. $>$ t-test $>$ Wilcoxon	t-test $\approx$ t-perm.
normal	t-test	t-test $\approx$ t-perm.
heavy tails (Laplace)	Wilcoxon $>$ t-test $\approx$ t-perm.[28]	Wilcoxon
very heavy tails (Cauchy)	m-perm. $>$ Wilcoxon $>$ t-test	m-perm.

Small samples are $m + n = 12$ and large samples $m + n = 36, 128$ (same conclusions in both cases). For very large samples the comparison can be summarized by the asymptotic relative efficiency. The ARE of the Wilcoxon–Mann–Whitney test relative to the two-sample t-test:

	ARE(Wilcox, t)
uniform	1
normal	$3/\pi \approx 0.955$
Laplace	1.5
exponential	3
Cauchy	∞

(The book prints 0.995 for the normal; the value is $3/\pi \approx 0.955$, as for the signed-rank test in Section 2.3.2. In the shift model the ARE is $12\sigma^2(\int f^2)^2$, giving these entries.) Except for the normal, the Wilcoxon rank-sum test has higher ARE, and even when normality holds its efficiency is not much lower than that of the t-test.

[28] "Wilcoxon $>$ t-test $\approx$ t-perm." means the Wilcoxon test is more powerful than the t-test, whose power is similar to that of the t-permutation test. The m-permutation test, not listed, is less powerful than the other tests.

4.6.2 When the two distributions differ under the null hypothesis

Suppose the means and variances of the two populations are (μ_1, σ_1^2), (μ_2, σ_2^2) and the variances cannot be assumed equal:

$$\sigma_1^2 \neq \sigma_2^2.$$

Then even when the null hypothesis $H_0 : \mu_1 = \mu_2$ about location is true, the two distributions differ, so neither the Wilcoxon–Mann–Whitney test nor any permutation test is guaranteed to control the type I error.

If the populations follow normality approximately, or the samples are large enough, Welch's two-sample t-test can be used. If the samples are not large enough or normality is badly violated, first try a transformation of the variables (e.g. $x \mapsto \log(x)$) so that the transformed samples approximately follow normality. As a rank-based approximate distribution-free method, a standardized Wilcoxon rank-sum statistic as in Chung and Romano (2016) can be used; comparative studies of the power of such methods are insufficient and are not discussed here.

Chapter 5. Association and Independence of Two Variables

Overview

Inference about the relation between two variables is a very important problem in statistics. E.g. to see whether university entrance exam scores are related to academic performance at university, we infer the association between exam scores and university GPA. When both variables are numerical, the **correlation coefficient** is the standard measure of their relation. This chapter covers nonparametric correlation coefficients that can replace the sample correlation, and (distribution-free) tests of independence of two variables based on them.

Example 5.1

The data below are the teacher's ranking and the reading-test scores of 10 children. We want to know whether the teacher's evaluation agrees with the test scores. Is the sample correlation a good measure of association for these data?

teacher's rank	1	2	3	4	5	6	7	8	9	10
test score	10	20	25	50	80	82	85	90	95	100

Example 5.2

Figure 5.1 of the book is a scatter plot of the happiness score (Score) and the GDP per capita of the 13 happiest countries of the 2020 World Happiness Report. Does happiness increase with GDP?

5.1 Correlation and simple linear regression

5.1.1 Pearson's correlation coefficient

The most standard measure of association between random variables X, Y is Pearson's correlation coefficient. With $\mu_X = E(X)$, $\sigma_X^2 = \mathrm{Var}(X)$, $\mu_Y = E(Y)$, $\sigma_Y^2 = \mathrm{Var}(Y)$,

$$\rho = \mathrm{Corr}(X, Y) = \frac{E\{(X - \mu_X)(Y - \mu_Y)\}}{\sigma_X \sigma_Y} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

measures the *linear* association of the two variables, and satisfies:

- $-1 \le \rho \le 1$;
- if X, Y are independent, $\rho = 0$;
- $\rho > 0$ means positive correlation (Y tends to increase with X), $\rho < 0$ negative correlation.

Pearson's ρ is determined by the population of the pair (X, Y) observed together (here the joint distribution of (X, Y)). If the population is bivariate normal,[29] $\rho = 0$ means the two variables are independent. If not, even a small ρ does not mean independence. E.g. if $Y = (X - \mu_X)^2$, Y is exactly determined by X (a very strong association), but the correlation, which measures only linear association, can be $\rho = 0$.

[29] If a random vector $\mathbf{x} = (X_1, \ldots, X_p)'$ is multivariate normal with mean $\mu = (E(X_1), \ldots, E(X_p))'$ and covariance matrix $\Sigma = (\sigma_{ij})_{i,j=1,\ldots,p}$, $\sigma_{ij} = \mathrm{Cov}(X_i, X_j)$, its joint density is $f(x_1, \ldots, x_p) = \frac{1}{\sqrt{|2\pi\Sigma|}} \exp\{-\frac{1}{2}(\mathbf{x} - \mu)'\Sigma^{-1}(\mathbf{x} - \mu)\}$.

From bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ the **sample correlation coefficient** is

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{S_{XY}}{\sqrt{S_{XX} S_{YY}}}.$$

Here $S_{XY} = \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$; divided by $n-1$ it is the sample covariance $\widehat{\mathrm{Cov}}(X, Y) = S_{XY}/(n-1)$, and S_{XX}, S_{YY} divided by $n-1$ are the sample variances. If $(X_i, Y_i) \overset{\text{i.i.d.}}{\sim} F$ and the population has Pearson correlation ρ, the sample correlation r estimates ρ.

Whether or not the population is normal, Pearson's ρ or the sample r is the most frequently used measure of association, mainly because it measures only linear association and so is easy and intuitive to interpret. "Correlation coefficient" refers to Pearson's.

5.1.2 Correlation and simple linear regression

To study the association of two variables, the relation of the pairs (X_i, Y_i) can be modeled by simple linear regression:

$$Y_i = \alpha + \beta X_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F, \quad i = 1, \ldots, n. \tag{5.1}$$

The least squares estimator $\hat{\beta}$ of the slope of the line $y = \alpha + \beta x$ is closely related to the sample correlation r:

$$\hat{\beta} = \frac{S_{XY}}{S_{XX}} = r\sqrt{\frac{S_{YY}}{S_{XX}}}.$$

If the error ϵ_i in (5.1) is normal and X_i is normal too, (X_i, Y_i) is bivariate normal. Under this normality, the slope β and the correlation ρ are related by $\beta = \rho\sigma_X/\sigma_Y$ (more precisely $\beta = \mathrm{Cov}(X, Y)/\sigma_X^2 = \rho\sigma_Y/\sigma_X$).

Drawing the fitted simple regression line on the scatter plot as a trend line, its slope is a good measure of correlation.

Example 5.3 (Example 5.2 continued)

Adding the fitted simple regression line to the data on happiness score (Score) and GDP per capita gives an increasing trend line, so there seems to be a positive correlation.

5.1.3 Spearman's rank correlation

Computing the sample correlation after rank-transforming the data gives **Spearman's rank correlation coefficient** r_s. Rank-transform the variable $(X_1, \ldots, X_n)$ to $(R_1, \ldots, R_n)$ and $(Y_1, \ldots, Y_n)$ to $(S_1, \ldots, S_n)$, and form the rank-transformed data

$$\{(R_1, S_1), \ldots, (R_n, S_n)\};$$

their sample correlation, Spearman's rank correlation, is

$$r_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2 \sum_{i=1}^n (S_i - \bar{S})^2}}.$$

It lies in $-1 \le r_s \le 1$; $r_s > 0$ means positive correlation and $r_s < 0$ negative. While the sample (Pearson) correlation measures only linear trends, Spearman's rank correlation measures any monotone increasing or decreasing trend, even nonlinear.

Example 5.4

To measure the association of "teacher's rank" and test score in Example 5.1, the sample correlation $r = 0.95$ shows a strong linear association. Spearman's rank correlation is $r_s = 1$: a perfect positive correlation (though not linear). The scatter plot (Figure 5.2 of the book) shows the relation clearly. Both are computed with R's `cor()`: `method = "pearson"` (or no argument) gives the sample correlation and `method = "spearman"` the rank correlation.

```
> ranks = 1:10
> score = c(10,20,25,50,80,82,85,90,95,100)
> cor(ranks,score)
[1] 0.948262
> cor(ranks,score, method = "spearman")
[1] 1
```

The sample correlation of the rank-transformed `ranks` and `score` is the rank correlation (`ranks` are already ranks):

```
> r = ranks
> (s = rank(score))
 [1]  1  2  3  4  5  6  7  8  9 10
> cor(r,s)
[1] 1
```

Spearman's rank correlation is a statistic computed from the data; there is no population parameter it corresponds to. If the two variables have a monotone trend (increasing or decreasing), $|r_s|$ is large; if they are independent or have no monotone trend, $r_s \approx 0$ tends to hold. Unlike the sample correlation, which applies only to numerical variables, it applies to ordinal categorical variables (e.g. very satisfied, satisfied, neutral, dissatisfied, very dissatisfied). Finally, the rank correlation is robust to outliers or influential points: in the scatter plot of Example 5.2, removing the last point on the right reveals a stronger monotone increase; if it is an outlier, the rank correlation is less affected by it.

Example 5.5

For the data of Example 5.2, the sample correlation is $r = 0.30$ and the rank correlation $r_s = 0.65$. The second variable `GDP` has ties; we check how they are handled.

```
> wh = read.csv("data/whappiness.csv")
> (r = rank(wh$Score))
 [1] 13 12 11 10  9  8  7  6  5  4  3  2  1
> (s = rank(wh$GDP.per.capita))
 [1]  8.5 11.0  8.5 12.0 10.0  7.0  2.0  6.0 13.0  4.0  5.0  3.0
[13]  1.0
> cor(r,s, method = "pearson")
[1] 0.6492441
> cor(wh$Score,wh$GDP.per.capita, method = "spearman")
[1] 0.6492441
```

5.2 Tests based on correlation coefficients

Consider tests of association of two variables using the sample correlation r or Spearman's r_s. The broadest hypotheses are

$$H_0 : \text{the two variables are independent} \quad \text{vs} \quad H_1 : \text{the two variables are related.}$$

If the population is bivariate normal, independence means $\rho = 0$, so the more specific hypotheses about ρ,

$$H_0 : \rho = 0 \quad \text{vs} \quad H_1 : \rho \neq 0, \tag{5.2}$$

are hypotheses of independence. One-sided alternatives $H_1 : \rho < 0$ or $H_1 : \rho > 0$ are also possible. Even if the population is not bivariate normal, (5.2) about ρ can be tested with the sample correlation.

5.2.1 When normality can be assumed

First the test based on the sample correlation r. When H_0 is true and normality holds, the increasing function of r

$$T = r\sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2} \tag{5.3}$$

is known to have exactly the t_{n-2} null distribution.

Proof (supplied). Condition on $X_1, \ldots, X_n$. Under bivariate normality with $\rho = 0$, $Y_1, \ldots, Y_n$ are i.i.d. normal and independent of the X's, i.e. the regression model (5.1) holds with normal errors and $\beta = 0$. In that model the usual t-statistic for $H_0 : \beta = 0$, $\hat{\beta}/\widehat{\text{se}}(\hat{\beta})$, is t_{n-2}-distributed. Since $\hat{\beta} = S_{XY}/S_{XX}$, the residual sum of squares is $SSE = S_{YY} - S_{XY}^2/S_{XX} = S_{YY}(1-r^2)$ and $\widehat{\text{se}}(\hat{\beta})^2 = SSE/((n-2)S_{XX})$, so

$$\frac{\hat{\beta}}{\widehat{\text{se}}(\hat{\beta})} = \frac{r\sqrt{S_{YY}/S_{XX}}}{\sqrt{S_{YY}(1-r^2)/((n-2)S_{XX})}} = r\sqrt{\frac{n-2}{1-r^2}} = T.$$

The conditional distribution t_{n-2} does not depend on the X's, so it is also the unconditional distribution. ■

So if the observed sample correlation is r_{obs}, the two-sided p-value for (5.2) is

$$p = P(|R| \geq |r_{\text{obs}}| \mid H_0) = P(|T| \geq |t_{\text{obs}}|) = 2\{1 - F_{t_{n-2}}(|t_{\text{obs}}|)\}, \tag{5.4}$$

where $F_{t_{n-2}}$ is the distribution function of t_{n-2} and $t_{\text{obs}} = r_{\text{obs}}\sqrt{\frac{n-2}{1-r_{\text{obs}}^2}}$.

Even if the population is not normal, for large n the test is approximately exact by the CLT. In this situation, however, the t-test is a test about the correlation ρ, not a test of independence of the two variables.

Example 5.6

Treat the teacher's ranks of Example 5.1 as if they were continuous observations, and test the hypotheses (5.2). In words, H_0 is "the teacher's rank and the test score are unrelated" and H_1 is "they are related." First compute the sample correlation r_{obs}:

```
> n = 10
> (r.obs = cor(ranks, score))
[1] 0.948262
```

Using (5.4), the value of the t-statistic and the two-sided p-value:

```
> (t = r.obs * sqrt((n-2)/(1-r.obs^2)))
[1] 8.447828
> 2*(1-pt(t, df = n-2))
[1] 2.944375e-05
```

A very small p-value: strong evidence that the teacher's rank and the test score are related. (But here the teacher's rank is treated as continuous, so the normality the t-test requires is not satisfied and this is not a good test.) `cor.test()` gives the same result:

```
> cor.test(ranks,score)

        Pearson's product-moment correlation

data:  ranks and score
t = 8.4478, df = 8, p-value = 2.944e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.7907589 0.9880014
sample estimates:
     cor
0.948262
```

The confidence interval in this output is an asymptotically approximate interval derived from (5.3).

5.2.2 Permutation tests

When the sample is not large enough for the CLT approximation, a permutation test with r as statistic can be performed. Under H_0: "the two variables are independent," the joint distributions of (X_1, Y_1) and (X_1, Y_2) are the same, so shuffling only one of the variables (e.g. Y) gives exchangeable observations. (If the variables are not independent, (X_1, Y_1) and (X_1, Y_2) have different distributions.) So for a permutation test of correlation, keep the "X" observations fixed, shuffle all n "Y" observations, and use the permutation distribution of the sample correlations r_i of the shuffles as the null distribution of r.

The table below shows, for the data of Example 5.1, two of the $n!$ possible shuffles and the statistics computed from them:

child	1	2	3	4	5	6	7	8	9	10	correlation
teacher's rank (x)	1	2	3	4	5	6	7	8	9	10	
test score (y)	10	20	25	50	80	82	85	90	95	100	$r_{\text{obs}} = 0.95$
shuffled scores	20	10	25	100	90	80	95	50	85	82	$r_1 = 0.64$
shuffled scores	80	10	100	85	90	25	50	82	20	95	$r_2 = -0.02$

The permutation test with r compares H_0: "the two variables are independent" with $H_1 : \rho \neq 0$ (or one-sided). The null is not "$\rho = 0$": if $\rho = 0$ but the variables are dependent, the shuffles above are not exchangeable. The alternative is stated specifically as "linear association" because r measures only linear correlation, so for nonlinear but related variables the power may be very low.

Example 5.7

With the data of Example 5.1, test independence of the two variables by the permutation test with the sample correlation r. All possible shuffles number $10! = 3628800$, too many, so we run a random permutation test. The R code repeats $M = 1999$ random permutations, shuffling only the values of `score` each time and recomputing the sample correlation under the null.

```
> M = 1999
> r.perm = replicate(M, {
+   perm.id = sample(1:n,n,replace = FALSE)
+   cor(ranks, score[perm.id])})
> (p.perm = 2*mean(c(r.obs,r.perm) >= r.obs))
[1] 0.001
```

(The book's figure shows the random permutation null distribution with the observed statistic `r.obs`.)

For large n the permutation distribution of r can be approximated by a normal. In the permutation distribution of

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{S_{XX} S_{YY}}},$$

the X_i are fixed and only the $(Y_1, \ldots, Y_n)$ are redrawn by random sampling without replacement from the finite population $\{Y_1, \ldots, Y_n\}$. Under any random draw (any shuffle) S_{XX} and S_{YY} do not change. The population mean of this finite population is $\bar{Y}$, so $E(Y_i) - \bar{Y} = 0$ and

$$E(r) = \frac{1}{\sqrt{S_{XX} S_{YY}}} \sum_{i=1}^n (X_i - \bar{X})\{E(Y_i) - \bar{Y}\} = 0.$$

Moreover, whatever the actual observations (X_i, Y_i), the variance of the permutation distribution is always

$$\mathrm{Var}(r) = \frac{1}{n-1}. \tag{5.5}$$

Proof of (5.5) (supplied). Let $a_i = X_i - \bar{X}$, $b_i = Y_i - \bar{Y}$ (so $\sum a_i = \sum b_i = 0$) and let π be a uniformly random permutation; $r = \sum_i a_i b_{\pi(i)} / \sqrt{S_{XX} S_{YY}}$. For each i, $b_{\pi(i)}$ is uniform on $\{b_1, \ldots, b_n\}$, so $\mathrm{Var}(b_{\pi(i)}) = \frac{1}{n} \sum b_k^2 = S_{YY}/n$; for $i \neq j$, $(b_{\pi(i)}, b_{\pi(j)})$ is uniform over ordered pairs of distinct indices, so $\mathrm{Cov}(b_{\pi(i)}, b_{\pi(j)}) = \frac{1}{n(n-1)} \sum_{k \neq l} b_k b_l = -\frac{S_{YY}}{n(n-1)}$. Using $\sum_{i \neq j} a_i a_j = -\sum_i a_i^2 = -S_{XX}$,

$$\mathrm{Var}\Big(\sum_i a_i b_{\pi(i)} \Big) = S_{XX} \frac{S_{YY}}{n} + S_{XX} \frac{S_{YY}}{n(n-1)} = \frac{S_{XX} S_{YY}}{n-1}.$$

Dividing by $S_{XX} S_{YY}$ gives (5.5). ■

These mean and variance are of the permutation distribution of r conditional on the observations (X_i, Y_i) (random shuffles of the order of the Y_i), not of the sampling distribution of r (with random (X_i, Y_i)).

For r following its permutation distribution, the finite-population CLT gives for large n

$$r\sqrt{n-1} \xrightarrow{d} N(0,1) :$$

the permutation distribution of r is approximately $N(0, \frac{1}{n-1})$, which gives a test. Figure 5.3 of the book compares the permutation distribution of Example 5.7 ($n = 10$) with the density of $N(0, 1/9)$: even for this very small sample the difference is hard to see by eye (in fact it is larger in the tails). With more observations the approximation is better.

5.2.3 Permutation tests with Spearman's rank correlation

Spearman's rank correlation is the sample correlation of the rank-transformed (R_i, S_i) of (X_i, Y_i), so tests proceed exactly like the permutation test with the sample correlation above. (Even the mean and variance of the large-sample approximation are the same.) The hypotheses here are

$$\begin{aligned} &H_0 : \text{the two variables are independent} \quad \text{vs} \\ &H_1 : \text{the two variables have a monotone increasing or decreasing correlation.} \end{aligned} \tag{5.6}$$

Correlation measures only linear association, while Spearman's rank correlation measures monotone relations, even nonlinear. A test with the rank correlation and its null distribution is unchanged by any monotone transformation of the observations (X_i, Y_i) (e.g. $x_i \mapsto x_i^2$, $y_i \mapsto \sqrt{y_i}$ when all x_i, y_i are positive), so it uses a distribution-free statistic and is a distribution-free test.

Example 5.8

For the data of Example 5.2, investigate with a permutation test based on the rank correlation whether the two variables are related. The null distribution is approximated by $M = 1999$ random permutations. The R code rank-transforms in advance, saving the work of ranking in every repetition.

```
> n = nrow(wh)
> r = rank(wh$Score)
> s = rank(wh$GDP.per.capita)
> (rs.obs = cor(r,s))
[1] 0.6492441
> rs.perm = replicate(1999, cor(r, s[sample(1:n,n,replace = F)])
  )
> (p.perm = 2*mean(c(rs.obs,rs.perm) >= rs.obs))
[1] 0.02
```

The p-value is 0.02, so at the 5% level we conclude the two variables are not independent. One may also use

```
cor.test(r,s, method = "spearman")
```

(these data have ties, so `cor.test()` cannot compute an exact p-value).

5.3 Kendall's tau

Another distribution-free measure of association of two variables is **Kendall's tau** (τ).

For bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ Kendall's tau is defined as follows. If two variables are positively correlated, then picking two observations (X_i, Y_i) and (X_j, Y_j), one point tends to be lower left and the other upper right: as one variable increases so does the other, and the pair is **concordant**. If they are negatively correlated, for two observations one point tends to be upper left and the other lower right: as one increases the other decreases, and the pair is **discordant**. In the data of Figure 5.2 all pairs of observations are concordant.

The average over all pairs of

$$U_{ij} = \begin{cases} +1, & (X_i - X_j)(Y_i - Y_j) > 0; \\ -1, & (X_i - X_j)(Y_i - Y_j) < 0; \\ 0, & (X_i - X_j)(Y_i - Y_j) = 0, \end{cases} \qquad 1 \le i < j \le n, \tag{5.7}$$

which is $+1$ for concordant and -1 for discordant pairs, is Kendall's tau computed from the sample,

$$\tau_n = \frac{1}{\binom{n}{2}} \sum_{i<j} U_{ij}. \tag{5.8}$$

With N_{Conc} concordant and N_{Disc} discordant pairs, $\sum_{i<j} U_{ij} = N_{\text{Conc}} - N_{\text{Disc}}$. If there are no "tied" pairs, i.e. none with $(X_i - X_j)(Y_i - Y_j) = 0$, then $N_{\text{Conc}} + N_{\text{Disc}} = \binom{n}{2}$, so

$$\tau_n = 2\frac{N_{\text{Conc}}}{\binom{n}{2}} - 1.$$

Example 5.9

Compute Kendall's τ_n for the data of Example 5.2. First generate the ids of the $\binom{13}{2} = 78$ pairs of observations, then compute the U_{ij}:

```
> n = nrow(wh)
> allpairs = gtools::combinations(n,2,1:n)
> head(allpairs,3)
     [,1] [,2]
[1,]    1    2
[2,]    1    3
[3,]    1    4
> U = apply(allpairs,1,function(pair.id){
+   sign( diff(wh$Score[pair.id]) * diff(wh$GDP.per.capita[pair.
    id]) )
+ })
```

There are $\#\{U_{ij} = 1\} = 58$ concordant pairs, $\#\{U_{ij} = -1\} = 19$ discordant pairs, and $\#\{U_{ij} = 0\} = 1$ tied pair. By (5.8):

```
> (tau = sum(U) / choose(n,2))
[1] 0.5
```

R's `cor()` with `method = "kendall"` gives a slightly different value. With ties, `cor()` computes Kendall's tau by a method of Kendall (1945) slightly different from (5.8); without ties the results agree.

```
> cor(wh$Score,wh$GDP.per.capita, method = "kendall")
[1] 0.5032363
```

Kendall's τ_n resembles the rank correlation r_s but differs. In common, both measure the monotone relation of two variables (on the same $[-1, 1]$ scale as Pearson's correlation) and are robust to outliers. The rank correlation is the sample correlation of the rank-transformed data, while Kendall's statistic uses sign-transformed data, so the values necessarily differ. The most important difference is that Kendall's tau has a corresponding parameter (the rank correlation does not). For two independent random vectors $(X, Y), (X', Y')$ drawn from a population, Kendall's tau parameter of the population is

$$\tau = P\{(X - X')(Y - Y') > 0\} - P\{(X - X')(Y - Y') < 0\} = 2P\{(X - X')(Y - Y') > 0\} - 1$$

(the second form for continuous populations). The parameter τ is interpreted as:

- $\tau > 0$: the proportion (probability) of concordant pairs exceeds that of discordant pairs, a positive correlation;
- $\tau < 0$: a negative correlation;
- $\tau = 0$: concordant and discordant pairs are equally likely, and the two variables are unrelated (in this sense).

The parameter τ and the statistic τ_n play a role like correlation but are generally not called correlation coefficients, only "Kendall's tau"; as a nonparametric measure of correlation, being robust, it is more widely used than the rank correlation. If the population is bivariate normal, τ is determined by the population parameter ρ (correlation):

$$\tau = \frac{2}{\pi} \arcsin(\rho). \tag{5.9}$$

Proof of (5.9) (supplied). $(U, V) = (X - X', Y - Y')$ is bivariate normal with mean 0 and correlation ρ; standardize so $U = Z_1$, $V = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$ with Z_1, Z_2 i.i.d. $N(0, 1)$. Write

$(Z_1, Z_2) = R(\cos\Theta, \sin\Theta)$ with Θ uniform on $[0, 2\pi)$, and $\rho = \sin\phi$, $\sqrt{1-\rho^2} = \cos\phi$, $\phi = \arcsin\rho \in [-\pi/2, \pi/2]$. Then $V = R\sin(\Theta + \phi)$, and $UV > 0$ iff $\cos\Theta$ and $\sin(\Theta+\phi)$ have the same sign. The set of such Θ is two arcs of total length $\pi + 2\phi$, so $P(UV > 0) = \frac{1}{2} + \frac{\arcsin\rho}{\pi}$ (Sheppard's formula), and $\tau = 2P(UV > 0) - 1 = \frac{2}{\pi}\arcsin\rho$. ■

If the data are normal but contain outliers, rather than estimating ρ by the sample correlation r, it is more robust to estimate τ by Kendall's τ_n and estimate ρ by $\hat{\rho} := \sin(\frac{\pi}{2}\tau_n)$.

5.3.1 Tests of independence with Kendall's tau

The null hypothesis "the two variables are independent" of (5.6) against the alternative "the two variables have a monotone increasing or decreasing relation" can be tested with Kendall's τ_n. Fixing one variable and shuffling the other (as in the permutation test with correlation), generating the permutation distribution of τ_n and computing p-values gives an exact permutation test. When H_0 is true (the variables are independent), $\tau = 0$, so if the test fails to reject H_0 we may accept $\tau = 0$; if it accepts the alternative, there is strong evidence that $\tau \neq 0$.

Without a permutation test, one can use as null distribution the sampling distribution of τ_n when H_0 (where only $\tau = 0$ is needed) is true; this sampling distribution is in fact the permutation distribution above.[30] Since τ_n depends only on the signs of differences of observations,[31] its sampling distribution (permutation distribution) is distribution-free, but for large n the exact sampling distribution is hard to obtain. So with many observations, the traditional test of independence with Kendall's tau uses the large-sample normal approximation

$$\tau_n \dot{\sim} N\Big(0, \frac{4n+10}{9n(n-1)}\Big) \tag{5.10}$$

as null distribution.[32]

Variance in (5.10) (supplied). Assume independence and continuous margins. Each U_{ij} is ± 1 with probability $1/2$, so $EU_{ij} = 0$ and $\operatorname{Var} U_{ij} = 1$. Pairs $\{i,j\}, \{k,l\}$ with no common index give independent U's. For pairs sharing one index, e.g. U_{12}, U_{13}, independence of X and Y gives $E(U_{12}U_{13}) = E[\operatorname{sgn}(X_1 - X_2)\operatorname{sgn}(X_1 - X_3)]\,E[\operatorname{sgn}(Y_1 - Y_2)\operatorname{sgn}(Y_1 - Y_3)]$. Conditional on X_1, $\operatorname{sgn}(X_1 - X_2)$ has mean $2F(X_1) - 1$, so the first factor is $E(2F(X_1) - 1)^2 = E(2V - 1)^2 = 1/3$ (V uniform), and likewise the second. So $\operatorname{Cov}(U_{12}, U_{13}) = 1/9$. There are $\binom{n}{2} \cdot 2(n-2)$ ordered pairs of pairs sharing exactly one index, so

$$\operatorname{Var}\Big(\sum_{i<j} U_{ij}\Big) = \binom{n}{2} + \binom{n}{2}\frac{2(n-2)}{9} = \binom{n}{2}\frac{2n+5}{9},$$

and $\operatorname{Var}(\tau_n) = \frac{2n+5}{9\binom{n}{2}} = \frac{2(2n+5)}{9n(n-1)} = \frac{4n+10}{9n(n-1)}$. Asymptotic normality follows from the theory of U-statistics (Hoeffding). ■

Figure 5.4 of the book compares, for $n = 10$, the null distribution of Kendall's statistic (Kendall's tau distribution, equal to the permutation distribution) with the approximating normal (5.10). The normal approximation is fairly good even for small samples, but the tails differ slightly.

Example 5.10

To test independence of the two variables of Example 5.2 with Kendall's tau, we run (1) a permutation test, (2) the test with Kendall's tau distribution and (3) the test with the normal approximation, computing from the ranks of the observations only.

[30] Without ties, the permutation distribution of τ_n is also called Kendall's tau distribution. R's `SuppDists` package gives its probability mass function via `dKendall()`.

[31] $(X_i - X_j)(Y_i - Y_j) > 0$ when the signs of $X_i - X_j$ and $Y_i - Y_j$ agree, so τ_n depends only on signs.

[32] For the computation of the variance of τ_n see Exercise 5.4 of the book; a derivation is supplied below.

```
> n = nrow(wh)
> r = rank(wh$Score)
> s = rank(wh$GDP.per.capita)
> (tau.obs = cor(r,s, method = "k"))
[1] 0.5032363
```

These data have ties, so the permutation test is the most accurate, but there are too many shuffles, so we run a random permutation test with $M = 1999$ shuffles (computing the p-value assuming the permutation distribution is symmetric). With T_n following Kendall's tau distribution, the p-value of the test with Kendall's tau distribution is $2P(T_n \geq \tau_n)$. With ties this p-value is not exact, but with only one tied pair the test is approximately exact. `pKendall()` of the `SuppDists` package gives the distribution function of Kendall's tau distribution. Finally the test with the normal approximation compares the observed τ_n with (5.10).

```
> # (1) random permutation
> tau.perm = replicate(1999, cor(r, s[sample(1:n,n,replace = F)
    ], method = "k"))
> p.perm = 2*mean(c(tau.obs,tau.perm) >= tau.obs)
>
> # (2) Kendall's tau distribution
> p.tau = 2*(1- SuppDists::pKendall(tau.obs,n))
>
> # (3) normal distribution
> p.z = 2*(1 - pnorm(tau,sd = sqrt(4*n +10)/sqrt(9*n*(n-1)) ))
>
> c(p.perm, p.tau, p.z)
[1] 0.02000000 0.01022864 0.01734336
```

Whichever test is used, at the 5% level H_0: "the two variables are independent" is rejected.

5.4 Linear association

The three measures of association above measure linear or monotone association, and tests of the presence of such association can be based on them. If the population is bivariate normal (Figure 5.5(a) of the book), only linear association exists between the variables, so the t-test for Pearson's correlation is the most appropriate and most efficient. Even when there is only linear correlation, if the marginal distributions have heavy tails (Figure 5.5(b)), tests with Spearman's rank correlation or Kendall's tau are more powerful than the t-test with the sample correlation. When the relation is not linear but monotone increasing (or decreasing) (Figure 5.5(c)), it is appropriate to quantify and test the trend with Spearman's rank correlation or Kendall's tau.

Example 5.11

The data of Figure 5.5 of the book were generated from $n = 100$ bivariate normal observations:

(a) (X, Y) bivariate normal with correlation $\rho = 0.9$;

(b) the data of (a) transformed to heavy tails, $(X, Y) \mapsto (X^3, Y^3)$ (the book prints X^4, Y^4; the plotted data take negative values and the rank statistics agree with (a), so the map is a monotone one such as the cube);

(c) the data of (a) transformed differently, with heavy tails, $(X, Y) \mapsto (\text{sgn}(X)X^2, -(Y - 4)^4)$;

(d) $X \sim N(0, 1)$, $Y = \sin(3X) + \epsilon$, $\epsilon \sim N(0, (0.2)^2)$.

For each data set we computed the sample correlation, the rank correlation and Kendall's statistic, and p-values of tests of independence with the large-sample approximation. Why are the rank correlation and Kendall's tau the same for data (a), (b), (c)? (All three maps are increasing in each coordinate on the range of the data, since $Y < 4$ there, so the ranks do not change.)

	sample correlation (p)	rank correlation (p)	Kendall's tau (p)
(a)	0.90 ($p = 0$)	0.89 ($p = 0$)	0.72 ($p = 0$)
(b)	0.82 ($p = 0$)	0.89 ($p = 0$)	0.72 ($p = 0$)
(c)	0.81 ($p = 0$)	0.89 ($p = 0$)	0.72 ($p = 0$)
(d)	0.005 ($p = 0.95$)	0.13 ($p = 0.2$)	0.09 ($p = 0.17$)

When the relation of the two variables is clear but not linear (Figure 5.5(d)), all three measures are near 0, so whichever statistic is used, the independence test fails to reject H_0: "the two variables are independent." Remember that here the failure to reject is not because H_0 is true but because all three tests lack power against nonlinear relations.

Chapter 6. Permutation Tests for Analysis of Variance

Overview

This chapter covers permutation tests in the analysis of variance (one-way and two-way layouts). Recall that distribution-free tests for comparing two populations were permutation tests after rank transformation; the distribution-free tests introduced here are also permutation tests after rank transformation. We introduce the Kruskal–Wallis test and Jonckheere test for the one-way layout, and the Friedman test, Page test and Skillings–Mack test for the two-way layout.[33]

6.1 One-way layout: comparing several populations

Consider comparing several populations or the effects of several treatments.

Example 6.1

The usage times of shared electric kickboards in service around Jamsil, Seoul, were surveyed among citizens who used one between 1 and 3 pm on a weekend.

company (brand)	usage time (min)	number of users
Kickgoing	10, 10, . . . , 50, 51	32
Lime	10, 10, 10, . . . , 103, 108	101
Gogossing	10, 10, . . . , 33, 44	22

Usage times under 10 minutes were recorded as 10 minutes. The data are `kickboard.csv`; their distributions are compared by box plots in the book. Do the usage times of the three companies' kickboards differ by service company?

Example 6.1 compares three populations of riders of three different brands of electric kickboards: a kind of population model as in Chapter 3.

Example 6.2

To shorten the time needed to bake a pizza, a student designed an experiment on the time sugar and milk take to activate the yeast. Four recipes (A, B, C, D) were assigned at random to 16 batches of dough, and the time until the dough rose was measured (`pizza.csv`).

recipe	time (s)
A	575, 395, 525, 450
B	280, 120, 175, 210
C	720, 554, 700, 650
D	200, 135, 250, 150

Do the four recipes differ? If so, which recipe activates fastest?

Example 6.2 compares the effects of four different treatments: a kind of randomization model. In experimental design such an experiment is called a **one-way layout** (one-way factorial design). There is one *factor* (here the recipe) affecting the result, and its values are one of A, B, C, D. Since one factor is laid out, it is called a one-way layout. In the population model of Example 6.1 each

[33] Implementing this chapter in R needs various statistics and shuffles. For simplicity of the examples several R functions are collected in `np_supp.R`; run `source("np_supp.R")` in R to use them.

population can be seen as a value of the factor "company (brand)," so it is the same model as a one-way layout. In statistics both situations are called one-way layouts.

The data of a one-way layout can be written as follows. Suppose the factor takes k distinct values, the *(factor) levels*; e.g. the factor "recipe" of Example 6.2 has $k = 4$ levels A, B, C, D. If there are n_1 observations $X_{11}, \ldots, X_{1n_1}$ at the first level, n_2 observations $X_{21}, \ldots, X_{2n_2}$ at the second, and so on, the data and their descriptive statistics are:

level	observations	sample size	sample mean	sample variance
1	$X_{11}, \ldots, X_{1n_1}$	n_1	$\bar{X}_1$	S_1^2
2	$X_{21}, \ldots, X_{2n_2}$	n_2	$\bar{X}_2$	S_2^2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
k	$X_{k1}, \ldots, X_{kn_k}$	n_k	$\bar{X}_k$	S_k^2

The total number of observations is $n = \sum_{i=1}^{k} n_i$. Assuming the observations are independent, model

$$X_{ij} = \mu_i + \epsilon_{ij}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, n_i. \tag{6.1}$$

The jth observation of level i has location μ_i and error ϵ_{ij}; all errors follow a distribution with mean 0 (or median 0). We test whether the location parameters μ_i are equal or different:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k \quad \text{vs} \quad H_1 : \mu_i \neq \mu_j \text{ for some } i, j.$$

The traditional one-way layout assumes normal errors with mean 0, but sometimes this normality is severely violated. E.g. the kickboard usage times of Example 6.1 have a long right tail with most users finishing within 10–20 minutes, a very skewed distribution, so normality cannot be assumed, and the CLT approximation applies only for very large samples.

We first look at the traditional method under normality, the analysis of variance (ANOVA), then the nonparametric (distribution-free) permutation test, and the rank-based permutation test, the Kruskal–Wallis test.

6.1.1 Classical analysis of variance

In classical ANOVA the different levels of the factor are called different **treatments** (e.g. recipes). ANOVA compares the variability of observations due to different factor levels (i.e. due to different treatments) with the variability due to error, using an F-statistic. To describe it simply, assume normality $X_{ij} \sim N(\mu_i, \sigma^2)$.

The variability due to different treatments is measured by the variability of the treatment means $\{\bar{X}_1, \ldots, \bar{X}_k\}$. Comparing them with the grand mean

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{k} \sum_{j=1}^{n_i} X_{ij} = \frac{1}{n} \sum_{i=1}^{k} n_i \bar{X}_i,$$

the treatment variability is

$$SS_{\text{Treatment}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (\bar{X}_i - \bar{X})^2 = \sum_{i=1}^{k} n_i (\bar{X}_i - \bar{X})^2. \tag{6.2}$$

Since $E(\bar{X}_i) = \mu_i$, $SS_{\text{Treatment}}$ tends to be large when the μ_i differ a lot. To standardize its size it is compared with the error variability

$$SS_{\text{Error}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X}_i)^2.$$

The ratio of the mean variabilities (each divided by its degrees of freedom) is the F-statistic

$$F = \frac{SS_{\text{Treatment}}/(k-1)}{SS_{\text{Error}}/(n-k)}. \tag{6.3}$$

When the alternative is true (μ_i differ), F tends to be large.

If $H_0 : \mu_1 = \cdots = \mu_k$ is true and the populations are normal (with equal variances), all observations X_{ij} are independent with the same normal distribution, and F is known to follow the F-distribution $F_{k-1,n-k}$ with degrees of freedom $k-1, n-k$. (I.e. the null distribution of F is $F_{k-1,n-k}$.) With observed F_{obs}, the ANOVA p-value is the probability of an F-statistic at least F_{obs}, $p = P(F_{k-1,n-k} \geq F_{\text{obs}})$. This test is the ANOVA F-test, or simply analysis of variance.

Why $F \sim F_{k-1,n-k}$ (supplied). Under H_0 all $X_{ij} \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$. Within level i, $\sum_j (X_{ij} - \bar{X}_i)^2/\sigma^2 \sim \chi^2_{n_i-1}$, independent of $\bar{X}_i$; the levels are independent, so $SS_{\text{Error}}/\sigma^2 \sim \chi^2_{n-k}$, independent of $(\bar{X}_1, \ldots, \bar{X}_k)$. The vector $(\sqrt{n_i}(\bar{X}_i - \mu)/\sigma)_i$ is standard normal in $\mathbb{R}^k$, and $SS_{\text{Treatment}}/\sigma^2$ is its squared length after projection onto the orthogonal complement of $(\sqrt{n_1}, \ldots, \sqrt{n_k})$, so it is χ^2_{k-1} (Cochran). The ratio of two independent χ^2's divided by their degrees of freedom is $F_{k-1,n-k}$. ■

Example 6.3

With the kickboard usage data of Example 6.1, let μ_i be the mean usage time of each of the three brands and run the F-test of $H_0 : \mu_1 = \mu_2 = \mu_3$. The $n = 155$ observations come at $k = 3$ factor levels; the treatment variability is $SS_{\text{Treatment}} = 739$, the error variability $SS_{\text{Error}} = 30287$, and the F-statistic 1.855. Compared with $F_{2,152}$ the p-value is 0.16: not enough evidence that mean usage time differs by brand. In R with `aov()`:

```
> kick = read.csv("data/kickboard.csv", header = TRUE)
> aov(time ~ brand, data = kick) %>% summary()
             Df Sum Sq Mean Sq F value Pr(>F)
brand         2    739   369.7   1.855   0.16
Residuals   152  30287   199.3
```

For reference, the total variability $SS_{\text{Total}} = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (X_{ij} - \bar{X})^2$ is the sum of treatment and error variability:

$$SS_{\text{Total}} = SS_{\text{Treatment}} + SS_{\text{Error}}. \tag{6.4}$$

Proof (supplied). Write $X_{ij} - \bar{X} = (X_{ij} - \bar{X}_i) + (\bar{X}_i - \bar{X})$, square and sum. The cross term is $2\sum_i (\bar{X}_i - \bar{X}) \sum_j (X_{ij} - \bar{X}_i) = 0$, since $\sum_j (X_{ij} - \bar{X}_i) = 0$ for each i. ■

Dividing each variability by its degrees of freedom gives sample variances, hence the name analysis of variance.

Permutation tests in ANOVA. When the populations resemble normal distributions, the ANOVA F-test controls the type I error well at the given level and has high power. But when the populations are very skewed, as in Example 6.1, normality is badly violated and the F-test cannot be said to control the type I error well. Then one can use the permutation test with the F-statistic (6.3). Recall that a permutation test, whatever statistic it uses, is exact (controls the type I error at the given level) as long as the exchangeability under the null holds.

In the one-way model (6.1), leave the distribution of ϵ_{ij} unspecified and assume only that it is a distribution with median 0; for Example 6.1 a right-skewed distribution seems the distribution of ϵ_{ij}. When $H_0 : \mu_1 = \cdots = \mu_k$ is true, the observations $\{X_{11}, \ldots, X_{kn_k}\}$ are independent with the same distribution, so the exchangeability needed for the permutation test holds. To consider all possible shuffles under the null, it is convenient to present the data as pairs (observation, factor level):

```
> pizza <- read.delim("data/pizza.txt")
> pizza
   Activation.Times Recipe
1               575      A
2               395      A
3               280      B
4               120      B
5               720      C
6               554      C
7               200      D
8               135      D
9               525      A
10              450      A
11              175      B
12              210      B
13              700      C
14              650      C
15              250      D
16              150      D
```

These are the data of Example 6.2, the numerical observations (X_{ij}; `Activation.Times`) split by the categorical factor ($i = A, B, C, D$; `Recipe`). When the null is true, the factor level of each observation is just the result of one random way of splitting the 16 observations into 4 subgroups of size 4, and any shuffle of `Recipe` would have arisen with the same probability under the null.

In other words, when the null is true and the data values X_{ij} are fixed, the classification of the observations by factor levels $1, \ldots, k$ is one of the

$$\binom{n}{n_1, \ldots, n_k} = \frac{n!}{n_1! \cdots n_k!}$$

ways of splitting the n observations into subgroups of sizes $n_1, \ldots, n_k$. So the null distribution of a statistic measuring how extreme the observed data are under the null can be generated from all $\binom{n}{n_1,\ldots,n_k}$ shuffles. In Example 6.2 there are only $n = 16$ observations, but the possible shuffles number about 63 million[34], quite many, so a random permutation test is recommended.

Example 6.4

For the pizza data of Example 6.2, approximating the permutation distribution (null distribution) of F with $M = 9999$ random shuffles and comparing with $F_{\text{obs}} = 44.7$ gives Figure 6.1 of the book.

```
> (Fobs = F.stat(Activation.Times, Recipe, data = pizza))
[1] 44.73917
> set.seed(0)
> M = 9999
> Fperm = replicate(M, {
+   perm.x = sample(pizza$Activation.Times, replace = FALSE)
+   F.stat(perm.x, pizza$Recipe)
+ })
```

(The function `F.stat()` is in `np_supp.R`.) In Figure 6.1 the histogram of the F-statistics f_i from 9999 random shuffles is the (approximate) permutation distribution of F. F_{obs} is very large, so the permutation p-value is

$$p = \frac{1 + \sum_{i=1}^{M} I(f_i \geq F_{\text{obs}})}{1 + M} = \frac{1 + 0}{1 + M} = 0.0001.$$

[34] $\binom{16}{4,4,4,4} = 63063000$.

```
> (p.val = mean(c(Fobs,Fperm) >= Fobs))
[1] 1e-04
```

Zooming in on $F \in [0, 5]$ in Figure 6.1 shows, in blue, the theoretical distribution $F_{3,12}$ of the statistic under normality; the permutation distribution and $F_{3,12}$ are very similar.

Since this permutation test uses the F-statistic, it is also called the permutation F-test.[35] When the populations are close to normal, the permutation F-test and the F-test (assuming normality) give roughly similar results. When normality is badly violated, the permutation test controls the type I error but the F-test is not exact.

6.1.2 The Kruskal–Wallis test

Permutation tests can use statistics other than F. We look at a rank-based permutation test.

Recall how the observations were rank-transformed in the Wilcoxon rank-sum test, the rank-based test for comparing two populations. When the null "the two populations are the same" is true, all observations were sampled at random, so the observations of the pooled sample were ranked from smallest to largest. (With ties of two or more observations, mid-ranks were given.)

For comparing two or more populations likewise, arrange all observations $X_{11}, \ldots, X_{kn_k}$ in order and rank them from the smallest, ignoring which population each came from. E.g. the ranks of the pizza data of Example 6.2:

```
> (pizza = pizza %>% mutate(rank = rank(Activation.Times)))
   Activation.Times Recipe rank
1               575      A   13
2               395      A    9
3               280      B    8
4               120      B    1
5               720      C   16
6               554      C   12
7               200      D    5
8               135      D    2
9               525      A   11
10              450      A   10
11              175      B    4
12              210      B    6
13              700      C   15
14              650      C   14
15              250      D    7
16              150      D    3
```

Kruskal and Wallis (Kruskal and Wallis, 1952) proposed the permutation test using as statistic the F-statistic computed from the rank-transformed values $X_{ij} \mapsto R_{ij}$, the **Kruskal–Wallis (one-way) test**. Concretely, with R_{ij} the rank of X_{ij}, the rank F-statistic is

$$F_R = \frac{\sum_{i=1}^{k} n_i(\bar{R}_i - \bar{R})^2/(k-1)}{\sum_{i=1}^{k}\sum_{j=1}^{n_i}(R_{ij} - \bar{R}_i)^2/(n-k)}, \tag{6.5}$$

the F-statistic computed as if the ranks $R_{11}, \ldots, R_{kn_k}$ were the observations. Kruskal and Wallis actually used the Kruskal–Wallis statistic (KW statistic), which gives the same test result as F_R:

$$KW = \frac{12}{n(n+1)} \sum_{i=1}^{k} n_i \Big(\bar{R}_i - \frac{n+1}{2}\Big)^2. \tag{6.6}$$

[35] Using $SS_{\text{Treatment}}$ or SS_{Error} instead of F gives the same permutation test, because by (6.4) SS_{Total} is the same for all shuffles: $F = \frac{n-k}{k-1} \frac{SS_{\text{Treatment}}}{SS_{\text{Total}} - SS_{\text{Treatment}}}$ is increasing in $SS_{\text{Treatment}}$.

Recalling that the ranks $\{1,\dots,n\}$ have mean $\bar{R} = \frac{1}{n}\sum_{i=1}^{n} i = \frac{n+1}{2}$ and sample variance $S_R^2 = \frac{1}{n-1}\sum_{i=1}^{n}(i-\bar{R})^2 = \frac{n(n+1)}{12}$, the KW statistic is the treatment variability of the ranks $\sum_{i=1}^{k} n_i(\bar{R}_i - \frac{n+1}{2})^2$ divided by the sample variance S_R^2 of the ranks. With tied observations, the sample variance of the ranks is not exactly $n(n+1)/12$, so to cover ties too the KW statistic is defined as

$$KW = \frac{\sum_{i=1}^{k} n_i\left(\bar{R}_i - \frac{n+1}{2}\right)^2}{S_R^2}, \tag{6.7}$$

where S_R^2 in (6.7) is the sample variance of the tie-corrected ranks $\{R_{11},\dots,R_{kn_k}\}$.

F_R and KW give the same test (supplied). For the ranks, $SS^R_{\text{Total}} = \sum_{i,j}(R_{ij}-\bar{R})^2 = (n-1)S_R^2$ is the same for every shuffle, and $SS^R_{\text{Treatment}} = \sum_i n_i(\bar{R}_i - \bar{R})^2 = S_R^2 \cdot KW$. By (6.4) applied to the ranks,

$$F_R = \frac{n-k}{k-1}\cdot\frac{SS^R_{\text{Treatment}}}{SS^R_{\text{Total}} - SS^R_{\text{Treatment}}} = \frac{n-k}{k-1}\cdot\frac{KW}{n-1-KW},$$

a strictly increasing function of KW on $[0, n-1)$. So the two statistics order the shuffles identically and give the same permutation p-value. ■

When the null is true there is no location difference between the populations, so the ranks R_{ij} take (without ties) one of $\{1,\dots,n\}$ at random. So the treatment mean ranks $\bar{R}_i$ are expected near the overall mean $\bar{R} = \frac{n+1}{2}$ and the KW statistic is expected to be small. When the alternative is true, e.g. if the third population lies to the right of the others, its mean rank $\bar{R}_i$ is expected to be large too: under the alternative the KW statistic tends to be large.

When the null is true, the null distribution of KW is obtained from its permutation distribution. The histogram in Figure 6.2 of the book approximates the null distribution of KW for the situation of Example 6.2, $n = 16$, $n_1 = n_2 = n_3 = n_4 = 4$, by $M = 9999$ random shuffles. (The overlaid density is its large-sample approximation, explained shortly.)

This permutation distribution has nothing to do with the exact values of the data X_{ij}, only with their relative positions. So it does not depend on the population distribution from which the data were sampled, and the test with the KW statistic is distribution-free. With observed KW statistic KW_{obs} the p-value is $p = P(K \geq KW_{\text{obs}})$ (with K following the permutation distribution of KW); with a random permutation test and shuffled values $k_1,\dots,k_M$ of KW,

$$p = \frac{1 + \sum_{i=1}^{M} I(k_i \geq KW_{\text{obs}})}{M+1}. \tag{6.8}$$

Example 6.5

For the pizza data of Example 6.2 the Kruskal–Wallis statistic is $KW_{\text{obs}} = 12.397$. Compared with the null distribution of KW (Figure 6.2), KW_{obs} is extreme, so the differences between recipes are statistically significant. The KW statistic can be computed by `KW.stat()` in `np_supp.R` or by R's built-in `kruskal.test()`:

```
> KW.stat(Activation.Times, Recipe, data = pizza)
[1] 12.39706
> kruskal.test(pizza$Activation.Times, pizza$Recipe)$statistic
Kruskal-Wallis chi-squared
                  12.39706
```

This statistic is called the Kruskal–Wallis "chi-squared" statistic; the next part explains why.

The Kruskal–Wallis chi-squared test. With several populations ($k \geq 3$), even a sample as small as $n = 16$ gives very many shuffles. Today, when computers make generating and approximating permutation distributions easy, using the (random) permutation distribution as the null

distribution is the right method; but in the 1950s, when the Kruskal–Wallis test was first developed, it was practically impossible. So Kruskal and Wallis devised approximating this permutation distribution for large n by the chi-squared distribution with $k-1$ degrees of freedom (χ^2_{k-1}). Using that under the null

$$KW \dot{\sim} \chi^2_{k-1},$$

the large-sample p-value is $p = P(\chi^2_{k-1} \geq KW_{\text{obs}})$.

To understand this approximation briefly: the numerator of the KW statistic (6.7) is the treatment variability with ranks as observations, $SS^R_{\text{Treatment}} := \sum_{i=1}^k n_i(\bar{R}_i - \bar{R})^2$. If the observations $X_{ij} \sim N(\mu, \sigma^2)$ were normal, each treatment mean $\bar{X}_i$ would also be normal $N(\mu, \sigma^2/n_i)$, and the treatment variability $SS_{\text{Treatment}} = \sum_{i=1}^k n_i(\bar{X}_i - \bar{X})^2$ is known to be exactly $\sigma^2\chi^2_{k-1}$.

Under the null, the ranks R_{ij} are not normal, but they are sampled uniformly from the finite population $\mathcal{A}_n := \{1, \ldots, n\}$, so by the (finite-population) CLT the $\bar{R}_i$ are asymptotically normal. So when the n_i are large, by the CLT, $SS^R_{\text{Treatment}}$ divided by the population variance σ^2_R of $\mathcal{A}_n$ is approximately χ^2_{k-1}; replacing σ^2_R by the sample variance S^2_R gives the KW statistic (6.7).

Sketch of $KW \xrightarrow{d} \chi^2_{k-1}$ (supplied). Let $Z_i = \sqrt{n_i}(\bar{R}_i - \bar{R})/S_R$. By Theorem 4.1 applied jointly (with the covariances of sampling without replacement), $\text{Var}(\bar{R}_i) = \frac{\sigma^2_R}{n_i}\frac{n-n_i}{n-1}$ and $\text{Cov}(\bar{R}_i, \bar{R}_l) = -\frac{\sigma^2_R}{n-1}$ for $i \neq l$, with $\sigma^2_R = \frac{n-1}{n}S^2_R$. Hence the covariance matrix of $(Z_1, \ldots, Z_k)$ tends to $I - \mathbf{u}\mathbf{u}'$ with $\mathbf{u} = (\sqrt{n_1/n}, \ldots, \sqrt{n_k/n})'$, a projection of rank $k-1$. By the multivariate finite-population CLT (Theorem 4.2 with the Cramér–Wold device), (Z_i) is asymptotically $N(0, I - \mathbf{u}\mathbf{u}')$, and $KW = \sum_i Z_i^2$ converges to the squared length of a standard normal vector projected onto a $(k-1)$-dimensional subspace, i.e. χ^2_{k-1}. ■

The blue curve in Figure 6.2 of the book ($n = 16$, $n_1 = \cdots = n_4 = 4$) is the density of the large-sample approximation χ^2_{4-1} of the permutation distribution of KW. A quantile plot comparing the permutation distribution with χ^2_3 (Figure 6.3 of the book) shows that the large-sample approximation is very inaccurate, especially in the tails. Note that for small samples the Kruskal–Wallis χ^2 test is not exact.

Figure 6.4 of the book compares the permutation distribution of KW with χ^2_2 when $(n_1, n_2, n_3) = (32, 101, 22)$ (exactly the sample sizes of Example 6.1). With many observations, the null distribution of KW (histogram) is well approximated by χ^2_2.

Example 6.6

For the data of Example 6.1 the KW statistic is $KW_{\text{obs}} = 2.67$, and the p-value of the Kruskal–Wallis chi-squared test against the large-sample distribution χ^2_2 is $p = P(\chi^2_2 \geq 2.67) = 0.2634$. This p-value is given directly by `kruskal.test()`:

```
> (KWobs = KW.stat(time, brand, data = kick))
[1] 2.668463
> (pvalue.chisq = 1 - pchisq(KWobs, df = 2))
[1] 0.2633605
> kruskal.test(time ~ brand, kick)

        Kruskal-Wallis rank sum test

data:  time by brand
Kruskal-Wallis chi-squared = 2.6685, df = 2, p-value = 0.2634
```

With many observations, the random permutation p-value and the large-sample p-value should be similar; the random permutation p-value $p = 0.2765$ leads to almost the same conclusion.

```
> set.seed(0)
> M = 9999
```

```
> KWperm = replicate(M, {
+   x.perm = sample(kick$time, replace = FALSE)
+   KW.stat(x.perm, kick$brand)
+ })
> (p.val = mean(c(KWobs,KWperm) >= KWobs))
[1] 0.2765
```

6.1.3 Ordered alternatives

When comparing several populations, the populations may follow some order, and we may hypothesize that the locations of observations increase or decrease accordingly.

Example 6.7

For 16 oak trees on Gwanak Mountain the annual increase of trunk diameter was measured, and each tree was classified by the slope of the place where it stands (`tree.csv`).

slope	annual increase of trunk diameter (cm)
slope I (steepest)	1.91, 1.53, 2.08, 1.71
slope II	2.44
slope III	2.45, 2.04, 1.60, 2.37
slope IV	2.52, 2.36, 2.73
slope V (no slope)	2.78, 2.88, 2.10, 1.66

For the data of Example 6.7, comparing five populations, we hypothesize that the steeper the slope, the smaller the growth of the oak. Regard the five slopes as five treatments and let $\mu_1, \ldots, \mu_5$ be the mean treatment effects. In the one-way model (6.1), against the null

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5,$$

the alternative can be set as

$$H_1 : \mu_1 < \mu_2 < \mu_3 < \mu_4 < \mu_5.$$

[36] When the alternative is this specific, a test with higher power than the F-test or the Kruskal–Wallis test, which consider broad alternatives, can be devised.

When H_1 is true, for any pair of populations $i < j$ implies $\mu_i < \mu_j$, so define a statistic T_{ij} that allows a one-sided test of the locations of the two populations. E.g. with $T_{ij} := \bar{X}_j - \bar{X}_i$, T_{ij} is expected to be large when $\mu_j > \mu_i$. When the alternative is true, all T_{ij} with $i < j$ are large, so

$$T = \sum_{i<j} T_{ij} \tag{6.9}$$

also tends to be large. When the null is true, the T_{ij} are around 0.

To make the statistic (6.9) distribution-free, Jonckheere (1954) proposed taking as pairwise statistic T_{ij} the Mann–Whitney U-statistic

$$T_{ij} = \sum_{\ell_i=1}^{n_i} \sum_{\ell_j=1}^{n_j} I(X_{i\ell_i} < X_{j\ell_j}).$$

[36] More generally, with F_i the distribution function of the ith population, one may set $H_0 : F_1(x) = F_2(x) = \cdots = F_k(x)$ and $H_1 : F_1(x) \geq F_2(x) \geq \cdots \geq F_k(x)$ (at least one inequality strict); then the kth population tends to have the largest values. Or, if the first population tends to have the largest values, $H_1 : F_1(x) \leq \cdots \leq F_k(x)$.

The permutation test with the statistic T (6.9) with Mann–Whitney U-statistics as T_{ij} is the **Jonckheere test** or **Jonckheere–Terpstra test**. In the Jonckheere–Terpstra test, using Wilcoxon rank-sum statistics instead of Mann–Whitney U-statistics gives the same result. Terpstra (1952) showed that the permutation test with T_{ij} is the same as the permutation test using Kendall's tau statistic τ_n, which measures nonparametrically the correlation between the observations X_{ij} and the population order $\{1, \ldots, k\}$.

The null distribution of the Jonckheere statistic can be obtained by random permutations; for large samples a large-sample approximation is also used. This approximation is the same as the large-sample approximation of Kendall's tau statistic, so the discussion is omitted here.

Example 6.8

In the hypothesis test about the oaks of Example 6.7, the Kruskal–Wallis chi-squared test of the general alternative H_1: "$\mu_i \neq \mu_j$ for some $i \neq j$" ("the growth rate of the trees differs by slope") has p-value 0.2016 and does not reject $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_5$. Setting the alternative more specifically as the ordered hypothesis "the lower the slope, the better the trees grow" ($H_1 : \mu_1 < \mu_2 < \mu_3 < \mu_4 < \mu_5$), the Jonckheere test has p-value 0.015 and rejects the null at the 5% level.

To run the Jonckheere test in R use `JonckheereTerpstraTest()` of the `DescTools` package. To use this function, the variable for the treatment must first be an "ordered" factor.

```
> tree = read.csv("data/tree.csv")
> head(tree)
  increment slope
1      1.91     1
2      1.53     1
3      2.08     1
4      1.71     1
5      2.44     2
6      2.45     3
> # this is needed for JT test
> (tree$slope = factor(tree$slope, ordered = TRUE))
 [1] 1 1 1 1 2 3 3 3 3 4 4 4 5 5 5 5
Levels: 1 < 2 < 3 < 4 < 5
```

The Jonckheere test implemented in `DescTools::JonckheereTerpstraTest()` gives the result of the exact permutation test generating all shuffles for $n \leq 100$, and the large-sample approximate p-value for $n > 100$.

```
> with(tree, JonckheereTerpstraTest(increment , slope, alternative =
    "increasing"))

        Jonckheere-Terpstra test

data:  increment and slope
JT = 73, p-value = 0.01534
alternative hypothesis: increasing
```

The oak data of Examples 6.7–6.8 have $n = 16$, so this p-value is that of the exact permutation test. To run a random permutation test, specify the argument `nperm`:

```
with(tree, JonckheereTerpstraTest(increment , slope, alternative = "
    increasing", nperm= 5000))
```

6.2 Inference in complete block designs without replication

So far we have compared several populations or several treatments. If, besides the treatment, another variable affects the observed effect, *blocking* is used to control it. Blocking compares treatments within homogeneous subgroups when the population to which the treatments are applied is not homogeneous. E.g. to compare the efficacy of a newly developed vaccine with a placebo, the effect may differ between healthy adults and adolescents; the treatments are vaccine/placebo and the block variable splits the whole population into the two subgroups adults/adolescents.

Example 6.9

Three diets A, B, C are tested on animals in four nutritional states. Three animals are obtained for each nutritional state, the diets are applied as in the design below, and the weight gains are recorded.

state I	state II	state III	state IV
A	C	B	A
B	A	C	C
C	B	A	B

A design like Example 6.9 is a (non-replicated) **complete block design** (CBD). Since the diets were assigned at random within each block (nutritional state), it is also called a randomized CBD. The measured weight gain mixes the gain due to the diet with the natural gain due to the nutritional state.

Example 6.10

An urban smart-farm entrepreneur growing lettuce wants to compare three cultivation methods (A, B, C). Growth may differ with the location of the smart farm, so with location (rooftop, inside a building, inside a subway station) as block variable, a randomized complete block design gave the observations below: production over 30 days (`smartfarm1.csv`).

	rooftop	inside building	inside subway station
A	85	105	95
B	95	96	110
C	100	112	117

Does production differ by cultivation method and by location of the smart farm?

In the two examples above, exactly one observation is obtained for each combination of treatment and block. Modeling the observations: with k treatments and b blocks, the observation X_{ij} of treatment i in block j is modeled as

$$X_{ij} = \mu + t_i + b_j + \epsilon_{ij}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, b. \tag{6.10}$$

Here μ is the effect unaffected by treatment or block, t_i the effect of the ith treatment, b_j the effect of the jth block, and ϵ_{ij} independent errors following a distribution with mean (or median) 0.

If several observations can be obtained for each treatment-block combination, it is a replicated complete block design, and can be seen as a somewhat more general two-way layout.

Example 6.11

For the smart-farm production of Example 6.10, several smart farms were installed at each location and more observations obtained. Does production differ by cultivation method and by location (`smartfarm.csv`)?

	rooftop	inside building	inside subway station
A	85, 88, 72	105, 100, 97	95, 69, 45
B	95, 100, 84	96, 87, 92	110, 94, 93
C	100, 105, 98	112, 109, 116	117, 121, 101

When repeated observations are possible, as in Example 6.11, not only the effects of treatments and blocks but also the treatment-block interaction can be modeled and estimated. The lth observation X_{ijl} of treatment i in block j is modeled as

$$X_{ijl} = \mu + t_i + b_j + (tb)_{ij} + \epsilon_{ijl}, \tag{6.11}$$

where $(tb)_{ij}$ is the interaction. The interaction could be modeled in (6.10) too, but with too few observations it cannot be estimated, so in complete block designs no interaction is assumed.

First consider tests of treatment effects in a complete block design with one observation per treatment and block, as in Examples 6.9 and 6.10. Assuming model (6.10), we compare the null

$$H_0 : t_1 = \cdots = t_k \tag{6.12}$$

(all treatment effects equal) with the alternative H_1: "not all treatment effects are equal." When the second variable is not a block but another kind of treatment, statistical inference proceeds the same way.

6.2.1 Classical analysis of variance

Writing the estimator of the effect of treatment i as $\bar{X}_{i\cdot} = \frac{1}{b}\sum_{j=1}^{b} X_{ij}$, of block j as $\bar{X}_{\cdot j} = \frac{1}{k}\sum_{i=1}^{k} X_{ij}$, and the grand mean as

$$\bar{X} = \bar{X}_{\cdot\cdot} = \frac{1}{kb}\sum_{i=1}^{k}\sum_{j=1}^{b} X_{ij} = \frac{1}{b}\sum_{j=1}^{b} \bar{X}_{\cdot j},$$

classical ANOVA writes the treatment variability as

$$SS_{\text{Treatment}} = \sum_{i=1}^{k}\sum_{j=1}^{b} (\bar{X}_{i\cdot} - \bar{X})^2 = b\sum_{i=1}^{k} (\bar{X}_{i\cdot} - \bar{X})^2.$$

In computing $\bar{X}_{i\cdot}$ and $\bar{X}$ the differences due to block effects cancel, so $SS_{\text{Treatment}}$ can be seen as treatment variability regardless of the block effects. Under the null (6.12), $SS_{\text{Treatment}}$ is expected to be small; when the alternative $t_i \neq t_{i'}$ is true it tends to be large. The F-statistic compares the treatment variability with the error variability SS_{Error}. In a complete block design the error is estimated after removing both treatment and block effects:

$$SS_{\text{Error}} = \sum_{i=1}^{k}\sum_{j=1}^{b} (X_{ij} - \bar{X}_{i\cdot} - \bar{X}_{\cdot j} + \bar{X})^2.$$

(The treatment effect $\bar{X}_{i\cdot}$ and block effect $\bar{X}_{\cdot j}$ both contain the grand mean $\bar{X}$, so to remove it only once it is added back.) The F-statistic is the ratio of the mean variabilities divided by their degrees of freedom:

$$F = \frac{SS_{\text{Treatment}}/(k-1)}{SS_{\text{Error}}/[(k-1)(b-1)]}. \tag{6.13}$$

When $H_0 : t_1 = \cdots = t_k$ is true and $\epsilon_{ij} \sim N(0, \sigma^2)$, F is known to follow the F-distribution with degrees of freedom $k-1$, $(k-1)(b-1)$, $F_{k-1,(k-1)(b-1)}$; so the p-value of F_{obs} is $p = P(F_{k-1,(k-1)(b-1)} \geq F_{\text{obs}})$.

Permutation tests in ANOVA. If normality cannot be assumed for the populations, or the sample is too small for the CLT approximation to an F-distribution, the null distribution of the F-statistic can be generated by shuffles. But here the permutation distribution is obtained not by shuffling the treatments of all observations, but by shuffling the treatments within each block. The reason is seen from the model when the null $H_0 : t_1 = \cdots = t_k = 0$ is true:

$$X_{ij} = \mu + b_j + \epsilon_{ij}. \tag{6.14}$$

If the ϵ_{ij} are independent with the same distribution, two observations in the first block, $X_{11} = \mu + b_1 + \epsilon_{11}$ and $X_{21} = \mu + b_1 + \epsilon_{21}$, are independent with the same distribution, hence exchangeable. But two observations in different blocks, X_{11} and X_{12}, have different distributions and are not exchangeable. The observations $X_{1j}, \ldots, X_{kj}$ within the jth block are exchangeable, but observations across blocks are not. So the observations within a block can be shuffled in $k!$ different ways, and doing this in all b blocks gives $(k!)^b$ possible shuffles.

The permutation F-test p-value is obtained as follows:

1. Compute F_{obs} from the observations.
2. Shuffle the levels of the treatment factor within each block, and compute the F-statistic f_i from the shuffled data.
3. Repeat step 2 for all $M = (k!)^b$ possible shuffles, or $M = 1999$ random shuffles, obtaining $f_1, \ldots, f_M$.
4. The permutation F-test p-value is $p = \frac{1+\sum_{i=1}^{M} I(f_i \geq F_{\text{obs}})}{1+M}$.

Example 6.12

For the (non-replicated) smart-farm data of Example 6.10, the observed F-statistic is $F_{\text{obs}} = 3.95$. The book compares the normal-theory null distribution $F_{2,4}$ with the exact permutation distribution computed from all $(k!)^b = (3!)^3 = 216$ shuffles (histogram: permutation distribution; curve: density of $F_{2,4}$). Under normality the p-value is $P(F_{2,4} \geq F_{\text{obs}}) = 0.113$, and the permutation F-test p-value is 0.139: both tests conclude there is not enough evidence that the three cultivation methods differ. The two functions `F.stat.CBD()` and `F.test2.exact()` in `np_supp.R` were used.

```
> farm = read.csv("data/smartfarm1.csv")
> # classical F-test
> (Fobs = F.stat.CBD(x, treatment, block, data = farm))
[1] 3.952191
> k = 3;b = 3
> (pval = 1-pf(Fobs,k-1, (k-1)*(b-1)))
[1] 0.1129032
>
> # exact permutation using F statistic (for b = 3 only)
> Fpermtest = F.test2.exact(x,treatment,block, data = farm)
> Fpermtest$pvalue
[1] 0.1388889
```

The permutation test with the F-statistic is exactly the same as the permutation test with the treatment variability $SS_{\text{Treatment}}$ as statistic.[37] This can be understood for the same reason that

[37] See Exercise 6.4 of the book.

comparing two populations, the permutation tests with the two-sample t-statistic and with the difference of means were the same, and comparing several populations, the permutation tests with F and with $SS_{\text{Treatment}}$ were the same. (Within each block the shuffles preserve the treatment-free total and block sums, so SS_{Total} and the block sum of squares are fixed and F is increasing in $SS_{\text{Treatment}}$.)

6.2.2 The Friedman test

The permutation test in a complete block design using as statistic the F-statistic computed after rank-transforming the observations, instead of the X_{ij}, is the distribution-free test proposed by Friedman (Friedman, 1937). Even under the null $H_0 : t_1 = \cdots = t_k$ the levels of the observations may differ by block, so the rank transformation is done separately for the k observations within each block: in the jth block, $X_{1j}, \ldots, X_{kj}$ are transformed to ranks $1, \ldots, k$.

Example 6.13

The rank transformation of the smart-farm data of Example 6.10; the ranks are assigned within blocks.

```
> farm = read.csv("data/smartfarm1.csv")
> as.data.frame(farm %>% arrange(block) %>% group_by(block) %>%
    mutate(ranks = rank(x)) )
    x treatment block ranks
1 105         A    in     2
2  96         B    in     1
3 112         C    in     3
4  85         A  over     1
5  95         B  over     2
6 100         C  over     3
7  95         A under     1
8 110         B under     2
9 117         C under     3
```

As the Kruskal–Wallis statistic did for comparing several populations without blocks, Friedman's statistic is the treatment variability of the (rank-transformed) ranks standardized by the sample variance of the ranks:

$$FM = \frac{12}{k(k+1)} b \sum_{i=1}^{k} (\bar{R}_{i\cdot} - \bar{R})^2.$$

The $\frac{k(k+1)}{12}$ in the denominator of Friedman's statistic is the sample variance of the ranks $1, \ldots, k$. With tied observations and mid-ranks, the denominator of Friedman's statistic is replaced by the average $\frac{1}{b}\sum_{j=1}^{b} S_j^2$ of the sample variances S_j^2 of the ranks in each block, and

$$FM = \frac{1}{\frac{1}{b}\sum_{j=1}^{b} S_j^2} \, b \sum_{i=1}^{k} (\bar{R}_{i\cdot} - \bar{R})^2$$

is Friedman's statistic. As for the Kruskal–Wallis statistic, it can be seen as standardized to use that for many observations, under the null,

$$FM \dot{\sim} \chi^2_{k-1}.$$

But in a complete block design with a fixed number of treatments k, the number of observations grows only as the number of blocks b grows, so in most cases where k and b are not large the large-sample null distribution is not accurate, and it is recommended to use the exact permutation distribution or a random permutation distribution as the null distribution.

Example 6.14

Testing the cultivation methods for the smart-farm data of Example 6.10, Friedman's statistic is $FM_{\text{obs}} = 4.67$, and compared with the large-sample χ^2_{3-1} the (approximate) p-value is 0.097. The function `FM.stat.CBD()` in `np_supp.R` computes Friedman's statistic.

```
> (FMobs = FM.stat.CBD(x,treatment,block,data = farm))
[1] 4.666667
> # pvalue (asymptotic)
> (pval1 = 1-pchisq(FMobs,k-1))
[1] 0.09697197
```

The large-sample p-value can also be computed with R's built-in `friedman.test()`:

```
> friedman.test(x ~ treatment | block,data = farm)

        Friedman rank sum test

data:  x and treatment and block
Friedman chi-squared = 4.6667, df = 2, p-value = 0.09697
```

The function `Friedman.test.exact()` in `np_supp.R` performs the exact permutation test:

```
> # pvalue (exact permutation test)
> Friedman.test.exact(x,treatment,block, data = farm)$pvalue
[1] 0.1944444
```

Compared with the exact null distribution, the permutation distribution, the observed statistic: of the $(3!)^3 = 216$ shuffles, 42 give statistics at least FM_{obs}, so the p-value is $42/216 = 0.194$, quite different from the large-sample p-value. This difference is because the permutation distribution of Friedman's statistic computed from ranks is very discrete. The quantile plot comparing the exact null distribution (permutation distribution) with χ^2_2 (in the book) shows that for these data with $n = 9$ the large-sample approximation of the null distribution is very inaccurate.

When considering all shuffles is hard, a random permutation test is used, generating the null distribution exactly; remember that the rank transformation and the shuffles are within blocks. The function `Friedman.test.randperm()` in `np_supp.R` implements the random permutation test with Friedman's statistic. Since the null distribution of the statistic is very discrete, the random shuffles converge slowly to the exact p-value.

```
> set.seed(0)
> Friedman.test.randperm(x, treatment, block, M = 9999, data = farm)$
    pvalue
[1] 0.1107
```

6.2.3 Tests of ordered alternatives

In the complete block model

$$X_{ij} = \mu + t_i + b_j + \epsilon_{ij}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, b, \tag{6.15}$$

if the treatments are expected to have effects in some order, an ordered alternative can be set: against the null $H_0 : t_1 = \cdots = t_k$,

$$H_1 : t_1 \le t_2 \le \cdots \le t_k \tag{6.16}$$

(with at least one inequality "$<$").

Without a block variable, the Jonckheere–Terpstra test tests hypotheses about the order of several populations, and was more powerful than the Kruskal–Wallis test (against general alternatives). Recall that the Jonckheere–Terpstra statistic measures nonparametrically the correlation between the observations X_{ij} and the population order $\{1, \ldots, k\}$.

With a block variable, the block effects are added to the observations X_{ij}. So one can compute within each block a statistic representing the correlation between the observations X_{ij} and the population order $\{1, \ldots, k\}$, and then simply add them up. If the ranks of the observations in block j are $R_{1j}, \ldots, R_{kj}$, then if the order (6.16) is true, $R_{1j} < \cdots < R_{kj}$ tends to hold, and $\sum_{i=1}^{k} iR_{ij}$ (related to Spearman's rank correlation) tends to be large. Page's (Page, 1963) L-statistic is the sum of these block statistics,

$$L = \sum_{j=1}^{b} \Big[\sum_{i=1}^{k} iR_{ij} \Big] = \sum_{i=1}^{k} iT(R_{i\cdot}), \tag{6.17}$$

where

$$T(R_{i\cdot}) = \sum_{j=1}^{b} R_{ij}$$

is the sum of the ranks of treatment i. Page proposed tests using as null distribution of this statistic the permutation distribution (generated by within-block shuffles) or its normal large-sample approximation, called **Page's trend test**. In complete block designs samples are usually small and the large-sample approximation[38] is very inaccurate, so a permutation test with L is recommended.

Example 6.15

The table below summarizes hypothetical data comparing four treatments in six blocks; the within-block ranks are in parentheses. Use Page's test to see whether the treatment effect is smallest for treatment 1 and largest for treatment 4.

	block 1	block 2	block 3	block 4	block 5	block 6	rank sum
treatment 1	128 (3)	128 (1)	160 (1)	142 (1)	157 (2)	179 (3)	11
treatment 2	122 (2)	137 (2)	177 (2)	177 (3)	160 (3)	138 (1)	13
treatment 3	207 (4)	188 (3)	181 (3)	164 (2)	155 (1)	175 (2)	15
treatment 4	120 (1)	208 (4)	199 (4)	194 (4)	177 (4)	195 (4)	21

For these data Page's L-statistic is $L = 1 \times 11 + 2 \times 13 + 3 \times 15 + 4 \times 21 = 166$, and the random permutation p-value is 0.018: the differences between treatment effects are significant. A Friedman test on these data gives $p = 0.113$, showing the lower power of Friedman's test. The function `Page.L.stat()` in `np_supp.R` computes L, and the random permutation test with this statistic is implemented in `Page.trend.randperm()`.

```
> dat = read.csv("data/toydat_page.csv")
> (Lobs = Page.L.stat(x,treatment,block, data =dat))
[1] 166
> set.seed(0)
> Page.trend.randperm(x,treatment,block, data = dat)$pvalue
[1] 0.018
```

The levels of the treatment must be ordered, so the categorical variable `treatment` must have levels in the numeric form $1, 2, \ldots, k$.

[38] For the large-sample null distribution of Page's L-statistic see Exercise 6.6 of the book. (Supplied: under H_0 the within-block rank vectors are independent uniform permutations, so $E(L) = \frac{bk(k+1)^2}{4}$ and $\mathrm{Var}(L) = \frac{bk^2(k+1)(k^2-1)}{144}$, and L is asymptotically normal as $b \to \infty$.)

6.3 General two-way layouts

Non-replicated complete block designs like Example 6.10 and replicated ones like Example 6.11 are special cases of the general two-way layout. Not only these two examples: any design without observations in every treatment-block cell, i.e. an incomplete block design, also with two factors (treatment and block, or two treatments), is called a two-way layout.

In a general two-way layout one sets up the model (6.11) with interaction, or, assuming no interaction effects, the simpler model

$$X_{ijl} = \mu + t_i + b_j + \epsilon_{ijl}, \qquad i = 1, \ldots, k, \quad j = 1, \ldots, b, \quad l = 1, \ldots, c. \tag{6.18}$$

Here the number of observations (replications) in each treatment-block cell is assumed to be c.

Without interaction as in (6.18), we compare the null $H_0 : t_1 = \cdots = t_k$ with the alternative H_1: "$t_i \neq t_j$ for some $i \neq j$." Assuming ϵ_{ijl} in (6.18) approximately normal, the F-test is the general two-way ANOVA method. If normality is hard to assume, a permutation test with the F-statistic is a distribution-free test. Then the kc observations within each block are exchangeable under the null, so, as without replication, the permutation distribution is generated by shuffling the observations within each block. (The number of all shuffles is $[(kc)!]^b$.)

As in the Friedman test, one can consider the permutation test using as statistic the F-statistic computed from the ranks of the observations within each block. The test considering rank transformation was first proposed by Skillings and Mack (Skillings and Mack, 1981) and is called the Skillings–Mack test. The Skillings–Mack statistic modifies the F-statistic slightly. Let R_{ijl} be the within-block ranks; the jth block has kc observations, so $R_{ijl} \in \{1, \ldots, kc-1, kc\}$. Dividing the rank-transformed treatment variability by the number of replications c,

$$SS_{\text{Treatment}}/c = b \sum_{i=1}^{k} (\bar{R}_{i\cdot\cdot} - \bar{R})^2,$$

the Skillings–Mack statistic standardizes it by the sample variance of the ranks:

$$SM = \frac{12}{kc(kc+1)} b \sum_{i=1}^{k} (\bar{R}_{i\cdot\cdot} - \bar{R})^2,$$

where $kc(kc+1)/12$ is the sample variance of the ranks $\{1, \ldots, kc-1, kc\}$. With ties, replacing it by the average $\frac{1}{b}\sum_{j=1}^{b} S_j^2$ of the within-block sample variances S_j^2 of the ranks,

$$SM = \frac{1}{\frac{1}{b}\sum_{j=1}^{b} S_j^2} \, b \sum_{i=1}^{k} (\bar{R}_{i\cdot\cdot} - \bar{R})^2$$

is the statistic. As for Friedman's statistic, when the null $H_0 : t_1 = \cdots = t_k$ is true, asymptotically $SM \sim \chi^2_{k-1}$, so the asymptotic p-value is $p = P(\chi^2_{k-1} \geq SM_{\text{obs}})$.

Example 6.16

With the replicated smart-farm data of Example 6.11, test whether production differs by cultivation method. The data have $k = 3$ factor levels, $b = 3$ blocks and $c = 3$ replications. The function `SM.stat()` in `np_supp.R` computes the Skillings–Mack statistic, and the random permutation test with it is implemented in `SM.perm.test()`.

```
> (SMobs = SM.stat(x,treatment,block,data = smartfarm))
[1] 4.489013
```

The test with the chi-squared large-sample approximation gives $p > 0.1$: the differences by cultivation method are not significant.

```
> (pvalue.asymp = 1 - pchisq(SMobs, df = 2))
[1] 0.1059798
```

With $n = 27$ observations, can the large-sample null distribution be trusted? The random permutation test gives a different result:

```
> SM.perm.test(x,treatment,block,M=9999,data = smartfarm)$pvalue
[1] 3e-04
```

Only 3 of the 10000 shuffles gave statistics at least the observed one, so the random permutation p-value is $p = 3/10000$: the difference by cultivation method is significant. In the code above the number of random shuffles $M = 9999$ was made rather large. Since the two tests conflict, the accuracy of the random permutation p-value needed to be increased, so the number of repetitions was made large.

In complete block designs where the number of replications per treatment-block cell is not constant, or in incomplete block designs, tests of treatment effects were developed in ways similar to the permutation tests, Friedman test and Skillings–Mack test above; see Chapter 7 of Hollander et al. (2013) for details.

Finally, when there is interaction, note that for tests of treatments or of interaction, rank transformation may remove the evidence of interaction present in the data. The interaction is a quadratic term, the product of treatment and block effects, and the nonlinear rank transformation may remove it. To make nonparametric tests in model (6.11) with interaction, it is recommended to use special rank transformations,[39] or not to rank-transform and instead regard the two-way model as a kind of linear regression model and test by resampling. (Resampling inference in linear regression is treated in Chapter 10.)

[39] The dangers of rank transformation with interaction and the special rank transformations mentioned in the text are described in detail in Section 9.2 of Higgins (2004).

Chapter 7. The Bootstrap

Overview

(Part II of the book, "The bootstrap and modern nonparametric statistics," begins with this chapter.)

The **bootstrap** is the standard resampling method for estimating the sampling distribution of a statistic. With it one obtains confidence intervals for the parameter the statistic estimates, the bias and standard error of the estimator, and tests about the parameter: a very general method. The bootstrap is a nonparametric tool that makes statistical inference possible for any population. Even under parametric model assumptions it has the advantage of replacing mathematical-statistical calculation by computation.

Briefly, the bootstrap approximates the population by suitable resampling from the given sample. Compared with the other standard resampling method, the permutation test: the permutation test generates the distribution under the null by suitable shuffles, while bootstrap resampling is a method for inference about the estimation of parameters rather than for tests of hypotheses about them. Two examples show when the bootstrap can be used.

Example 7.1

The coefficient of variation of a population (standard deviation divided by the mean) is often used as an index of inequality. The scholarship amounts $x_1, \ldots, x_n$ of $n = 100$ randomly chosen Seoul National University students were surveyed and the estimate $\widehat{cv} = s/\bar{x}_n$ of the coefficient of variation computed. Is this estimator $\widehat{cv}$ biased compared with the population coefficient of variation cv? How large is $\mathrm{Var}(\widehat{cv})$?

Example 7.2

Kendall's tau $\tau = 2P\{(X - X')(Y - Y') > 0\} - 1$, the population measure of association of two variables (X, Y), can be estimated by Kendall's statistic τ_n, and the null distribution of τ_n under $H_0 : \tau = 0$ can be approximated by shuffles. If $\tau \neq 0$ appears to hold, can the distribution of τ_n be known without knowing the value of τ? If we want a confidence interval for τ, what method can be used?

In Example 7.1, with population mean and variance μ, σ^2, the population coefficient of variation is $cv = \sigma/\mu$, and we are interested in the properties of its estimator $\widehat{cv} = s/\bar{X}_n$. Among the properties of an estimator, the bias $E(\widehat{cv}) - cv$ and the standard error $\mathrm{se}(\widehat{cv}) := \sqrt{\mathrm{Var}(\widehat{cv})}$ are good measures of its uncertainty. The mean $E(\widehat{cv})$ and variance $\mathrm{Var}(\widehat{cv})$ can in principle be found by mathematical-statistical calculation under a parametric model, but their form is very complicated or (for normal populations[40]) has not yet been computed theoretically. Remember also that this bias and standard error depend on the distribution of the population. The bootstrap estimates the bias and standard error of a statistic from whatever population.

Kendall's statistic τ_n in Example 7.2 has a sampling distribution depending only on the parameter τ. Whatever the actual distribution, populations with the same value of τ give the same sampling distribution of τ_n, but τ is unknown, so the sampling distribution of τ_n is hard to specify. Even then, the bootstrap gives the sampling distribution of τ_n and confidence intervals for τ.

[40] Hendricks and Robey (1936) computed the approximate density of $\widehat{cv}$ for $X_i \sim N(\mu, \sigma^2)$, but its form is very complicated. In this paper of almost 100 years ago, to verify or test whether the computed density is correct, 10240 coin tosses were (actually) carried out; to approximate a normal variable, the number of heads in 10 coin tosses was used, approximately normal. By comparison, today one can draw random samples directly from the normal distribution, and repetitions in the millions rather than thousands are easy.

7.1 Sampling distributions and standard errors of statistics

The sampling distribution and the standard error of a statistic are essential for quantifying its uncertainty. From a population with distribution F, compute from n independent observations $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ a statistic $\hat{\theta}_n = T(\mathbb{X}_n) = T(X_1, \ldots, X_n)$, regarded as an estimator of a parameter $\theta (= \theta(F)$, a value depending on $F)$. The sampling distribution of $\hat{\theta}_n$ is its distribution under the given population, and its standard error is the standard deviation of that distribution. The sampling distribution or standard error answers (probabilistically) how close the estimator $\hat{\theta}_n$ is to θ.

First, sampling distributions and standard errors in a hypothetical situation. The data `flights2.RData` record the scheduled departure and actual departure times of all passenger flights departing from all US airports in 2015, giving the departure delay. In 2015 there were 5,819,079 flights in all; regard them as the population. (It is a finite population, but with very many units, so treat it as infinite.) Figure 7.1 of the book visualizes 5000 of the five million records chosen at random.

A delay below 0 means the flight departed earlier than scheduled. If what we want to know from these data is the population median θ, we have the whole population, so we know $\theta = -2$ (minutes). Indeed more than half the flights departed earlier than scheduled. Note that in ordinary data analysis the parameter value is never known.

We have all flights, but suppose we could survey only $n = 100$ flights: the observed data are 100 delays $X_1, \ldots, X_n$ chosen at random from the five million of the population. How good an estimator of the population median θ is the sample median $\hat{\theta}_n = \text{median}\{X_1, \ldots, X_n\}$?

Since the population is accessible, the effect of the randomness of the sample on the sample median can be checked by simulation. Drawing a random sample of n from the population and computing its sample median $\hat{\theta}_n$ gives one observation of the sample median. If we sample again, how different will this observation be? Since the population is known, we can sample many times and compute the sample medians $\hat{\theta}_n^{(1)}, \hat{\theta}_n^{(2)}, \hat{\theta}_n^{(3)}, \ldots$ of each. If we could repeat this infinitely and see the distribution of $\{\hat{\theta}_n^{(i)} : i = 1, 2, 3, \ldots, \infty\}$, that is exactly the distribution of the sample median computed from n samples (Figure 7.2 of the book).

Without infinite repetition, repeating random sampling from the population sufficiently many times ($M = 100000$) and computing $\hat{\theta}_n$ approximates its sampling distribution (Figure 7.3 of the book, a histogram). If we observed a single sample, its sample median would be one value of this sampling distribution. (The sampling distribution shows that the sample median is very likely to be -2 or -1.) In the same way the standard error $\text{se}(\hat{\theta}_n)$ can be approximated: with $\hat{\theta}_n^{(i)}$ the median of each sample and $\bar{\hat{\theta}}_n^{(\cdot)}$ their mean,

$$\text{se}(\hat{\theta}_n) \approx \widehat{\text{se}}_M(\hat{\theta}_n) = \sqrt{\frac{1}{M-1}\sum_{i=1}^{M}\left(\hat{\theta}_n^{(i)} - \bar{\hat{\theta}}_n^{(\cdot)}\right)^2} = 0.759.$$

Now suppose the population is inaccessible, and we obtained only $n = 100$ observations with sample median $\hat{\theta}_n = \text{median}\{x_1, \ldots, x_n\} = -1$. (Compared with the sampling distribution, $\hat{\theta}_n = -1$ is a quite plausible value.) If the sampling distribution of $\hat{\theta}_n$ were approximately normal, the standard error $\text{se}(\hat{\theta}_n) = 0.759$ would give the 95% interval for θ

$$\hat{\theta}_n \pm 1.96\,\text{se}(\hat{\theta}_n) \approx [-2.49, 0.49].$$

In reality the sampling distribution is neither normal nor is the standard error $\text{se}(\hat{\theta}_n)$ known. Without access to the population, how can we know the sampling distribution of $\hat{\theta}_n$? And how to estimate the standard error $\text{se}(\hat{\theta}_n)$?[41] To answer this, the bootstrap regards the given sample as a (hypo-

[41] For some parametric models and statistics the sampling distribution is easy to obtain. E.g. if $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and the statistic is $T(\mathbb{X}_n) = \bar{X}_n$ (the sample mean), normal theory gives $\bar{X}_n \sim N(\mu, \sigma^2/n)$, so the sampling distribution is $N(\mu, \frac{\sigma^2}{n})$ and the standard error $\text{se}(\bar{X}_n) = \sigma/\sqrt{n}$. If σ is unknown, plugging in the sample standard deviation s gives an approximate standard error.

thetical) population and follows exactly the approximation process for the sampling distribution above.

7.2 Bootstrap samples and estimating sampling distributions

We saw that the sampling distribution of a statistic $T(\mathbb{X}_n)$ can be approximated by repeated sampling from the population. In practice the population is unavailable, so the bootstrap estimates the population distribution from the observations and performs repeated resampling with this estimated distribution as a hypothetical population: these are bootstrap samples.

With observations $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$ from a population whose form F cannot be specified, the best estimator of the population distribution function $F(x) = P(X \leq x)$ is the empirical distribution function

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} I(X_i \leq x).$$

The larger the sample, the closer the empirical distribution function is to the actual one (the difference between $F(x)$ and $F_n(x)$ is about $1/\sqrt{n}$). The empirical distribution F_n is the discrete distribution with weight $\frac{1}{n}$ on each observation X_i, so if the X_i have no ties, F_n is the discrete uniform distribution on $\mathbb{X}_n = \{X_1, \ldots, X_n\}$. To draw n independent observations from this distribution F_n, sample with replacement from $\mathbb{X}_n$.

The bootstrap sampling method that approximates the sampling distribution of a statistic $\hat{\theta}_n = T(\mathbb{X}_n)$ from observations $\mathbb{X}_n = \{x_1, \ldots, x_n\}$ of one population:

Nonparametric bootstrap resampling:

1. Draw n observations from $\mathbb{X}_n = \{x_1, \ldots, x_n\}$ by simple sampling with replacement. These observations $\mathbb{X}_n^* = \{x_1^*, \ldots, x_n^*\}$ are called a bootstrap sample.
2. Compute the statistic $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ from the bootstrap sample.
3. Repeat steps 1–2 a predetermined number B of times, obtaining $\hat{\theta}_n^{*(b)}$ for $b = 1, \ldots, B$.
4. The empirical distribution of $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ is the **bootstrap sampling distribution** of $\hat{\theta}_n$, an estimate of the sampling distribution of $\hat{\theta}_n$.

Figure 7.4 of the book shows this schematically. The values $\hat{\theta}_n^{*(b)}$ computed from bootstrap samples are also called bootstrap replicates.

The bootstrap sampling distribution has two errors: the error of approximating the population distribution $F(x)$ by $F_n(x)$ (of order $1/\sqrt{n}$), and the error of approximating the distribution of $\hat{\theta}_n^*$ by the B bootstrap replicates $\hat{\theta}_n^{*(b)}$ (of order $1/\sqrt{B}$). B can be made large enough, so the performance of the bootstrap is governed more by reducing the first error (i.e. increasing the sample size) than the second.

Example 7.3

Suppose we have data `dat` of $n = 10$ flight delays. Two bootstrap samples drawn at random in R:

```
> dat
 [1]  -5  12   9  35  -5 -10   1  -1  -3  -6
> (bootstrap.dat.1 = sample(dat,n, replace = TRUE))
 [1]  -3  -1 -10  12  -3  -1 -10   9 -10  -1
> (bootstrap.dat.2 = sample(dat,n, replace = TRUE))
 [1]  -6   1   9  -5   1 -10  -1  -6  35  -5
```

The original data `dat` contain only one observation -1, but in a bootstrap sample drawn with replacement this observation can be drawn several times. The median of the original data is $\hat{\theta}_n = -2$, and in the bootstrap samples $\hat{\theta}_n^{*(1)} = -2$, $\hat{\theta}_n^{*(2)} = -3$.

Example 7.4

A sample size $n = 10$ is too small, so take $n = 100$: repeating the bootstrap resampling $B = 2000$ times gives the empirical distribution of $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$, the bootstrap sampling distribution, shown in the book. Compare with the sampling distribution of Figure 7.3.

```
n = 100
dat = sample(delay, n) # delay: population of 5 million departure delays
B = 2000
med.boot = replicate(B, median( sample(dat,n, replace = TRUE)))
```

7.3 Bootstrap estimates of bias and standard error

Call the sampling distribution of a statistic $\hat{\theta}_n$ G (density g). (This sampling distribution is determined by the parameter θ, and more broadly by the population F and the sample size n.) Denote its estimate, the bootstrap sampling distribution, by G^* (density g^*). The properties of the statistic $\hat{\theta}_n$ relative to the target parameter θ are summarized by the bias

$$\text{bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta = \int \hat{\theta}_n \, dG - \theta = \int \hat{\theta}_n g(\hat{\theta}_n) \, d\hat{\theta}_n - \theta,$$

and the variance

$$\text{Var}(\hat{\theta}_n) = \int (\hat{\theta}_n - \mu_{\hat{\theta}_n})^2 \, dG = \int (\hat{\theta}_n - \mu_{\hat{\theta}_n})^2 g(\hat{\theta}_n) \, d\hat{\theta}_n, \qquad \mu_{\hat{\theta}_n} = E(\hat{\theta}_n),$$

or standard error $\text{se}(\hat{\theta}_n) = (\text{Var}(\hat{\theta}_n))^{1/2}$. To estimate the bias and standard error, substitute the bootstrap sampling distribution G^* for the unknown sampling distribution G in these formulas: these are the bootstrap estimates of bias and standard error.

First, the bootstrap estimate of the standard error:

$$\widehat{\text{se}}_{\text{Boot}}(\hat{\theta}_n) = \Big(\frac{1}{B-1} \sum_{i=1}^{B} (\hat{\theta}_n^{*(b)} - \mu_{\hat{\theta}_n^*})^2 \Big)^{1/2} \approx \Big(\int (\hat{\theta}_n^* - \mu_{\hat{\theta}_n^*})^2 \, dG^* \Big)^{1/2}, \tag{7.1}$$

where

$$\mu_{\hat{\theta}_n^*} = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}_n^{*(b)} = \int \hat{\theta}_n^* \, dG^*.$$

The first expression in (7.1) (the sample standard deviation) is the one mainly used.[42]

To compute the bootstrap estimate of the bias, we must replace not only the sampling distribution G but also the parameter θ. Just as G was replaced by G^*, the parameter $\theta = \theta(F)$, defined through the distribution of the population, can be replaced by $\hat{\theta}_n = \theta(F_n)$, defined on $\mathbb{X}_n$ (the hypothetical population). So the bootstrap estimate of the bias is

$$\widehat{\text{bias}}_{\text{Boot}}(\hat{\theta}_n) = \frac{1}{B} \sum_{b=1}^{B} \hat{\theta}_n^{*(b)} - \hat{\theta}_n = \mu_{\hat{\theta}_n^*} - \hat{\theta}_n.$$

[42] The sample variance of the bootstrapped statistics is used because the bootstrap resamples only B times, not infinitely often, and dividing by $B - 1$ reduces the bias (of order $1/B$).

Example 7.5

From the sample of $n = 100$ delays of Example 7.4, the bootstrap estimates of the standard error and bias of the median $\hat{\theta}_n$:

```
> sd(med.boot)
[1] 0.8294599
> (bias = mean(med.boot) - median(dat))
[1] 0.2405
```

Compared with the actual standard error and bias of the median, $\text{se}(\hat{\theta}_n) \approx 0.73$ and $\text{bias}(\hat{\theta}_n) \approx 0.48$,[a] each estimate still varies. With more than $n = 100$ observations the estimates become more accurate.

[a] The sample median is in general a biased estimator. In the delay example the population distribution is not continuous but a heavily right-skewed discrete distribution, so the bias of the sample median $\hat{\theta}_n$ is not 0. (The sample median $\hat{\theta}_n$ satisfies $\text{median}\{\hat{\theta}_n\} - \theta = 0$, i.e. it is median-unbiased.)

7.4 Parametric bootstrap resampling

If a specific parametric model can be assumed for the population from which the observations were sampled, using this fact increases accuracy. E.g. if the observations are success/failure or presence/absence, the population can only be a Bernoulli distribution with success probability p, so there is no way not to assume this parametric model. Or if the population is believed approximately normal, one can assume it is in the normal family $\{N(\mu, \sigma^2) : -\infty < \mu < \infty,\ \sigma^2 > 0\}$ and bootstrap.

A new data set for illustration:

Example 7.6

The March wind speeds (m/s) in Seoul, when spring cold spells hit, observed over ten years, are summarized below (Yoo and Kim, 2012). The data were originally continuous values but are recorded as discrete wind speeds (all speeds between 0 and 1 recorded as 0.5).

wind speed (m/s)	0.5	1.5	2.5	3.5	4.5	5.5	6.5	7.5	8.5	9.5
frequency	6421	14246	11953	7463	2322	659	120	11	3	0

A small wind generator is to be installed in Seoul, and its mean power output θ in March estimated. At wind speed v the output is known to be approximately $p(v) = 1 - 2v + v^2 - 0.05v^3$. What are the estimate of θ and its standard error?

With the wind speed a random variable $V \sim F$ (F the population of March wind speeds in Seoul), the parameter is

$$\theta = E(p(V)) = E(1 - 2V + V^2 - 0.05V^3),$$

and with observations v_i $(i = 1, \ldots, n)$ the natural estimator of θ is the moment estimator

$$\hat{\theta}_n = \frac{1}{n}\sum_{i=1}^{n} p(v_i) = 1.827462. \tag{7.2}$$

(Here n is not the number of speed categories, 10, but the total of all frequencies, $n = 43198$.) The estimator is a mean, so its standard error can be estimated by the formula

$$\text{se}(\hat{\theta}_n) = \Big(\frac{1}{n}\,\text{Var}\{p(V)\}\Big)^{1/2} \approx \Big(\frac{1}{n(n-1)}\sum_{i=1}^{n}\{p(v_i) - \overline{p(v.)}\}^2\Big)^{1/2} = \widehat{\text{se}}(\hat{\theta}_n). \tag{7.3}$$

Of course the standard error can also be estimated by nonparametric bootstrap resampling.

Wind speed is known to be modeled well by the Weibull distribution, so the population can be modeled by the Weibull family $\mathcal{W}$. The Weibull family $\mathcal{W} = \{f(\cdot; k, \lambda) : k > 0, \lambda > 0\}$ consists of the Weibull distributions $W(k, \lambda)$ with density

$$f(v; k, \lambda) = \frac{k}{\lambda}\Big(\frac{v}{\lambda}\Big)^{k-1} e^{-(v/\lambda)^k}, \qquad v \geq 0.$$

The mean and variance (i.e. bias and standard error) of the estimator $\hat{\theta}_n$ (7.2) can then be computed mathematically, but at a great cost of time and effort. In that case, or when mathematical calculation is impossible, the parametric bootstrap is used.

From the data one obtains the maximum likelihood estimators $(\hat{k}, \hat{\lambda})$ of the two Weibull parameters,[43] and with them $W(\hat{k}, \hat{\lambda})$, a better estimate of the population distribution than the empirical distribution function. Samples can be drawn freely from this parametric distribution $W(\hat{k}, \hat{\lambda})$: these are parametric bootstrap samples.

Parametric bootstrap resampling:

1. Assume the observations follow a parametric distribution $f(\cdot; \eta)$, where η is a parameter θ or a vector of parameters. Obtain from the data $\mathbb{X}_n$ the estimate $\hat{\eta}$ of the parameter η of this distribution. The statistic of interest is $\hat{\theta}_n = T(\mathbb{X}_n)$.
2. Draw n random observations from $f(\cdot; \hat{\eta})$. These observations $\mathbb{X}_n^* = \{x_1^*, \ldots, x_n^*\}$ are called a (parametric) bootstrap sample.
3. Compute the statistic $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ from the bootstrap sample.
4. Repeat steps 2–3 a predetermined number B of times, obtaining $\hat{\theta}_n^{*(b)}$ for $b = 1, \ldots, B$.
5. The empirical distribution of $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ is the bootstrap sampling distribution of $\hat{\theta}_n$, an estimate of its sampling distribution.

When the parametric assumption holds, parametric bootstrap resampling approximates the unknown population more accurately and so estimates the standard error of a statistic more accurately. Whether parametric or nonparametric bootstrap resampling is done, the results are the bootstrap replicates $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$, so whichever resampling is used, the estimation of the standard error, and the confidence interval methods introduced later, are the same.

Example 7.7

For the data of Example 7.6, the maximum likelihood estimates of the two Weibull parameters are $(\hat{k}, \hat{\lambda}) = (1.94, 2.49)$, and the estimated parametric distribution $W(\hat{k}, \hat{\lambda})$ fits the data very well (figure in the book). The R code below carries out parametric bootstrap resampling for the moment estimator $\hat{\theta}_n$. With the observations in a vector `v` of length $n = 43198$, first estimate the two Weibull parameters, then resample bootstrap samples from $W(\hat{k}, \hat{\lambda})$.

```
> mle=weibull.mle(v, threshold = 0, interval = c(0,3))
> k.hat = mle$shape
> lambda.hat = mle$scale
>
> # theta.hat
> theta.hat = mean(1 - 2 * v + v^2 - 0.05 * v^3)
>
> set.seed(1)
> theta.boot = replicate(B,{
+   v.boot = rweibull(n, k.hat, lambda.hat)
+   pv.boot = 1 - 2 * v.boot + v.boot^2 - 0.05 * v.boot^3
```

[43] The maximum likelihood estimators of the two Weibull parameters have no closed-form solution, so they are found by numerical optimization; here `weibull.mle()` of R's `weibullness` package was used.

```
+   mean(pv.boot)
+ })
> head(theta.boot)
[1] 1.814683 1.842158 1.825138 1.830239 1.799006 1.799625
>
> se.boot.par = sd(theta.boot)
> bias.boot.par = mean(theta.boot) - theta.hat
```

The table compares (a) the standard error of $\hat{\theta}_n$ by formula (7.3), (b) the standard error and bias of $\hat{\theta}_n$ from the nonparametric bootstrap, and (c) those from the parametric bootstrap.

	formula (7.3)	nonparametric bootstrap	parametric bootstrap
standard error estimate	0.0122	0.0123	0.0125
bias estimate		0.0001	−0.009

The three standard error estimates for $\hat{\theta}_n$ are very similar. The estimator is a mean, hence theoretically unbiased, and the bootstrap bias estimates are also very close to 0.

When the parametric assumption is true, the parametric bootstrap gives more accurate estimates than the nonparametric bootstrap. In comparison, the nonparametric bootstrap can be used over a very wide range of populations.

7.5 Various bootstrap resampling schemes

The principle of bootstrap resampling is to reproduce the generating process of the original data as faithfully as possible. So far the observations from one population were assumed to be a simple random sample, so bootstrap samples were also obtained by simple random sampling from the empirical distribution F_n or an estimated parametric distribution $f(\cdot;\hat{\eta})$. Applying this principle to various situations gives bootstrap resampling suited to each. Two examples follow.

7.5.1 Bootstrap resampling when comparing several populations

Suppose the observations from k populations are

$$\begin{aligned} X_{11},\ldots,X_{1n_1} &\overset{\text{i.i.d.}}{\sim} F_1,\\ X_{21},\ldots,X_{2n_2} &\overset{\text{i.i.d.}}{\sim} F_2,\\ &\vdots\\ X_{k1},\ldots,X_{kn_k} &\overset{\text{i.i.d.}}{\sim} F_k. \end{aligned} \tag{7.4}$$

When the statistic $T(\mathbb{X}_n)$ computed from the sample $\mathbb{X}_n = \{X_{11},\ldots,X_{1n_1},\ldots,X_{k1},\ldots,X_{kn_k}\}$ is of interest, the bootstrap approximates the sampling distribution and variance of $T(\mathbb{X}_n)$. Here bootstrap resampling is done separately within each population. In nonparametric bootstrap resampling, the k different populations of (7.4) are replaced by the empirical distributions $\hat{F}_j$ of the samples from each, and bootstrap samples are drawn. Step 1 of resampling is:

1. Draw n_1 observations from the first population's sample $\{X_{11},\ldots,X_{1n_1}\}$ by simple sampling with replacement, $x^*_{11},\ldots,x^*_{1n_1}$. Likewise for each $j=2,\ldots,k$, draw n_j observations from the jth population's sample $\{X_{j1},\ldots,X_{jn_j}\}$ by sampling with replacement, $x^*_{j1},\ldots,x^*_{jn_j}$. The resampled observations $\{x^*_{11},\ldots,x^*_{1n_1},\ldots,x^*_{k1},\ldots,x^*_{kn_k}\}$ are the bootstrap sample.

The remaining steps 2–4 proceed exactly as steps 2–4 of nonparametric bootstrap resampling to obtain the bootstrap sampling distribution of $T(\mathbb{X}_n)$.

For parametric bootstrap resampling, each population F_j is expressed by a parametric model. E.g. under normality with equal variances, $F_j = N(\mu_j, \sigma^2)$, and with the maximum likelihood estimates of the parameters $\mu_1, \ldots, \mu_k, \sigma^2$ we obtain $\hat{F}_j = N(\hat{\mu}_j, \hat{\sigma}^2)$. Replace the k different populations F_j of (7.4) by $\hat{F}_j$, and for $j = 1, \ldots, k$ draw n_j new observations at random from $\hat{F}_j$ as the bootstrap sample. Then proceed with steps 3–5 of parametric bootstrap resampling as before to obtain the bootstrap sampling distribution of $T(\mathbb{X}_n)$.

Example 7.8

The acute leukemia gene expression data of Example 4.1 in Chapter 4 recorded gene expression scores of $n_1 = 47$ patients sampled from population ALL and $n_2 = 25$ from population AML. We estimate the standard error $\text{se}(D)$ of the statistic $D = \bar{X}_1 - \bar{X}_2$ by bootstrap resampling. With the observations in vectors `ALL` and `AML` of lengths 47 and 25, nonparametric bootstrap resampling:

```
md.boot = replicate(B,
          mean(ALL[sample(1:n1, n1, replace = TRUE)]) -
          mean(AML[sample(1:n2, n2, replace = TRUE)])
          )
```

The observed statistic is $D = -67.28$ and the bootstrap estimate of the standard error is `sd(md.boot)` $= 28.3$. The bootstrap sampling distribution of this statistic is shown as a histogram in the book.

7.5.2 Bootstrap resampling for bivariate data

The generating process of bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$ can be viewed in roughly one of two ways. The first, as introduced in Chapter 5, views them as data sampled at random from a population F following a bivariate distribution:

$$(X_1, Y_1), \ldots, (X_n, Y_n) \overset{\text{i.i.d.}}{\sim} F. \tag{7.5}$$

The second, a model often assumed for inference in simple regression, views the "X" variable of the two as given in advance or fixed, and only the "Y" variable as random. Under model (7.5) this can be seen as the conditional distribution of Y_i after fixing the observations X_i, but consider the special case of the linear regression model

$$Y_i = \alpha + \beta x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F_\epsilon. \tag{7.6}$$

More complex regression models can be assumed, e.g. a regression curve $\alpha + \beta x_i + \gamma x_i^2$ instead of the line $\alpha + \beta x_i$, or errors ϵ_i whose distribution varies with x_i instead of identically distributed ϵ_i; here assume the simplest simple regression model (7.6) is true.

Both models (7.5) and (7.6) generate bivariate data $\{(X_1, Y_1), \ldots, (X_n, Y_n)\}$, but in resampling, the bootstrap sampling must differ according to which model is assumed to have generated the data. The nonparametric bootstrap resampling for each data-generating model:

- **Bootstrapping pairs (cases):** for data (7.5) sampled at random from a population following a bivariate distribution, with the sample written as vectors $\mathbf{x}_i = (X_i, Y_i)$, $\mathbf{x}_1, \ldots, \mathbf{x}_n$, draw n vectors with replacement from them to get the bootstrap sample $\{\mathbf{x}_1^*, \ldots, \mathbf{x}_n^*\}$ and do bootstrap resampling.
- **Residual bootstrap:** for data (7.6) where the "X" variable is given in advance and the "Y" variables do not have the same distribution, the ϵ_i, which have the same distribution, are

resampled. The errors ϵ_i are not observable, so the residuals of the fitted regression line,

$$e_i = Y_i - \hat{\alpha} - \hat{\beta}x_i, \qquad i = 1, \ldots, n,$$

are resampled instead. I.e. draw n residuals with replacement from $\{e_1, \ldots, e_n\}$ to get bootstrap residuals $\{e_1^*, \ldots, e_n^*\}$, and collect the bootstrap-resampled

$$\begin{pmatrix} Y_i^* \\ x_i^* \end{pmatrix} = \begin{pmatrix} \hat{\alpha} + \hat{\beta}x_i + e_i^* \\ x_i \end{pmatrix}, \qquad i = 1, \ldots, n,$$

to generate the bootstrap sample $\{(x_1^*, Y_1^*), \ldots, (x_n^*, Y_n^*)\}$. (In every bootstrap sample $x_i^* = x_i$ is fixed.)

The remaining steps 2–4 of the nonparametric bootstrap are carried out as is to obtain the bootstrap sampling distribution of the statistic of interest.

Which data-generating model fits better in which case? Both models leave the shapes of the distributions F and F_ϵ unrestricted, so both can be thought of as nonparametric models. But model (7.6) also has a "parametric model" $E(Y_i) = \alpha + \beta x_i$ besides the nonparametric distribution of ϵ_i, so it is called a semi-parametric model. The residual bootstrap, assuming model (7.6), may be inaccurate when the parametric assumption (simple linearity) does not hold. By comparison, the pairs bootstrap, assuming the nonparametric model (7.5) without specific parameters, can be used in most cases.

Recall that inference on the slope β in regression always assumes the model $E(Y_i) = \alpha + \beta x_i$ is true. Also, most inference in regression assumes the explanatory "X" variable is fixed, i.e. model (7.6). So for the bootstrap sampling distribution and standard error of the regression slope estimator $\hat{\beta}$, the residual bootstrap is the standard. Nonparametric inference in regression is explained in Chapter 10.

For bivariate or multivariate statistics in contexts other than regression, assume model (7.5) and use the pairs bootstrap.

Example 7.9

With the GDP and happiness data of the World Happiness Report seen in Chapter 5 (Example 5.2), infer Kendall's tau τ between the two variables. For these data Kendall's statistic is $\tau_n = 0.503$, and the permutation test of $H_0 : \tau = 0$ against $H_1 : \tau > 0$ gives $p < 0.01$: $\tau > 0$ very significantly (R code below). The permutation distribution of τ_n used in the permutation test is shown in Figure 7.5 of the book; it is the distribution when the null $\tau = 0$ is true.
What, then, is the sampling distribution and standard error of τ_n when $\tau \neq 0$? To answer, compute the bootstrap sampling distribution and estimate the standard error by bootstrap resampling of pairs of observations. In the code below for the pairs bootstrap, note that the ids of the observations are resampled, so the same observations are drawn for the two variables `gdp` and `score`.

```
> wh = read.csv("data/whappiness.csv")
> score = wh$Score; gdp = wh$GDP.per.capita
> (k.obs = cor(gdp,score, method = "k"))
[1] 0.5032363
> B = 1999
> k.boot = replicate(B, {
+   id.boot = sample(1:n,n, replace = TRUE)
+   cor(gdp[id.boot], score[id.boot], method = "k")
+ })
> sd(k.boot)
[1] 0.2096958
```

The bootstrap estimate of the standard error is 0.21. The distribution of the bootstrap replicates τ_n^* (`k.boot`) is the bootstrap sampling distribution, an estimate of the sampling distribution of τ_n in the population (estimated to have $\tau \neq 0$); see Figure 7.5 of the book. The random permutation distribution from shuffles, contrasted with bootstrap resampling, was generated with this R code:

```
> # random permutation test
> M = 1999
> n = length(gdp)
> k.perm = replicate(M, {
+   id.perm = sample(1:n, n, replace = FALSE)
+   cor(gdp,score[id.perm], method = "k")
+ })
> (pvalue.perm = mean(c(k.obs,k.perm) >= k.obs))
[1] 0.0075
```

Instead of sampling with replacement (`replace = TRUE`) as in the bootstrap, the permutation test samples without replacement (`replace = FALSE`). The bootstrap draws the pairs of observations (x_i, y_i) together, preserving the correlation between the two variables, while the permutation test keeps x_i fixed and shuffles only y_i, (deliberately) destroying any correlation between the two variables.

Chapter 8. Bootstrap Confidence Intervals

Overview

Of the methods using bootstrap resampling, the most frequently used and most developed is the estimation of confidence intervals. The best-known way to obtain a confidence interval for a parameter θ uses an estimator $\hat{\theta}_n$ of θ. A bootstrap confidence interval uses the bootstrap sampling distribution of the estimator $\hat{\theta}_n$ of θ to obtain a confidence interval for θ. This chapter explains them with the following example.

Example 8.1 (Team Fortress 2 data)

Figure 8.1 of the book is a histogram of the Team Fortress 2 play time X_i of $n = 2322$ randomly chosen gamers who bought "Team Fortress 2" on the game platform Steam (`TeamFortress2.csv`). To estimate the mean μ of all Steam users who enjoy Team Fortress 2, the sample mean $\bar{X}_n$ is computed and used to obtain a confidence interval for μ.

When $X_1, \ldots, X_n$ are a random sample from a population F, the sample mean $\bar{X}_n$ has the following properties. With population mean μ and variance σ^2,

$$E(\bar{X}_n) = \mu, \qquad \mathrm{Var}(\bar{X}_n) = \frac{\sigma^2}{n} \approx \frac{s^2}{n},$$

where s^2 is the sample variance. And by the central limit theorem,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \approx \frac{\bar{X}_n - \mu}{s/\sqrt{n}} \dot{\sim} N(0,1).$$

So the 95% interval by the normal approximation is $\mu \in \bar{X}_n \pm z_{0.025} s/\sqrt{n}$, with asymptotically approximately 95% confidence, or coverage probability 0.95: $P(\mu \in \bar{X}_n \pm z_{0.025} s/\sqrt{n}) \approx 0.95$. But when the population is very skewed, as in Example 8.1, convergence to normality by the CLT is very slow, so the normal-approximation interval cannot be guaranteed to have coverage close to 95% at the given sample size n.

Without using the asymptotic distribution of $\bar{X}_n$, estimating its sampling distribution directly by bootstrap resampling gives a more accurate interval. More generally: when the estimator of a parameter $\theta = \theta(F)$ of the population F is $\hat{\theta}_n = T(\mathbb{X}_n)$, we look at how to obtain a $1-\alpha$ interval for θ from the bootstrap sampling distribution of $\hat{\theta}_n$ (or the B bootstrap replicates $\{\hat{\theta}_n^{*b} : b = 1, \ldots, B\}$) and the bootstrap estimate $\widehat{\mathrm{se}}_{\mathrm{Boot}}$ of the standard error. Unlike the bootstrap estimate $\widehat{\mathrm{se}}_{\mathrm{Boot}}$ of the standard error, of which there was only one, there are several methods of bootstrap interval estimation. We look at six representative bootstrap intervals:

- the normal-approximation bootstrap interval;
- the basic bootstrap interval;
- the bootstrap-t interval, or studentized pivotal interval;
- the bootstrap percentile interval, or quantile interval;
  - the bias-corrected (BC) percentile interval;
  - the bias-corrected and accelerated (BCa) percentile interval.

There are more interval methods,[44] but the bootstrap-t and "BCa" methods are the most accurate, and percentile methods are very easy to implement generally, so knowing these methods suffices for applications.

[44] See Davison and Hinkley (1997) and Carpenter and Bithell (2000).

8.1 Normal-approximation bootstrap intervals

First the simplest, the normal-approximation bootstrap interval. In many situations the standardized estimator $\hat{\theta}_n$ is approximately normal: $(\hat{\theta}_n - \theta)/\operatorname{se}(\hat{\theta}_n) \dot{\sim} N(0,1)$. If the estimator $\hat{\theta}_n$ has a non-negligible bias $\operatorname{bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta$, it must be adjusted:

$$\frac{\hat{\theta}_n - \operatorname{bias}(\hat{\theta}_n) - \theta}{\operatorname{se}(\hat{\theta}_n)} \dot{\sim} N(0,1). \tag{8.1}$$

When this can be accepted (when the normal approximation can be assumed accurate), the asymptotically approximate $1-\alpha$ interval is $\theta \in \hat{\theta}_n - \operatorname{bias}(\hat{\theta}_n) \pm z_{\alpha/2} \operatorname{se}(\hat{\theta}_n)$. The standard error $\operatorname{se}(\hat{\theta}_n)$ and bias $\operatorname{bias}(\hat{\theta}_n)$ are unknown, so substituting the bootstrap estimates $\widehat{\operatorname{se}}_{\text{Boot}}$ and $\widehat{\operatorname{bias}}_{\text{Boot}}$,

$$\theta \in \hat{\theta}_n - \widehat{\operatorname{bias}}_{\text{Boot}}(\hat{\theta}_n) \pm z_{\alpha/2} \widehat{\operatorname{se}}_{\text{Boot}}(\hat{\theta}_n),$$

is the **normal-approximation bootstrap interval**.

Note that this method uses only two values of the bootstrap resampling results, $\widehat{\operatorname{se}}_{\text{Boot}}$ and $\widehat{\operatorname{bias}}_{\text{Boot}}$: of the bootstrap sampling distribution, which potentially contains a great deal of information, it uses only the mean and variance. If the shape of the bootstrap sampling distribution of $\hat{\theta}_n$ is similar to a normal, all the remaining information is contained in the normal approximation, and the normal-approximation bootstrap interval is asymptotically exact. Conversely, if the bootstrap sampling distribution does not resemble a normal, there is no guarantee that this interval is exact.

Example 8.2

For the Team Fortress 2 data of Example 8.1, the sample mean is $\bar{X}_n = 70.64$, and its bootstrap sampling distribution is in Figure 8.2 of the book. The standard error estimate is $\widehat{\operatorname{se}}_{\text{Boot}} = 5$ and the bias estimate $\widehat{\operatorname{bias}}_{\text{Boot}} = 70.56 - 70.64 = -0.08$, so the 95% normal-approximation bootstrap interval for μ is

$$\mu \in \bar{X}_n - \widehat{\operatorname{bias}}_{\text{Boot}}(\bar{X}_n) \pm z_{\alpha/2} \widehat{\operatorname{se}}_{\text{Boot}}(\bar{X}_n) = 70.64 + 0.08 \pm 1.96 \times 5 = (60.9, 80.5).$$

In R:

```
> TFtime = read.csv("data/TeamFortress2.csv")$x
> (hat.theta = mean(TFtime))
[1] 70.64311
> n = length(TFtime)
> B = 1999
> set.seed(1)
> theta.boot = replicate(B, mean(sample(TFtime, n, replace =
    TRUE)))
> (se.est = sd(theta.boot))
[1] 5.003227
> (bias.est = mean(theta.boot) - hat.theta)
[1] -0.07757111
> (norm.app.boot.CI = hat.theta - bias.est + se.est * qnorm
    (0.025) *c(1,-1))
[1] 60.91453 80.52683
```

As Figure 8.2 shows, the bootstrap sampling distribution is approximately normal in shape, so this interval can be trusted.

The normal-approximation bootstrap interval is exact only when the normal approximation (8.1) holds. If the sample is small and the normal approximation of the standardized statistic does not occur, or the asymptotic distribution of the statistic is not normal even for large samples, this interval cannot be used.

8.2 The basic bootstrap interval

Rewriting the assumption (8.1) behind the normal-approximation interval gives $\hat{\theta}_n - \theta \dot{\sim} N(\text{bias}, \text{se}^2)$. Here, instead of restricting the distribution of $\hat{\theta}_n - \theta$ to be normal, suppose freely that it has some (continuous) distribution H:

$$\hat{\theta}_n - \theta \sim H. \tag{8.2}$$

This distribution H has mean bias and variance se^2, but otherwise its shape is unspecified: a nonparametric model. With the quantile function $Q_H(p) = H^{-1}(p)$ of H,

$$\begin{aligned} 1 - \alpha &= P(Q_H(\tfrac{\alpha}{2}) \le \hat{\theta}_n - \theta \le Q_H(1 - \tfrac{\alpha}{2})) \\ &= P(\hat{\theta}_n - Q_H(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - Q_H(\tfrac{\alpha}{2})). \end{aligned} \tag{8.3}$$

If this distribution H does not depend on θ, i.e. if $\hat{\theta}_n - \theta$ is a **pivot**,[45] the probability (8.3) holds for every θ, and so $\hat{\theta}_n - Q_H(1 - \frac{\alpha}{2}) \le \theta \le \hat{\theta}_n - Q_H(\frac{\alpha}{2})$ is a $1 - \alpha$ interval for θ.

Call the bootstrap-resampled $\hat{\theta}_n$ the bootstrap replicate $\hat{\theta}_n^*$; the distribution of $\hat{\theta}_n^*$ is the bootstrap sampling distribution G^*. The **basic bootstrap interval** (basic bootstrap C.I.) is the interval approximating the distribution H (8.2) of $\hat{\theta}_n - \theta$ by the distribution of $\hat{\theta}_n^* - \hat{\theta}_n$ (call it H^*):

$$\hat{\theta}_n - Q_{H^*}(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - Q_{H^*}(\tfrac{\alpha}{2}). \tag{8.4}$$

The distribution H^* is the distribution G^* of $\hat{\theta}_n^*$ shifted left by $\hat{\theta}_n$, i.e. for $p \in (0, 1)$

$$Q_{H^*}(p) = Q_{G^*}(p) - \hat{\theta}_n,$$

so (8.4) can be rewritten

$$2\hat{\theta}_n - Q_{G^*}(1 - \tfrac{\alpha}{2}) \le \theta \le 2\hat{\theta}_n - Q_{G^*}(\tfrac{\alpha}{2}). \tag{8.5}$$

A worked example of the basic bootstrap interval is given in Example 8.3, after the bootstrap-t interval.

The basic bootstrap interval relies on the bootstrap approximation that for all θ

$$H^* \sim \hat{\theta}_n^* - \hat{\theta}_n \overset{d}{\approx} \hat{\theta}_n - \theta \sim H. \tag{8.6}$$

This method makes no specific assumption about the distribution of $\hat{\theta}_n$ and can be trusted, but it requires the questionable assumption (8.2) that $\hat{\theta}_n - \theta$ is a pivot. If this assumption fails, the convergence of H^* to H[46] is slow, so the accuracy of the basic bootstrap interval may be low.

8.3 The bootstrap-t interval

The bootstrap-t interval (bootstrap-t C.I.) is also called the studentized pivotal interval. Even when the distribution H of $\hat{\theta}_n - \theta$ and the distribution H^* of $\hat{\theta}_n^* - \hat{\theta}_n$ are not approximated as in (8.6), the following bootstrap approximation using the studentized pivot can be more accurate:

$$H^* \sim \frac{\hat{\theta}_n^* - \hat{\theta}_n}{\widehat{\text{se}}^*} \overset{d}{\approx} \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \sim H. \tag{8.7}$$

If the population is $N(\theta, \sigma^2)$ and $\hat{\theta}_n = \bar{x}_n$, then $(\hat{\theta}_n - \theta)/\widehat{\text{se}} = \sqrt{n}(\bar{x}_n - \theta)/s \sim t_{n-1}$, so here the distribution H is the t-distribution with $n - 1$ degrees of freedom. Without normality, H cannot

[45] For a parameter θ of a family $\mathcal{F}$ and statistics T_1, T_2, a random variable $S = S(T_1, T_2, \theta)$ whose distribution is the same for every population $F \in \mathcal{F}$ is called a pivot (for θ). If S is a pivot, its distribution is the same for every population in the family, so it does not depend on the parameter θ nor on the other parameters η of the family.

[46] Convergence here means that as the sample size n grows, the distribution H^* of $\hat{\theta}_n^* - \hat{\theta}_n$ converges to the distribution H of $\hat{\theta}_n - \theta$.

be assumed to be a t-distribution, so the key idea is to approximate H by H^* using the bootstrap. (The name "bootstrap-t" comes from this.)

On the right side of (8.7), $\widehat{\text{se}} = \widehat{\text{se}}(\hat{\theta}_n) \approx \sqrt{\text{Var}(\hat{\theta}_n \mid F)}$ is the standard error estimate of $\hat{\theta}_n = T(\mathbb{X}_n)$ computed from $\mathbb{X}_n$ sampled from the population F. This estimate is given by the given sample $\mathbb{X}_n = \{x_1, \ldots, x_n\}$ (or the empirical distribution F_n). E.g. with the bootstrap standard error estimate,

$$\widehat{\text{se}} = \widehat{\text{se}}_{\text{Boot}}. \tag{8.8}$$

When the standard error is easy to compute, e.g. if $\hat{\theta}_n = \bar{x}_n$,

$$\widehat{\text{se}} = s/\sqrt{n} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x}_n)^2}{n(n-1)}}. \tag{8.9}$$

With a parametric model, the maximum likelihood estimate of the population standard deviation may be used instead of the sample standard deviation s. Either way, $\widehat{\text{se}}$ is determined by the given sample.

Similarly, on the left of (8.7), $\widehat{\text{se}}^* = \widehat{\text{se}}^*(\hat{\theta}_n^*) \approx \sqrt{\text{Var}(\hat{\theta}_n^* \mid F_n)}$ is the standard error estimate of $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ estimated from the bootstrap sample $\mathbb{X}_n^* = \{x_1^*, \ldots, x_n^*\}$ resampled from F_n (the empirical distribution). Note that it is computed from the bootstrap sample $\mathbb{X}_n^*$, not the original sample $\mathbb{X}_n$: the $\widehat{\text{se}}$ computed from the original sample and the $\widehat{\text{se}}^*$ computed from the bootstrap sample differ. When the standard error estimate is easy to compute, e.g. if $\hat{\theta}_n = \bar{x}_n$ then $\hat{\theta}_n^* = \bar{x}_n^*$ and $\widehat{\text{se}}^* = s^*/\sqrt{n} = \sqrt{\frac{\sum_{i=1}^n (x_i^* - \bar{x}_n^*)^2}{n(n-1)}}$.

Otherwise (when $\widehat{\text{Var}}(\hat{\theta}_n)$ is hard to express by a formula), $\hat{\theta}_n^* = T(\mathbb{X}_n^*)$ is bootstrap-resampled to compute $\widehat{\text{se}}^*_{\text{Boot}}$. This bootstrap resampling resamples again from the bootstrap-resampled $\mathbb{X}_n^*$, so it is called **second-level bootstrap resampling**. Figure 8.3 of the book shows schematically the first-level resampling done B times and the second-level resampling done C times.

The studentized pivotal interval using the bootstrap approximation (8.7) is

$$\hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(\tfrac{\alpha}{2}), \tag{8.10}$$

where $\widehat{\text{se}}$ is given by (8.8) or (8.9), and $Q_{H^*}(p)$ is the p-quantile (the $p \times B$-th smallest value) of the bootstrap-resampled studentized pivots

$$\frac{\hat{\theta}_n^{*(1)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(1)}}, \ \frac{\hat{\theta}_n^{*(2)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(2)}}, \ \ldots, \ \frac{\hat{\theta}_n^{*(B)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(B)}}$$

when bootstrap resampling is repeated B times. With $B = 2000$ bootstrap resamples and second-level resampling C times to approximate $\widehat{\text{se}}^*$, a total of $B \times C$ bootstrap resamples is needed, so a large C takes very long computation. Second-level resampling of about $C = 25$ is recommended. For intervals, $B = 1999$ or more bootstrap resamples are recommended.

The studentized pivotal interval (8.10) is more accurate the better the bootstrap approximation (8.7) holds. Its coverage probability $P(\theta \in \text{CI})$ is asymptotically $1 - \alpha$:

$$\begin{aligned}
P(\theta \in \text{CI}) &= P\big(\hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - \widehat{\text{se}} \times Q_{H^*}(\tfrac{\alpha}{2})\big) \\
&\approx P\big(\hat{\theta}_n - \widehat{\text{se}} \times Q_H(1 - \tfrac{\alpha}{2}) \le \theta \le \hat{\theta}_n - \widehat{\text{se}} \times Q_H(\tfrac{\alpha}{2})\big) \\
&= P\Big(Q_H(\tfrac{\alpha}{2}) \le \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \le Q_H(1 - \tfrac{\alpha}{2})\Big) \\
&= 1 - \alpha.
\end{aligned}$$

Example 8.3 (Bootstrap-t and basic bootstrap intervals)

For the Team Fortress 2 data, obtain the basic bootstrap interval and the bootstrap-t interval for the population mean θ. We use the nonparametric bootstrap resampling example. Since the statistic is $\hat{\theta}_n = \bar{x}_n$, $\widehat{\text{se}}$ and $\widehat{\text{se}}^*$ can be computed by (8.9) without assuming a parametric model for the population. For each bootstrap sample $\mathbb{X}_n^{*(b)} = \{x_1^{*(b)}, \ldots, x_n^{*(b)}\}$ compute

$$\hat{\theta}_n^{*(b)} = \bar{x}_n^{*(b)}, \qquad \widehat{\text{se}}^{*(b)} = \sqrt{\frac{\sum_{i=1}^n (x_i^{*(b)} - \bar{x}_n^{*(b)})^2}{n(n-1)}}, \qquad b = 1, \ldots, B, \tag{8.11}$$

and from them the studentized pivots

$$t^{*(b)} = \frac{\hat{\theta}_n^{*(b)} - \hat{\theta}_n}{\widehat{\text{se}}^{*(b)}}, \qquad b = 1, \ldots, B.$$

Sorting these $t^{*(b)}$ to order statistics $t^*_{(1)} \le t^*_{(2)} \le \cdots \le t^*_{(B)}$, $Q_{H^*}(p) = t^*_{(\lfloor p(B+1) \rfloor)}$.[a] With $B = 1999$ and $\alpha = 0.05$, $Q_{H^*}(\alpha/2) = t^*_{(50)}$ and $Q_{H^*}(1-\alpha/2) = t^*_{(1950)}$: the 50th and 1950th smallest $t^{*(b)}$. The R code below carries out the bootstrap resampling.

```
> B = 1999
> boot.out = replicate(B, {
+             TFtime.b = sample(TFtime, n, replace = TRUE)
+             c(mean(TFtime.b), sd(TFtime.b)/sqrt(n))
+   }) %>% t()
> colnames(boot.out) = c("theta*","se*")
> head(boot.out)
       theta*      se*
[1,] 63.77382 4.428347
[2,] 64.20879 4.390289
[3,] 65.45922 4.332700
[4,] 69.65388 4.917695
[5,] 74.98221 5.243905
[6,] 77.79625 5.621729
> hat.theta = mean(TFtime)
> t.pivot = sort( (boot.out[,"theta*"] - hat.theta) / boot.out[,
    "se*"] )
```

By (8.10) the 95% bootstrap-t interval for θ follows; here $\widehat{\text{se}}$ is not the bootstrap estimate of the standard error but the estimate by formula (8.11).

```
> se.hat.theta = sd(TFtime) / sqrt(n)
> hat.theta - se.hat.theta * c(t.pivot[1950], t.pivot[50])
[1] 61.82523 82.17272
```

The quantiles can also be obtained with `quantile()`:

```
> hat.theta - se.hat.theta * quantile(t.pivot, probs= c
    (1-0.025,0.025),type = 1)
   97.5%     2.5%
61.82523 82.17272
```

By (8.5) the basic bootstrap interval is:

```
> boot.dist = sort(boot.out[,"theta*"])
> 2*hat.theta - c(boot.dist[1950], boot.dist[50])
[1] 60.78312 80.57468
```

[a] The bootstrap-approximated sampling distribution H^* or G^* is a discrete distribution with mostly B values. For a discrete distribution the quantile functions Q_{H^*} or Q_{G^*} can be defined exactly, but in practice, considering that this discrete distribution approximates a continuous (bootstrap) distribution, various

approximations of the quantile function are used. E.g. R's `quantile()`, which computes quantiles of given data, offers 9 different kinds of quantile computation. Its argument `type = 1` computes the quantile as defined here; the default is `type = 7`.

Example 8.4 (Second-level bootstrap)

When the standard error estimate $\widehat{\text{se}}$ of the statistic $\hat{\theta}$ cannot be given by a formula, a second-level bootstrap is needed. With $C = 25$ repetitions, the second-level bootstrap is carried out additionally for each bootstrap sample as below. (It takes a very long time.)

```
> B = 1999
> C = 25
> boot.out = replicate(B, {
+   TFtime.b = sample(TFtime, n, replace = TRUE)
+   theta.boot = mean(TFtime.b)
+   theta.boot.se = sd( replicate(C, mean (sample(TFtime.b, n,
    replace = TRUE)) ))
+   c(theta.boot, theta.boot.se)
+ }) %>% t()
> colnames(boot.out) = c("theta*","se*")
> head(boot.out)
       theta*      se*
[1,] 70.63376 4.071472
[2,] 56.81318 6.023104
[3,] 74.89871 5.378420
[4,] 65.11792 4.644929
[5,] 74.22063 5.994275
[6,] 67.25078 4.730485
> hat.theta = mean(TFtime)
> t.pivot = sort( (boot.out[,"theta*"] - hat.theta) / boot.out[,
    "se*"] )
```

By (8.10) the 95% bootstrap-t interval for θ follows, here with the bootstrap estimate of the standard error as $\widehat{\text{se}}$.

```
> se.hat.theta.boot = sd(boot.out[,"theta*"])
> hat.theta - se.hat.theta * c(t.pivot[1950], t.pivot[50]) #
    bootstrap-t
[1] 61.11097 81.99433
```

The bootstrap-t interval performs best when the studentized statistic $t := (\hat{\theta}_n - \theta)/\widehat{\text{se}}$ is a pivot. If the pivot of the statistic $\hat{\theta}_n$ and the parameter θ has another form, the resulting interval is necessarily less accurate. E.g. if the population is $N(\mu, \sigma^2)$ and $\theta = \sigma^2$, the pivot of its estimator $\hat{\theta}_n = S^2$ is

$$(n-1)\frac{S^2}{\sigma^2} \sim \chi^2_{n-1},$$

so $(S^2 - \sigma^2)/\widehat{\text{se}}$ cannot be said to be a pivot. Then a transformation of $\hat{\theta}_n$ can improve the accuracy of the basic or bootstrap-t interval. E.g. if $(\log(\hat{\theta}_n) - \log(\theta))/\widehat{\text{se}}\{\log(\hat{\theta}_n)\}$ is approximately a pivot, the bootstrap-t interval for the parameter $\log(\theta)$,

$$L \le \log(\theta) \le U,$$

transforms to the interval for θ

$$\exp(L) \le \theta \le \exp(U).$$

(Supplied: since exp is increasing, $\{L \le \log\theta \le U\} = \{e^L \le \theta \le e^U\}$, so the coverage probability is unchanged.)

8.4 The bootstrap percentile interval

The bootstrap percentile (quantile) interval (percentile C.I.) is very simple to compute. Given an estimator $\hat{\theta}_n$ of θ and its bootstrap replicates $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ (or the bootstrap sampling distribution G^* of $\hat{\theta}_n$), the bootstrap percentile interval at level $1 - \alpha$ is given by the $\frac{\alpha}{2}$- and $(1 - \frac{\alpha}{2})$-quantiles of G^*:

$$\theta \in (Q_{G^*}(\tfrac{\alpha}{2}),\ Q_{G^*}(1 - \tfrac{\alpha}{2})). \tag{8.12}$$

With $B = 1999$ and $\alpha = 0.05$, the lower and upper limits are the 50th and 1950th of the ordered bootstrap replicates. Writing the sorted $\hat{\theta}_n^{*(b)}$ as order statistics $\hat{\theta}^*_{(1)} \leq \cdots \leq \hat{\theta}^*_{(B)}$,

$$\theta \in (\hat{\theta}^*_{(50)},\ \hat{\theta}^*_{(1950)})$$

is the 95% percentile interval.

Example 8.5

The sampling distribution G^* obtained by $B = 1999$ bootstrap resamples of $\hat{\theta}_n = \bar{X}_n$ for the Team Fortress 2 data is shown in Figure 8.4 of the book. The 95% percentile interval for the population mean θ is $(\hat{\theta}^*_{(50)}, \hat{\theta}^*_{(1950)}) = (60.7, 80.5)$, computed as follows:

```
> set.seed(1)
> sort.b = replicate(B, mean(sample(TFtime, n, replace = TRUE)))
    %>% sort()
> (perc.int = c(sort.b[50],sort.b[1950]))
[1] 60.71154 80.50310
```

This percentile interval is marked in Figure 8.4.

The 95% percentile interval is the "middle" 95% of the bootstrap sampling distribution, so it can easily be computed by any bootstrap resampling. Why can this interval be trusted? Inferring the location of the parameter θ from the sampling distribution of a statistic may look natural, but this understanding, as the statistician Peter Hall said, amounts to "looking up the wrong statistical tables backwards" (Hall, 2013, p. 36): it finds the approximate answer only by accident.

The percentile interval rests on the following assumption. When introducing the basic and bootstrap-t intervals, we saw that one may transform to make $(\hat{\theta}_n - \theta)/\widehat{\text{se}}$ a pivot, e.g. the log transform $(\hat{\theta}_n \mapsto \varphi(\hat{\theta}_n) := \log(\hat{\theta}_n))$. But in a given problem such a transform can be hard to find. The percentile interval assumes that there exists some monotone increasing transform φ such that the transformed values form a pivot following the standard normal distribution, i.e.

$$\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1). \tag{8.13}$$

(Or approximately a pivot.) Transforming the bootstrap replicates $\hat{\theta}_n^*$ too,

$$\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n) \overset{d}{\approx} \varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1), \tag{8.14}$$

so the distribution of $\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n)$ is also approximately standard normal. The left side $\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n) \sim N(0, 1)$ of this formula is about the distribution of the bootstrap replicates $\varphi(\hat{\theta}_n^*)$, with $\varphi(\hat{\theta}_n)$ an already observed, fixed value. Remember that on the right side of (8.14), or in (8.13), $\varphi(\hat{\theta}_n)$ is not a fixed value but a statistic, a random variable.

If the transform φ were known, (8.13) would give the exact $1 - \alpha$ interval $\varphi(\theta) \in (\varphi(\hat{\theta}_n) + \Phi^{-1}(\frac{\alpha}{2}),\ \varphi(\hat{\theta}_n) + \Phi^{-1}(1 - \frac{\alpha}{2}))$, and transforming it the exact $1 - \alpha$ interval for θ:

$$\theta \in \left(\varphi^{-1}\{\varphi(\hat{\theta}_n) + \Phi^{-1}(\tfrac{\alpha}{2})\},\ \varphi^{-1}\{\varphi(\hat{\theta}_n) + \Phi^{-1}(1 - \tfrac{\alpha}{2})\}\right) \tag{8.15}$$

(possible since the transform φ is monotone increasing).

In interval (8.15), $\varphi(\hat{\theta}_n) + \Phi^{-1}(\frac{\alpha}{2})$, expressing the lower limit, is a random variable on the right of (8.14) but a fixed value with respect to the left side. This fixed value is the expectation $\varphi(\hat{\theta}_n)$ of the bootstrap replicates

$$\varphi(\hat{\theta}_n^*) \sim N(\varphi(\hat{\theta}_n), 1)$$

shifted by $\Phi^{-1}(\frac{\alpha}{2})$. I.e. using the bootstrap approximation (8.14), $\varphi(\hat{\theta}_n) + \Phi^{-1}(\frac{\alpha}{2})$ is exactly the $\frac{\alpha}{2}$-quantile of the distribution $N(\varphi(\hat{\theta}_n), 1)$ of the random variable $\varphi(\hat{\theta}_n^*)$.[47] Likewise the upper limit $\varphi(\hat{\theta}_n) + \Phi^{-1}(1 - \frac{\alpha}{2})$ is approximately the $(1 - \frac{\alpha}{2})$-quantile of the distribution of $\varphi(\hat{\theta}_n^*)$.

For simplicity take $B = 1999$, $\alpha = 0.05$, and the bootstrap replicates arranged in order $\hat{\theta}^*_{(1)} \le \cdots \le \hat{\theta}^*_{(1999)}$. Since φ is monotone increasing,

$$\varphi(\hat{\theta}^*_{(1)}) \le \cdots \le \varphi(\hat{\theta}^*_{(1999)}), \tag{8.16}$$

and the $\frac{\alpha}{2}$-quantile of the distribution of $\varphi(\hat{\theta}_n^*)$ is $\varphi(\hat{\theta}^*_{(50)})$, the $(1 - \frac{\alpha}{2})$-quantile $\varphi(\hat{\theta}^*_{(1950)})$. So the interval (8.15) is approximated by the percentile interval:

$$\begin{aligned}\theta &\in \left(\varphi^{-1}\{\varphi(\hat{\theta}^*_{(50)})\},\ \varphi^{-1}\{\varphi(\hat{\theta}^*_{(1950)})\}\right)\\ &= (\hat{\theta}^*_{(50)},\ \hat{\theta}^*_{(1950)})\\ &= (Q_{G^*}(\tfrac{\alpha}{2}),\ Q_{G^*}(1 - \tfrac{\alpha}{2})).\end{aligned}$$

The transform φ does not affect the computation of the percentile interval, so it need not be known. The bootstrap percentile interval has the advantage of great simplicity and is used often.

8.5 Adjusted percentile intervals

Whether the percentile interval succeeds depends on the existence of a transform φ satisfying assumption (8.13), and in general such a transform cannot be said always to exist. Adjusted percentile intervals weaken this transform assumption so that approximation is possible over a wider range of populations. Among adjusted percentile intervals we look at the most representative, the bias-corrected and accelerated (BCa) interval, and its special case, the bias-corrected interval.

Suppose we have an estimator $\hat{\theta}_n$ of the parameter θ and its bootstrap replicates $\{\hat{\theta}_n^{*(b)} : b = 1, \ldots, B\}$ (or the bootstrap sampling distribution G^* of $\hat{\theta}_n$). In the percentile interval we assumed a monotone increasing transform φ with $\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1)$; but $\varphi(\hat{\theta}_n)$ may be a biased estimator of $\varphi(\theta)$, and the standard error $\mathrm{se}(\varphi(\hat{\theta}_n))$ may depend on the value of $\varphi(\theta)$. Taking all this into account, assume that for an unknown b (bias-correction factor), a (acceleration factor, or skewness-correction factor) and an unknown monotone increasing transform φ,

$$\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(-b\sigma_\varphi, \sigma_\varphi^2), \qquad \sigma_\varphi = 1 + a\varphi(\theta). \tag{8.17}$$

If $a = 0$, then $\mathrm{se}(\varphi(\hat{\theta}_n)) = \sigma_\varphi = 1$ for every value of $\varphi(\theta)$. If $b = 0$, $\varphi(\hat{\theta}_n)$ is an unbiased estimator of $\varphi(\theta)$. Why the unknown b is called the bias-correction factor is obvious, but why the unknown a is called the acceleration or skewness-correction factor rather than a "variance-correction factor" is hard to explain simply at the level of this book; just accept it for now. The form $\sigma_\varphi = 1 + a\varphi(\theta)$ of the standard error can be understood as a first-order approximation of a possibly more complicated standard error. As with the percentile interval, if a transform φ satisfying (8.17) merely exists, the

[47] In $\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1)$, $\varphi(\hat{\theta}_n)$ is a random variable and $\varphi(\theta)$ a fixed parameter. So the limits of interval (8.15) are random variables under $\varphi(\hat{\theta}_n) - \varphi(\theta) \sim N(0, 1)$, while under $\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n) \sim N(0, 1)$ (the left side of (8.14)) they are fixed values. A fixed value $\varphi(\hat{\theta}_n) + \Phi^{-1}(p)$ is the p-quantile of the distribution $N(\varphi(\hat{\theta}_n), 1)$ of $\varphi(\hat{\theta}_n^*)$, which is not hard to see.

adjusted percentile interval can be obtained even without knowing φ. The two unknowns b and a must be estimated from the data, but first suppose b and a are known.

The model (8.17) gives the exact $1-\alpha$ interval for the parameter $\varphi(\theta)$

$$\varphi(\theta) \in \left(\frac{\varphi(\hat{\theta}_n) + b + \Phi^{-1}(\frac{\alpha}{2})}{1 - a\{b + \Phi^{-1}(\frac{\alpha}{2})\}},\ \frac{\varphi(\hat{\theta}_n) + b + \Phi^{-1}(1-\frac{\alpha}{2})}{1 - a\{b + \Phi^{-1}(1-\frac{\alpha}{2})\}} \right) := (L, U). \tag{8.18}$$

Coverage of (8.18) (supplied; Exercise 8.3 of the book). Write $\hat{\varphi} = \varphi(\hat{\theta}_n)$, $\phi = \varphi(\theta)$, and assume $1 + a\phi > 0$. By (8.17), $Z := \frac{\hat{\varphi} - \phi}{1 + a\phi} + b \sim N(0,1)$. Solving for ϕ,

$$\phi = g(Z) := \frac{\hat{\varphi} - (Z - b)}{1 + a(Z - b)}, \qquad g'(Z) = -\frac{1 + a\hat{\varphi}}{(1 + a(Z-b))^2} < 0$$

(when $1 + a\hat{\varphi} > 0$). So g is decreasing, and with $z_l = \Phi^{-1}(\frac{\alpha}{2})$, $z_u = \Phi^{-1}(1-\frac{\alpha}{2}) = -z_l$,

$$g(-z_l) = \frac{\hat{\varphi} + b + z_l}{1 - a(b + z_l)} = L, \qquad g(-z_u) = \frac{\hat{\varphi} + b + z_u}{1 - a(b + z_u)} = U.$$

Hence $\phi \in [L, U] \iff Z \in [-z_u, -z_l] = [z_l, z_u]$, which has probability $1 - \alpha$. ■

Since the transform φ is monotone increasing, the interval for θ is obtained with the inverse transform of φ: $\theta \in (\varphi^{-1}(L), \varphi^{-1}(U))$ is the exact $1-\alpha$ interval for θ. We do not know φ, so this interval will be approximated by the bootstrap.

Rewriting (8.17) as a pivot following the standard normal:

$$\frac{\varphi(\hat{\theta}_n) - \varphi(\theta)}{1 + a\varphi(\theta)} + b \sim N(0,1).$$

As in the percentile bootstrap, if a transform φ satisfying (8.17) exists, the bootstrap-approximated pivot also (approximately) follows the standard normal:

$$\frac{\varphi(\hat{\theta}_n^*) - \varphi(\hat{\theta}_n)}{1 + a\varphi(\hat{\theta}_n)} + b \overset{d}{\approx} \frac{\varphi(\hat{\theta}_n) - \varphi(\theta)}{1 + a\varphi(\theta)} + b \sim N(0,1).$$

Looking at the left side of this formula (again: $\varphi(\hat{\theta}_n)$ is a fixed value on the left and a random variable on the right), the transformed bootstrap replicates $\varphi(\hat{\theta}_n^*)$ approximately follow the normal $N(\mu_*, \sigma_*^2)$ with

$$\mu_* = \varphi(\hat{\theta}_n) - b\{1 + a\varphi(\hat{\theta}_n)\}, \qquad \sigma_*^2 = \{1 + a\varphi(\hat{\theta}_n)\}^2.$$

The limits (L, U) of the interval (8.18) for $\varphi(\theta)$ are no longer random variables but fixed values, so they can be expressed by the quantile function Q^* of $\varphi(\hat{\theta}_n^*) \sim N(\mu_*, \sigma_*^2)$ (Figure 8.5 of the book). The interval (8.18) for $\varphi(\theta)$ is expressed by the p_l- and p_u-quantiles of the distribution of $\varphi(\hat{\theta}_n^*)$:

$$(L, U) = (Q^*(p_l),\ Q^*(p_u)). \tag{8.19}$$

By the same logic as in the percentile interval,

$$\theta \in (Q_{G^*}(p_l),\ Q_{G^*}(p_u))$$

is the bootstrap approximation of the exact interval $\theta \in (\varphi^{-1}(L), \varphi^{-1}(U))$. Here the two probabilities p_l, p_u, obtained by solving (8.19), are

$$\begin{aligned} p_l &= \Phi\Big(\frac{L - \mu_*}{\sigma_*}\Big) = \Phi\Big(\frac{b + \Phi^{-1}(\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(\frac{\alpha}{2}))} + b\Big), \\ p_u &= \Phi\Big(\frac{U - \mu_*}{\sigma_*}\Big) = \Phi\Big(\frac{b + \Phi^{-1}(1-\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(1-\frac{\alpha}{2}))} + b\Big). \end{aligned} \tag{8.20}$$

Derivation of (8.20) from (8.19) (supplied; Exercise 8.4 of the book). Write $\hat{\varphi} = \varphi(\hat{\theta}_n)$, $w = b + \Phi^{-1}(\frac{\alpha}{2})$. From (8.18), $L(1-aw) = \hat{\varphi} + w$, so $L - \hat{\varphi} = w + aLw = w(1 + a\frac{\hat{\varphi}+w}{1-aw}) = \frac{w(1+a\hat{\varphi})}{1-aw}$. Hence

$$\frac{L-\mu_*}{\sigma_*} = \frac{L - \hat{\varphi} + b(1+a\hat{\varphi})}{1+a\hat{\varphi}} = \frac{w}{1-aw} + b,$$

and $p_l = P(\varphi(\hat{\theta}_n^*) \le L) = \Phi((L-\mu_*)/\sigma_*)$ is as stated. The same computation with $\Phi^{-1}(1-\frac{\alpha}{2})$ gives p_u. ■

Substituting the two probabilities (8.20) into (8.19), the $1-\alpha$ **bias-corrected and accelerated (BCa) percentile interval** is

$$\theta \in \left(Q_{G^*}\Big[\Phi\Big\{\frac{b + \Phi^{-1}(\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(\frac{\alpha}{2}))} + b\Big\}\Big],\ Q_{G^*}\Big[\Phi\Big\{\frac{b + \Phi^{-1}(1-\frac{\alpha}{2})}{1 - a(b + \Phi^{-1}(1-\frac{\alpha}{2}))} + b\Big\}\Big] \right).$$

If $a = 0$ (or one assumes so), a simpler expression is obtained; the interval

$$\theta \in (Q_{G^*}[\Phi\{2b + \Phi^{-1}(\tfrac{\alpha}{2})\}],\ Q_{G^*}[\Phi\{2b + \Phi^{-1}(1-\tfrac{\alpha}{2})\}])$$

contains only the bias-correction factor b, so it is called the **bias-corrected (BC) percentile interval**.

8.5.1 Estimating the bias-correction and acceleration factors

As in the percentile interval, the transform φ satisfying (8.17) need not be known, but the bias-correction factor b and acceleration factor a are estimated from the data as follows.

Bias-correction factor: $\hat{b} = \Phi^{-1}(G^*(\hat{\theta}_n)) = \Phi^{-1}(\#\{\hat{\theta}_n^{*(b)} \le \hat{\theta}_n\}/(B+1))$. If $\hat{\theta}_n$ is the median of the bootstrap sampling distribution, $\hat{b} = 0$ (then it is called median-unbiased).

Why this estimates b (supplied)

Under (8.17) and its bootstrap version, $P(\varphi(\hat{\theta}_n^*) \le \varphi(\hat{\theta}_n)) = \Phi((\hat{\varphi} - \mu_*)/\sigma_*) = \Phi(b)$. Since φ is increasing, the left side is $P(\hat{\theta}_n^* \le \hat{\theta}_n) = G^*(\hat{\theta}_n)$. So $b = \Phi^{-1}(G^*(\hat{\theta}_n))$.

Acceleration factor: to estimate the acceleration or skewness-correction factor a, a resampling independent of the bootstrap resampling is carried out. Recall that the statistic $\hat{\theta}_n$ is computed from all n observations, $\hat{\theta}_n = T(\mathbb{X}_n)$. The resampling used here deletes the observations one at a time and is repeated n times. With $\mathbb{X}_{(i)} := \{X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n\}$ the data without the ith observation X_i, the recomputed statistic $\hat{\theta}_{(i)} = T(\mathbb{X}_{(i)})$, and their mean $\hat{\theta}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^n \hat{\theta}_{(i)}$,

$$\hat{a} = \frac{1}{6} \frac{\sum_{i=1}^n \left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^3}{\left[\sum_{i=1}^n \left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^2\right]^{3/2}} \tag{8.21}$$

estimates the acceleration factor a. This expression has the form of an estimator of skewness divided by 6, which hints why it is called the "skewness-correction factor."

The resampling used to estimate the acceleration factor is called the **jackknife** (Quenouille, 1949). The name jackknife (a pocket knife) was later used by Tukey (1958) when he employed it to estimate the bias and standard error of estimators. The bootstrap method can be seen as an extension of the jackknife.

Example 8.6 (Bias-corrected percentile interval)

Using nonparametric bootstrap resampling for $\hat{\theta}_n = \bar{X}_n$ of the Team Fortress 2 data, compute the 95% bias-corrected percentile interval for the population mean θ. First, to obtain the bias-correction factor $\hat{b}$, comparing the bootstrap sampling distribution or replicates $\hat{\theta}_n^{*(b)}$ with $\hat{\theta}_n$ gives $\#\{\hat{\theta}_n^{*(b)} \leq \hat{\theta}_n\}/(B+1) = 0.509$: the median of this distribution is (slightly) smaller than $\hat{\theta}_n$, a downward bias.

```
> hat.theta = mean(TFtime)
> n = length(TFtime); B = 1999
> theta.boot = replicate(B, mean(sample(TFtime, n, replace =
    TRUE)))
> b.factor = qnorm(sum(theta.boot <= hat.theta)/(B+1))
> b.factor
[1] 0.02381522
```

To correct the bias (towards smaller values) of the estimate, the lower and upper limits are adjusted slightly "upward."

```
> (BC.int.probs = pnorm(2*b.factor + c(qnorm(0.025), qnorm
    (1-0.025))))
[1] 0.02791671 0.97765680
```

Where the percentile interval used the 2.5% and 97.5% percentiles, the bias-corrected interval uses the 2.8% and 97.8% percentiles.

```
> (BC.interval = quantile(theta.boot, BC.int.probs))
2.791671% 97.76568%
 60.96237  80.75818
```

Example 8.7 (Bias-corrected and accelerated percentile interval)

Next, the 95% BCa percentile interval for θ. The bias-correction factor was already computed, so carry out the jackknife resampling (8.21) for the acceleration factor $\hat{a}$.

```
> theta.jack = vector(length = n)
> for (i in 1:n){theta.jack[i] = mean(TFtime[-i])}
> acc = sum((mean(theta.jack) - theta.jack)^3)/
+   (6 * sum(( mean(theta.jack) - theta.jack)^2)^(3/2))
> acc
[1] 0.02325921
```

Compared with the bias-corrected interval, the interval is adjusted further "upward."

```
> bz = b.factor + c(qnorm(0.025), qnorm(1-0.025))
> (BCa.int.probs = pnorm( (bz)/(1 - acc * (bz)) + b.factor ))
[1] 0.03370732 0.98229140
> (BCa.interval = quantile(theta.boot, BCa.int.probs))
3.370732% 98.22914%
 61.34667  81.40983
```

8.6 Comparing confidence intervals

We have looked at several ways of resampling the estimator $\hat{\theta}_n$ by the bootstrap to obtain intervals for the parameter θ. In particular, with the "Team Fortress 2" data, observations from a population

with a very heavy right tail, we computed intervals for the population mean θ using $\hat{\theta}_n = \bar{X}_n$. The table summarizes the seven 95% intervals; "BC" means the bias-corrected and "BCa" the bias-corrected and accelerated percentile interval.

	lower	upper
normal-approximation interval	60.95384	80.33238
normal-approximation bootstrap interval	60.91453	80.52683
basic bootstrap interval	60.78312	80.57468
bootstrap-t interval	**61.82523**	**82.17272**
percentile interval	60.71154	80.50310
BC percentile interval	60.96237	80.75818
BCa percentile interval	**61.34667**	**81.40983**

The sample is large enough here, so the normal approximation by the CLT is somewhat accurate and the normal-approximation interval

$$\bar{X}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} = (60.9, 80.3) \tag{8.22}$$

is also somewhat accurate. Why use bootstrap intervals anyway? The answer has two parts.

The first is when the standard error or sampling distribution of the estimator cannot be derived: when the asymptotic distribution of the estimator is unknown, or it cannot be asserted to converge to a normal, or it converges to a normal but the sample is not large enough to use the normal approximation. Then a bootstrap interval, approximating the unknown sampling distribution from the data, is a good answer.

The second is the accuracy of intervals. When the population distribution cannot be specified, the coverage probability $P(\theta \in \text{CI})$ of all intervals introduced so far cannot be asserted to equal the given level $1 - \alpha$. But it is known that as the sample grows, the coverage converges to the given level. If the population is not normal but the estimator or pivot converges to a normal, the coverage of a normal interval like (8.22) is known to be

$$P(\theta \in \bar{X}_n \pm z_{\alpha/2} s/\sqrt{n}) = 1 - \alpha + O(n^{-1/2}).^{48}$$

That is, the difference between the coverage probability and the level converges to 0 at the rate $n^{-1/2}$. Among bootstrap intervals, the bootstrap-t and the bias-corrected and accelerated (BCa) percentile intervals are known to converge to the level faster:

$$\begin{aligned} P(\theta \in \text{"Bootstrap-}t\text{ CI"}) &= 1 - \alpha + O(n^{-1}), \\ P(\theta \in \text{"BCa CI"}) &= 1 - \alpha + O(n^{-1}). \end{aligned}$$

Being more accurate than the normal interval, they are said to have "second-order accuracy." By comparison the normal-approximation bootstrap interval (when the normal approximation holds), the basic bootstrap interval, the percentile interval and the bias-corrected interval have difference $O(n^{-1/2})$ between coverage and level, "first-order accuracy." In the table above, the second-order accurate bootstrap-t and BCa intervals are similar to each other and differ from the other, first-order accurate intervals.

For the basic and bootstrap-t intervals, whether the "pivot" holds is very important. So one needs to find a suitable transform, build the interval with an approximate pivot, and convert it back to an interval for the original parameter. Also, when the standard error is hard to compute, the

[48] A sequence a_n is $O(n^{-1/2})$ if for some $0 < M < \infty$ and all n, $\frac{|a_n|}{n^{-1/2}} = n^{1/2}|a_n| \le M$: a_n converges to 0 as fast as $n^{-1/2}$ or faster. E.g. comparing $a_n = n^{-1/2}$ and $b_n = n^{-1}$, b_n converges to 0 faster than a_n; then $a_n = O(n^{-1/2})$, $b_n = O(n^{-1})$.

bootstrap-t interval requires second-level bootstrap resampling, so the computation time becomes very long.

Percentile methods are more convenient since they need no effort to find a transform to a pivot. Among them, the BCa percentile interval, which guarantees second-order accuracy, is widely recommended. However, when bootstrap sampling does not take place within one population (several populations, regression, etc.), there is no guarantee that the estimate of the acceleration factor by (8.21) is accurate.

Chapter 9. Bootstrap Tests

Overview

The bootstrap approximates an unknown population by the given sample to approximate the sampling distribution of a statistic or estimator: for a population F and a parameter $\theta = \theta(F)$, it approximates the distribution $P(\hat{\theta}_n \le x \mid F)$ of the statistic $\hat{\theta}_n$ by bootstrap resampling, and this bootstrap sampling distribution generally depends on the actual value of θ. In contrast, in a hypothesis test comparing $H_0 : \theta = \theta_0$ with $H_1 : \theta \ne \theta_0$, the interest is the distribution of the statistic $\hat{\theta}_n$ when the null is true, i.e. when $\theta = \theta_0$ (the null distribution). In Part I we studied permutation tests, which obtain the null distribution of a test statistic by shuffling the observations in various situations.

Can the null distribution be approximated by bootstrap resampling? The bootstrap distribution approximates the actual population distribution F, so is it impossible when the population F_0 under the null differs from the actual F? Under a completely nonparametric model (no restriction on the form of the population), bootstrap approximation under F_0 is generally impossible. This chapter confirms that in some special situations, particularly semiparametric models with some modeling added, the null distribution of a statistic can be approximated by the bootstrap, and looks at test methods using this.

To aid understanding, we first look at bootstrap tests under a parametric model (e.g. the normal model), then explain bootstrap tests in nonparametric or semiparametric models with examples of tests about the location of one population and comparing two populations.[49]

9.1 Parametric bootstrap tests

A parametric bootstrap test is a test using parametric bootstrap resampling under the assumption that the population of the observations follows some parametric model (e.g. the normal family, the exponential family). Since a parametric model is assumed, one might think the bootstrap is unnecessary, but it is a test method used a lot in practice.

9.1.1 Resampling with a pivot

To explain the idea, assume observations $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ in the familiar normal model, and consider testing $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$, with the variance σ^2 unknown. The optimal test is the t-test with the statistic

$$T_0 = \frac{\bar{X}_n - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \quad (\text{when } H_0 : \mu = \mu_0 \text{ is true}). \tag{9.1}$$

Since T_0 has the form of a pivot, even when the null is not true, e.g. when $\mu = \mu_1$,

$$T_1 = \frac{\bar{X}_n - \mu_1}{s/\sqrt{n}} \sim t_{n-1} \quad (\text{when } \mu = \mu_1 \text{ is true}) \tag{9.2}$$

follows the same t_{n-1} distribution. As explained in Section 7.4, parametric bootstrap resampling substitutes the estimates $\hat{\mu} = \bar{X}_n$, $\hat{\sigma}^2 = s^2$ of the parameters μ, σ^2 and repeatedly generates bootstrap samples $\mathbb{X}_n^* = \{X_1^*, \dots, X_n^*\}$ from the bootstrap "population" $F^* = N(\hat{\mu}, \hat{\sigma}^2)$. Then

$$X_1^*, \dots, X_n^* \overset{\text{i.i.d.}}{\sim} N(\hat{\mu}, \hat{\sigma}^2),$$

[49] Bootstrap tests can also be used in more complex situations including regression; bootstrap tests in regression are treated in Chapter 10. For more general bootstrap test methods see Davison and Hinkley (1997).

so with the bootstrap-replicated sample mean $\bar{X}_n^* = \frac{1}{n}\sum_{i=1}^n X_i^*$ and sample standard deviation

$$s^* = \sqrt{\frac{1}{n-1}\sum_{i=1}^n (X_i^* - \bar{X}_n^*)^2},$$

the t-statistic is

$$T^* = \frac{\bar{X}_n^* - \hat{\mu}}{s^*/\sqrt{n}} \sim t_{n-1}. \tag{9.3}$$

Note that the distributions of (9.1), (9.2) and (9.3) all coincide. Since the test statistic T_0 is a pivot, its null distribution equals the sampling distribution of the pivot $T = (\bar{X}_n - \mu)/(s/\sqrt{n})$, where μ is the population mean μ of the actual population. This sampling distribution can be bootstrap-approximated.[50] The key of the parametric bootstrap test is to approximate this sampling distribution (the null distribution of T) by the bootstrap sampling distribution, i.e. the distribution of the replicates T^*, using the parametric bootstrap.

9.1.2 Resampling under the null hypothesis

When a pivot containing the test statistic for the given hypothesis is hard to find in a given parametric model, a statistic distinguishing null and alternative (even if not a pivot) can still be used. Continue assuming a normal population $N(\mu, \sigma^2)$ and testing $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$. Even without deep knowledge of the normal distribution, the statistic $D = \bar{X}_n - \mu_0$ obviously tends to be larger when the alternative is true, so take it as test statistic. The null distribution of this (roughly chosen) D is

$$D = \bar{X}_n - \mu_0 \sim N(0, \sigma^2) \quad (\text{when } H_0 : \mu = \mu_0 \text{ is true}). \tag{9.4}$$

(More precisely $N(0, \sigma^2/n)$.) Accepting the null $H_0 : \mu = \mu_0$, the distribution of the observations is $X_i \sim N(\mu_0, \sigma^2)$. If the still unknown σ^2 is estimated by its maximum likelihood estimate under the null,

$$\hat{\sigma}_0^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu_0)^2,$$

the most plausible "population" under the null is $F_0^* = N(\mu_0, \hat{\sigma}_0^2)$, and parametric bootstrap resampling can be done from this distribution F_0^*. With bootstrap samples $\{X_1^{0*}, \ldots, X_n^{0*}\} \overset{\text{i.i.d.}}{\sim} N(\mu_0, \hat{\sigma}_0^2)$, the bootstrap replicates are

$$D^* = \bar{X}_n^{0*} - \mu_0 \sim N(0, \hat{\sigma}_0^2/n). \tag{9.5}$$

When $\hat{\sigma}_0^2 \approx \sigma^2$ (when the null is true and the sample is large), the distribution (9.5) of D^* is expected to approximate the null distribution (9.4) of D well. So taking the distribution of D^* generated by the bootstrap as the null distribution of D is the bootstrap test with resampling under the null.

If the pivot T_0 (9.1) is taken as the test statistic, the bootstrap replicates under resampling under the null (here the bootstrap sample is $X_i^{0*} \sim N(\mu_0, \hat{\sigma}_0^2)$) are

$$T_0^* = \frac{\bar{X}_n^{0*} - \mu_0}{s^{0*}/\sqrt{n}} \sim t_{n-1}, \tag{9.6}$$

so the (bootstrapped) distribution of T_0^* equals the distribution t_{n-1} (9.3) of the pivot-resampled T^*. (Supplied, Exercise 9.1 of the book: under $X_i^{0*} \overset{\text{i.i.d.}}{\sim} N(\mu_0, \hat{\sigma}_0^2)$, T_0^* is the one-sample t-statistic of a normal sample with true mean μ_0, so it is exactly t_{n-1} whatever $\hat{\sigma}_0^2$.) I.e. when the test statistic is a pivot, resampling under the null and resampling with the pivot are approximately equivalent. When the test statistic is a pivot the bootstrap approximation is more accurate, so it is better to take a (or an approximately) pivotal test statistic. But when a pivot for the parameter to be tested is hard to find, resampling under the null may be the only bootstrap test method.

[50] Under normality all distributions are t-distributions, so the bootstrap is not needed in this example. The bootstrap approximation is used when the sampling distribution of the statistic T is unknown.

9.1.3 Bootstrap tests in general parametric models

Summarizing the two methods explained for the normal distribution for a general parametric model: in a parametric model $F(\cdot;\theta,\eta)$ we test the null $H_0:\theta=\theta_0$. The parameter θ is the one whose value the null fixes, and η the parameter the null does not fix. E.g. in the normal model $N(\mu,\sigma^2)$ with $\theta=\mu$, $\eta=\sigma^2$. If an exponential model $\text{Exp}(\theta)$ is assumed and a hypothesis about the mean set, there is no parameter η, so it can be ignored. Given data $\mathbb{X}_n=\{x_1,\ldots,x_n\}$, the test statistic is taken as a pivot (or approximate pivot) $S=S(\mathbb{X}_n,\theta_0)$.[51] (Supplied check: for Laplace(μ,b), $|X_i-\mu|/b\sim\text{Exp}(1)$, so $2\sum_i|X_i-\mu|/b\sim\chi^2_{2n}$; with $b=2$ this is $\sum_i|X_i-\mu|$.) The test rejects the null when the observed value S_{obs} of S is large.

1. **Resampling under the null:** assume the null $\theta=\theta_0$ is true, find from the data the maximum likelihood estimate $\hat\eta_0$ of the unknown parameter η, carry out parametric bootstrap resampling from $F(\cdot;\theta_0,\hat\eta_0)$, and obtain the bootstrap replicates $S_0^{*(b)}=S(\mathbb{X}_n^{*(b)},\theta_0)$ of the statistic S. The null distribution of S is taken as the distribution of $\{S_0^{*(b)}:b=1,\ldots,B\}$, and the p-value computed:

$$p_0=\frac{1+\sum_{b=1}^B I(S_0^{*(b)}\geq S_{\text{obs}})}{B+1}.$$

2. **Resampling with a pivot:** obtain from the given data the maximum likelihood estimates $\hat\theta,\hat\eta$, carry out parametric bootstrap resampling from $F(\cdot;\hat\theta,\hat\eta)$, and obtain the bootstrap replicates $S^{*(b)}=S(\mathbb{X}_n^{*(b)},\hat\theta)$ of the statistic S. ($\hat\theta$ is used in bootstrap resampling instead of θ_0.) Since S is a pivot, its null distribution is approximated by the distribution of $\{S^{*(b)}:b=1,\ldots,B\}$. So the p-value is

$$p_{\text{pivot}}=\frac{1+\sum_{b=1}^B I(S^{*(b)}\geq S_{\text{obs}})}{B+1}.$$

Two examples compare the resampling methods and the choice of test statistic.

Parametric bootstrap test for the location of one population. Using the gene expression levels $x_1,\ldots,x_n$ of the $n=25$ AML patients among the leukemia data seen in Example 4.1 of Chapter 4, we test the hypothesis $H_0:\theta=166$ against $H_1:\theta>166$ about the location (mean) θ of this population. Figure 9.1 of the book shows these data.

Assume the variation of gene expression in this population can be modeled by the Laplace distribution, i.e. $X_i\overset{\text{i.i.d.}}{\sim}\text{Laplace}(\theta,b)$. Then the maximum likelihood estimates from the data are $\hat\theta=\text{median}\{X_i\}$, $\hat b=\frac{1}{n}\sum_{i=1}^n|X_i-\hat\theta|$, and under the null ($\theta=\theta_0=166$) the maximum likelihood estimate of b is $\hat b_0=\frac{1}{n}\sum_{i=1}^n|X_i-\theta_0|$. For the following three test statistics we carry out bootstrap tests with the two kinds of resampling (six bootstrap tests in all):

- t-statistic: $T=(\bar X_n-\theta_0)/(s/\sqrt{n})$;
- difference of means: $D=\bar X_n-\theta_0$;
- ratio of medians: $M=\text{median}\{X_i\}/\theta_0=\hat\theta/\theta_0$.

The observed values of the statistics:

[51] The pivot for μ in the normal $N(\mu,\sigma^2)$ is $S(\mathbb{X}_n,\mu_0)=(\bar X_n-\mu_0)/(s/\sqrt{n})\sim t_{n-1}$, and the pivot for σ^2 is $S(\mathbb{X}_n,\sigma_0^2)=(n-1)s^2/\sigma_0^2\sim\chi^2_{n-1}$. Assuming a parametric model does not mean a pivot can always easily be found. In the Laplace distribution Laplace$(\mu,2)$ the pivot for μ is $S(\mathbb{X}_n,\mu_0)=\sum_{i=1}^n|X_i-\mu_0|\sim\chi^2_{2n}$, but in Laplace$(\mu,b)$ a pivot for μ is not easy to find. Then the approximate pivot $S(\mathbb{X}_n,\mu_0)=(\bar X_n-\mu_0)/(s/\sqrt{n})\overset{\cdot}{\sim}N(0,1)$ from the CLT can be used.

```
> leukemia <- read.csv("data/leukemia4.csv")
> x <- leukemia$ge[leukemia$label == "AML"]
> # hypothesis H0: mu =mu0, H1: mu > mu0
> mu0 = 166
> # test statistic (two test stats)
> t0 = sqrt(n) * (mean(x) - mu0 ) / sd(x)
> d0 = mean(x) - mu0
> m0 = median(x) / mu0
> c(t0,d0,m0)
[1]  2.722207 67.120000  1.451807
```

Resampling under the null. Bootstrap samples are drawn from Laplace$(\theta_0, \hat{b}_0)$, and from each the mean $\bar{X}_n^{0*}$, standard deviation s^{0*} and median median$\{X_i^{0*}\}$ are computed; the distributions of the bootstrap replicates

$$T_0^* = (\bar{X}_n^{0*} - \theta_0)/(s^{0*}/\sqrt{n}), \qquad D_0^* = \bar{X}_n^{0*} - \theta_0, \qquad M_0^* = \text{median}\{X_i^{0*}\}/\theta_0 \tag{9.7}$$

are taken as the null distributions of the respective statistics. (In the superscript "$0*$" of statistics computed from bootstrap samples, "0" means a sample under the null model and "$*$" a bootstrap sample.)

Resampling with a pivot. The method proceeds assuming each statistic is an approximate pivot. (By the CLT, T and D are easily seen to be approximate pivots, but $M(=\text{median}\{X_i\}/\theta)$ has asymptotically mean 1 and variance proportional to θ^{-2}, so it is not even an approximate pivot.) Bootstrap samples are drawn from Laplace$(\hat{\theta}, \hat{b})$, and from each the mean $\bar{X}_n^*$, standard deviation s^* and median median$\{X_i^*\}$ are computed; the distributions of the bootstrap replicates

$$T^* = (\bar{X}_n^* - \hat{\theta})/(s^*/\sqrt{n}), \qquad D^* = \bar{X}_n^* - \hat{\theta}, \qquad M^* = \text{median}\{X_i^*\}/\hat{\theta} \tag{9.8}$$

are taken as the null distributions. The bootstrap-approximated null distributions in each case are obtained by:

```
# assume Laplace dist'n, and get MLEs
mu.hat = median(x); b.hat = mean(abs(x - mu.hat)) # mle (general)
b.hat0 = mean(abs(x-mu0))  # mle under H0

# parametric bootstrap for (T, D, M)
set.seed(1)
boot.out = replicate(3999, {
  x.boot = rlaplace(n, mu = mu.hat, sigma = b.hat) # sample under the
    best-fitting model
  x.boot0 = rlaplace(n, mu = mu0, sigma = b.hat0) # sample under null
    model
  c(mean(x.boot), sd(x.boot), mean(x.boot0), sd(x.boot0), median(x.
    boot), median(x.boot0))
}) %>% t()

t.boot0 = sqrt(n) * (boot.out[,3] - mu0) / boot.out[,4]
t.boot = sqrt(n) *(boot.out[,1] - mu.hat) / boot.out[,2]
d.boot0 = boot.out[,3] - mu0
d.boot = boot.out[,1] - mu.hat
m.boot0 = boot.out[,6] / mu0
m.boot = boot.out[,5] / mu.hat
```

Each null distribution is compared with the observed statistic to get the p-value. Figure 9.2 of the book shows the bootstrap-approximated null distributions and p-values in all six cases; every test chose the alternative $\theta > 166$ at the 5% level (p-values: T 0.0035 / 0.0055, D 0.0205 / 0.0145, M 0.00475 / 0.00025 for null / pivot resampling). Comparing the test methods:

1. The two resampling methods do not give very different results. But the statistic M is not a pivot, so "pivot" resampling is not recommended for it.
2. The p-value of the bootstrap test with D is higher than with the other statistics. As under normality, the test using the studentized T is more powerful than the test using D. Building test statistics in the "studentized difference" form like T was recommended by Hall and Wilson (1991).
3. The median, the maximum likelihood estimator of the parameter θ, is the best estimator of the parameter, so the test using the median, M, is more powerful. For reference, taking the more familiar form $M' = \text{median}\{X_i\} - \theta_0$ as statistic, the result with resampling under the null is very similar to that with M, but the result with pivot resampling differs slightly. (Here M' is a pivot.)
4. Since the (possibly wrong) assumption that the population follows the Laplace distribution was made, there is no guarantee that every test controls the type I error. Assuming a more appropriate parametric model than the Laplace (e.g. the normal model) controls the type I error more exactly while raising the power.

Hypothesis test for the locations of two populations. Using all the gene expression data of the leukemia patients, we run a parametric bootstrap test for the location difference of the two populations. Take the expression levels of the $m = 47$ ALL patients as observations $X_1, \ldots, X_m$ of the first population and those of the $n = 25$ AML patients as observations $Y_1, \ldots, Y_n$ of the second, and assume the parametric model

$$X_i \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu_1, b), \qquad Y_i \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu_2, b). \tag{9.9}$$

We bootstrap-test the null $H_0 : \mu_1 = \mu_2$ against $H_1 : \mu_1 < \mu_2$, or $H_0 : \theta = 0$ against $H_1 : \theta > 0$ with $\theta = \mu_2 - \mu_1$.

Write the maximum likelihood estimates of the parameters of model (9.9) as $\hat{\mu}_1 = \text{median}\{X_i\}$, $\hat{\mu}_2 = \text{median}\{Y_i\}$, $\hat{b} = \frac{1}{m+n}\big(\sum_{i=1}^m |X_i - \hat{\mu}_1| + \sum_{i=1}^n |Y_i - \hat{\mu}_2|\big)$, and under the null $\hat{\mu}_0 := \hat{\mu}_1^{(0)} = \hat{\mu}_2^{(0)} = \text{median}\{X_i, Y_j\}$ (the median of all observations), $\hat{b}_0 = \frac{1}{m+n}\big(\sum_{i=1}^m |X_i - \hat{\mu}_0| + \sum_{i=1}^n |Y_i - \hat{\mu}_0|\big)$. Both parameters μ_1, μ_2 are medians and means, so we take the following two statistics as test statistics and consider bootstrap tests with the two kinds of resampling:

- studentized difference of means: $T = (\bar{Y}_n - \bar{X}_m)/\widehat{\text{se}} = (\bar{Y}_n - \bar{X}_m - \theta_0)/\widehat{\text{se}}$ (here $\widehat{\text{se}}$ is the standard error estimate of $\bar{Y}_n - \bar{X}_m$ assuming equal variances, and θ_0 the value of θ under the null, $\theta_0 = 0$);
- difference of medians: $M = \text{median}\{Y_i\} - \text{median}\{X_i\} = \text{median}\{Y_i\} - \text{median}\{X_i\} - \theta_0$.

Resampling under the null. When the null is true, all observations were sampled at random from Laplace(μ_0, b), so bootstrap samples are drawn as

$$X_1^{0*}, \ldots, X_m^{0*}, Y_1^{0*}, \ldots, Y_n^{0*} \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{\mu}_0, \hat{b}_0).$$

(Here under the null the two populations are not distinguished; if the scale b of the Laplace distribution differed by population, the bootstrap samples would be drawn from each sample.) From these bootstrap samples the replicates

$$T_0^* = (\bar{Y}_n^{0*} - \bar{X}_m^{0*})/s_p^{0*}, \qquad M_0^* = \text{median}\{Y_i^{0*}\} - \text{median}\{X_i^{0*}\}$$

are computed, and their distributions used as the null distributions of the statistics to compute p-values.

Resampling with a pivot. Both statistics are approximate pivots, so pivot resampling proceeds as follows. With the estimates $(\hat{\mu}_1, \hat{\mu}_2, \hat{b})$ of each population's parameters, bootstrap samples

$$X_1^*, \ldots, X_m^* \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{\mu}_1, \hat{b}), \qquad Y_1^*, \ldots, Y_n^* \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\hat{\mu}_2, \hat{b})$$

are drawn, and the distributions of the replicates

$$T_0^* = (\bar{Y}_n^* - \bar{X}_m^* - (\hat{\mu}_2 - \hat{\mu}_1))/s_p^*, \qquad M_0^* = \text{median}\{Y_i^*\} - \text{median}\{X_i^*\} - (\hat{\mu}_2 - \hat{\mu}_1)$$

are used as the null distributions of the statistics to compute p-values. The R code implementing this follows; the function `pooled.se()` is defined in `np_supp.R`.

```
source("np_supp.R")

leukemia <- read.csv("data/leukemia4.csv")
y = leukemia$ge[leukemia$label == "AML"]; m = length(x)
x = leukemia$ge[leukemia$label == "ALL"]; n = length(y)

# test statistic (two test stats)
t0 = (mean(y) - mean(x)) / pooled.se(x,y) #t.test(y,x, var.equal = T)$
    statistic
m0 = median(y) - median(x)

# assume Laplace, get MLE
mu1.hat = median(x); mu2.hat = median(y)
b.hat = mean( abs( c(x - mu1.hat, y - mu2.hat))) # mle (general)
mu0.hat = median(c(x,y))
b.hat0 = mean(abs(c(x,y) - mu0.hat))  # mle under H0

# parametric bootstrap for both (T,M)
set.seed(1)
boot.out = replicate(3999, {
  x.boot = rlaplace(m, mu = mu1.hat, sigma = b.hat)
  y.boot = rlaplace(n, mu = mu2.hat, sigma = b.hat)

  x.boot0 = rlaplace(m, mu = mu0.hat, sigma = b.hat0)
  y.boot0 = rlaplace(n, mu = mu0.hat, sigma = b.hat0)

  t.boot0 = (mean(y.boot0) - mean(x.boot0)) / pooled.se(x.boot0,y.
    boot0)
  t.boot = (mean(y.boot) - mean(x.boot) - (mu2.hat - mu1.hat)) /
    pooled.se(x.boot,y.boot)

  m.boot0 = median(y.boot0) - median(x.boot0)
  m.boot = median(y.boot) - median(x.boot)- (mu2.hat - mu1.hat)

  c(t.boot0, t.boot, m.boot0, m.boot)
}) %>% t()
```

(In the book's code, `m` and `n` are assigned before `x` is redefined; with $m = \#\text{ALL} = 47$, $n = \#\text{AML} = 25$ the lengths should be `m = length(x)` after defining `x` as ALL and `n = length(y)`.) The null distributions and p-values in each case are in Figure 9.3 of the book (T: 0.0095 / 0.007; M: 0.0162 / 0.016 for null / pivot resampling). All tests chose the alternative $H_1 : \mu_1 < \mu_2$ significantly at the 5% level. (The two-sample t-test, the permutation test with the t-statistic, and the distribution-free Wilcoxon–Mann–Whitney test give the same result.)

As in the one-population case, the two resampling methods hardly differ. Resampling with a pivot is recommended by Hall and Wilson (1991) among others, but in bootstrap tests in complex parametric models, resampling under the null may be the only feasible bootstrap method.

9.2 Nonparametric bootstrap tests

A nonparametric or distribution-free test is a test whose performance (control of the type I error and high power) is guaranteed under the assumption of a broad family of populations. The permutation test, which obtains the null distribution of a carefully defined statistic by shuffles, is the representative nonparametric test. Permutation tests have the disadvantage that they must satisfy the exchangeability assumption and cannot be applied when it fails. If the permutation test is understood as a resampling method that samples the observations without replacement, one can consider applying bootstrap sampling (sampling with replacement) instead of sampling without replacement.

A simple example first. With observations from two populations $X_1, \ldots, X_m \overset{\text{i.i.d.}}{\sim} F_1$, $Y_1, \ldots, Y_n \overset{\text{i.i.d.}}{\sim} F_2$, we test the hypotheses

$$H_0 : F_1 = F_2 \quad \text{vs} \quad H_1 : F_1 \neq F_2. \tag{9.10}$$

If the locations of the two populations are expected to differ, the two-sample t-statistic or the Mann–Whitney U-statistic can be the test statistic; more generally the Kolmogorov–Smirnov statistic K can be used, and tests by permutation or large-sample approximation were treated in Chapter 3. Can bootstrap resampling be used instead of resampling by shuffles? The method of resampling under the null seen above can be applied to this situation.

Under the null, all observations were sampled from one population $F = F_1 = F_2$, so bootstrap resampling assuming this null is done by random sampling from the empirical distribution approximating this common population F,

$$F_N(x) = \frac{1}{m+n}\Big\{\sum_{i=1}^{m} I(X_i \le x) + \sum_{i=1}^{n} I(Y_i \le x)\Big\}, \tag{9.11}$$

or by sampling $N = m + n$ data with replacement from $\{X_1, \ldots, X_m, Y_1, \ldots, Y_n\}$. The distribution of the test statistic bootstrap-resampled in this way is the null distribution of that statistic. (Unless the samples are too small, the results of the permutation test and of this bootstrap test are similar.)

Example 9.1

With the leukemia data, carry out the nonparametric bootstrap test of hypotheses (9.10) using (9.11). The bootstrap samples $x_1^*, \ldots, x_m^*$ are drawn at random from F_N (9.11), and $y_1^*, \ldots, y_n^*$ also from F_N; so $x_1^*, \ldots, x_m^*, y_1^*, \ldots, y_n^* \overset{\text{i.i.d.}}{\sim} F_N$. The R code shows the bootstrap test with the two-sample t-statistic:

```
> t0 = (mean(y) - mean(x)) / pooled.se(x,y)
> set.seed(1)
> B = 3999
> boot.out = replicate(B, {
+   boot.id = sample(1:(m+n),(m+n),replace = TRUE)
+   x.boot = c(x,y)[boot.id[1:m]]
+   y.boot = c(x,y)[boot.id[(m+1):(m+n)]]
+   (mean(y.boot) - mean(x.boot)) / pooled.se(x.boot,y.boot)
+ })
> boot.p = 2* min( c( mean(c(t0,boot.out) >=t0),
+                     mean(c(t0,boot.out) <=t0)))
> boot.p
[1] 0.011
```

The bootstrap p-value is 0.011. For comparison, the permutation test (with the two-sample t-statistic) on the same data gives a random permutation p-value of 0.016: the bootstrap test and the permutation test reach the same conclusion. In this situation the only difference between the permutation test and the bootstrap test procedures is: resampling the observa-

tions without replacement gives the permutation test, with replacement the bootstrap test. Compare the procedures and results in the R code below.

```
> perm.out = replicate(B, {
+   perm.id = sample(1:(m+n),(m+n),replace = FALSE)
+   x.perm = c(x,y)[perm.id[1:m]]
+   y.perm = c(x,y)[perm.id[(m+1):(m+n)]]
+   (mean(y.perm) - mean(x.perm)) / pooled.se(x.perm,y.perm)
+ })
> perm.p = 2* min( c( mean(c(t0,perm.out) >=t0),
+                     mean(c(t0,perm.out) <=t0)))
> perm.p
[1] 0.016
```

If we want to test the location difference of two populations, the null of (9.10) (the two distributions are completely identical) may be too narrow. E.g. to allow the dispersions of the two populations to differ even when the null of equal locations is true, neither the ordinary permutation test nor the bootstrap test just explained can be used. Bootstrap resampling is possible within each population, unlike the permutation test, which shuffles the two populations, so tests by bootstrap resampling are possible even when the dispersions of the two populations differ. For this a semiparametric model, applicable even under the null, is needed.

We look at nonparametric bootstrap tests using a representative semiparametric model for hypothesis tests about the locations of populations.

9.2.1 Bootstrap tests for the locations of two populations

For the gene expression data of leukemia patients, when a parametric model is hard to assume, the following semiparametric location model can be assumed:

$$X_i = \mu_1 + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F, \quad i = 1, \ldots, m,$$
$$Y_i = \mu_2 + e_i, \qquad e_i \overset{\text{i.i.d.}}{\sim} G, \quad i = 1, \ldots, n.$$

Here F and G are distributions with mean or median 0 and need not be equal. The location shift $\Delta = \mu_2 - \mu_1$ of the two populations is defined regardless of the unknown distributions F, G. We are interested in testing the hypotheses about the locations of the two populations

$$H_0 : \mu_1 = \mu_2 \quad \text{vs} \quad H_1 : \mu_2 > \mu_1. \tag{9.12}$$

(The book prints $H_1 : \mu_1 > \mu_2$; the test below and Example 9.2 are upper-tailed in $\Delta = \mu_2 - \mu_1$, i.e. $H_1 : \Delta > 0$.) (Even when the null is true the two populations may differ, so the null of (9.12) contains more distributions than the null of (9.10).)

If the distributions F, G have known forms, the parametric bootstrap test explained above can be used. E.g. if $X_i \sim \text{Laplace}(\mu_1, b_1)$ and $Y_i \sim \text{Laplace}(\mu_2, b_2)$, the (approximate) pivot for $\Delta = \mu_2 - \mu_1$,

$$T = T_\Delta = \frac{\bar{Y}_n - \bar{X}_m - \Delta}{\sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}}, \tag{9.13}$$

can be used. Substitute the null value $\Delta = \Delta_0 = 0$ for the parameter Δ in (9.13) and the observations for X_i, Y_i to get the observed T_{obs} as test statistic, and obtain the null distribution of this statistic by pivot resampling.

Even in the nonparametric situation where the forms of F, G cannot be specified, assuming the statistic T of (9.13) is an approximate pivot, the nonparametric bootstrap can be used. Writing the estimate of Δ as $\hat{\Delta} = \bar{Y}_n - \bar{X}_m$ and its standard error $\widehat{\text{se}} = \sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}$, $T = (\hat{\Delta} - \Delta)/\widehat{\text{se}}$.

Nonparametric resampling with a pivot. The approximate distribution H of the pivot T is bootstrap-approximated as

$$\begin{aligned} T &= \frac{\bar{Y}_n - \bar{X}_m - \Delta}{\sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}} = \frac{\hat{\Delta} - \Delta}{\widehat{\text{se}}} \dot{\sim} H \\ \overset{d}{\approx} T^* &= \frac{\bar{Y}_n^* - \bar{X}_m^* - (\bar{Y}_n - \bar{X}_m)}{\sqrt{\frac{1}{m}(s_1^*)^2 + \frac{1}{n}(s_2^*)^2}} = \frac{\hat{\Delta}^* - \hat{\Delta}}{\widehat{\text{se}}^*} \dot{\sim} H^*. \end{aligned} \tag{9.14}$$

Here the bootstrap samples are: $x_1^*, \ldots, x_m^*$ drawn as m simple random samples with replacement from the observations $\{X_1, \ldots, X_m\}$, and $y_1^*, \ldots, y_n^*$ as n from $\{Y_1, \ldots, Y_n\}$. Since the distributions of the two populations differ under both null and alternative, bootstrap resampling is done separately. The distribution of the null test statistic T_0 is approximated by H, which is again bootstrap-approximated by H^*; so the nonparametric bootstrap test proceeds as follows (Efron and Tibshirani, 1994):

1. Draw $x_1^*, \ldots, x_m^*$ as m simple random samples with replacement from $\{X_1, \ldots, X_m\}$ and $y_1^*, \ldots, y_n^*$ as n from $\{Y_1, \ldots, Y_n\}$, and compute the bootstrap replicate T^*.
2. Repeat step 1 a predetermined number B of times, obtaining T^{*b} for $b = 1, \ldots, B$.
3. The p-value for hypotheses (9.12) is $\hat{p} = (1 + \sum_{b=1}^{B} I\{T^{*b} \geq T_{\text{obs}}\})/(B+1)$.

That this p-value is approximately exact can be seen from:

$$p = P(T_{\Delta_0} \geq T_{\text{obs}} \mid H_0) = P(T_\Delta \geq T_{\text{obs}}) \approx P(T^* \geq T_{\text{obs}}) \approx \hat{p}.$$

The first equality is the definition of the p-value and the second is given by the assumption that $T = T_\Delta$ is a pivot; the last two approximations are given by the bootstrap approximation.

Example 9.2

For hypotheses (9.12) on the leukemia data, carry out the nonparametric bootstrap test with the procedure of Efron and Tibshirani (1994). Comparing the null $H_0 : \Delta = \mu_2 - \mu_1 = 0$ with the alternative $H_1 : \Delta > 0$, obtain the null distribution of the statistic (9.13) by the nonparametric bootstrap and compute the p-value:

```
> d0 = mean(y) - mean(x)
> t0 = (mean(y) - mean(x)) / pooled.se(x,y)
> set.seed(1)
> B = 3999
> boot.out = replicate(B, {
+   x.boot = sample(x, m, replace = TRUE)
+   y.boot = sample(y, n, replace = TRUE)
+   t.boot = (mean(y.boot)  - mean(x.boot) - d0) / pooled.se(x,y
    )
+ })
> boot.p = mean(c(t0,boot.out) >= t0)
> boot.p
[1] 0.0065
```

(The book's code standardizes T^* with `pooled.se(x,y)` of the original data rather than the bootstrap sample; (9.14) uses $\widehat{\text{se}}^*$.) Compared with the p-value of Example 9.1, the p-value using the pivot is slightly smaller. When (9.14) holds approximately, the bootstrap test using the pivot is more powerful.

Remark

When the distributions of the two populations differ, resampling under the null is not recommended. Suppose the null $\Delta = \mu_2 - \mu_1 = 0$ is true; using the estimate $\hat{\mu}_0 = (m\bar{X}_m + n\bar{Y}_n)/(m+n)$ of $\mu_0 = \mu_1 = \mu_2$, consider resampling from each sample $X_i - \hat{\mu}_0$ and $Y_i - \hat{\mu}_0$. Then the bootstrap replicates of T are

$$T_0^* = \frac{\bar{Y}_n^* - \bar{X}_m^* - (\hat{\mu}_0 - \hat{\mu}_0)}{\sqrt{\frac{1}{m}(s_1^*)^2 + \frac{1}{n}(s_2^*)^2}} = \frac{\bar{Y}_n^* - \bar{X}_m^*}{\sqrt{\frac{1}{m}(s_1^*)^2 + \frac{1}{n}(s_2^*)^2}},$$

and the distribution of the T_0^* is the bootstrap sampling distribution of $T_0 = (\bar{Y}_n - \bar{X}_m)/\sqrt{\frac{1}{m}s_1^2 + \frac{1}{n}s_2^2}$, not its null distribution. (Centering at a common $\hat{\mu}_0$ does not remove the observed difference $\bar{Y}_n - \bar{X}_m$ from the resampled data, so T_0^* is centered near T_0 rather than near 0.)

9.2.2 Bootstrap tests for the location of one population

To test the location parameter of one population, assume the observations $X_1, \ldots, X_n$ follow the semiparametric location model

$$X_i = \theta + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F, \quad i = 1, \ldots, n.$$

With $\hat{\theta}_n$ an estimator of the location parameter θ, if $\hat{\theta}_n - \theta$ or the studentized $(\hat{\theta}_n - \theta)/\widehat{\text{se}}$ is a pivot, the distribution of this pivot can be bootstrap-approximated as

$$H^* \sim \hat{\theta}_n^* - \hat{\theta}_n \overset{d}{\approx} \hat{\theta}_n - \theta \sim H \tag{9.15}$$

or

$$H^* \sim \frac{\hat{\theta}_n^* - \hat{\theta}_n}{\widehat{\text{se}}^*} \overset{d}{\approx} \frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \sim H. \tag{9.16}$$

Here $\hat{\theta}_n^*$ and $\widehat{\text{se}}^*$ are bootstrap replicates obtained by nonparametric bootstrap resampling, as for the bootstrap-t interval.

Suppose we test the null $H_0 : \theta = \theta_0$ against the alternative $H_1 : \theta > \theta_0$. With (9.15) the rejection region is $\hat{\theta}_n > c$ for some value c, so the p-value is

$$\begin{aligned} p_1 &= P(\hat{\theta}_n \geq \hat{\theta}_{\text{obs}} \mid H_0) \\ &= P(\hat{\theta}_n - \theta_0 \geq \hat{\theta}_{\text{obs}} - \theta_0 \mid H_0) \\ &= P(\hat{\theta}_n - \theta \geq \hat{\theta}_{\text{obs}} - \theta_0) \qquad \text{(for every } \theta, \text{ since } \hat{\theta}_n - \theta \text{ is a pivot)} \\ &\approx P(\hat{\theta}_n^* - \hat{\theta}_{\text{obs}} \geq \hat{\theta}_{\text{obs}} - \theta_0) \\ &\approx \frac{1 + \sum_{b=1}^{B} I(\hat{\theta}_n^{*b} - \hat{\theta}_{\text{obs}} \geq \hat{\theta}_{\text{obs}} - \theta_0)}{B+1} =: \hat{p}_1. \end{aligned} \tag{9.17}$$

With (9.16) the rejection region is

$$\frac{\hat{\theta}_n - \theta_0}{\widehat{\text{se}}} > c$$

for some value c, so the p-value is

$$\hat{p}_2 = \frac{1 + \sum_{b=1}^{B} I\{(\hat{\theta}_n^{*b} - \hat{\theta}_{\text{obs}})/\widehat{\text{se}}^* \geq (\hat{\theta}_{\text{obs}} - \theta_0)/\widehat{\text{se}}\}}{B+1}.$$

Justification of $\hat{p}_2$ (supplied; Exercise 9.4 of the book). Repeat (9.17) with the studentized pivot. With $\widehat{\text{se}}$ computed from the observed data,

$$p_2 = P\Big(\frac{\hat{\theta}_n - \theta_0}{\widehat{\text{se}}} \geq \frac{\hat{\theta}_{\text{obs}} - \theta_0}{\widehat{\text{se}}_{\text{obs}}} \;\Big|\; H_0\Big) = P\Big(\frac{\hat{\theta}_n - \theta}{\widehat{\text{se}}} \geq \frac{\hat{\theta}_{\text{obs}} - \theta_0}{\widehat{\text{se}}_{\text{obs}}}\Big)$$

for every θ, since $(\hat{\theta}_n - \theta)/\widehat{\text{se}}$ is a pivot. By (9.16) this is approximately $P((\hat{\theta}^*_n - \hat{\theta}_{\text{obs}})/\widehat{\text{se}}^* \geq (\hat{\theta}_{\text{obs}} - \theta_0)/\widehat{\text{se}}_{\text{obs}})$, which the B bootstrap replicates estimate by $\hat{p}_2$. ■

If the alternative is two-sided, $H_1 : \theta \neq \theta_0$ (using (9.16)), the rejection region is $|\frac{\hat{\theta}_n - \theta_0}{\widehat{\text{se}}}| > c$, and the p-value (as in (9.17)) is

$$\hat{p}_{\text{both}} = \frac{1 + \sum_{b=1}^{B} I\{|(\hat{\theta}^{*b}_n - \hat{\theta}_{\text{obs}})/\widehat{\text{se}}^*| \geq |(\hat{\theta}_{\text{obs}} - \theta_0)/\widehat{\text{se}}|\}}{B+1}.$$

Example 9.3

As in Section 9.1.3, test $H_0 : \theta = 166$ against $H_1 : \theta > 166$ for the mean θ of the gene expression levels of the AML patients among the leukemia data. The p-value of the bootstrap test using the pivot (9.16) is computed as follows:

```
> x <- leukemia$ge[leukemia$label == "AML"]
> n = length(x)
>
> # hypothesis H0: mu =mu0, H1: mu > mu0
> mu0 = 166
>
> # test statistic
> hat.mu = mean(x)
> t0 = sqrt(n) * (mean(x) - mu0 ) / sd(x)
>
> # nonparametric bootstrap
> set.seed(1)
> boot.out = replicate(3999, {
+   x.boot = sample(x, n, replace = TRUE)
+   sqrt(n) * (mean(x.boot) - hat.mu) / sd(x.boot)
+ })
> boot.p = mean(c(t0,boot.out) >= t0)
> boot.p
[1] 0.01225
```

9.3 Duality of bootstrap tests and bootstrap confidence intervals

Bootstrap hypothesis tests and bootstrap confidence intervals are closely related. We know that in some parametric models the complement of the rejection region of a level-α test coincides with the level-$(1-\alpha)$ confidence interval: if a value θ_0 of a parameter lies in the 95% interval, the test of the null $H_0 : \theta = \theta_0$ about that parameter does not reject the null at level 5%. Conversely, if θ_0 does not lie in the interval, the test rejects the null.

Likewise, if the p-value of the two-sided test using the pivot (9.15) is larger than $\alpha = 0.05$ (fails to reject $H_0 : \theta = \theta_0$), the null value θ_0 lies in the 95% basic bootstrap interval constructed assuming (9.15) is a pivot. Conversely, if θ_0 is contained in (or not contained in) the 95% basic bootstrap interval for θ, the null $H_0 : \theta = \theta_0$ fails to be rejected (or is rejected) at level 5%. Similarly, the test using the pivot (9.16) corresponds to the bootstrap-t interval.

Sketch (supplied). Using (9.15), the two-sided level-α test fails to reject $H_0 : \theta = \theta_0$ iff $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta}_{\text{obs}} - \theta_0 \leq Q_{H^*}(1 - \frac{\alpha}{2})$, i.e. iff $\hat{\theta}_{\text{obs}} - Q_{H^*}(1 - \frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta}_{\text{obs}} - Q_{H^*}(\frac{\alpha}{2})$, which is the basic bootstrap interval (8.4). The same argument with the studentized pivot gives the bootstrap-t interval (8.10). ■

In the same context, a hypothesis test can be made by "inverting" the percentile interval (or adjusted percentile interval). The percentile interval was developed assuming the existence of some monotone transform, and since that monotone transform is unknown, a test method corresponding to the percentile interval was not developed separately.

Chapter 10. Linear Regression

Overview

This chapter treats nonparametric inference in linear regression. Linear regression, which models the relation among variables to infer it or to predict the response from explanatory variables, is the most basic method. We first study parametric and nonparametric inference for the regression coefficients via least squares, the most basic estimation method; then, briefly, robust estimators usable when the regression model is right but there are outliers or the error distribution deviates badly from normality. Nonparametric regression with general regression curves (not lines) is treated in Chapter 12.

10.1 Simple linear regression

Given bivariate observations $(x_1, y_1), \ldots, (x_n, y_n)$, the simple linear regression model is the simplest model to explain and predict the response ("Y") by the explanatory variable ("X"); it assumes the relation of the two variables is "linear":

$$y_i = \alpha + \beta x_i + \epsilon_i, \qquad i = 1, \ldots, n. \tag{10.1}$$

Apart from the error ϵ_i, y_i is the linear function $\alpha + \beta x_i$ of x_i; the errors ϵ_i are uncorrelated and all follow a distribution F_ϵ with mean 0 and variance σ^2. The values $x_1, \ldots, x_n$ may be randomly sampled from some population, observed by chance, or fixed in advance by the data collector. (In the last case resampling the "X" variable may not be right.) In any case, ordinary inference in linear regression is conditional on the observed $x_1, \ldots, x_n$.

The errors of (10.1) are assumed only to have equal mean and variance; the shape of their distribution is unspecified. If the errors are assumed normal, (10.1) is a parametric model; without assumptions on $\epsilon_i \sim F_\epsilon$ it is a semiparametric model: the regression part is parametric and the error distribution nonparametric.

Example 10.1

Using World Bank data on 153 countries in 2020, we model the happiness score by the linear relation with GDP (`worldbank2020.csv`). With GDP as explanatory variable and happiness (Score) as response, the scatter plot (Figure 10.1 of the book) shows the relation is roughly explained by the simple linear model (10.1).

10.1.1 Least squares estimators

The most basic estimators of the coefficients α, β are the least squares estimators

$$\hat\beta = \frac{S_{XY}}{S_{XX}} = \frac{\sum_{i=1}^n (x_i - \bar x)(y_i - \bar y)}{\sum_{i=1}^n (x_i - \bar x)^2}, \qquad \hat\alpha = \bar y - \hat\beta \bar x.$$

The estimate of the regression line $y = \alpha + \beta x$ is $\hat y = \hat\alpha + \hat\beta x$, and the residuals at the observations (x_i, y_i) are $e_i = y_i - \hat y_i \equiv y_i - (\hat\alpha + \hat\beta x_i)$. The estimator of the error variance σ^2 is

$$\hat\sigma^2 = s^2 = \frac{\sum_{i=1}^n e_i^2}{n-2}.$$

Inference on the unknown coefficient β is usually based on the slope estimate $\hat\beta$. When the model (10.1) is true, $E(\hat\beta) = \beta$ and $\mathrm{Var}(\hat\beta) = \sigma^2/S_{XX}$; writing only mean and variance,

$$\hat\beta \sim (\beta, \sigma^2/S_{XX}). \tag{10.2}$$

Proof (supplied). $\hat\beta = \sum_i w_i y_i$ with $w_i = (x_i - \bar x)/S_{XX}$, $\sum_i w_i = 0$, $\sum_i w_i x_i = 1$, $\sum_i w_i^2 = 1/S_{XX}$. So $E\hat\beta = \sum_i w_i(\alpha + \beta x_i) = \beta$ and, the ϵ_i being uncorrelated with variance σ^2, $\operatorname{Var}\hat\beta = \sigma^2 \sum_i w_i^2 = \sigma^2/S_{XX}$. ■

To approximate the standard error $\operatorname{se}(\hat\beta) = \sigma/\sqrt{S_{XX}}$ the estimated standard error

$$\widehat{\operatorname{se}}(\hat\beta) = \frac{s}{\sqrt{S_{XX}}} \tag{10.3}$$

is used. (10.2) says nothing about the distribution of $\hat\beta$; if the errors are normal, $\hat\beta \sim N(\beta, \sigma^2/S_{XX})$ is known.

Errors and residuals. When model (10.1) is true, the errors satisfy $\epsilon_i \sim (0, \sigma^2)$, uncorrelated, but the residuals satisfy $e_i \sim (0, \sigma^2(1-h_i))$, weakly correlated, where $h_i = n^{-1} + (x_i - \bar x)^2/S_{XX}$ is the *leverage* of the ith observation.[52] When n is small or the observations are not spread evenly, the h_i can differ a lot, so the **adjusted residuals**

$$r_i = e_i/\sqrt{1-h_i}, \qquad i = 1, \dots, n,$$

are also used.

$\operatorname{Var}(e_i) = \sigma^2(1-h_i)$ **and** $\operatorname{Var}(r_i) = \sigma^2$ **(supplied; Exercise 10.2).** With H the hat matrix, $\mathbf{e} = (I-H)\mathbf{y} = (I-H)\boldsymbol\epsilon$ since $(I-H)X = 0$. So $E\mathbf{e} = 0$ and $\operatorname{Var}(\mathbf{e}) = \sigma^2(I-H)(I-H)' = \sigma^2(I-H)$, H being a symmetric projection. In simple regression $H_{ii} = \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{XX}} = h_i$, so $\operatorname{Var}(e_i) = \sigma^2(1-h_i)$, and $Er_i = 0$, $\operatorname{Var}(r_i) = \sigma^2$. The off-diagonal entries $-\sigma^2 H_{ij}$ give the weak correlations. ■

10.1.2 Inference under normality

If model (10.1) is (approximately) true and the errors are (approximately) normal, or the sample is large enough for the CLT normal approximation, normal theory gives inference on β. Under the model with normal errors,

$$\hat\beta \sim N(\beta, \sigma^2/S_{XX}), \qquad t = \frac{\hat\beta - \beta}{\widehat{\operatorname{se}}(\hat\beta)} = \frac{\hat\beta - \beta}{s/\sqrt{S_{XX}}} \sim t_{n-2}, \tag{10.4}$$

so the $1-\alpha$ interval for β is exactly

$$\beta \in \hat\beta \pm t_{\alpha/2, n-2}\frac{s}{\sqrt{S_{XX}}}, \tag{10.5}$$

and the p-value for $H_0 : \beta = 0$ against $H_1 : \beta > 0$ is exactly $p = P(T_{n-2} \ge t_0)$, $t_0 = \frac{\hat\beta - 0}{\widehat{\operatorname{se}}(\hat\beta)}$.

Example 10.2

For the GDP–happiness regression of Example 10.1, test $H_0 : \beta = 0$. From the data $\hat\beta = 2.31$, $\widehat{\operatorname{se}}(\hat\beta) = 0.15$, $t_0 = 15.08$; this value is very large, so even without the t_{n-2} distribution ($n = 153$) a tiny p-value is expected. For the p-value to be accurate, the error distribution must not depend on the explanatory variable (GDP) and must be approximately normal. To check this, compute the adjusted residuals and visualize them by a scatter plot and histogram:

```
h = read.csv("data/worldbank2020.csv")
y = h$Score
x = h$GDP.per.capita
```

[52] The leverage h_i is large when x_i is far from the mean $\bar x$; then, like a heavy weight far from the fulcrum of a lever, the regression line changes much. Large leverage indicates a possible outlier or influential point, which needs attention in interpreting the fitted model.

```
(lm.out = lm(y ~ x))
adjusted.res = lm.out$residuals / sqrt(1 - hatvalues(lm.out))
```

The errors are roughly homoscedastic, but their distribution has a slightly longer left tail; normality is roughly satisfied.

If model (10.1) is (approximately) true but the error distribution is very different from normal, the accuracy of intervals and tests based on normal theory is not guaranteed. Alternatives are resampling-based permutation tests and the bootstrap.

10.1.3 Permutation tests for the regression coefficient

In simple regression, $H_0 : \beta = 0$ against $H_1 : \beta > 0$ for the slope can be tested by a permutation test, assuming the errors $\{\epsilon_1, \ldots, \epsilon_n\}$ are exchangeable (satisfied if they are independent with the same distribution); the test statistic can simply be $\hat{\beta}$. Without normality the null distribution of $\hat{\beta}$ cannot be obtained from (10.4).

Simple regression and its least squares estimator are closely related to correlation, in particular the sample correlation (Chapter 5). When the "X" values are chosen and fixed in advance, the population correlation ρ estimated by the sample correlation is hard to define. But whether the explanatory variable is random or fixed in model (10.1), if $H_0 : \beta = 0$ is true,

$$y_i = \alpha + \epsilon_i, \qquad i = 1, \ldots, n, \tag{10.6}$$

and the explanatory and response variables are unrelated. The errors $\{\epsilon_1, \ldots, \epsilon_n\}$ are exchangeable, so the responses $\{y_1, \ldots, y_n\}$ are too. As bivariate observations, (x_1, y_1) and (x_1, y_2) have the same distribution, so the permutation distribution of $\hat{\beta}$ can be generated by fixing the "X" values and shuffling only the "Y" values. (The same shuffles were used in the permutation test of correlation.) There are $n!$ shuffles, very many, so a random permutation test is recommended:

1. Compute the least squares estimate $\hat{\beta}_{\text{obs}}$ of β in model (10.1) from $\{(x_1, y_1), \ldots, (x_n, y_n)\}$.
2. Randomly shuffle the order of the responses to get a shuffled sample $\{(x_1, y_1^*), \ldots, (x_n, y_n^*)\}$, where $(y_1^*, \ldots, y_n^*)$ is drawn by sequential simple sampling without replacement from $\{y_1, \ldots, y_n\}$.
3. Compute $\hat{\beta}^*$ from the shuffled sample.
4. Repeat steps 2–3 $M = 1999$ times to generate the permutation distribution of $\hat{\beta}$ as the empirical distribution of $\{\hat{\beta}^{*(0)}, \ldots, \hat{\beta}^{*(M)}\}$ ($\hat{\beta}^{*(0)} := \hat{\beta}_{\text{obs}}$).
5. For the upper-tailed $H_1 : \beta > 0$, $p \approx \sum_{i=1}^{M} I(\hat{\beta}^{*(i)} \geq \hat{\beta}_{\text{obs}})/(M+1)$.

For lower-tailed or two-sided tests change the definition of extreme. $M = 1999$ is usual (more for accuracy). Since exchangeability of the responses under the null was used, this permutation test is exact conditionally on the observations.

Above the statistic was $\hat{\beta}^* = S_{XY^*}/S_{XX}$ ("$*$" = value after shuffling). This permutation test coincides with permutation tests using any of:

- $S_{XY^*} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i^* - \bar{y})$ or $\sum_{i=1}^{n} x_i y_i^*$;
- $t^* = \hat{\beta}^*/\widehat{\text{se}}(\hat{\beta}^*)$, where $\widehat{\text{se}}(\hat{\beta}^*) = [\sum_{i=1}^{n}(e_i^*)^2/\{(n-2)S_{XX}\}]^{1/2}$, $e_i^* = y_i^* - \hat{\alpha}^* - \hat{\beta}^* x_i$.

As in two-sample permutation tests (sum of one sample, difference of means, two-sample t all giving the same test), S_{XY^*} and t^* are monotone increasing functions of $\hat{\beta}^*$, so any of them gives the same test.

Why t^* is increasing in $\hat{\beta}^*$ (supplied; Exercise 10.3). S_{XX} and $S_{YY} = \sum(y_i^* - \bar{y})^2$ are the same for every shuffle, and $\sum(e_i^*)^2 = S_{YY} - (\hat{\beta}^*)^2 S_{XX}$, so

$$\widehat{\mathrm{se}}^2(\hat{\beta}^*) = \frac{S_{YY} - (\hat{\beta}^*)^2 S_{XX}}{(n-2)S_{XX}}, \qquad t^* = \frac{\hat{\beta}^* \sqrt{(n-2)S_{XX}}}{\sqrt{S_{YY} - (\hat{\beta}^*)^2 S_{XX}}},$$

which is strictly increasing in $\hat{\beta}^*$ on $(\hat{\beta}^*)^2 S_{XX} < S_{YY}$. Hence $\{\hat{\beta}^* \geq \hat{\beta}\} = \{t^* \geq t\}$ for every shuffle. ■

Example 10.3

For the happiness–GDP regression, the permutation test of $H_0 : \beta = 0$ against $H_1 : \beta > 0$ with $\hat{\beta}$ as statistic. The random permutation test below gave $p = 1/2000$; compared with the observed $\hat{\beta}_{\text{obs}} = 2.31$, the permutation distribution under the null almost never produces such a value, so the linear relation between the variables is very significant.

```
> n = length(x)
> Sxx = sum( (x - mean(x))^2 )
> Sxy = sum( (x - mean(x)) * (y - mean(y)) )
> beta.hat = Sxy / Sxx
>
> M = 1999
> beta.hat.perm = replicate(M, {
+   yperm = sample(y,n, replace = FALSE)
+   sum( (x - mean(x)) * (yperm - mean(yperm))) / Sxx
+ })
> (p = mean( c(beta.hat, beta.hat.perm) >= beta.hat ))
[1] 5e-04
```

10.1.4 Bootstrap confidence intervals for the regression coefficient

Permutation tests are exact, but they are hard to use for confidence intervals because of the computation. For nonparametric interval estimation of β the bootstrap is used. Chapter 7 gave two bootstraps of bivariate data; given $(x_1, y_1), \ldots, (x_n, y_n)$ and the linear model $y_i = \alpha + \beta x_i + \epsilon_i$:

- **Case (pairs) bootstrap:** resample the n pairs $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ by simple sampling with replacement. Unlike shuffling in permutation tests, which destroys the correlation between "X" and "Y", the bootstrap always draws the pairs (x_i, y_i) together, preserving the linear relation.
- **Residual bootstrap:** assuming the linear model is true, the errors $\epsilon_i \sim F_\epsilon$ in $y_i = \alpha + \beta x_i + \epsilon_i$ carry the randomness of sampling from the population. To approximate F_ϵ from the data, use the residuals $\{e_1, \ldots, e_n\}$ computed from $\hat{\alpha}, \hat{\beta}$ and resample them with replacement. Since the residual variances $\sigma^2(1 - h_i)$ differ slightly with the leverage, resampling the adjusted residuals $\{r_1, \ldots, r_n\}$ is preferred. With bootstrap residuals $r_1^*, \ldots, r_n^*$, the bootstrap sample is

$$(x_i^*, y_i^*) = (x_i, \hat{\alpha} + \hat{\beta} x_i + r_i^*). \tag{10.7}$$

  In the residual bootstrap the explanatory values are not resampled: each bootstrap sample has the same "X" values as the original.

Whichever resampling is used, the bootstrap replicates of $\hat{\beta}$ are obtained by:

1. Get a bootstrap sample $D^* = \{(x_1^*, y_1^*), \ldots, (x_n^*, y_n^*)\}$ by the case or residual bootstrap.

2. From D^* compute $\hat{\beta}^* = S_{X^*Y^*}/S_{X^*X^*}$ and $t^* = (\hat{\beta}^* - \hat{\beta})/\widehat{\text{se}}(\hat{\beta}^*)$, $\widehat{\text{se}}(\hat{\beta}^*) = [\sum_{i=1}^n (e_i^*)^2/\{(n-2)S_{X^*X^*}\}]^{1/2}$.
3. Repeat steps 1–2 B times to get $\{(\hat{\beta}^{*b}, \widehat{\text{se}}(\hat{\beta}^{*b}), t^{*b}) : b = 1, \ldots, B\}$.

(For the residual bootstrap, $S_{X^*X^*} = S_{XX}$, which saves computation.) The empirical distribution of $\{\hat{\beta}^{*b}\}$ is the bootstrap sampling distribution of $\hat{\beta}$, giving bootstrap estimates of bias and standard error. In linear regression $\hat{\beta}$ is unbiased and its standard error is (10.3), so these are not really needed; but when normal errors are hard to assume or the sample is small, bootstrap intervals are more accurate than normal-theory ones.

From the replicates one gets the basic, bootstrap-t and (adjusted) percentile intervals of Chapter 8. For regression coefficients, of the two second-order accurate intervals (bootstrap-t and BCa), the bootstrap-t is recommended. It is not used often in general because the standard error needs a second-level bootstrap; in regression the standard error of $\hat{\beta}$ has a formula, so no second level is needed. That is why $\widehat{\text{se}}(\hat{\beta}^*)$ and t^* were computed above along with $\hat{\beta}^*$.

Example 10.4

The 95% bootstrap-t interval for β in the happiness–GDP regression. With $Q_H^*(\alpha/2)$, $Q_H^*(1-\alpha/2)$ the 2.5% and 97.5% quantiles of the replicates $t^{*1}, \ldots, t^{*B}$, the interval is

$$\beta \in (\hat{\beta} - \widehat{\text{se}}(\hat{\beta}) Q_H^*(1 - \tfrac{\alpha}{2}),\ \hat{\beta} - \widehat{\text{se}}(\hat{\beta}) Q_H^*(\tfrac{\alpha}{2})). \tag{10.8}$$

The code records only t^{*b} from $B = 1999$ case-bootstrap resamples; computing t^* is lengthy, so a function `t.pivot()` is defined.

```
t.pivot = function(x,y, beta0){
  n = length(x)
  Sxx = sum( (x - mean(x))^2 )
  Sxy = sum( (x - mean(x)) * (y - mean(y)) )
  beta.hat = Sxy / Sxx
  res = y - mean(y) - beta.hat * ( x - mean(x))
  se.beta.hat = sqrt( sum( res^2 ) / (n - 2) ) / sqrt(Sxx)
  t = (beta.hat - beta0)  / se.beta.hat
  t
}
set.seed(1)
B = 1999
boot.t = replicate(B, {
    boot.id = sample(1:n,n, replace = TRUE) # case resampling
    xboot = x[boot.id]
    yboot = y[boot.id]
    t.pivot(xboot, yboot, beta.hat)
})
res = y - mean(y) - beta.hat * ( x - mean(x))
s = sqrt( sum( res^2 ) / (n - 2) )
se.beta.hat = s / sqrt(Sxx)
boot.t.ci = beta.hat - se.beta.hat * quantile(boot.t, probs = c
    (1-0.025, 0.025))
```

Compared with the normal-theory interval (10.5):

```
> boot.t.ci
   97.5%     2.5%
2.024506 2.597004
> (norm.appr.ci = beta.hat + se.beta.hat * qt(c(0.025, 1-0.025),
    df = n-2))
[1] 2.012202 2.618830
```

With the residual bootstrap the coding is more efficient (S_{XX} is common to all bootstrap samples). Ignoring that, the residual-bootstrap interval, continuing Example 10.2:

```
> set.seed(1)
> boot.t.res = replicate(B, {
+   boot.res = sample(adjusted.res,n, replace = TRUE) # residual
    resampling
+   yboot = lm.out$fitted.values + boot.res
+   t.pivot(x,yboot, beta.hat)
+ })
> (res.boot.t.ci = beta.hat - se.beta.hat * quantile(boot.t.res,
    probs = c(1-0.025, 0.025)))
   97.5%     2.5%
2.024342 2.611618
```

The residual-bootstrap interval $\beta \in (2.02, 2.61)$ is almost the same as the normal-theory interval.

The case and residual bootstraps differ, and the residual-bootstrap interval is closer to the normal-theory one. Recall the derivation of (10.5): the explanatory variable is treated as fixed, and the interval is valid when the linear model is true; i.e. it is inference conditional on the observed explanatory values. The residual bootstrap also fixes the explanatory values and resamples, so it gives results closer to the usual normal theory. In Example 10.1 the errors are approximately normal, so the intervals hardly differ; when errors are far from normal the bootstrap intervals are more accurate.

The case bootstrap allows accurate estimation of the parameter β even if the linear model is wrong (the relation is not linear). Then β is no longer a coefficient of a linear model but can be understood as the value minimizing the mean squared error $E(Y-\alpha-\beta X)^2$. The case bootstrap gives inference robust to model misspecification, but when the model is true it is less efficient than the residual bootstrap. In bivariate data from a designed experiment, the case bootstrap changes the explanatory values (the design information that should be fixed) in every resample, which can also be a drawback (Davison and Hinkley, 1997).

10.1.5 Bootstrap tests for the regression coefficient

Chapter 9 tested parameters by bootstrapping their estimates; tests of regression coefficients proceed the same way. Assuming model (10.1) and testing $H_0 : \beta = 0$ against $H_1 : \beta > 0$, the statistic is

$$t_0 = \frac{\hat{\beta} - 0}{\widehat{\text{se}}(\hat{\beta})}, \tag{10.9}$$

rejecting H_0 when t_0 is large. Bootstrap resampling for the test can be resampling under the null or resampling with a pivot.

Nonparametric bootstrap resampling under the null. Bootstrap samples are drawn assuming the null is true. If $\beta = 0$,

$$y_i = \alpha + \epsilon_i, \qquad i = 1, \ldots, n, \tag{10.10}$$

and x and y are unrelated (independent). Two bootstrap resamplings are possible:

1. **Case bootstrap.** Draw $x_1^*, \ldots, x_n^*$ by simple sampling with replacement from $\{x_1, \ldots, x_n\}$ and, independently, $y_1^*, \ldots, y_n^*$ from $\{y_1, \ldots, y_n\}$. From the bootstrap sample compute $t^* = \hat{\beta}^*/\widehat{\text{se}}(\hat{\beta}^*)$; the empirical distribution of $\{t^{*1}, \ldots, t^{*B}\}$ over B repetitions is the null distribution of t_0 (10.9). (Under the null the two variables are independent, so the distribution of (X, Y)

is the product of the distribution F_X of X and $F_Y(y) = F_\epsilon(y-\alpha)$ of Y; it is approximated by the replicates.)

2. **Residual bootstrap.** Used when the explanatory values are fixed or inference is conditional on them. The null model (10.10) still has the unknown α, so plug in its null estimate $\hat{\alpha}^0 = \bar{y}$ to get residuals
$$e_i^0 = y_i - \bar{y}, \qquad i = 1, \ldots, n, \tag{10.11}$$
and adjusted residuals $r_i^0 = e_i^0/\sqrt{1-\frac{1}{n}}$ (superscript "0" = value in the null model). Resample them with replacement as $r_1^{0*}, \ldots, r_n^{0*}$, form $(x_i^*, y_i^*) = (x_i, \hat{\alpha} + r_i^{0*})$, and compute $t^* = \hat{\beta}^*/\widehat{\text{se}}(\hat{\beta}^*)$. The empirical distribution of $\{t^{*1}, \ldots, t^{*B}\}$ is the null distribution of t_0 (10.9).

Example 10.5

Testing whether $\beta > 0$ in the happiness–GDP regression, p-values by the case and residual bootstraps under the null; `t.pivot()` of Example 10.4 is reused.

```
# 1. case bootstrap, resampling under H0
set.seed(1)
t0 = t.pivot(x,y,0)
B = 1999
boot.t = replicate(B, {
  xboot = sample(x,n, replace = TRUE)
  yboot = sample(y,n, replace = TRUE)
  t.pivot(xboot,yboot, 0)
})
p.null.case = mean(c(t0,boot.t ) >= t0)
```

The statistic $t_0 = 15.1$ is very large and no replicate in `boot.t` exceeds it (Figure 10.2 of the book), so $p = 1/2000$. The residual bootstrap under the null gives $p = 1/2000$ as well:

```
# 2. residual bootstrap, resampling under H0
adjusted.res = (y - mean(y))/ sqrt( 1- 1/n)
boot.t = replicate(B, {
  boot.res = sample(adjusted.res, n, replace = TRUE)
  yboot = mean(y) + boot.res
  t.pivot(x,yboot, 0)
})
p.null.res = mean(c(t0,boot.t ) >= t0)
```

Bootstrap resampling with a pivot. If the general form of t_0,
$$t = (\hat{\beta} - \beta)/\widehat{\text{se}}(\hat{\beta}), \tag{10.12}$$
is assumed a pivot, the distribution of t_0 under $\beta = \beta_0 = 0$ (the null distribution) equals that of the pivot t. It is approximated by the bootstrap distribution of
$$t^* = (\hat{\beta}^* - \hat{\beta})/\widehat{\text{se}}(\hat{\beta}^*) \tag{10.13}$$
and used as the null distribution of t_0 (10.9). Resampling is as for bootstrap intervals: case bootstrap (resampling the pairs) or residual bootstrap (resampling the linear-model residuals $e_i = y_i - \hat{\alpha} - \hat{\beta}x_i$). Pivot resampling is the same as inverting the bootstrap-t interval.

Example 10.6

Testing whether $\beta > 0$ in the happiness–GDP regression with the pivot and the case bootstrap:

```
# 3. pivot, case bootstrap
boot.t = replicate(B, {
  boot.id = sample(1:n,n, replace = TRUE)
  xboot = x[boot.id]
  yboot = y[boot.id]
  t.pivot(xboot,yboot, beta.hat)
})
p.pivot.case = mean(c(t0,boot.t ) >= t0)
```

The replicates `boot.t` are exactly those of the bootstrap-t interval with the case bootstrap. The residual bootstrap with the pivot follows the same resampling as the residual-bootstrap interval:

```
# 4. pivot, residual bootstrap
boot.t = replicate(B, {
boot.res = sample(adjusted.res,n, replace = TRUE)
  yboot = lm.out$fitted.values + boot.res
  t.pivot(x,yboot, beta.hat)
})
p.pivot.res = mean(c(t0,boot.t ) >= t0)
```

The four bootstrap null distributions of Examples 10.5 and 10.6 are shown as histograms in Figure 10.2 of the book; all are similar and close to the t_{151} (or standard normal) distribution.

10.2 Multiple linear regression

The multiple linear regression model explains a response Y by a linear combination of several explanatory variables $X_1, \ldots, X_p$. With n observations,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n. \tag{10.14}$$

The errors ϵ_i are uncorrelated with mean 0 and variance σ^2, distribution F_ϵ. Regression analysis estimates the coefficients β_i $(i = 0, \ldots, p)$, gives their confidence intervals, and tests various hypotheses about them. As in simple regression, the explanatory values are assumed fixed and inference is conditional. When F_ϵ is not $N(0, \sigma^2)$, resampling-based inference can be considered, and these methods extend the resampling methods of simple regression naturally.

Example 10.7

Using World Bank 2020 data for 153 countries (`worldbank2020.csv`), fit a linear model of the happiness score (Score) on six explanatory variables: (1) GDP, (2) Social (social support), (3) Life (healthy life expectancy), (4) Freedom, (5) Generosity, (6) Corruption, social-health indices. The scatter plot matrix with sample correlations (Figure 10.3 of the book) gives a quick view. With $X_1, \ldots, X_p$ $(p = 6)$ in this order, we want intervals and significance for each coefficient. The scatter plots suggest the last two variables (Generosity, Corruption) have weak linear association with happiness. Should we use the model with all variables or with only the first four?

10.2.1 Least squares estimators

Least squares is again basic. Stacking (10.14) over $i = 1, \ldots, n$,

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}, \qquad \text{or} \qquad \mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}. \tag{10.15}$$

The matrix X is the design (model) matrix; the parameters are the $p+1$ coefficients $\boldsymbol{\beta}$ and σ^2. The least squares estimator is

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y},$$

the fitted values $\hat{\mathbf{y}} = X\hat{\boldsymbol{\beta}} = X(X'X)^{-1}X'\mathbf{y}$, and $H = X(X'X)^{-1}X'$ $(n \times n)$ is the hat matrix ($\hat{\mathbf{y}} = H\mathbf{y}$). With I_n the identity, the residual vector is $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (I_n - H)\mathbf{y}$, and

$$\hat{\sigma}^2 = s^2 = \frac{\sum_{i=1}^n e_i^2}{n-p-1} = \frac{\mathbf{e}'\mathbf{e}}{n-p-1} = \frac{\mathbf{y}'(I_n - H)\mathbf{y}}{n-p-1}. \tag{10.16}$$

When model (10.14) is true,

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \qquad \mathrm{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2(X'X)^{-1}, \tag{10.17}$$

and $\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) = s^2(X'X)^{-1}$. If the errors are normal, $\hat{\boldsymbol{\beta}}$ is multivariate normal with mean and covariance (10.17).

Proof of (10.17) (supplied). $\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'(X\boldsymbol{\beta} + \boldsymbol{\epsilon}) = \boldsymbol{\beta} + (X'X)^{-1}X'\boldsymbol{\epsilon}$, so $E\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}$ and $\mathrm{Var}\,\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$. Also $E(\mathbf{e}'\mathbf{e}) = E\,\boldsymbol{\epsilon}'(I-H)\boldsymbol{\epsilon} = \sigma^2\mathrm{tr}(I-H) = \sigma^2(n-p-1)$, so s^2 is unbiased. ■

Errors and residuals. As in simple regression, when (10.14) is true the errors $\epsilon_i \sim (0, \sigma^2)$ are uncorrelated with the same distribution, but the residuals are weakly correlated with $e_i \sim (0, \sigma^2(1-h_i))$, with different variances (h_i is the ith diagonal element of H). So the adjusted residuals

$$r_i = e_i/\sqrt{1-h_i}, \qquad i = 1, \ldots, n, \tag{10.18}$$

are also used.

10.2.2 Inference on the coefficients under normality

The coefficient β_j and its estimate $\hat{\beta}_j$ do not simply reflect the linear relation between the jth explanatory variable and the response; they are the coefficient explaining linearly by the jth explanatory variable the variation of the response not explained by the other explanatory variables. So inference on β_j is not about the relation between the jth variable and the response but about the role of the jth variable when the multiple regression model is true. (This is why older papers and textbooks call the β_j partial regression coefficients.) Under normality, intervals and tests for β_j are exact.

With normal errors, $\hat{\boldsymbol{\beta}}$ is multivariate normal with mean and covariance (10.17). The estimate $\hat{\beta}_j$ has the marginal distribution

$$\hat{\beta}_j \sim N(\beta_j, \sigma^2(X'X)^{-1}_{jj}), \tag{10.19}$$

where $(X'X)^{-1}_{jj}$ is the $(j+1)$th diagonal element of the $(p+1)\times(p+1)$ matrix $(X'X)^{-1}$. With $\mathrm{se}(\hat{\beta}_j) = \{\sigma^2(X'X)^{-1}_{jj}\}^{1/2}$ and its estimate $\widehat{\mathrm{se}}(\hat{\beta}_j) = \{s^2(X'X)^{-1}_{jj}\}^{1/2}$, the studentized pivot satisfies

$$t_j = \frac{\hat{\beta}_j - \beta_j}{\widehat{\mathrm{se}}(\hat{\beta}_j)} \sim t_{n-p-1}. \tag{10.20}$$

So the $1-\alpha$ interval for β_j is exactly

$$\beta_j \in \hat{\beta}_j \pm t_{\alpha/2,n-p-1}\widehat{\text{se}}(\hat{\beta}_j), \tag{10.21}$$

and for $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$ the observed $t_{j(0)} = (\hat{\beta}_j - 0)/\widehat{\text{se}}(\hat{\beta}_j)$ gives exactly $p = P(|T_{n-p-1}| \geq |t_{j(0)}|)$.

Example 10.8

Fit the multiple regression with all $p = 6$ variables of Example 10.7:

```
h = read.csv("data/worldbank2020.csv")
dat = h %>% dplyr::select(4:9,3)
n = nrow(dat); p = 6
lm.out = lm(Score ~ . , data = dat)
```

The design matrix X is obtained by `model.matrix()`; its first column is all 1 and it has $p+1$ columns.

```
> y = dat$Score
> X = model.matrix(lm.out)
> head(X)
  (Intercept)   GDP Social  Life Freedom Generosity Corruption
1           1 1.285  1.500 0.961   0.662      0.160      0.478
2           1 1.327  1.503 0.979   0.665      0.243      0.495
3           1 1.391  1.472 1.041   0.629      0.269      0.408
```

The estimated standard errors of the $\hat{\beta}_j$:

```
> s = sqrt( sum ( (lm.out$residuals)^2 ) / (n - p - 1) )
> (se.hat = s *  ( solve(t(X) %*% X) %>% diag() %>% sqrt() ) )
(Intercept)         GDP      Social        Life     Freedom
  0.2272244   0.2648616   0.2799800   0.3604369   0.4151976
 Generosity  Corruption
  0.5095199   0.4878042
```

For Generosity, the estimate is $\hat{\beta}_j = 0.6203$; its 95% interval contains 0, so $H_0 : \beta_j = 0$ is not rejected.

```
> beta.hat = solve(t(X) %*% X) %*% t(X) %*% y
> beta.hat[6] + se.hat[6] * qt(1-0.025, n-p-1) * c(-1,1)
[1] -0.3867092  1.6272659
```

10.2.3 Comparing two linear models under normality

Beyond inference on each coefficient, multiple regression allows comparing models. E.g. in Example 10.7, choose between the reduced model with only the $k = 4$ explanatory variables (dropping Generosity and Corruption) and the full model with all $p = 6$. With the full model (10.14), the reduced model with the first k variables is

$$\begin{aligned} y_i &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad (i = 1, \ldots, n) \\ &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + 0 \cdot x_{i,k+1} + \cdots + 0 \cdot x_{ip} + \epsilon_i, \end{aligned}$$

the special case $\beta_{k+1} = \cdots = \beta_p = 0$ of the full model.[53] So model selection becomes the test

$$H_0 : \beta_{k+1} = \cdots = \beta_p = 0 \quad \text{vs} \quad H_1 : \text{at least one of } \beta_{k+1}, \ldots, \beta_p \text{ is not } 0. \tag{10.22}$$

[53] More general hypotheses on coefficients, e.g. $H_0 : C\boldsymbol{\beta} = 0$, also have normal-theory tests, and in such general cases the null distribution can be approximated by normal theory, permutation or bootstrap. The principle and process are the same as comparing reduced and full models, so they are not discussed here.

In testing (10.22) or comparing two models, note that the variables $X_1, \ldots, X_k$ not being tested are always in the model. The alternative is not simply "$X_{k+1}, \ldots, X_p$ are related to Y" but

H_1 : "a linear combination of $X_{k+1}, \ldots, X_p$ explains variation of Y not explained by $X_1, \ldots, X_k$",

and the null is

H_0 : "the variation of Y explained by $X_{k+1}, \ldots, X_p$ is already explained by $X_1, \ldots, X_k$".

Write the reduced (null) model as

$$\mathbf{y} = X_0\boldsymbol{\beta}_0 + \boldsymbol{\epsilon} = \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix} \tag{10.23}$$

(subscript "0" = null model). Then the full model (10.15) is

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon} = X_0\boldsymbol{\beta}_0 + X_1\boldsymbol{\beta}_1 + \boldsymbol{\epsilon}, \qquad X_1\boldsymbol{\beta}_1 = \begin{pmatrix} x_{1,k+1} & \cdots & x_{1p} \\ \vdots & \ddots & \vdots \\ x_{n,k+1} & \cdots & x_{np} \end{pmatrix} \begin{pmatrix} \beta_{k+1} \\ \vdots \\ \beta_p \end{pmatrix}, \tag{10.24}$$

with $q = p - k$ added (tested) variables and $X = [X_0, X_1]$. Let $H = X(X'X)^{-1}X'$ and $H_0 = X_0(X_0'X_0)^{-1}X_0'$. In the reduced model (10.23),

$$\hat{\boldsymbol{\beta}}_{0(0)} = (X_0'X_0)^{-1}X_0'\mathbf{y}, \qquad \mathbf{e}_0 = \mathbf{y} - X_0\hat{\boldsymbol{\beta}}_{0(0)} = (I_n - H_0)\mathbf{y}.$$

In the full model (10.24),

$$\hat{\boldsymbol{\beta}} = (X'X)^{-1}X'\mathbf{y} = \begin{pmatrix} \hat{\boldsymbol{\beta}}_0 \\ \hat{\boldsymbol{\beta}}_1 \end{pmatrix} = \begin{pmatrix} \hat{\boldsymbol{\beta}}_0 \\ \{X_1'(I_n - H_0)X_1\}^{-1}X_1'(I_n - H_0)\mathbf{y} \end{pmatrix}$$

(in general $\hat{\boldsymbol{\beta}}_0 \neq \hat{\boldsymbol{\beta}}_{0(0)}$), with residuals $\mathbf{e} = (I_n - H)\mathbf{y}$. Comparing the residual sizes tests whether $\boldsymbol{\beta}_1$ is significantly nonzero. The reduced-model residual sum of squares is

$$SSE_0 = \sum_{i=1}^{n} e_{i(0)}^2 = \mathbf{e}_0'\mathbf{e}_0 = \mathbf{y}'(I_n - H_0)\mathbf{y},$$

the full-model one $SSE = \mathbf{e}'\mathbf{e} = \mathbf{y}'(I_n - H)\mathbf{y}$, and the gain from additionally estimating $\boldsymbol{\beta}_1$ given $\boldsymbol{\beta}_0$ is

$$SSR_{\beta_1|\beta_0} = SSE_0 - SSE = \mathbf{y}'(I_n - H_0)\mathbf{y} - \mathbf{y}'(I_n - H)\mathbf{y} = \mathbf{y}'(H - H_0)\mathbf{y}.$$

Standardizing by the full-model residual sum of squares, each divided by its degrees of freedom, the statistic for (10.22) is

$$F = \frac{SSR_{\beta_1|\beta_0}/q}{SSE/(n-p-1)}. \tag{10.25}$$

When the null is true (and errors normal), $F \sim F_{q,n-p-1}$, the F-statistic. With observed F_{obs}, $p = P(F_{q,n-p-1} \geq F_{\text{obs}})$.

Why $F \sim F_{q,n-p-1}$ (supplied). H and H_0 are orthogonal projections with $\text{col}(X_0) \subset \text{col}(X)$, so $HH_0 = H_0$, $H - H_0$ is a projection of rank q, and $(H - H_0)(I - H) = 0$. Under H_0, $\mathbf{y} = X_0\boldsymbol{\beta}_0 + \boldsymbol{\epsilon}$ and $(H - H_0)X_0 = 0$, $(I - H)X_0 = 0$, so $SSR_{\beta_1|\beta_0} = \boldsymbol{\epsilon}'(H - H_0)\boldsymbol{\epsilon}$ and $SSE = \boldsymbol{\epsilon}'(I - H)\boldsymbol{\epsilon}$. For $\boldsymbol{\epsilon} \sim N(0, \sigma^2 I)$ these are independent $\sigma^2\chi_q^2$ and $\sigma^2\chi_{n-p-1}^2$ (Cochran), so $F \sim F_{q,n-p-1}$. ■

Example 10.9

Continuing Example 10.8, test $H_0 : \beta_5 = \beta_6 = 0$ (the coefficients of Generosity and Corruption) under normality. The reduced design X_0 is the first 5 columns of X, and X_1 the remaining two.

```
> X0 = X[,1:5]
> X1 = X[,6:7]
> q = 2
```

The observed F (10.25) (`drop()` turns the 1×1 matrix into a scalar):

```
> Hat = X %*% solve(t(X) %*% X) %*% t(X)
> Hat0 = X0 %*% solve(t(X0) %*% X0) %*% t(X0)
> SSR = t(y) %*% (Hat - Hat0) %*% y
> SSE = sum(y^2) - t(y) %*% Hat %*% y
> (F.obs = drop(SSR/SSE * (n-p-1) / q))
[1] 3.599035
```

Compared with the null distribution $F_{2,153-6-1}$:

```
> (p.theory = 1- pf(F.obs, q, n-p-1))
[1] 0.02980354
```

Remark

When the reduced model omits a single variable of the full model, $q = 1$ and the F-statistic (10.25) is the square of the t-statistic $t_{j(0)}$ (10.20). Conversely, in a test of β_j, the null model $H_0 : \beta_j = 0$ is the reduced model with the $k = p - 1$ explanatory variables other than the jth.

10.3 Nonparametric inference in multiple regression

Multiple regression involves two kinds of inference: intervals and tests for one coefficient β_j, and tests of $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$ for several coefficients. When normal errors cannot be assumed even approximately, permutation tests and bootstrap intervals and tests replace normal-theory t- and F-tests.

10.3.1 Permutation tests

For the general hypothesis (10.22) by permutation, the null distribution is obtained by shuffling for the statistic F (10.25). If the number of coefficients tested is $q = p - k = 1$, the test is about that coefficient and the t-statistic t_j (10.20) can be used. For a two-sided test of one coefficient the permutation tests with F and with t_j agree, but the one-sided $H_1 : \beta_j > 0$ needs the t-statistic.

Permutation tests in regression require exchangeable errors ϵ_i. In simple regression, under the null the responses $y_1, \ldots, y_n$ were exchangeable and we shuffled them. In multiple regression under $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad (i = 1, \ldots, n), \tag{10.26}$$

the errors ϵ_i are exchangeable but, unfortunately, the observations y_i are not.[54] So the advantage of permutation tests, "(conditionally) exact inference," is lost. The exchangeable errors cannot be

[54] To see this simply, let $k = 1$, $\beta_1 = 1$: then $E(y_1) = \beta_0 + x_{11} \neq \beta_0 + x_{21} = E(y_2)$, so the means of y_1, y_2 differ with the explanatory values and they are not exchangeable.

shuffled (they are unobservable), and shuffling the non-exchangeable observations is unsatisfactory, so various permutation tests have been proposed. None is exact, but they are asymptotically exact as n grows, and for very large n they agree. They were developed after Efron's bootstrap and resemble bootstrap tests. With p explanatory variables, write the ith observation $(x_{i1}, \dots, x_{ip}, y_i) = (\mathbf{x}_i, y_i)$ and the full-model data $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$. We look at (1) shuffling the observations, (2) shuffling the residuals under the null, (3) shuffling the residuals of the full model.

Shuffling the observations. The simplest: treat the (non-exchangeable) responses $\{y_1, \dots, y_n\}$ as exchangeable and shuffle them to generate the permutation distribution of F (Manly, 2006). The shuffled sample is $\mathcal{D}^* = \{(\mathbf{x}_1, y_1^*), \dots, (\mathbf{x}_n, y_n^*)\}$, with $(y_1^*, \dots, y_n^*)$ drawn by sequential simple sampling without replacement from $\{y_1, \dots, y_n\}$. The random permutation test:

1. Compute F_{obs} from $\mathcal{D}$.
2. For $i = 1, \dots, M$ $(M = 1999)$ generate a randomly shuffled sample $\mathcal{D}^{*i}$ and compute F^{*i}.
3. The p-value for (10.22) is $p \approx \sum_{i=0}^{M} I(F^{*i} \geq F_{\text{obs}})/(M+1)$ $(F^{*0} := F_{\text{obs}})$.

Shuffling the residuals under the null. Shuffle the residuals $e_{i(0)}$ (or adjusted residuals $r_{i(0)}$) instead of the exchangeable but unobservable errors (Freedman and Lane, 1983). Assume the reduced model (10.26) under the null; with its estimates $\hat{\boldsymbol{\beta}}_{0(0)} = (\hat\beta_{0(0)}, \dots, \hat\beta_{k(0)})$ and residuals $(e_{1(0)}, \dots, e_{n(0)})' \equiv \mathbf{e}_0 = (I_n - H_0)\mathbf{y}$, build the most plausible null model

$$y_i = \hat\beta_{0(0)} + \hat\beta_{1(0)} x_{i1} + \cdots + \hat\beta_{k(0)} x_{ik} + e_{i(0)}, \qquad i = 1, \dots, n. \tag{10.27}$$

In this approximate model, shuffle the (approximately exchangeable) residuals $\{e_{1(0)}, \dots, e_{n(0)}\}$ by sampling without replacement to $(e^*_{1(0)}, \dots, e^*_{n(0)})$ and form the shuffled responses

$$y_i^* = \hat\beta_{0(0)} + \hat\beta_{1(0)} x_{i1} + \cdots + \hat\beta_{k(0)} x_{ik} + e^*_{i(0)}, \qquad i = 1, \dots, n, \tag{10.28}$$

and finally $\mathcal{D}^* = \{(\mathbf{x}_1, y_1^*), \dots, (\mathbf{x}_n, y_n^*)\}$. The procedure is the same as steps 1–3 above; only the generation of $\mathcal{D}^*$ differs.

Shuffling the residuals of the full model. The third method is the permutation version of bootstrap resampling with a pivot. The two methods above produce shuffled samples plausible under the null; this one produces samples plausible under the alternative (when it is true). It shuffles the full-model residuals, whose variance is smaller than that of the reduced-model residuals, aiming at a less dispersed null distribution of F (i.e. higher power) (Ter Braak, 1992). Get the full-model residuals $(e_1, \dots, e_n)' \equiv \mathbf{e} = (I_n - H)\mathbf{y}$ (adjusted residuals r_i (10.18) may be used), shuffle them without replacement to $(e_1^*, \dots, e_n^*)$, add them to the full-model fitted values $\hat y_i$:

$$y_i^* = \hat\beta_0 + \hat\beta_1 x_{i1} + \cdots + \hat\beta_p x_{ip} + e_i^* = \hat y_i + e_i^*, \qquad i = 1, \dots, n, \tag{10.29}$$

and form $\mathcal{D}^*_{\text{full}} = \{(\mathbf{x}_1, y_1^*), \dots, (\mathbf{x}_n, y_n^*)\}$. This is a sample from the full (alternative) model: the least squares estimate $\hat{\boldsymbol{\beta}}^*$ from $\mathcal{D}^*_{\text{full}}$ lies "near" $\hat{\boldsymbol{\beta}}$, like a bootstrap replicate. (From samples shuffled under the null, $\hat\beta^*_{k+1}, \dots, \hat\beta^*_p$ are near 0.) So with this shuffling the statistic must be expressible as a pivot.

The usual F follows the F-distribution only when $\boldsymbol{\beta}_1 = 0$ (under normality), so expressing it as a pivot for $\boldsymbol{\beta}_1 \neq 0$ seems hard. But knowing that $SSR_{\beta_1|\beta_0}$ in (10.25) equals $\hat{\boldsymbol{\beta}}_1'\{X_1'(I_n - H_0)X_1\}\hat{\boldsymbol{\beta}}_1$,[55]

$$F = \frac{SSR_{\beta_1|\beta_0}/q}{SSE/(n-p-1)} = \frac{(\hat{\boldsymbol{\beta}}_1 - 0)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1 - 0)}{q \cdot s^2}$$

[55] Accept this at the level of the book. (Supplied: by the Frisch–Waugh–Lovell theorem, $\hat{\boldsymbol{\beta}}_1$ is the least squares coefficient of $(I - H_0)\mathbf{y}$ on $\tilde X_1 = (I - H_0)X_1$, and $H - H_0$ is the projection onto $\text{col}(\tilde X_1)$, so $\mathbf{y}'(H - H_0)\mathbf{y} = \hat{\boldsymbol{\beta}}_1'\tilde X_1'\tilde X_1\hat{\boldsymbol{\beta}}_1$.)

(s^2 the full-model error variance estimate (10.16)). So

$$F_{\beta_1} = \frac{(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1)}{q \cdot s^2} \tag{10.30}$$

follows the same $F_{q,n-p-1}$ distribution even when $\boldsymbol{\beta}_1 \neq 0$: a pivot. Even when F_{β_1} is not $F_{q,n-p-1}$ (normality cannot be assumed), if it is a pivot, the permutation test with the shuffled samples $\mathcal{D}^*_{\text{full}}$:

1. Compute F_{obs} (10.25) from $\mathcal{D}$.
2. For $i = 1, \ldots, M$ ($M = 1999$) generate a randomly shuffled sample $\mathcal{D}^{*i}_{\text{full}}$, compute $\hat{\boldsymbol{\beta}}_1^{*i}$ and $(s^{*i})^2$, and
$$F^{*i}_{\beta_1} = \frac{(\hat{\boldsymbol{\beta}}_1^{*i} - \hat{\boldsymbol{\beta}}_1)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1^{*i} - \hat{\boldsymbol{\beta}}_1)}{q \cdot (s^{*i})^2}.$$
3. The p-value for (10.22) is $p \approx \sum_{i=0}^{M} I(F^{*i}_{\beta_1} \geq F_{\text{obs}})/(M+1)$ ($F^{*0}_{\beta_1} := F_{\text{obs}}$).

Remark

A hypothesis on one coefficient can also be tested with F. But for the one-sided test

$$H_0 : \beta_j = 0 \quad \text{vs} \quad H_1 : \beta_j > 0 \tag{10.31}$$

the t-statistic $t_j = (\hat{\beta}_j - \beta_j)/\widehat{\text{se}}(\hat{\beta}_j)$ is better, and all three permutation methods above can be used. E.g. with the pivot t_j and full-model residual shuffling:

1. Compute $t_{j(0)} = (\hat{\beta}_j - 0)/\widehat{\text{se}}(\hat{\beta}_j)$ from $\mathcal{D}$.
2. For $i = 1, \ldots, M$ ($M = 1999$) generate $\mathcal{D}^{*i}_{\text{full}}$ (full-model residual shuffling), compute $\hat{\beta}_j^*$, $\widehat{\text{se}}(\hat{\beta}_j^*)$ and
$$t_j^{*i} = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\text{se}}(\hat{\beta}_j^*), \qquad i = 1, \ldots, M. \tag{10.32}$$
3. The p-value for (10.31) is $p \approx \sum_{i=0}^{M} I(t_j^{*i} \geq t_{j(0)})/(M+1)$ ($t_j^{*0} := t_{j(0)}$).

Example 10.10

For $H_0 : \beta_5 = \beta_6 = 0$ of Example 10.9, the three random permutation tests. First a function `F.pivot()` computes F in the pivot form (10.30):

```
> X10 = X1 - Hat0 %*% X1
> XX10 = t(X10) %*% X10
> XXX10 = solve( t(X10) %*% X10) %*% t(X10)
> F.pivot = function(y.perm, beta1){
+   beta1.hat = XXX10 %*% y.perm
+   SSE = sum(y.perm^2) - t(y.perm) %*% Hat %*% y.perm
+   SSR = t(beta1.hat - beta1) %*% XX10 %*% (beta1.hat - beta1)
+   as.vector(SSR / SSE / q * (n-p-1))
+ }
```

The design matrix is fixed under every shuffle, so for speed H, H_0, X_0, $X_{10} := (I_n - H_0)X_1$, $X_{10}'X_{10} = X_1'(I_n - H_0)X_1$ and $(X_{10}'X_{10})^{-1}X_{10}'$ are computed once outside the function. The observed F_{obs} uses the original response vector and the null value $(\beta_5, \beta_6) = (0, 0)$; as in Example 10.9 it is 3.599.

```
> (F.obs = F.pivot(y, c(0,0)))
[1] 3.599035
```

(1) Shuffling the observations: sample the response vector without replacement.

```
F.perm.raw = replicate(M, {
  yperm = sample(y,n, replace = FALSE)
  F.pivot(yperm, c(0,0))
})
p.raw = mean(c(F.obs,F.perm.raw) >= F.obs)
```

(2) Shuffling the null residuals needs the reduced-model fitted values $\hat{\mathbf{y}}_0$ and residuals $\mathbf{e}_0$.

```
yhat0 = Hat0 %*% y
res0 = y - yhat0
F.perm.null = replicate(M, {
  yperm = yhat0 + sample(res0,n, replace = FALSE)
  F.pivot(yperm, c(0,0))
})
p.null = mean(c(F.obs,F.perm.null) >= F.obs)
```

(3) Shuffling the full-model residuals needs the fitted values $\hat{\mathbf{y}}$, residuals $\mathbf{e}$, and, for the pivot, the full-model $\hat{\boldsymbol{\beta}}_1$.

```
yhat = Hat %*% y
res = y - yhat
beta.hat1 = XXX10 %*% y
F.perm.full = replicate(M, {
  yperm = yhat + sample(res,n, replace = FALSE)
  F.pivot(yperm, beta.hat1)
})
p.full = mean(c(F.obs,F.perm.full) >= F.obs)
```

The p-values are all similar:

```
> c(p.theory, p.raw, p.null, p.full)
[1] 0.02980354 0.03150000 0.02850000 0.03300000
```

For these data the three permutation distributions are also nearly the same (histograms in the book, compared with the theoretical $F_{q,n-p-1}$ and F_{obs}).

Comparing the three methods. All three use the explanatory values X (in particular X_0 of the reduced model) as they are, so all are tests conditional on the explanatory values. Shuffling the observations is known not to control the type I error in small samples (type I error above α). It actually generates the permutation distribution under $H_0 : \beta_1 = \cdots = \beta_p = 0$, not $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$, so it is not appropriate. The null-residual and full-residual shuffles draw residuals by simple sampling without replacement; drawing with replacement instead gives bootstrap resampling. (Two of the bootstrap tests in Section 10.3.3 are these.) Finally, the full-model resampling hinges on the existence of a pivot.

The three methods are asymptotically equivalent for large n; for finite n simulations show Freedman and Lane's (1983) shuffling of the null (reduced-model) residuals is best.[56]

10.3.2 Bootstrap confidence intervals for the coefficients

Bootstrap intervals for β_j in multiple regression proceed almost as in simple regression. Bivariate bootstrap could be done by the case or residual bootstrap. Multiple regression data have $p + 1$ variables, but the p explanatory variables play the role of "X" in bivariate data, so the extension is natural:

[56] See Anderson and Robinson (2001); Anderson and Legendre (1999).

- **Case bootstrap:** resample the n pairs $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ with replacement.
- **Residual bootstrap:** from the residuals $e_i = y_i - \hat{y}_i$ or adjusted residuals $\{r_1, \ldots, r_n\}$ (10.18), draw $r_1^*, \ldots, r_n^*$ with replacement and form $(\mathbf{x}_i^*, y_i^*) = (\mathbf{x}_i, \hat{y}_i + r_i^*)$.

Pros and cons are as in simple regression, but with the residual bootstrap the design matrix X (and H) is the same in every bootstrap sample, so the computational advantage is larger. With bootstrap replicates of $\hat{\beta}_j$ the bias, standard error and intervals of β_j can be estimated. When n is much larger than p, bootstrap samples approximate the population well and the estimates are reliable.

The code below uses the residual bootstrap to compute X, $(X'X)^{-1}$ (`XX`) and H (`Hat`) once and gives 95% bootstrap-t intervals for all β_j; the pivot is $t_j = (\hat{\beta}_j - \beta_j)/\widehat{\text{se}}(\hat{\beta}_j)$ (10.20).

Example 10.11 (Residual bootstrap, bootstrap-t intervals)

Continuing Example 10.10, to get bootstrap-t intervals for the coefficients, compute $t_j^* = (\hat{\beta}_j^* - \hat{\beta}_j)/\widehat{\text{se}}(\hat{\beta}_j^*)$ for all $j = 0, 1, \ldots, p$ in each bootstrap sample; define `t.pivot()` first:

```
XX = solve(t(X) %*% X)
Hat = X %*% XX %*% t(X)
t.pivot = function(yboot, beta){
          yhat = Hat %*% yboot
          res = yboot - yhat
          beta.hat  = XX %*% t(X) %*% yboot
          se.beta.hat = sqrt( sum(res^2) / (n-p-1) * diag(XX))
          as.vector( (beta.hat - beta)  / se.beta.hat )
}
```

The bootstrap replicates t_j^*: except for using adjusted residuals, this is almost the same as the shuffling of full-model residuals in the permutation test; the only difference is the `replace` argument of `sample()` (`TRUE` for the bootstrap, `FALSE` for permutation).

```
yhat = lm.out$fitted.values
adjusted.res = lm.out$residuals / sqrt(1 - hatvalues(lm.out))
beta.hat = lm.out$coefficients
B = 1999
t.boot.res = replicate(B, {
        yboot = yhat + sample(adjusted.res, n, replace = TRUE)
        t.pivot(yboot, beta.hat)
}) %>% t()
```

The 95% bootstrap-t interval for β_5 (Generosity) is below; these data roughly satisfy normality, so it is almost the same as the normal interval of Example 10.8.

```
> ci.beta5 = beta.hat[6] - se.hat[6] * quantile(t.boot.res[,6],
    probs = c(1-0.025,0.025))
> ci.beta5
     97.5%      2.5%
-0.3737554  1.6320302
```

10.3.3 Bootstrap tests

In simple regression there were four bootstrap tests for β: two resamplings (case and residual bootstrap) under two scenarios (resampling under the null, resampling with a pivot). Tests in multiple regression proceed the same way. For $H_0 : \beta_{k+1} = \cdots = \beta_p = 0$ use the pivot F (10.30);

for $H_0 : \beta_j = 0$ the pivot t (10.20). The observed statistics are

$$F_{\text{obs}} = \frac{SSR_{\beta_1|\beta_0}/q}{SSE/(n-p-1)} = \frac{(\hat{\boldsymbol{\beta}}_1 - 0)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1 - 0)}{q \cdot s^2} \tag{10.33}$$

or

$$t_{j(\text{obs})} = (\hat{\beta}_j - 0)/\widehat{\text{se}}(\hat{\beta}_j). \tag{10.34}$$

To approximate their null distributions by resampling under the null:

- **Resampling under the null, case bootstrap.** Draw the explanatory bootstrap sample $\mathbf{x}_1^*, \ldots, \mathbf{x}_n^*$ from $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ with replacement and, independently, the responses $y_1^*, \ldots, y_n^*$ from $\{y_1, \ldots, y_n\}$; form $\{(\mathbf{x}_1^*, y_1^*), \ldots, (\mathbf{x}_n^*, y_n^*)\}$.
- **Resampling under the null, residual bootstrap.** With the null (reduced) model fitted values $\hat{y}_0$ and residuals $r_{i(0)}$, draw the residuals with replacement and form $y_i^* = \hat{y}_{i(0)} + r_{i(0)}^*$, giving $\{(\mathbf{x}_1, y_1^*), \ldots, (\mathbf{x}_n, y_n^*)\}$. (The resampled observations are almost those of shuffling the null residuals (10.28); only sampling is with replacement.)

In each bootstrap sample ($b = 1, \ldots, B$) compute

$$F^{*b} = \frac{(\hat{\boldsymbol{\beta}}_1^{*b} - 0)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1^{*b} - 0)}{q \cdot (s^{*b})^2}$$

or $t_j^{*b} = (\hat{\beta}_j^{*b} - 0)/\widehat{\text{se}}(\hat{\beta}_j^{*b})$, and the two-sided p-values ($F^{*0} := F_{\text{obs}}$, $t_j^{*0} = t_{j(\text{obs})}$)

$$p_F^* = \sum_{b=0}^{B} I(F^{*b} \geq F_{\text{obs}})/(B+1), \qquad p_t^* = \sum_{b=0}^{B} I(|t_j^{*b}| \geq |t_{j(\text{obs})}|)/(B+1). \tag{10.35}$$

To approximate the null distribution with a pivot:

- **Pivot, case bootstrap.** Resample the n pairs $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)\}$ with replacement (as for intervals).
- **Pivot, residual bootstrap.** From the residuals $e_i = y - \hat{y}_i$ or adjusted residuals $\{r_1, \ldots, r_n\}$ (10.18) draw $r_1^*, \ldots, r_n^*$ with replacement and form $(\mathbf{x}_i^*, y_i^*) = (\mathbf{x}_i, \hat{y}_i + r_i^*)$ (as for intervals; also as in shuffling the full-model residuals (10.29) but with replacement).

In each bootstrap sample compute

$$F^{*b} = \frac{(\hat{\boldsymbol{\beta}}_1^{*b} - \hat{\boldsymbol{\beta}}_1)'\{X_1'(I_n - H_0)X_1\}(\hat{\boldsymbol{\beta}}_1^{*b} - \hat{\boldsymbol{\beta}}_1)}{q \cdot (s^{*b})^2}$$

or $t_j^* = (\hat{\beta}_j^{*b} - \hat{\beta}_j)/\widehat{\text{se}}(\hat{\beta}_j^{*b})$, and the two-sided p-value by (10.35). The pivot-based bootstrap test of $H_0 : \beta_j = 0$ is the inversion of the bootstrap-t interval for β_j.

Example 10.12

In Example 10.11 the 95% bootstrap-t interval for β_5 (Generosity) contained 0. Test $H_0 : \beta_5 = 0$ against $H_1 : \beta_5 \neq 0$ by the pivot-based residual bootstrap; the replicates (10.32) were already stored in `t.boot.res` in Example 10.11, so:

```
> t5 = t.pivot(y, rep(0,p+1))[6]
> (pvalue = mean(abs( c(t5, t.boot.res[,6])) >= abs(t5)))
[1] 0.215
```

Compared with the 95% bootstrap-t interval $(-0.39, 1.614)$ from `ci.beta5`: the null value 0

is in the interval, so H_0 is not rejected at the 5% level.

10.4 Robust linear regression

So far, permutation tests and bootstrap resampling in simple and multiple regression assumed least squares estimation of the coefficients. The least squares estimator coincides with the maximum likelihood estimator when the errors are independent and identically normal, so in the two situations

- the error distribution has much heavier or skewed tails than normal,
- the observations contain outliers,

least squares loses efficiency.[57] Then nonparametric or robust estimators are preferred.

This section covers nonparametric, robust estimators of regression coefficients. "Nonparametric" means estimators using ranks and signs in the traditional sense; "robust" means estimators like the median and trimmed means whose value does not change much when part of the data is contaminated. Estimators using ranks and signs keep the same ranks and signs even if the largest value (e.g. 10) is contaminated to 10^{100}, so they are a kind of robust estimator. We first introduce robust estimators of the location of one population to understand the concept, then apply them to regression.

10.4.1 Robust estimation of the mean

Introducing permutation tests, we used not only the difference of sample means but also of medians or trimmed means as statistics for the location difference, and saw that the efficiency (power) depends on the statistic. For asymmetric or heavy-tailed populations, medians and trimmed means were more powerful than means. Heavy-tailed distributions often produce values far from the center, which are often regarded as outliers. (Conversely, data with outliers can be viewed as sampled from a very heavy-tailed distribution.) For the location (the population mean, when it is defined), the sample mean is very sensitive to outliers, a "non-robust" statistic; the median and trimmed mean are representative robust statistics. There are many measures of robustness; here we use a high *breakdown point* as the criterion.

Breakdown point. For a statistic $T = T(\mathbb{X}_n)$ computed from $\mathbb{X}_n = \{x_1, \ldots, x_n\}$: if sending one observation x_i to infinity sends T to infinity ($\lim_{x_i \to \infty} T(\mathbb{X}_n) = \infty$), the statistic "breaks down." If sending any $b-1$ observations to infinity does not break it, but sending some b observations $x_{i_1}, \ldots, x_{i_b}$ to infinity makes it infinite, it breaks when b or more observations are contaminated. The breakdown point of T is the smallest fraction of contaminated observations that breaks it: increasing the number of contaminations from 1, if for any $b-1$ observations

$$\lim_{x_{i_1}, \ldots, x_{i_{b-1}} \to \infty} |T(\mathbb{X}_n)| < \infty$$

while for some b observations

$$\lim_{x_{i_1}, \ldots, x_{i_b} \to \infty} |T(\mathbb{X}_n)| = \infty,$$

[57] Loss of efficiency of $\hat{\beta}$ means all of: (1) $\hat{\beta}$ has large MSE, variance or standard error; (2) intervals for β based on $\hat{\beta}$ are long; (3) tests have low power. If normality fails, normal-theory tests do not control the type I error, but permutation (or bootstrap) tests based on $\hat{\beta}$ do (approximately in multiple regression); their power can still be low. (In two-sample comparisons, the permutation test with the two-sample t was less powerful than with the difference of medians.) Likewise, bootstrap intervals based on $\hat{\beta}$ are long but (approximately) correct.

the breakdown point is

$$BP(T) = \frac{b}{n}.$$

For the sample mean, median and 10% trimmed mean, roughly

$$BP(\bar{X}_n) = \frac{1}{n} \approx 0, \qquad BP(\text{median}\{X_i\}) = \frac{1}{2}, \qquad BP(10\% \text{ trimmed mean}) = 10\%,$$

easily seen. The median, with breakdown point 50%, is the most robust estimator. (Nothing is free: most robust estimators lose efficiency. Example 2.16 showed the ARE of the sample median to the sample mean is 63.7%.)

The simplest way to devise robust estimators is to "ignore" observations far from the "center." The median is computed from the middle one or two observations, ignoring all others. The 10% trimmed mean removes the 10% of observations at each extreme and averages the remaining 80%. Replacing them instead by the 10% and 90% sample quantiles and averaging gives the Winsorized mean; the 10% Winsorized mean has breakdown point 10%.

Example 10.13

Suppose $n = 11$ observations $-5, -3, -3, -3, -1, 0, 1, 2, 3, 4, 1000$ are sampled from a population; the last, $x_{11} = 1000$, is a contaminated outlier. The mean, 10% trimmed mean and 10% Winsorized mean (R package `WRS2`):

```
> x = c(-5,-3,-3,-3,-1,0,1,2,3,4,1000)
> c(mean(x), mean(x, tr = 0.1), WRS2::winmean(x, tr = 0.1))
[1] 90.45454545  0.00000000  0.09090909
```

The trimmed mean averages the 9 middle observations after removing -5 and 1000. The Winsorized mean replaces -5 by the 10% sample quantile -3 and 1000 by the 90% quantile 4, then averages all 11 (Winsorized) observations. Substituting 10^{100} for the outlier 1000 does not change the trimmed or Winsorized means.

M-estimators. Trimmed and Winsorized means remove the same fraction of observations at both extremes. If, as in Example 10.13, the outliers are only on the right, there is no reason to remove or replace the normal observation -5. The M-estimators introduced here take the distribution of the observations into account and remove or adjust only the effect of the outliers, wherever they are.

First another expression for $\bar{x}_n$. If $x_i = \mu + \epsilon_i$, $i = 1, \ldots, n$, with $\epsilon_i \overset{\text{i.i.d.}}{\sim} F_\epsilon$ of mean 0, the estimator $\bar{x}_n$ of μ is the least squares estimator:

$$\bar{x}_n = \arg\min_\mu \sum_{i=1}^n (x_i - \mu)^2 = \arg\min_\mu \sum_{i=1}^n \epsilon_i^2. \tag{10.36}$$

($\arg\min_\mu$ is the μ minimizing $\sum (x_i - \mu)^2$.) Differentiating in μ, the minimizer solves $\sum_{i=1}^n (x_i - \mu) = 0$, i.e. $\hat{\mu} = \bar{x}_n$. The trimmed and Winsorized means are least squares estimators computed after removing some observations with large $|x_i - \mu|$ (trimmed) or replacing some x_i by other values (Winsorized), on the grounds that observations with large $|x_i - \mu|$ are likely outliers. The sample mean is not robust because for a large outlier x_i its squared error $(x_i - \mu)^2$ is huge, and reducing it pulls $\bar{x}_n$ strongly toward the outlier. The key idea of M-estimators of μ is to use, instead of $(x_i - \mu)^2$, a function that adjusts the weight of the error so that an observation with large $|x_i - \mu|$ can be suitably "ignored."

Remark

If the errors are normal, $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, a parametric model, and the log-likelihood is

$$\ell(\mu; \mathbb{X}_n) = -\frac{1}{2}\sum_{i=1}^{n}\Big(\frac{x_i - \mu}{\sigma}\Big)^2 + \text{const.} \tag{10.37}$$

With σ^2 known, the maximum likelihood estimator of μ,

$$\bar{x}_n = \arg\max_{\mu} \ell(\mu; \mathbb{X}_n) = \arg\min_{\mu} \sum_{i=1}^{n}\Big(\frac{x_i - \mu}{\sigma}\Big)^2, \tag{10.38}$$

is the least squares estimator. The "M" of M-estimator means an estimator of "MLE type." It may also be read as "an estimator of minimum (or maximum) type."

An M-estimator replaces the square function $y = x^2$ by some function $y = \Psi(x)$:

$$\hat{\mu}^{\Psi} = \arg\min_{\mu} \sum_{i=1}^{n} \Psi\Big(\frac{x_i - \mu}{\tau}\Big),$$

where τ is an estimator of the population standard deviation σ, possibly estimated from the data. Equivalently, the M-estimator $\hat{\mu}^{\Psi}$ is the solution of

$$\sum_{i=1}^{n} \psi\Big(\frac{x_i - \mu}{\tau}\Big) = 0$$

with $\psi(x) = \frac{d}{dx}\Psi(x)$ replacing the derivative $\frac{d}{dx}x^2 = 2x$ of the square function. The M-estimator is determined by Ψ (or ψ). Representative examples:

1. The sample mean: the M-estimator with $\Psi(x) = x^2/2$, i.e. $\psi(x) = x$.
2. The sample median: the M-estimator with $\Psi(x) = |x|$, $\psi(x) = \text{sgn}(x)$.
3. Huber's (1964) Ψ: for some $k > 0$, $\psi(x) = \max\{-k, \min\{k, x\}\}$, and

$$\Psi(x) = \begin{cases} x^2/2, & |x| \le k; \\ k|x| - \frac{k^2}{2}, & |x| > k. \end{cases}$$

   The constant k can be set to various values, but generally $k = 1.28$, the 90% quantile of the standard normal, is used.

4. Tukey's "biweight" function: for some $c > 0$,

$$\psi(x) = \begin{cases} x(1 - (\frac{x}{c})^2)^2, & |x| \le c; \\ 0, & |x| > c. \end{cases}$$

(Figure 10.4 of the book plots these ψ.) Except for the sample mean, $|\psi(x)|$ is always below a fixed constant. Since the input x of ψ is the standardized error $(x_i - \mu)/\tau$, a large error does not make $|\psi(x)|$ large, so very large errors are suitably ignored. Huber's Ψ has the form of the mean's Ψ in the "center" ($|x| \le k$) and of the median's in the "periphery"; this form was designed to have both the high efficiency of the mean and the high robustness of the median. Huber's M-estimator is the solution in μ of

$$\sum_{i=1}^{n} \psi\Big(\frac{x_i - \mu}{\hat{\tau}}\Big) = 0, \qquad \psi(x) = \max\{-k, \min\{k, x\}\}, \tag{10.39}$$

where $\hat{\tau} = \text{MAD}\{x_1, \ldots, x_n\} \times 1.4826$ is a robust estimator of σ under the normal $N(\mu, \sigma^2)$.[58] (Supplied: for $N(\mu, \sigma^2)$, MAD $\to \sigma\Phi^{-1}(3/4) \approx 0.6745\sigma$, and $1/0.6745 \approx 1.4826$.) Equation (10.39) has no analytic solution, so Huber's M-estimator $\hat{\mu}^{\Psi}$ is found numerically. The M-estimator with Huber's Ψ is the most representative robust location estimator.

Example 10.14

The data of Example 10.13 seem to have the outlier 1000. For Huber's M-estimator of location use `mest()` of the `WRS2` package ("M-estimation"); `bend = 1.28` sets $k = 1.28$.

```
> WRS2::mest(x,bend = 1.28)
[1] 0.0693184
```

The standardized form $(1000 - \hat{\mu}^{\Psi})/\hat{\tau}$ of the last observation was Winsorized; the other observations were used as they are in computing the mean.

The breakdown point of an M-estimator is always 50% when $|\psi(x)|$ is bounded. The MAD used in Huber's M-estimator also has breakdown point 50%, so $BP(\hat{\mu}^{\Psi}) = 50\%$. The efficiency of Huber's M-estimator depends on the choice of k: in normal populations it increases with k, and at $k = 1.35$ the ARE of Huber's M-estimator to the sample mean is known to be about 90%.

10.4.2 Inference on location with outliers

When outliers are suspected, how should intervals for a location parameter be estimated and hypotheses tested? Assume observations $X_i = \mu + \epsilon_i$, $E(\epsilon_i) = 0$, but with contaminated data.

Example 10.15

The `phones` data of R's `MASS` package record the number of telephone calls (millions) over the 24 years 1950–1973 (Figure 10.5 of the book). At least 6 of the 24 observations look like outliers. Using these data we want an interval for the mean number of calls in the 50s–70s and a test of whether the mean is 15 (million).

The hypotheses of Example 10.15 are $H_0 : \mu = 15$ and $H_1 : \mu \neq 15$. From Figure 10.5, removing the 6 apparent outliers one would expect not to reject $\mu = 15$. If the uncontaminated observations X_i are assumed symmetric, the sign test and signed-rank test of Chapter 2 are good methods. The next example confirms that traditional distribution-free methods work for contaminated data.

Example 10.16

With the data of Example 10.15, the t-statistic $\sqrt{n}(\bar{X}_n - \mu_0)/s$ is pulled up by the outliers (the sample mean $\bar{X}_n$ is large), and $H_0 : \mu = 15$ is rejected at 5%. The t-statistic is not free of the influence of outliers, so it is a poor test here.

```
> x = MASS::phones$calls
> mean(x)
[1] 49.99167
> t.test(x, mu = 15)$p.value
[1] 0.0154505
```

The large-sample p-value of the signed-rank test (Section 2.2) is 0.1573, not significant at 5%, and the large-sample interval for μ is $(12.7, 81.6)$, containing 15. With the signed-rank test $H_0 : \mu = 15$ is not rejected.

```
> wilcox.test(x,mu = 15, alternative = "two.sided",
```

[58] R's `mad()` computes not $\text{MAD}\{x_1, \ldots, x_n\}$ but $\text{MAD}\{x_1, \ldots, x_n\} \times 1.4826$.

```
+             exact = F, conf.int = T)

        Wilcoxon signed rank test with continuity correction

data:  x
V = 200, p-value = 0.1573
alternative hypothesis: true location is not equal to 15
95 percent confidence interval:
 12.74998 81.59997
sample estimates:
(pseudo)median
        24.306
```

The exact p-value of the sign test (Section 2.1) is 0.84, and the sign-test interval is $(8.1, 43)$: the same conclusion as the signed-rank test.

```
> s = sign(x-15)
> n = sum(s != 0)
> (Sobs = sum(s > 0))
[1] 12
> (pvalue.sign.test = 2*(1- pbinom(Sobs,n,1/2)))
[1] 0.8388197
>
> alpha = 0.05
> l = qbinom(alpha/2, size = n, prob = 1/2); u = n+ 1 - l
> (ci.sign.test.x = sort(x)[c(l,u)]) # CI by sign test
[1]  8.1 43.0
```

(The book's two-sided sign p-value code computes $2(1-P(S \leq 12))$; the exact two-sided value is $2\min\{P(S \geq 12), P(S \leq 12)\}$, which leads to the same conclusion.) The Hodges–Lehmann estimate of the signed-rank test, the median of the Walsh averages, 24.306, is closer to the sign test's Hodges–Lehmann estimate, the sample median 15.5, than to the (contaminated) sample mean 49.99. The breakdown point of the sample mean is 0, of the median of the Walsh averages about 29%, and of the median 50%; the lower the breakdown point, the less robust, and the interval from a low-breakdown statistic is wider.

Breakdown point of the Hodges–Lehmann estimator (supplied)

If m observations are sent to infinity, the Walsh averages involving them also go to infinity; their number is about $n^2/2-(n-m)^2/2$ of $\approx n^2/2$. The median of the Walsh averages breaks down when this fraction exceeds $1/2$, i.e. $1-(1-m/n)^2 > 1/2$, i.e. $m/n > 1-1/\sqrt{2} \approx 0.293$.

Next, intervals from Huber's M-estimator. M-estimators including Huber's are asymptotically normal for large samples, so if the standard error can be estimated, normal-approximation intervals and tests are possible.[59] Below, from Huber's M-estimator, the 95% interval for μ by (1) the normal-approximation bootstrap interval with bootstrap estimates of standard error and bias (Section 8.1), and (2) the percentile bootstrap interval (Section 8.4).

Example 10.17

From the data of Example 10.15, Huber's M-estimator is $\hat{\mu}^{\Psi} = 20.84$.

```
> (mest.obs = WRS2::mest(x))
[1] 20.84105
```

[59] Estimating the standard error of M-estimators requires the influence function, a concept not introduced in this book, so it is not treated in detail. Instead the bootstrap can be used to estimate the standard error.

Bootstrap approximation of its bias and standard error:

```
> set.seed(1)
> B = 1999
> mest.boot = replicate(B,WRS2::mest(sample(x, length(x),
    replace = T)))
> (se.est = sd(mest.boot))
[1] 11.46062
> (bias.est = mean(mest.boot) - mest.obs)
[1] 2.514652
```

For large samples the M-estimator is normal, so

$$\frac{\hat{\mu}^{\Psi} - \text{bias}(\hat{\mu}^{\Psi}) - \mu}{\text{se}(\hat{\mu}^{\Psi})} \dot{\sim} N(0,1);$$

plugging in the bootstrap estimates of bias and standard error gives the 95% normal-approximation bootstrap interval:

```
> (norm.app.boot.CI = mest.obs - bias.est + se.est * qnorm
    (0.025) *c(1,-1))
[1] -4.136005 40.788792
```

The interval for μ, which must be positive, contains negative values: the sample is small and the normal approximation is poor. The percentile bootstrap interval seems more accurate:

```
> (boot.perc.ci.mest = quantile(mest.boot, probs = c
    (0.025,1-0.025)))
    2.5%    97.5%
11.24107 53.34024
```

For reference, the 95% BCa interval for μ from the same bootstrap resampling is $(12.1, 67.1)$.

10.4.3 Robust estimators of regression coefficients

On the same principle as robust estimators of location for contaminated data, various robust estimators have been developed for regression coefficients.

Rank tests and the Theil–Sen estimator. The most traditional robust estimator of regression coefficients using ranks and signs is the Theil–Sen estimator (Theil, 1950; Sen, 1968), defined for the coefficients α, β of the simple regression model

$$y_i = \alpha + \beta x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{i.i.d.}}{\sim} F_\epsilon. \tag{10.40}$$

Theil defined it with ranks; Sen showed it is related to Kendall's tau. From each pair of observations (x_i, y_i), (x_j, y_j) of $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ compute the "slope"

$$\beta_{ij} := \frac{y_i - y_j}{x_i - x_j}, \qquad 1 \le i < j \le n; \tag{10.41}$$

their median

$$\hat{\beta}^{TS} = \text{median}\{\beta_{ij} : 1 \le i < j \le n\} \tag{10.42}$$

is the Theil–Sen estimator of β (pairs with $x_i = x_j$ excluded). This resembles the median of the Walsh averages (from the signed-rank test for one location) or of the Mann–Whitney–Wilcoxon differences used as estimators. The Theil–Sen estimator of α is

$$\hat{\alpha}^{TS} = \text{median}\{y_i - \hat{\beta}^{TS} x_i\}, \tag{10.43}$$

or the median of the Walsh averages of the $y_i - \hat{\beta}^{TS} x_i$ (here simply (10.43)).

To see the relation to Kendall's tau, recall the test with Kendall's tau statistic. If model (10.40) is true and we test $H_0 : \beta = 0$ against $H_1 : \beta \neq 0$, under the null the two variables are independent, so the test with Kendall's τ_n (exact permutation test or large-sample approximation) is possible. With $K_n(\cdot)$ the distribution function of the null distribution of Kendall's tau, $p = 2\min\{1 - K_n(\tau_n), K_n(\tau_n)\}$. How do we test, for a given slope β_0,

$$H_0 : \beta = \beta_0 \quad \text{vs} \quad H_1 : \beta \neq \beta_0? \tag{10.44}$$

Under the null $y_i = \alpha + \beta_0 x_i + \epsilon_i$, so the pairs $(\alpha + \epsilon_i, x_i) = (y_i - \beta_0 x_i, x_i)$ are independent, and Kendall's statistic is $\tau_n(\beta_0) = \sum_{i<j} U_{ij}(\beta_0) / \binom{n}{2}$ with

$$\begin{aligned} U_{ij}(\beta_0) &= \operatorname{sgn}[(x_i - x_j)\{(y_i - \beta_0 x_i) - (y_j - \beta_0 x_j)\}] \\ &= \operatorname{sgn}[(x_i - x_j)\{(y_i - y_j) - \beta_0 (x_i - x_j)\}] \\ &= \operatorname{sgn}[(x_i - x_j)^2 (\beta_{ij} - \beta_0)] \\ &= \operatorname{sgn}(\beta_{ij} - \beta_0), \end{aligned}$$

with β_{ij} as in (10.41). Two facts: first, if $H_0 : \beta = \beta_0$ is true, the null distribution of $\tau_n(\beta_0)$ is again K_n, so the test at a given level is possible. Second, Kendall's statistic is

$$\tau_n(\beta_0) = \frac{1}{\binom{n}{2}} \big[\#\{\beta_{ij} - \beta_0 > 0\} - \#\{\beta_{ij} - \beta_0 < 0\} \big], \tag{10.45}$$

so the result depends on the position of β_0 relative to the pairwise slopes $\{\beta_{ij}\}$. By the same logic as the interval from the order statistics of the Walsh averages for the signed-rank test, the interval for β from Kendall's tau uses the order statistics $\beta_{(1)} < \cdots < \beta_{(N)}$ ($N = \binom{n}{2}$) of the slopes. Under the null the distribution of $\tau_n(\beta_0)$ is always Kendall's tau distribution, so with $Q_K(p; n)$ its p-quantile, the $1 - \alpha$ interval for β is $(\beta_{(l)}, \beta_{(u)})$ with

$$l = \binom{n}{2} \cdot \frac{Q_K(\alpha/2, n) + 1}{2}, \qquad u = \binom{n}{2} \cdot \frac{Q_K(1 - \alpha/2, n) + 1}{2}. \tag{10.46}$$

Justification of (10.46) (supplied; Exercise 10.6). With no ties, $N\tau_n(\beta_0) = \#\{\beta_{ij} > \beta_0\} - \#\{\beta_{ij} < \beta_0\} = 2\#\{\beta_{ij} > \beta_0\} - N$. The level-$\alpha$ test accepts $H_0 : \beta = \beta_0$ iff $Q_K(\frac{\alpha}{2}) \leq \tau_n(\beta_0) \leq Q_K(1 - \frac{\alpha}{2})$ (with the usual care at the discrete boundary), i.e. iff

$$N \frac{1 + Q_K(\frac{\alpha}{2})}{2} \leq \#\{\beta_{ij} > \beta_0\} \leq N \frac{1 + Q_K(1 - \frac{\alpha}{2})}{2}.$$

$\#\{\beta_{ij} > \beta_0\}$ is a decreasing step function of β_0 that drops by one at each $\beta_{(m)}$, so the acceptance set is an interval of consecutive order statistics of the slopes. Its endpoints are the order statistics with indices given by (10.46) (up to the symmetric relabelling $m \leftrightarrow N + 1 - m$, using the symmetry of K_n). Since $\tau_n(\beta) \sim K_n$ at the true β, the coverage is at least $1 - \alpha$. ■

As the level of this interval is lowered, what remains is the median of the pairwise slopes, the Theil–Sen slope estimator $\hat{\beta}^{TS}$ (10.42). The Theil–Sen estimator is intuitive and easy to explain, has a high breakdown point $BP(\hat{\beta}^{TS}) = 29.3\%$, and is known to have about 96% ARE to least squares under normal errors (Wilcox, 2011; Peng et al., 2008).

Example 10.18 (Theil–Sen estimator in simple regression)

For the GDP and happiness data of 13 countries in Example 5.2, the least squares estimates are $\hat{\alpha} = 6.5$, $\hat{\beta} = 0.6$: the relation between GDP and happiness does not look strong. Under normality the usual t-test of $H_0 : \beta = 0$ has $p = 0.311$, so the relation is not significant.

But one data point has an unusually large GDP that seems to distort the overall trend, and the independence test with Kendall's tau (large-sample approximation) gives $p = 0.02$: significant. The Theil–Sen test of the same null obviously gives the same $p = 0.02$. The R code below tests independence by Kendall's tau:

```
> wh = read.csv("data/whappiness.csv")
> x = wh$GDP.per.capita
> y = wh$Score
> cor.test(x,y, method = "k")$p.value
[1] 0.01713551
```

Now compare the least squares line with the Theil–Sen line. The Theil–Sen estimates:

```
> n = length(x)
> pairids = gtools::combinations(n,2)
> pairbetas = data.frame(i = pairids[,1], j = pairids[,2],
+                        betaij = rep(0,nrow(pairids)))
> for (i in 1:nrow(pairids)){
+   betaijs = (y[pairids[i,1]] - y[pairids[i,2]]) /
+     (x[pairids[i,1]] - x[pairids[i,2]])
+   pairbetas[i,3] = betaijs
+ }
> pairbeta_rm = pairbetas %>% filter(betaij != Inf)
> (betaTS = median(pairbeta_rm$betaij))
[1] 2.113208
> (alphaTS = median(y - betaTS * x))
[1] 4.55934
```

Computing the pairwise slopes for all $\binom{13}{2} = 78$ pairs of countries shows one pair with equal GDP (X) values. Excluding that pair, the median of the slopes gives $\hat{\beta}^{TS} = 2.11$, $\hat{\alpha}^{TS} = 4.55$: a much steeper slope than least squares (Figure 10.6 of the book compares the two lines). For reference, `mblm()` of the `mblm` package, `zyp.sen()` of `zyp` and `tsreg()` of `WRS` also compute the Theil–Sen estimates in simple regression:

```
> mblm::mblm(y ~ x, repeated = FALSE)

Call:
mblm::mblm(formula = y ~ x, repeated = FALSE)

Coefficients:
(Intercept)            x
      4.559        2.113
```

Finally the 95% interval for β from (10.46): compute the orders l, u from Kendall's tau distribution, then $(\beta_{(l)}, \beta_{(u)})$.

```
> (l = (1 + SuppDists::qKendall(0.025,n)) * choose(n,2)/2)
[1] 23
> (u = (1 + SuppDists::qKendall(1-0.025,n)) * choose(n,2)/2)
[1] 55
> (TS.CI = sort(pairbeta_rm$betaij)[c(l,u)])
[1] 0.3395639 4.1333333
```

Consistent with the test by Kendall's tau (rejecting $\tau = 0$ or $\beta = 0$), the interval for β does not contain 0.

The Theil–Sen estimator is not easily defined in multiple regression; various different estimation methods have been proposed. One using backfitting is briefly introduced. Assume the multiple

regression model

$$y_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n. \tag{10.47}$$

Backfitting estimates β_j assuming $\hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p$ are already given. Of course none is given at first, so backfitting needs initial values; given them, β_j is estimated sequentially for $j = 0, 1, \ldots, p$, repeatedly.

1. For $j = 1, \ldots, p$ compute the Theil–Sen estimate $\hat\beta_j^{(0)}$ in the simple regression $y_i = \beta_0 + \beta_j x_{ij} + \epsilon_i$ (superscript "(0)" = initial value). The intercept is set to $\hat\beta_0^{(0)} = \text{median}\{y_i - \sum_{j=1}^p \hat\beta_j^{(0)} x_{ij}\}$.
2. For $k = 1, 2, 3, \ldots$, for each $j = 1, \ldots, p$, with response $z_i^{(k)} := y_i - \sum_{l \neq j} \beta_l^{(k-1)} x_{il}$ and explanatory variable x_{ij}, compute the Theil–Sen estimate in the simple regression $z_i^{(k)} = \beta_0 + \beta_j x_{ij} + \epsilon_i$ as $\hat\beta_j^{(k)}$. Finally $\hat\beta_0^{(k)} = \text{median}\{y_i - \sum_{j=1}^p \beta_j^{(k)} x_{ij}\}$.
3. Repeat step 2 until $\hat\beta_j^{(k)} - \hat\beta_j^{(k-1)}$ is very small for all j.

Example 10.19

The Theil–Sen estimator in multiple regression uses the backfitting algorithm above. Here we obtain the Theil–Sen estimates for the multiple regression explaining happiness of the 153 countries of Example 10.7 by 6 explanatory variables. The backfitting algorithm is implemented in `tsreg()` of the R package `WRS`.[a]

```
> h = read.csv("data/worldbank2020.csv")
> dat = h %>% dplyr::select(3:9)
> colnames(dat) = c("Score", "GDP", "Social",
+                   "Life", "Freedom", "Generosity", "Corruption
    ")
>
> ts.out = WRS::tsreg(as.matrix(dat[,2:7]), dat$Score,iter =
    100)
> ts.out$coef
[1] 1.5819695 0.7203707 1.4465492 1.0861433 1.0267905 1.2907304
    1.1652263
```

(`iter = 30` would force 30 iterations of backfitting.) Compared with least squares:

```
> lm(Score ~ ., data = dat)$coefficients
(Intercept)         GDP      Social        Life     Freedom
  1.8870774   0.7380488   1.1529984   0.9821926   1.4832835
 Generosity  Corruption
  0.6202784   0.9719126
```

Not a big difference, but most coefficient estimates went up a little. In the scatter plot matrix of Figure 10.3 several observations look like outliers; the Theil–Sen estimator removed their influence.

[a] The `WRS` package is not on CRAN; install it directly from https://github.com/nicebread/WRS.

Inference on coefficients with the Theil–Sen estimator in multiple regression has no particularly good method, but resampling (bootstrap or shuffling) can be used. The bootstrap, however, artificially produces many tied observations, so the accuracy of the bootstrap sampling distribution is not guaranteed. Permutation tests can be done without this tie problem.

Least trimmed squares estimators. The least squares estimator of the coefficients in the multiple regression model (10.47) is

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \mathbf{x}_i'\boldsymbol{\beta})^2 = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (e_i(\boldsymbol{\beta}))^2. \tag{10.48}$$

Like removing large values at both ends in a trimmed mean, removing the observations with large squared errors $e_i^2(\boldsymbol{\beta})$ in regression and taking the $\boldsymbol{\beta}$ with the smallest sum of squared errors gives the least trimmed squares (LTS) estimator. More precisely, with $e_{(i)}^2(\boldsymbol{\beta})$ the order statistics of the squared errors for each $\boldsymbol{\beta}$, the 10%-LTS estimator is

$$\hat{\boldsymbol{\beta}}^{LTS} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{h} e_{(i)}^2(\boldsymbol{\beta}), \qquad h = \lfloor n - 0.1n \rfloor. \tag{10.49}$$

The LTS estimator with $h = \lfloor n/2 \rfloor + \lfloor (p+1)/2 \rfloor$ trimmed observations is often recommended (Wilcox, 2011).

Several other robust estimators similar to LTS using squared errors have been proposed, e.g. the least median of squares (LMS) estimator minimizing the median of the squared errors. These have high breakdown points but low efficiency and are not used often.

Huber's M-estimator in regression. Replacing the "square" of the errors in the least squares estimator (10.48) of the multiple regression model by Huber's Ψ,

$$\hat{\boldsymbol{\beta}}^{\Psi} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \Psi\Big(\frac{y_i - \mathbf{x}_i'\boldsymbol{\beta}}{\hat{\tau}(\boldsymbol{\beta})}\Big), \tag{10.50}$$

is Huber's M-estimator in regression, where the error standard deviation estimate $\hat{\tau}(\boldsymbol{\beta})$ is

$$\hat{\tau}(\boldsymbol{\beta}) = \text{median}\{|y_i - \mathbf{x}_i'\boldsymbol{\beta}|\} \times 1.4826, \tag{10.51}$$

the MAD assuming the median of the $\{y_i - \mathbf{x}_i'\boldsymbol{\beta}\}$ is 0.

Huber's M-estimator is robust and highly efficient. Its relative efficiency to least squares (under normality) depends on the boundary value k of Ψ; at $k = 1.28$ it is known to be about 90%. The robustness of this M-estimator admits another interpretation: Huber's M-estimator is robust when the response has outliers but not when the explanatory variables do (high-leverage points). To see this, simply let $p = 1$, $\hat{\tau} = 1$; the derivative of $\Psi(y_i - \alpha - \beta x_i)$ in β is

$$\frac{d}{d\beta}\Psi(y_i - \alpha - \beta x_i) = -x_i \psi(y_i - \alpha - \beta x_i). \tag{10.52}$$

Even if y_i goes to infinity this derivative stays bounded, but as x_i grows it diverges. Because $|\psi(x)|$ is bounded, robustness holds, but not with respect to x_i. Yohai's MM-estimator, next, uses Tukey's biweight as $\psi(x)$ to be robust in the explanatory variables too.

Yohai's MM-estimator. Yohai (1987) devised an estimator keeping the merit of M-estimators (high efficiency) while robust to any outliers in explanatory and response variables. First, instead of Huber's function with $\psi(x) = \pm k$ for large x, use Tukey's biweight as $\psi(x)$. Tukey's biweight increases through 0 and then decreases again (Figure 10.4). Such functions are called redescending. In (10.52), as x_i grows the error $|y_i - \alpha - \beta x_i|$ grows and $\psi(y_i - \alpha - \beta x_i)$ becomes 0, so an observation with outlying x_i is removed. Yohai's MM-estimator in multiple regression is the solution of the system

$$\sum_{i=1}^{n} \psi^{\text{biweight}}\Big(\frac{y_i - \mathbf{x}_i'\boldsymbol{\beta}}{\hat{\tau}}\Big) x_{ij} = 0, \qquad j = 0, \ldots, p, \tag{10.53}$$

where $x_{i0} = 1$ and $\hat{\tau}$ is fixed as one of the robust estimators of the error standard deviation. Yohai allowed any high-breakdown estimator as $\hat{\tau}$: the MAD as in Huber's M-estimator, or another robust estimator such as the scale M-estimator or the S-estimator.[60] The MM-estimator also has no analytic solution and must be computed numerically; R's `MASS::rlm()` implements it with the S-estimator as $\hat{\tau}$. Tukey's biweight also needs a boundary value c; $c = 4.685$ is known to give 95% relative efficiency under normal errors, so this value is used (Yu and Yao, 2017). Yohai's MM-estimator, with breakdown point 50%, is one of the most robust estimators.

Huber's M-estimator and Yohai's MM-estimator both apply in multiple regression, but to see their difference clearly we use a simple regression example.

Example 10.20

For the simple regression of happiness on GDP of 13 countries, Huber's M-estimator and Yohai's MM-estimator via `rlm()` of `MASS`:

```
> wh = read.csv("data/whappiness.csv")
> x = wh$GDP.per.capita; y = wh$Score
> MASS::rlm(y ~ x, method = "M")$coefficients
(Intercept)           x
  6.4250015   0.6929371
> MASS::rlm(y ~ x, method = "MM")$coefficients
(Intercept)           x
  6.3118128   0.7783051
```

Compared with Theil–Sen, the slope estimates differ little from least squares: the observation judged an "outlier" is not an outlier from the M-estimator's viewpoint, nor far from the others for the MM-estimator. To see the difference more clearly, adjust that outlier's GDP to a very large 100 and re-estimate:

```
> x[9] = 100
> MASS::rlm(y ~ x, method = "M")$coefficients
 (Intercept)            x
 7.360972591 -0.001222253
> MASS::rlm(y ~ x, method = "MM")$coefficients
(Intercept)           x
   4.649281    2.056968
```

Figure 10.7 of the book compares the fitted lines in both cases with the least squares line.

10.4.4 Inference on coefficients with M-estimators

Standard errors of robust estimators in regression can be expressed only asymptotically, and are valid only for very large samples. But M- or MM-estimates of coefficients are computed by minimizing a weighted sum of squares, so the standard error approximation obtained as a byproduct of the weighted least squares computation can be used as the standard error estimate. The next example compares this approximation with the bootstrap estimate.

Example 10.21

Multiple regression with the data of Example 10.19. First the standard error estimates approximated by weighted least squares:

```
> MASS::rlm(Score ~ ., data = dat, method = "MM") %>% summary()

Call: rlm(formula = Score ~ ., data = dat, method = "MM")
```

[60] The discussion would be long, so robust scale estimators and S-estimators are not treated.

```
Residuals:
     Min       1Q   Median       3Q      Max
-1.81988 -0.33616  0.02929  0.32349  1.36173

Coefficients:
            Value  Std. Error t value
(Intercept) 1.9314 0.2221     8.6961
GDP         0.7363 0.2589     2.8441
Social      1.0480 0.2737     3.8296
Life        0.9553 0.3523     2.7115
Freedom     1.7037 0.4058     4.1979
Generosity  0.5601 0.4980     1.1246
Corruption  1.3417 0.4768     2.8138
```

The standard errors are in the `Std. Error` column. To bootstrap them, $B = 1999$ resamples:

```
> n = nrow(dat)
> set.seed(1)
> B = 1999
> MM.boot = replicate(B, {
+   boot.id = sample(1:n,n,replace = TRUE)
+   dat.boot = dat[boot.id,]
+   MASS::rlm(Score ~ ., data = dat.boot, method = "MM")$
    coefficients}
+ ) %>% t()
> apply(MM.boot,2,sd)
(Intercept)         GDP      Social        Life     Freedom
  0.3819560   0.2476741   0.4911845   0.4050225   0.4693110
 Generosity  Corruption
  0.6501763   0.5714038
```

The bootstrap estimates are more accurate, but roughly similar to the weighted-least-squares approximations.

Both the M- and MM-estimators of coefficients have asymptotically computable standard error estimates, so in principle the approximate-pivot permutation tests and bootstrap-t intervals of Section 10.3 can be used.

Example 10.22 (Bootstrap-t intervals with Yohai's MM-estimator, case bootstrap)

With the MM-estimator $\hat{\beta}_i$ of the multiple regression on the data of Example 10.19, the 95% bootstrap-t interval for β_i is

$$(\hat{\beta}_i - \widehat{\text{se}}(\hat{\beta}_i)Q^*(1-\alpha/2),\ \hat{\beta}_i - \widehat{\text{se}}(\hat{\beta}_i)Q^*(\alpha/2)), \tag{10.54}$$

where Q^* is the quantile function of the bootstrap replicates $(\hat{\beta}_i^* - \hat{\beta}_i)/\widehat{\text{se}}(\hat{\beta}_i^*)$, computed for all $i = 0, \ldots, p$:

```
> MM.out = summary(rlm(Score ~ ., data = dat, method = "MM"))
    [[4]]
> beta.hat = MM.out[,1]
> beta.hat.se = MM.out[,2]
```

These are the MM-estimates $\hat{\beta}_i$ and standard errors $\widehat{\text{se}}(\hat{\beta}_i)$. The code for Q^*:

```
> B = 1999
> set.seed(1)
```

```
> # resampling t
> MM.boot = replicate(B, {
+   boot.id = sample(1:n,n,replace = TRUE)
+   dat.boot = dat[boot.id,]
+   MM.out.boot = summary(rlm(Score ~ ., data = dat.boot, method
    = "MM"))[[4]]
+   (MM.out.boot[,1] - beta.hat)/MM.out.boot[,2]}
+ ) %>% t()
> ci = matrix(NA, nrow = 7, ncol = 2); colnames(ci) = c("2.5%","
    97.5%")
> row.names(ci) = colnames(MM.boot)
> for (i in 1:7){
+   ci[i,] = beta.hat[i] - beta.hat.se[i] * quantile(MM.boot[,i
    ],probs=c(1-0.025,0.025))
+ }
> ci
                 2.5%    97.5%
(Intercept)  1.26065294 2.816903
GDP          0.25383080 1.249724
Social       0.00938614 1.975257
Life         0.07427750 1.761928
Freedom      0.68511127 2.634962
Generosity  -0.82880966 1.785123
Corruption   0.34732311 2.611980
```

Except for the coefficient β_5 of Generosity, all 95% intervals exclude 0.

Example 10.23 (Permutation tests of coefficients with Yohai's MM-estimator, shuffling observations)

With the MM-estimates of the multiple regression on the data of Example 10.19, random permutation tests at level 5% of

$$H_0 : \beta_i = 0 \quad \text{vs} \quad H_1 : \beta_i \neq 0 \tag{10.55}$$

for each coefficient β_i $(i = 0, \ldots, p)$. The code shuffles the observations to approximate the null distribution of $T_i = (\hat{\beta}_i - 0)/\widehat{\text{se}}(\hat{\beta}_i)$ and computes the p-values:

```
> (t.stats = beta.hat / beta.hat.se)
(Intercept)         GDP      Social        Life     Freedom
   8.696273    2.844139    3.829285    2.711510    4.198152
 Generosity  Corruption
   1.124606    2.814131
> set.seed(1)
> M = 1999
> MM.perm = replicate(M,{
+   perm.id = sample(1:n,n, replace = FALSE)
+   dat.perm = dat
+   dat.perm$Score = dat$Score[perm.id]
+   MM.out.perm = summary(rlm(Score ~ ., data = dat.perm, method
    = "MM"))[[4]]
+   (MM.out.perm[,1])/MM.out.perm[,2]}
+ ) %>% t()
> perm.pvalues = vector(length = 7)
> for (i in 1:7){
+   perm.pvalues[i] = mean( abs(c(t.stats[i],MM.perm[,i])) >=
    abs(t.stats[i]))
+ }
```

```
> perm.pvalues
[1] 1.0000 0.0070 0.0010 0.0110 0.0005 0.2480 0.0060
```

Only the hypothesis on β_5 (Generosity) has $p = 0.248$ and is not rejected; all other coefficients differ significantly from 0 at 5%. This agrees with the conclusion of Example 10.22. (The intercept's permutation p-value is 1 because shuffling the response does not alter the intercept test's null behavior in a useful way: shuffling the observations tests $\beta_1 = \cdots = \beta_p = 0$ jointly, not $\beta_0 = 0$, as noted in Section 10.3.1.)

Chapter 11. Smoothing and Density Estimation

Overview

This chapter and Chapter 12 discuss nonparametric function estimation by **smoothing**. So far the parameter to be estimated or tested was a single value (location, scale, a regression coefficient) or a vector (the coefficient vector in multiple regression); the parameter in these chapters is an infinite-dimensional function. The most representative cases of statistical inference where a function is the target are density estimation and nonparametric regression.

11.1 Density estimation and histograms

Given data $X_1, \ldots, X_n$ sampled from a population with density f, estimating f is **density estimation**. To estimate the distribution itself, the empirical distribution F_n estimates the distribution function F, but examining the density f, the derivative of F, is more intuitive. The empirical distribution function F_n is a step function without derivative, so it is hard to estimate the density directly from it.

Various methods have been developed for density estimation. The simplest way to visualize the empirical distribution of data as a density is the histogram, and so far we have visualized distributions by histograms. Turning the histogram into a smooth form (*smoothing* it) to get a density is the basic idea of density estimation.

Example 11.1

From data on the thickness of 485 stamps produced in 1872 in Hidalgo, a state of Mexico (`stamp.csv`), we want to estimate the number of stamp factories in Hidalgo. Under the assumption that the stamps produced by one factory have a fixed thickness, the question reduces to how many peaks (modes) the density f of the thickness of Hidalgo stamps of 1872 has.

For a rough view of the density of the Hidalgo stamp thickness data, a histogram can be used. A histogram divides the range of the data (here $(x_{\min}, x_{\max}) = (0.06, 0.131)$) into intervals and visualizes the number of observations in each, so the picture depends on the size and position of the intervals. Figure 11.1 of the book shows histograms of the same data with interval size h decreasing ($h = 0.015, 0.010, 0.005, 0.002$). How many "peaks" are visible?

The number of peaks seen varies with the interval size, so very different conclusions are possible. The human eye and brain easily ignore very small peaks (i.e. smooth naturally), so small interval sizes are better for exploring data. E.g. with the small $h = 0.002$, the very small peak between $x = 0.07$ and $x = 0.08$ is easily ignored.

Not only the interval size but also the position matters. Figure 11.2 of the book shows histograms with the interval size fixed and only the "starting point" moved slightly left; the shape changes slightly with position. The changes due to position are less marked when the intervals are small.

So if a (normalized to integrate to 1) histogram is used as a density estimator, very different estimates result depending on interval size and position. A good way to ignore the differences due to position is to average histograms, making them smooth (flat). The kernel density estimator introduced next is sometimes called a smoothed histogram.

11.2 The kernel density estimator

Kernel density estimation is the most representative method of density estimation. Before explaining it, first see how "distributions are estimated" in simpler situations:

- **Nonparametric estimation of a probability mass function.** If observations $X_1, \ldots, X_n$ follow a discrete distribution $f(x)$ ($x \in \mathcal{X} = \{0, 1, 2, \ldots, 10\}$), it suffices to examine the probability mass function at the 11 values $x \in \mathcal{X}$; for each $x \in \mathcal{X}$ the value $f(x) = P(X = x)$ can be estimated by
$$\hat{p}(x) = \hat{P}(X = x) = \frac{1}{n}\sum_{i=1}^{n} I(X_i = x). \tag{11.1}$$
This $\hat{p}(x)$ is a nonparametric estimator; since there are only finitely many parameters $\{p_x = p(x) : x \in \mathcal{X}\}$, it is very simple.
- **Nonparametric estimation of the distribution function.** Without a specific parametric model, the empirical distribution function F_n is the natural estimator of the distribution function F. It is a step function increasing by $1/n$ at each observation.

To estimate the density value $f(x)$ at a position x, there are no observations there (for continuous distributions $P(X = x) = 0$), or at most one. So we cannot count observations as in probability mass estimation. But if the density is smooth, increasing or decreasing x by small Δ_x,
$$f(x - \Delta_x) \approx f(x) \approx f(x + \Delta_x),$$
its value changes little. Kernel density estimation estimates $f(x)$ at a given x not only from observations with $X_i = x$ (which may not exist) but from all observations around x, $X_i \in (x - \Delta_x, x + \Delta_x)$. A "kernel" is a function that determines "how" the observations around a given x are used. E.g. the Gaussian kernel[61]
$$K(\Delta_x) = \frac{1}{\sqrt{2\pi}} \exp\Big(-\frac{\Delta_x^2}{2}\Big) \tag{11.2}$$
is designed so that when Δ_x is small (i.e. X_i is close to x) the kernel value is large and has high influence, and when far the kernel value is small and has negligible influence.

For a given kernel $K(\cdot)$ and a given width (*bandwidth*, analogous to the size of a histogram interval) $h > 0$, the kernel density estimator is
$$\hat{f}(x) = \frac{1}{n}\sum_{i=1}^{n} \frac{1}{h} K\Big(\frac{x - X_i}{h}\Big), \qquad x \in (-\infty, \infty). \tag{11.3}$$

As the density is defined for all $x \in (-\infty, \infty)$, so is its estimate, but usually $\hat{f}(x)$ is computed for x in the range of the data.

Example 11.2

To estimate the density of the population from the sodium intake data of Example 2.2 in Chapter 2, compute the kernel density estimate with the Gaussian kernel (11.2) and $h = 500$. The solid curve in Figure 11.3 of the book is the graph of $\hat{f}(x)$ (11.3) for $x \in (500, 7000)$. This density estimate is the sum of the scaled Gaussian kernels
$$\frac{1}{nh} K\Big(\frac{x - X_i}{h}\Big), \qquad i = 1, \ldots, n.$$
In Figure 11.3 the kernel functions (dashed) at $X_i = 3800, 2300$ can be found. Each kernel is

[61] The Gaussian kernel has the form of the standard normal density ("Gaussian distribution" is another name of the normal distribution). The kernel itself is not used as a distribution, so to avoid confusion the term "Gaussian kernel" is used.

highest at $x = X_i$ and decreases as x moves away from X_i. The shape of the kernel function K is chosen, and the width h decides how far away observations are used to estimate $f(x)$. E.g. the influence of $X_i = 2300$ on the estimate $\hat{f}(4000)$ is very small, judged by the very low height at $x = 4000$ of the kernel centered at that observation.

Example 11.3

Kernel density estimates for the Hidalgo stamp data, again with the Gaussian kernel, comparing two widths $h = 0.001$ and 0.004. The simplest R function for the kernel density estimator is `density()`. With the data in the vector `thickness`,

```
density(thickness)
```

computes it. (The width $h \approx 0.004$ was set roughly by the "rule of thumb"; more precisely (11.12) was used; the discussion of bandwidth selection is postponed.) To make the width four times smaller than the default use

```
density(thickness, adjust = 1/4)
```

The book compares the two kernel density estimates with the histogram.

In kernel density estimation, the shape of the estimate changes greatly with the choice of the width h. Large h gives a smooth estimate; small h gives a very rough and spiky one. To explain the choice of h we must first understand the properties of the estimator determined by each value of h. Before that, kernel functions.

11.2.1 Kernels

The examples above used the Gaussian kernel, but other kernel functions can define kernel density estimates. In density estimation, a function $K : (-\infty, \infty) \to (0, \infty)$ satisfying the following is used as a kernel:

1. $K(x) \geq 0$, $\int K(x)\,dx = 1$;
2. $\lim_{|x|\to\infty} K(x) = 0$;
3. $K(x) = K(-x)$.

The first property must be satisfied by every kernel used to estimate a density; because of it the density estimator $\hat{f}(\cdot)$ has $\hat{f}(x) \geq 0$ and

$$\int \hat{f}(x)\,dx = 1,$$

so it is a density.

Proof (supplied; Exercise 11.2). $\hat{f} \geq 0$ since $K \geq 0$. Substituting $u = (x - X_i)/h$, $\int \frac{1}{h} K(\frac{x - X_i}{h}) dx = \int K(u)\,du = 1$ for each i, so $\int \hat{f} = \frac{1}{n}\sum_i 1 = 1$. ■

The second property makes the kernel large at the center (origin $x = 0$) and smaller farther from it; the third means the kernel is symmetric about the origin.[62] There are very many kinds of kernels, but the following are frequently used:

- Gaussian kernel: $K(x) = \frac{1}{\sqrt{2\pi}} \exp(-\frac{x^2}{2})$;

[62] A kernel need not satisfy all three properties. E.g. to estimate the "derivative" of a density, kernels taking negative values ($K(x) < 0$) can be used; depending on the form of the parameter distribution, asymmetric kernels ($K(x) \neq K(-x)$) can also be used.

- Epanechnikov kernel: $K(x) = \frac{3}{4}(1 - x^2)I(|x| \le 1)$;
- Uniform kernel: $K(x) = \frac{1}{2}I(|x| \le 1)$.

Figure 11.4 of the book shows these three kernels. All have the largest value at the center and smaller values away from it. The value or height of the kernel acts as a "weight": in computing $\hat{f}(x)$, observations near x get high weights and distant ones small (or 0) weights. The width h, which decides near and far, affects the estimator far more than the shape of the kernel.[63] So from now on we simply assume the Gaussian kernel.

11.2.2 Is the kernel density estimator a good estimator?

To see whether the kernel density estimator $\hat{f}(\cdot)$ (11.3) is good, we can apply the existing criteria for (numerical, not function-valued) estimators. E.g. for the estimator $\hat{\theta}_n = \bar{X}_n$ of θ with $X_i \overset{\text{i.i.d.}}{\sim} (\theta, \sigma^2)$, we looked at the bias $\text{bias}(\hat{\theta}_n) = E(\hat{\theta}_n) - \theta$ and variance $\text{Var}(\hat{\theta}_n)$ (or standard error $\text{se}(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)}$), and to see both at once the mean squared error $\text{MSE}(\hat{\theta}_n) = E(\hat{\theta}_n - \theta)^2$.[64] For a general estimator $\hat{\theta}_n$ the MSE splits into bias and variance:

$$\text{MSE}(\hat{\theta}_n) = E(\hat{\theta}_n - \theta)^2 = \{\text{bias}(\hat{\theta}_n)\}^2 + \text{Var}(\hat{\theta}_n). \tag{11.4}$$

Proof (supplied; Exercise 11.4). Write $\hat{\theta}_n - \theta = (\hat{\theta}_n - E\hat{\theta}_n) + (E\hat{\theta}_n - \theta)$ and square. The cross term $2(E\hat{\theta}_n - \theta)E(\hat{\theta}_n - E\hat{\theta}_n)$ vanishes, leaving $\text{Var}(\hat{\theta}_n) + \text{bias}(\hat{\theta}_n)^2$ (assuming $E\hat{\theta}_n^2 < \infty$). ■

If both bias and standard error decrease as the sample grows ($n \to \infty$), the MSE also decreases. If $\lim_{n\to\infty} \text{MSE}(\hat{\theta}_n) = 0$, the estimator $\hat{\theta}_n$ is **consistent** for θ (a consistent estimator of θ). E.g. $\hat{\theta}_n = \bar{X}_n$ has bias 0 and variance σ^2/n, so

$$\lim_{n\to\infty} \text{MSE}(\bar{X}_n) = 0^2 + \sigma^2 \lim_{n\to\infty} \frac{1}{n} = 0,$$

and $\bar{X}_n$ is a consistent estimator of $\theta = E(X_1)$.

The kernel density estimator $\hat{f}(x)$ at a given x is an estimator of the density value $f(x)$ at x. So (fixing x) the properties of $\hat{f}(x)$ as an estimator can be seen through how fast the bias $\text{bias}(\hat{f}(x))$, the variance $\text{Var}(\hat{f}(x))$ and $\text{MSE}(\hat{f}(x))$ converge to 0 as the sample grows. To see how well the "function" $\hat{f}(\cdot)$ estimates the function density $f(\cdot)$, "sum" $(\hat{f}(x) - f(x))^2$ over all x: the mean integrated squared error (MISE),

$$\text{MISE}(\hat{f}) = E\Big(\int (\hat{f}(x) - f(x))^2 \, dx \Big).$$

Bias and variance of the kernel density estimator. First, how well $\hat{f}(x)$ estimates the density value $f(x)$ at a given point x: we need the bias and variance of $\hat{f}(x)$, which are not simply expressed. So we derive approximate expressions assuming $n \to \infty$ and simultaneously $h \to 0$. Why is the width h assumed to shrink as the sample grows (as in large-sample approximation)? The kernel density estimator $\hat{f}(\cdot)$ (11.3) depends greatly on the choice of h. With large h, many observations are "included" in computing $\hat{f}(x)$ at a given x; with small h, few. If the sample size n grows, the number of observations included in a kernel of fixed width grows too, so h is designed to shrink as n grows.

First assume the population density $f(x)$ is a smooth function:

[63] When the sample is large enough, the effects of the width h and of the kind of kernel can be compared through Theorems 11.1, 11.2 and 11.3.

[64] The mean squared error is the expectation (risk) of the loss function $L(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$: $\text{MSE}(\hat{\theta}) = R(\hat{\theta}, \theta) = E\{L(\hat{\theta}, \theta)\}$. Papers and books often say "risk" instead of mean squared error.

Assumption 1. The second derivative f'' of the density is absolutely continuous, and the third derivative satisfies $\int (f'''(x))^2\,dx < \infty$.

Under Assumption 1, near a point x the density f can be approximated by a quadratic function. By Taylor expansion, for t close to x,

$$f(t) = f(x) + f'(x)(t-x) + \frac{f''(x)}{2}(t-x)^2 + R(t-x),$$

where the last term $R(t-x) = o((t-x)^2)$, i.e. $\lim_{t\to x} R(t-x)/(t-x)^2 = 0$.[65] Writing t near x as $t = x - hu$,

$$f(x-hu) = f(x) - hf'(x)u + \frac{h^2 f''(x)}{2}u^2 + o((hu)^2). \tag{11.5}$$

Theorem 11.1

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} f$, with the population density f satisfying Assumption 1. At a point x with $f(x) > 0$, the bias and variance of the kernel density estimator $\hat{f}(x)$, as $n \to \infty$, $h \to 0$, are

$$\operatorname{bias}(\hat{f}(x)) = \frac{f''(x)\sigma_K^2}{2}h^2 + o(h^2), \tag{11.6}$$

$$\operatorname{Var}(\hat{f}(x)) = \frac{1}{nh}f(x)\int K^2(u)\,du + o\Big(\frac{1}{nh}\Big). \tag{11.7}$$

The bias is large when f is very complicated, i.e. when $|f''(x)|$ is large. To reduce the bias one can make the width h small, but then the variance becomes large. Conversely, enlarging h to make the estimator flatter reduces the variance but increases the bias. This phenomenon, the **bias–variance trade-off**, occurs not only in density estimation but commonly in many estimation problems. Remember that, as in (11.4), the squared bias plus the variance is the MSE; Figure 11.5 of the book illustrates the trade-off.

Proof of Theorem 11.1. First the bias:

$$\operatorname{bias}(\hat{f}(x)) = E(\hat{f}(x)) - f(x) = \frac{1}{n}\sum_{i=1}^{n} E\Big[\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big] - f(x). \tag{11.8}$$

Regarding the kernel as a density, its first and second moments (mean and variance) can be computed. By the third property of kernels, "symmetry,"

$$\mu_K = \int xK(x)\,dx = 0, \qquad \sigma_K^2 = \int (x-0)^2 K(x)\,dx.$$

Using this and the approximation (11.5) of the density,

$$\begin{aligned} E\Big[\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big] &= \int \frac{1}{h}K\Big(\frac{x-t}{h}\Big)f(t)\,dt = \int K(u)f(x-hu)\,du \\ &= \int K(u)\Big[f(x) - hf'(x)u + \frac{h^2 f''(x)}{2}u^2 + o((hu)^2)\Big]du \\ &= f(x) + \frac{f''(x)\sigma_K^2}{2}h^2 + o(h^2), \end{aligned}$$

and substituting into (11.8) gives (11.6).[66]

[65] A sequence a_n is $a_n = o(n^{-1})$ ("little o") if $\lim_{n\to\infty} a_n/n^{-1} = 0$, i.e. a_n converges to 0 faster than n^{-1}. More generally, functions g, h satisfy $g(x) = o(h(x))$ as $x \to \infty$ if $\lim_{x\to\infty} g(x)/h(x) = 0$.

[66] Integrating "little o" or "big O" does not always give the same term again: $\int o(h^2u^2)\,du = o(h^2)$ is not always true. To be exact, the error of the second-order approximation of $f(x)$ must be expressed exactly with Taylor's theorem, and the order of lim and $\int$ exchanged using the dominated convergence theorem or similar. (Supplied: under Assumption 1, f'' is bounded and uniformly continuous on compact sets, and with $\int u^2K(u)\,du < \infty$ the remainder $\int K(u)[f(x-hu) - f(x) + huf'(x) - \frac{h^2u^2}{2}f''(x)]du$ is $h^2\int K(u)u^2\,\epsilon(hu)\,du$ with $\epsilon(s) \to 0$ and bounded; dominated convergence gives $o(h^2)$.)

The variance is computed similarly. For simplicity use "big O" notation for $n \to \infty$, $h \to 0$; e.g. $h^2 + o(h^2) = O(h^2)$, $\frac{1+h+h^2}{n} = O(\frac{1}{n})$.

$$\begin{aligned}\operatorname{Var}(\hat{f}(x)) &= \frac{1}{n}\operatorname{Var}\Big[\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big] \\ &= \frac{1}{n}\Big[E\Big\{\frac{1}{h}K\Big(\frac{x-X_i}{h}\Big)\Big\}^2 - f^2(x) + O(h^2)\Big] \\ &= \frac{1}{nh}f(x)\int K^2(u)\,du + O\Big(\frac{1}{n}\Big).\end{aligned}$$

(Supplied detail: $E\{\frac{1}{h^2}K^2(\frac{x-X_i}{h})\} = \frac{1}{h}\int K^2(u)f(x-hu)\,du = \frac{1}{h}f(x)\int K^2 + O(1)$, and the squared mean is $f(x)^2 + O(h^2)$.) Since $h \to 0$, $O(\frac{1}{n})$ is $o(\frac{1}{nh})$, which gives (11.7). ■

To reduce the MSE both bias and variance must shrink: as the sample grows ($n \to \infty$), the width must shrink ($h \to 0$), but not faster than the sample grows ($nh \to \infty$).

Theorem 11.2

When $n \to \infty$, $h \to 0$, $nh \to \infty$, the mean squared error and mean integrated squared error of the kernel density estimator $\hat{f}(x)$ are

$$\begin{aligned}\operatorname{MSE}(\hat{f}(x)) &= \frac{(f''(x)\sigma_K^2)^2}{4}h^4 + \frac{f(x)}{nh}\int K^2(u)\,du + o(h^4) + o\Big(\frac{1}{nh}\Big), \\ \operatorname{MISE}(\hat{f}) &= \frac{\sigma_K^4\int(f''(x))^2\,dx}{4}h^4 + \frac{1}{nh}\int K^2(u)\,du + o(h^4) + o\Big(\frac{1}{nh}\Big).\end{aligned} \tag{11.9}$$

Proof (supplied). The first line is (11.4) with Theorem 11.1. For the second, $\operatorname{MISE}(\hat{f}) = \int \operatorname{MSE}(\hat{f}(x))\,dx$ by Fubini; integrating the leading terms uses $\int f = 1$, and the remainders integrate to $o(h^4) + o(\frac{1}{nh})$ under Assumption 1 (by the same dominated-convergence argument applied uniformly in x). ■

By Theorem 11.2, when all three conditions $n \to \infty$, $h \to 0$, $nh \to \infty$ hold, $\operatorname{MSE}(\hat{f}(x)) \to 0$, so $\hat{f}(x)$ is a consistent estimator of $f(x)$; also $\operatorname{MISE}(\hat{f}) \to 0$, so the kernel density estimator $\hat{f}$ is a consistent estimator of the density f.

11.2.3 Choosing the bandwidth

In the expression (11.9) of the MISE, given the kernel K, the density f to be estimated and the sample size n, the value of h minimizing $\operatorname{MISE}(\hat{f})$ can be found. Ignoring the two asymptotically negligible terms $o(h^4) + o(\frac{1}{nh})$ and regarding $\operatorname{MISE}(\hat{f})$ as a function of h, the point where its derivative is 0 (the value of h where the MSE in Figure 11.5 is smallest) is

$$\frac{d}{dh}\operatorname{MISE}(\hat{f}) = 0 \iff h^* = \Big(\frac{\int K^2(u)\,du}{\sigma_K^4\int(f''(x))^2\,dx}\cdot\frac{1}{n}\Big)^{\frac{1}{5}}.$$

(Supplied: $\frac{d}{dh}[\frac{A}{4}h^4 + \frac{B}{nh}] = Ah^3 - \frac{B}{nh^2} = 0$ gives $h^5 = \frac{B}{An}$, with $A = \sigma_K^4\int(f'')^2$ and $B = \int K^2$; the second derivative is positive, so this is the minimum.) This h^* is the asymptotically optimal bandwidth. This theoretical width depends on the kernel K and the second derivative f'' of the population density, but as the sample size n grows it shrinks at roughly the rate $h^* \approx n^{-1/5}$.[67] If K is the Gaussian kernel,

$$\int K^2(u)\,du = \frac{1}{2\sqrt{\pi}}, \qquad \sigma_K^2 = 1,$$

so

$$h^* = \Big(\frac{1}{2\sqrt{\pi}\int(f''(x))^2\,dx}\cdot\frac{1}{n}\Big)^{\frac{1}{5}}, \tag{11.10}$$

[67] At $n = 100$, $n^{-1/5} = 0.39$; at $n = 10000$, $n^{-1/5} = 0.16$: the width shrinks much more slowly than the sample size grows.

but this value can be known only if the unknown parameter f is known. Various methods have been proposed depending on how $\int (f''(x))^2\,dx$ is estimated; three representative ones are introduced here.

Normal reference rule. If $f(x) = \frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$, i.e. the population is normal $N(\mu, \sigma)$, it is known that

$$\int (f''(x))^2\,dx = \frac{3}{8\sqrt{\pi}\sigma^5}. \tag{11.11}$$

Proof of (11.11) (supplied). For $\sigma = 1$, $\phi''(x) = (x^2-1)\phi(x)$ and $\phi(x)^2 = \frac{1}{2\sqrt{\pi}} \cdot \frac{1}{\sqrt{2\pi}\cdot 2^{-1/2}} e^{-x^2}$ is $\frac{1}{2\sqrt{\pi}}$ times the $N(0, \frac{1}{2})$ density. So $\int(\phi'')^2 = \frac{1}{2\sqrt{\pi}} E(Z^2-1)^2$ with $Z \sim N(0, \frac{1}{2})$: $EZ^4 - 2EZ^2 + 1 = \frac{3}{4} - 1 + 1 = \frac{3}{4}$, giving $\frac{3}{8\sqrt{\pi}}$. For general σ, $f''(x) = \sigma^{-3}\phi''((x-\mu)/\sigma)$, so $\int(f'')^2 = \sigma^{-5}\int(\phi'')^2$. ■

Substituting (11.11) into (11.10) gives $h^* = (4/3)^{1/5} n^{-1/5}\sigma$, and replacing the population variance σ^2 by its estimate gives Scott's (Scott, 2015) normal reference rule (nrd):

$$h_n^{\text{nrd}} = 1.06 n^{-\frac{1}{5}}\hat{\sigma}, \qquad \hat{\sigma} = \min\Big\{s, \frac{\text{IQR}}{1.34}\Big\},$$

where the estimate $\hat{\sigma}$ of the population standard deviation is the smaller of the sample standard deviation s and the robust estimator IQR/1.34. The IQR is the distance between the 75% quantile $Q_n(3/4)$ and 25% quantile $Q_n(1/4)$ of the data, IQR $= Q_n(3/4) - Q_n(1/4)$. If the population is $N(\mu, \sigma)$, IQR/1.34 is a consistent estimator of σ. (Supplied: the IQR of $N(\mu, \sigma^2)$ is $2 \cdot 0.6745\sigma = 1.349\sigma$.)

"Rule of thumb." For populations not following a normal distribution, h_n^{nrd} from the normal reference rule tends to be a little too large and is generally inefficient, so the width is adjusted smaller:

$$h_n^{\text{nrd0}} = 0.9 n^{-\frac{1}{5}}\hat{\sigma}, \qquad \hat{\sigma} = \min\Big\{s, \frac{\text{IQR}}{1.34}\Big\}. \tag{11.12}$$

This is Silverman's (Silverman, 1986) rule-of-thumb method. Silverman's rule of thumb is fast and easy to compute and the most used, but it also assumes a density f similar to the normal.

Plug-in estimation. This method does not assume the form of the density f in advance but directly estimates the unknown parameter $A(f) \equiv \int (f''(x))^2\,dx$. Sheather and Jones (1991) proposed estimating $A(f)$ directly using another kernel density estimate and plugging it in. With this estimate $\hat{A}(f)$,[68] the Sheather–Jones plug-in estimator is

$$h_n^{\text{SJ}} = \Big(\frac{1}{2\sqrt{\pi}\hat{A}(f)} \cdot \frac{1}{n}\Big)^{\frac{1}{5}}.$$

Besides estimating and plugging in the unknown $A(f)$, there are various other bandwidth selection methods, e.g. using cross-validation. Silverman's rule of thumb is easy to understand and fast, so in most cases it is the first choice. Generally the Sheather–Jones plug-in estimate is recommended as the width h of kernel density estimation. In R, the normal reference width h_n^{nrd} is computed by `bw.nrd()`, the rule-of-thumb width h_n^{nrd0} by `bw.nrd0()`, and the plug-in width h_n^{SJ} by `bw.SJ()`. The width can also be specified inside `density()`, which computes density estimates. The next example compares the three methods.

[68] The Sheather–Jones estimate $\hat{A}(f)$ is expressed by an integral of another kernel estimate of the rth derivative $f^{(r)}$ of f ($r = 1, 2, 3, 4$). Estimating derivatives of a density is harder than estimating the density, and those estimates are more variable.

Example 11.4

Kernel density estimates for the Hidalgo stamp data with the three widths above. As in the code below, specify the argument `bw = "nrd0"`; the results are in Figure 11.6 of the book. The density of these data has several peaks and differs greatly from a normal density, so the Sheather–Jones plug-in width $h_n^{\mathrm{SJ}} = 0.0012$ seems the most appropriate.

```
> bw.SJ(thickness)
[1] 0.001205108
> kde.nrd = density(thickness, bw = "nrd")
> kde.nrd0 = density(thickness, bw = "nrd0") # default bw = "
    nrd0"
> kde.SJ = density(thickness, bw = "SJ")
```

11.3 Multivariate density estimation

A multivariate or joint density f is a function expressing the joint distribution of p continuous random variables $X_1, \ldots, X_p$ in the form

$$P((X_1, \ldots, X_p) \in A) = \int_A f(x_1, \ldots, x_p)\, dx_1 \cdots dx_p.$$

Given vector observations $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{i.i.d.}}{\sim} f$ (here $\mathbf{X}_i = (X_{i1}, \ldots, X_{ip})'$), we want to estimate the joint density f from them. The multivariate kernel density estimator is defined by directly extending the univariate one.

As in one dimension, to estimate $f(\mathbf{x})$ at a given $\mathbf{x} = (x_1, \ldots, x_p)'$ we use observations around $\mathbf{x}$. This neighborhood is part of p-dimensional space, so the univariate kernel is extended to a multivariate kernel. The most representative Gaussian kernel has the form of the multivariate standard normal density:

$$K(\mathbf{x}) = (2\pi)^{-\frac{p}{2}} \exp\big(-\tfrac{1}{2}\mathbf{x}'\mathbf{x}\big).$$

The multivariate kernel is highest at the origin $\mathbf{0}$ and symmetric about it, and its size can be changed by a bandwidth matrix H. E.g. from the multivariate Gaussian kernel K, defining

$$K_H(\mathbf{x}) := |H|^{-1/2} K(H^{-1/2}\mathbf{x})$$

gives the joint density of the normal $N_p(\mathbf{0}, H)$. To see the effect of the bandwidth matrix H (with the Gaussian kernel, the covariance matrix of the normal) in a multivariate kernel, Figure 11.7 of the book shows the Gaussian kernel for $p = 2$ with different H:

$$H_a = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad H_b = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad H_c = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad H_d = \begin{pmatrix} 1 & -.8 \\ -.8 & 1 \end{pmatrix}.$$

E.g. with the kernel bandwidth matrix H_b, more smoothing is done in the direction of the first variable x_1 and less in the direction of x_2.

Besides the Gaussian kernel, other kernels can be used; e.g. the uniform kernel generalizes to the box-shaped kernel

$$K^{\mathrm{Unif}}(\mathbf{x}) = K^{\mathrm{Unif}}(x_1, \ldots, x_p) = \tfrac{1}{2} I(|x_1| \le 1) \times \cdots \times \tfrac{1}{2} I(|x_p| \le 1).$$

Given the kernel K and bandwidth matrix H, the multivariate kernel density estimator from the sample $\mathbf{X}_1, \ldots, \mathbf{X}_n$ is

$$\hat{f}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^{n} K_H(\mathbf{x} - \mathbf{X}_i). \tag{11.13}$$

11.3.1 Multivariate density estimation with the R package `ks`

To do multivariate density estimation with the kernel estimator (11.13), the kernel bandwidth matrix H must be set. This section shows an example of kernel density estimation with the R package `ks`, which computes the plug-in estimate of the optimal bandwidth matrix[69] and provides several auxiliary visualization tools. (The same `ks` package can of course also estimate univariate densities.)

Example 11.5

Using data on the size of several penguin species in the Palmer Archipelago of Antarctica (`penguins.csv`), estimate the joint density of two variables: bill length and flipper length. The scatter plot of the two variables, colored by the three species surveyed (Adelie, Chinstrap, Gentoo), is in Figure 11.8 of the book.

Ignoring species information, we estimate the density of bill and flipper length for all penguins. First, `ks::Hpi()` gives the plug-in estimate $\hat{H}$ of the bandwidth matrix:

```
> penguins = read.csv("data/penguins.csv")
> penguins.b = penguins %>%
+   select(bill_length_mm, flipper_length_mm)
> library(ks)
> H <- ks::Hpi(penguins.b)
> H
          [,1]      [,2]
[1,] 2.618165  3.454826
[2,] 3.454826 13.736760
```

The bandwidth matrix $\hat{H}$ roughly matches the dispersion (covariance matrix) of all the data. The kernel density estimate (11.13) with this bandwidth matrix is computed by `ks::kde()`; its result can be visualized with the `plot()` function redefined in the `ks` package. Two examples of visualization are in Figure 11.9 of the book.

```
kde.p <- ks::kde(x = penguins.b, H = H)
# contour of density estimate.
plot(kde.p, display = "filled.contour", cont = seq(5, 95, by = 10),
     xlab = "bill length", ylab = "flipper length")
points(kde.p$x, pch = 1, cex = 0.5)
# 3D plot
plot(kde.p, display = "persp", xlab = "bill_length", ylab = "flipper
    length")
```

11.3.2 Multivariate density estimation and the curse of dimensionality

Smoothing is a very powerful way to estimate parameters (here densities) in parameter spaces of uncountably high dimension, and the key to its success is that as the kernel width $h \to 0$, the number of observations included in the kernel K, $nh \to \infty$, does not decrease but grows. The problem that commonly arises when applying smoothing to multivariate data is precisely the **curse of dimensionality**.

Why it arises, by a simple example. First suppose univariate observations from Uniform$(0, 1)$. With the uniform kernel of width 0.05, the number of observations in the kernel is on average $0.1 \times n = n/10$. If the dimension increases by one and bivariate observations come from the

[69] This book does not derive the optimal bandwidth matrix in multivariate density estimation. The bandwidth matrix H^* minimizing asymptotically the MSE of Theorem 11.3 is the optimal one, and as in Section 11.2.3 the plug-in estimator can be derived by similar logic (Wand and Jones, 1994).

uniform distribution on $(0,1)\times(0,1)$, the average number of observations in the uniform kernel with bandwidth matrix $H=(0.05)^2 I$ is $(0.1)^2 n = n/100$. If 10-dimensional observations come from the uniform distribution on $(0,1)^p$ ($p=10$), the average number of observations in the uniform kernel with $H=(0.05)^2 I$ is

$$(0.1)^p n = \frac{n}{10000000000},$$

so ten billion observations are needed for the computation of a 10-dimensional kernel density estimate $\hat{f}(\mathbf{x})$ (11.13) to include on average one observation. Nonparametric estimation of multivariate data needs a very large number of observations. This is the statistical curse of dimensionality.[70]

Consider the influence of the curse of dimensionality on multivariate density estimation. Suppose the ith diagonal element of the bandwidth matrix H is h_i^2 and all other elements are 0 (this restriction is often used in practice):

$$H = \begin{pmatrix} h_1^2 & 0 & \cdots & 0 \\ 0 & h_2^2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & h_p^2 \end{pmatrix},$$

and

$$K_H(\mathbf{x}) = \frac{1}{h_1} K\Big(\frac{x_1}{h_1}\Big)\cdots\frac{1}{h_p} K\Big(\frac{x_p}{h_p}\Big), \tag{11.14}$$

where $K(x)$ on the right is the kernel defined in one dimension, or the multivariate kernel with $p=1$. The influence of the dimension p on the MSE of the multivariate kernel density estimator at each point $\mathbf{x}$:

Theorem 11.3 (García-Portugués (2021), Theorem 3.2)

With a kernel satisfying (11.14), the MSE of $\hat{f}(\mathbf{x})$ is

$$\mathrm{MSE}(\hat{f}(\mathbf{x})) = \Big[\frac{\sigma_K^2}{2}\sum_{j=1}^{p} h_j^2 A_j(f)\Big]^2 + \frac{(\int K^2(u)\,du)^p}{nh_1\cdots h_p} + o\Big(\sum_j h_j^4\Big) + o\Big(\frac{1}{nh_1\cdots h_p}\Big), \tag{11.15}$$

where $A_j(f) = \frac{\partial^2}{\partial x_j \partial x_j} f(\mathbf{x})$ (the book writes $\int(\frac{\partial^2}{\partial x_j \partial x_j} f(\mathbf{x}))^2 d\mathbf{x}$, which belongs to the integrated version), and $K(u)$ is the one-dimensional kernel of (11.14).

Sketch (supplied). As in Theorem 11.1, $E\hat{f}(\mathbf{x}) = \int \prod_j K(u_j) f(\mathbf{x} - (h_1u_1,\dots,h_pu_p)')\,d\mathbf{u}$. A second-order Taylor expansion in p variables, with $\int u_j K(u_j) = 0$, $\int u_j^2 K(u_j) = \sigma_K^2$ and the cross terms $u_j u_k$ ($j\neq k$) integrating to 0 by independence of coordinates under the product kernel, gives bias $\frac{\sigma_K^2}{2}\sum_j h_j^2 \partial_{jj} f(\mathbf{x}) + o(\sum_j h_j^2)$. The variance is $\frac{1}{n}\mathrm{Var}\, K_H(\mathbf{x}-\mathbf{X}_1) = \frac{f(\mathbf{x})}{nh_1\cdots h_p}(\int K^2)^p + O(\frac{1}{n})$, since $\int K_H^2 = (h_1\cdots h_p)^{-1}(\int K^2)^p$. (The variance term carries the factor $f(\mathbf{x})$, omitted in the book's display.) Squaring the bias and adding the variance gives (11.15). ■

For simplicity take all widths equal ($h_1=\cdots=h_p=h$). With the dimension p fixed, the optimal width minimizing $\mathrm{MSE}(\hat{f}(\mathbf{x}))$ is $h^* = O(n^{-1/(4+p)})$. Substituting it back into (11.15),

$$\mathrm{MSE}(\hat{f}(\mathbf{x})) = O(n^{-\frac{4}{4+p}}),$$

so when the dimension p is high the MSE decreases to 0 very slowly. (Supplied: balancing h^4 against $\frac{1}{nh^p}$ gives $h^{4+p} \asymp \frac{1}{n}$, and then both terms are $\asymp n^{-4/(4+p)}$.) That is, when the dimension is high, the sample size needed to ensure the accuracy of multivariate density estimation is very large. Density estimation for one- or two-dimensional observations is often used, but for three or more dimensions its practicality drops greatly.

[70] The curse of dimensionality appears in various forms in multivariate statistical analysis. Wasserman (2006) called "the statistical curse of dimensionality" situations where the required sample size grows geometrically with the dimension, or where the sample looks large but the dimension is so high that estimates are very uncertain.

11.4 Confidence intervals for the density

So far we have looked at density estimation and its asymptotic properties; in particular, with suitable choice of kernel and width, density estimation is asymptotically accurate. Then, when the sample is not infinitely large, what statistical inference is possible about the uncertainty of the estimated density? This question can be answered by giving confidence intervals for the estimated density. Here we look at intervals for the univariate kernel density estimator.

The density f has a value $f(x)$ at each point x. A $1-\alpha$ confidence interval for the density value $f(x)$ is an interval $C_{1-\alpha}(x)$ computable from the data such that the probability that the actual density value $f(x)$ lies in $C_{1-\alpha}(x)$ is at least $1-\alpha$. The parameter f is a function with values at infinitely many x, so two definitions are used:

- **Pointwise confidence interval:** for a given position x, the interval $C_{1-\alpha}(x)$ satisfies
$$P(f(x) \in C_{1-\alpha}(x)) \geq 1-\alpha. \tag{11.16}$$
- **Confidence band:** the intervals $C_{1-\alpha}(x)$ satisfy
$$P(f(x) \in C_{1-\alpha}(x),\ \forall x) \geq 1-\alpha. \tag{11.17}$$

The pointwise interval guarantees coverage locally at a given position x; the band guarantees coverage globally, for all x or the whole function $f(\cdot)$. In nonparametric models with few restrictions on the form of f an exact interval for the density is nearly impossible, so asymptotically exact intervals are sought. Also, every kernel density estimator $\hat{f}(x)$ has bias. In univariate density estimation, e.g., even with the optimal width $h^* = O(n^{-1/5})$ the bias is $O(n^{-2/5})$, and the standard error of the estimator is also $O(n^{-2/5})$, a non-negligible size. Methods for bias-adjusted (removed) intervals have been developed,[71] but they exceed the level of this book; here we estimate not the original density $f(x)$ but the smoothly mollified density

$$f_h(x) = E(\hat{f}(x)),$$

where h is the width of the kernel density estimator. This width is generally estimated from the data, but once set it is treated as given.

Estimating the mollified density is actually much used, especially effective when the density f has a very wiggly shape, for grasping the shape and topological features of the distribution (Figure 11.10 of the book).

11.4.1 Pointwise confidence intervals

The kernel density estimator $\hat{f}$ has the form of a mean of about $2nh$ observations included in the kernel (exactly a mean with the uniform kernel, a weighted mean with the Gaussian kernel). If $nh \to \infty$, by the CLT

$$\sqrt{nh}(\hat{f}(x) - f_h(x)) \dot{\sim} N(0, \sigma_f^2(x)), \tag{11.18}$$

where $\sigma_f^2(x) = f(x)a_K$, $a_K = \int K^2(u)\,du$. With the Gaussian kernel $a_K = 1/(2\sqrt{\pi})$, with the uniform kernel $a_K = 1/4$. The large-sample interval for the mollified density value $f_h(x)$ at x is obtained from (11.18); the unknown $\sigma_f^2(x)$ or $f(x)$ is replaced by its estimate.

Plug-in large-sample interval. Substituting $\hat{f}(x)$ for $f(x)$,

$$f_h(x) \in \hat{f}(x) \pm z_{\alpha/2}\sqrt{\frac{\hat{f}(x)a_K}{nh}}$$

is an asymptotically exact pointwise interval.

[71] Bias-removed intervals are most easily introduced in Chen (2017).

Example 11.6

For the Hidalgo stamp data, using the kernel density estimate $\hat{f}(x)$ with width $h = h_n^{\mathrm{SJ}} = 0.0012$, compute plug-in large-sample pointwise intervals at each point x of $f(x)$. The code below uses `ks::kde()` for the estimate `ks.out$estimate`, then the plug-in large-sample intervals. `ks.out$estimate` is the vector of $\hat{f}(x_i)$ at the points $x_1, \ldots, x_M$ defined in `ks.out$eval.points`, so `ci.lower` and `ci.upper` below are vectors of the same length.

```
> h = bw.SJ(thickness)
> n = length(thickness)
> ks.out = ks::kde(thickness, h = h)
> f.hat = ks.out$estimate
> ci.lower = f.hat - qnorm(1-0.025) * sqrt( f.hat / (2*sqrt(pi))
    / (n*h))
> ci.upper = f.hat + qnorm(1-0.025) * sqrt( f.hat / (2*sqrt(pi))
    / (n*h))
> data.frame(x = ks.out$eval.points, ci.low = ci.lower, ci.up =
    ci.upper) %>% head()
          x      ci.low      ci.up
1 0.05554110 -0.03660399 0.03810936
2 0.05574090 -0.04897191 0.05170534
3 0.05594069 -0.06449555 0.06932493
4 0.05614048 -0.08357777 0.09187994
5 0.05634028 -0.10651872 0.12040574
6 0.05654007 -0.13344939 0.15605133
```

This kernel density estimate and pointwise 95% intervals are shown in Figure 11.11 of the book.

Bootstrap plug-in interval. Estimate $\sigma_f^2(x)$ by the bootstrap. Rewriting (11.18),

$$\hat{f}(x) \dot{\sim} N(f_h(x), \mathrm{se}^2(\hat{f}(x))),$$

so compute the nonparametric bootstrap estimate of $\mathrm{se}(\hat{f}(x))$. Given a grid $\mathcal{X} = \{x_1, \ldots, x_M\}$ finely dividing the interval containing the observations, the statistics $\hat{f}(x)$ at $x \in \mathcal{X}$ are all estimates from one sample $X_1, \ldots, X_n$, so bootstrap resampling can be done once for all $x_i \in \mathcal{X}$. Generate for each x_i the bootstrap replicates

$$\hat{f}^{*1}(x_i), \ldots, \hat{f}^{*B}(x_i), \tag{11.19}$$

take their sample standard deviation as the bootstrap estimate of the standard error,

$$\widehat{\mathrm{se}}^{\mathrm{Boot}}(\hat{f}(x)) = \Big(\frac{\sum_{b=1}^{B} (\hat{f}^{*b}(x_i) - \overline{\hat{f}^{*}(x_i)})^2}{B-1} \Big)^{1/2},$$

and the $1-\alpha$ bootstrap plug-in interval is

$$f_h(x) \in \hat{f}(x) \pm z_{\alpha/2} \widehat{\mathrm{se}}^{\mathrm{Boot}}(\hat{f}(x)).$$

Example 11.7

With the data of Example 11.6, the bootstrap plug-in interval:

```
x = ks.out$eval.points
kde0 = ks.out$estimate
n = length(thickness)

B = 1999
```

```
kde.boot = replicate(B,{
          ks::kde(sample(thickness,n, replace = TRUE),
          h = h,           # same bandwidth h
          eval.points = x # same points x
          )$estimate       # compute kde(x)
})
se.fhat = apply(kde.boot,1,sd)
ci.lower = f.hat - qnorm(1-0.025) * se.fhat
ci.upper = f.hat + qnorm(1-0.025) * se.fhat
```

The result is shown in Figure 11.11 of the book.

Most of the bootstrap intervals of Chapter 8 apply, but the number of points to examine ($M = \#\mathcal{X}$) is large, so the computation grows. The simplest, the basic bootstrap interval, is often used.

Basic bootstrap interval. From the bootstrap replicates (11.19) at each point x, compute the $1-\frac{\alpha}{2}$ and $\frac{\alpha}{2}$ quantiles $Q_{\hat{f}^*}(1-\frac{\alpha}{2};x)$, $Q_{\hat{f}^*}(\frac{\alpha}{2};x)$; the $1-\alpha$ basic bootstrap interval is

$$f_h(x) \in \big(2\hat{f}(x) - Q_{\hat{f}^*}(1-\tfrac{\alpha}{2};x),\ 2\hat{f}(x) - Q_{\hat{f}^*}(\tfrac{\alpha}{2};x)\big).$$

It approximates the distribution of $\hat{f}(x) - f_h(x)$ by the bootstrap $\hat{f}^*(x) - \hat{f}(x)$.

Absolute-deviation bootstrap interval. If the distribution of the deviation $\hat{f}(x) - f_h(x)$ can be assumed symmetric (asymptotically true), the distribution of the absolute deviation $|\hat{f}(x) - f_h(x)|$ can be bootstrap-approximated. The bootstrap replicates are

$$\Delta^{*1}(x) = |\hat{f}^{*1}(x) - \hat{f}(x)|,\ \ldots,\ \Delta^{*B}(x) = |\hat{f}^{*B}(x) - \hat{f}(x)|,$$

and with their $1-\alpha$ quantile $Q_{\Delta^*}(1-\alpha;x)$, the $1-\alpha$ absolute-deviation bootstrap interval is

$$f_h(x) \in \big(\hat{f}(x) - Q_{\Delta^*}(1-\alpha;x),\ \hat{f}(x) + Q_{\Delta^*}(1-\alpha;x)\big).$$

(Note the $1-\alpha$ quantile is used, not $1-\frac{\alpha}{2}$.) All intervals introduced so far are asymptotically exact; e.g. the absolute-deviation bootstrap interval, as $n \to \infty$, $h \to 0$, $nh \to \infty$, $B \to \infty$, satisfies

$$P\big[f_h(x) \in \big(\hat{f}(x) - Q_{\Delta^*}(1-\alpha;x),\ \hat{f}(x) + Q_{\Delta^*}(1-\alpha;x)\big)\big] = 1 - \alpha + o(1).$$

Why the $1-\alpha$ quantile (supplied; Exercise 11.6). If $|\hat{f}(x) - f_h(x)| \sim G$ with G continuous and invertible, then $P(|\hat{f}(x) - f_h(x)| \le G^{-1}(1-\alpha)) = 1-\alpha$, i.e. $P(f_h(x) \in \hat{f}(x) \pm G^{-1}(1-\alpha)) = 1-\alpha$. The absolute value already folds both tails, so the $1-\alpha$ (not $1-\frac{\alpha}{2}$) quantile is used. If the bootstrap distribution G^* of $|\hat{f}^*(x) - \hat{f}(x)|$ equals G, the bootstrap interval is exact; asymptotically $G^* \to G$. ■

Example 11.8

Basic bootstrap intervals from the replicates `kde.boot` of Example 11.7:

```
kde.quantiles = apply(kde.boot, 1,
                      function(x){quantile(x,probs = c
                      (1-0.025,0.025))})
ci.lower = 2*f.hat - kde.quantiles[1,]
ci.upper = 2*f.hat - kde.quantiles[2,]
```

For the absolute-deviation bootstrap interval, first compute the bootstrap replicates of the absolute deviations at all $x \in \mathcal{X}$:

```
abs.dev.boot = apply(kde.boot, 2, function(x){ abs(x - f.hat) })
abs.dev.boot.quantiles = apply(abs.dev.boot, 1,
                         function(x){quantile(x,probs =
                         1-0.05)})
ci.lower = f.hat - abs.dev.boot.quantiles
ci.upper = f.hat + abs.dev.boot.quantiles
```

Figure 11.11 of the book shows the four pointwise intervals computed so far. The plug-in large-sample interval is a little longer, but they are broadly similar.

The intervals of Figure 11.11 are (asymptotically) exact at each point x, but the graph of the density f cannot be asserted to lie inside the intervals at the same level: (11.16) is satisfied but (11.17) may not be. Bands satisfying (11.17) are treated next.

Remark

To simplify the standard error estimate so that no additional computation is needed,

$$\frac{\hat f(x) - f_h(x)}{\widehat{\mathrm{se}}(\hat f(x))}, \qquad \widehat{\mathrm{se}}(\hat f(x)) = \hat\sigma_f(x)/\sqrt{nh} = \sqrt{\frac{\hat f(x) a_K}{nh}}, \tag{11.20}$$

is asymptotically a pivot, so the standard bootstrap-t interval can be used. The distribution of the pivot (11.20) is bootstrap-approximated by

$$t^*(x) \equiv \frac{\hat f^*(x) - \hat f(x)}{\sqrt{\frac{\hat f^*(x) a_K}{nh}}} \mathrel{\dot\sim} \frac{\hat f(x) - f_h(x)}{\widehat{\mathrm{se}}(\hat f(x))}.$$

With $Q_{t^*}(p; x)$ the p-quantile of the bootstrap replicates $t^*(x)$, the bootstrap-t interval is

$$f_h(x) \in \Big(\hat f(x) - Q_{t^*}(1-\tfrac{\alpha}{2}; x)\sqrt{\frac{\hat f(x) a_K}{nh}},\ \hat f(x) - Q_{t^*}(\tfrac{\alpha}{2}; x)\sqrt{\frac{\hat f(x) a_K}{nh}}\Big).$$

However, if $\hat f^*(x) \approx 0$ then $\widehat{\mathrm{se}}(\hat f^*(x)) \approx 0$, so often $t^* = (\hat f^*(x) - \hat f(x))/\widehat{\mathrm{se}}(\hat f^*(x)) \approx \pm\infty$ and the interval may be far too wide.

11.4.2 Confidence bands

A confidence band controls at the confidence level the probability that the value $f(x)$ lies in the band at every position x, so it uses not the distribution of $\hat f(x) - f(x)$ but that of $\Delta = \sup_x |\hat f(x) - f(x)|$.[72] The distribution of Δ is hard to know exactly, and its asymptotic distribution is not normal either, so only fully nonparametric bootstrap bands are obtained.

The distribution of Δ is bootstrap-approximated from the bootstrap replicates $\hat f^*(x)$ of the kernel density estimate $\hat f(x)$ as follows. The function values are generally computed only on a grid $\mathcal{X} = \{x_1, \ldots, x_M\}$ finely dividing the interval containing the observations, so $\sup_x$ of Δ is approximated by $\max_{x\in\mathcal{X}}$. With the bootstrap replicates of Δ

$$\Delta^{*1} = \max_{x\in\mathcal{X}} |\hat f^{*1}(x) - \hat f(x)|, \ \ldots, \ \Delta^{*B} = \max_{x\in\mathcal{X}} |\hat f^{*B}(x) - \hat f(x)|$$

and their $1-\alpha$ quantile $Q_{\Delta^*}(1-\alpha)$, the bootstrap band is, for all x,

$$f_h(x) \in (\hat f(x) - Q_{\Delta^*}(1-\alpha),\ \hat f(x) + Q_{\Delta^*}(1-\alpha)).$$

[72] As multiple testing of finitely many hypotheses at once uses the distribution of the maximum of the test statistics to control the overall error rate, here the band is built from the distribution of the maximum of $|\hat f(x) - f(x)|$ over infinitely many $x \in \mathbb{R}$.

As $n \to \infty$, $h \to 0$, $nh \to \infty$, $B \to \infty$, this band is asymptotically exact:

$$P\big[f_h(x) \in (\hat{f}(x) - Q_{\Delta^*}(1-\alpha),\ \hat{f}(x) + Q_{\Delta^*}(1-\alpha)),\ \forall x\big] = 1 - \alpha + o(1).$$

Example 11.9

A band using only bootstrapped absolute deviations. In the code below, `abs.dev.boot` is the $M \times B$ matrix computed for the absolute-deviation bootstrap intervals in Example 11.8.

```
Delta = apply(abs.dev.boot,2,max)
q.Delta = quantile(Delta, 1-0.05)
conf.band.lower = f.hat - q.Delta
conf.band.upper = f.hat + q.Delta
```

The band is generally much wider than the pointwise intervals. Figure 11.12 of the book shows the bootstrap band together with the pointwise absolute-deviation bootstrap intervals.

Sometimes the lower limit of a band or pointwise interval is negative. A density cannot be negative, so replacing the lower limit $l(x)$ of a band (or interval) by $\max\{0, l(x)\}$ does not change the coverage. E.g. the 95% band of Figure 11.12 can be adjusted as in Figure 11.13 of the book.

Remark

When the interval at each x of the band is very wide, even if $\hat{f}(x)$ is near 0 and the pointwise interval is narrow, a very wide band results. To prevent this, one may take the maximum not of $|\hat{f}^{*b}(x) - \hat{f}(x)|$ but of the studentized $|\hat{f}^{*b}(x) - \hat{f}(x)|/\widehat{\text{se}}(\hat{f}^{*b}(x))$; but the variance of the bootstrap replicates is very large (consider $\hat{f}(x) \approx 0$), so this is not recommended. In practice, pointwise intervals are used more often than "simultaneous" bands covering all $x \in \mathcal{X}$. Some authors call the pointwise intervals at all $x \in \mathcal{X}$ a "confidence band"; take care.

Chapter 12. Nonparametric Regression

Overview

Nonparametric regression extends the "linear" regression line of the linear model $Y = \beta_0 + \beta_1 X + \epsilon$ to a regression curve of free form. The nonparametric regression model is

$$Y = m(X) + \epsilon,$$

where the regression curve $m(x) = E(Y \mid X = x)$ belongs to a space $\mathcal{H}$ of smooth functions. Nonparametric regression aims to estimate $m(\cdot)$ in the very large, infinite-dimensional space $\mathcal{H}$ and to quantify its uncertainty.

Example 12.1

The results of opinion polls on the favorability of Joe Biden during his second term as vice president of the US (from 2009) are shown in Figure 12.1 of the book. To model the flow of Biden's favorability over time, the linear model cannot capture the complex change of favorability well; the red curve of Figure 12.1 captures the flow more accurately. This curve is the nonparametrically estimated regression curve.

In the data of Example 12.1 (Figure 12.1), favorability decreases around $x = 2013$ and increases in 2016. Narrowing the time range, increases and decreases alternate between 2013 and 2016: overall there is an increasing trend, but locally increases and decreases repeat. Methods using kernels to detect such local trends are the most basic tools of nonparametric regression. This chapter explains estimation of the regression curve mainly by kernel-based methods. There are also spline methods, penalized estimation, and nonparametric regression with several explanatory variables, which are not treated.

12.1 Kernel regression estimation

Assume the data $\{(x_1, y_1), \ldots, (x_n, y_n)\}$ follow the regression model

$$y_i = m(x_i) + \epsilon_i, \qquad i = 1, \ldots, n.$$

The errors ϵ_i have mean 0 and are independent. The regression curve $m(\cdot)$ is assumed continuous and smooth. Being a nonparametric model with unspecified form of the curve, estimating $m(x)$ at every x in the domain $\mathcal{X}$ of the "X" variable gives an estimate of $m(\cdot)$. Since $m(x) = E(Y \mid X = x)$, at $x = x_i$

$$m(x_i) = E(y_i),$$

so the most natural estimator of $m(x_i)$ is y_i; but a single observation varies too much, and at $x \neq x_i$ there is no corresponding y value. The kernel regression estimator at a point x uses the weighted mean of the observations near x as the estimate. Specify the width h deciding how close an observation is, and the shape of the kernel K giving higher weights to closer observations and lower to farther ones:

$$\hat{m}(x) = \sum_{i=1}^{n} \frac{K_h(x - x_i)}{\sum_{j=1}^{n} K_h(x - x_j)} y_i = \sum_{i=1}^{n} w_i^0(x)\, y_i \tag{12.1}$$

is the kernel regression estimator of $m(x)$. Here the kernel $K_h(x) = \frac{1}{h} K(\frac{x}{h})$ is the same as in kernel density estimation; the Gaussian kernel is most common. This kernel regression estimator is also called the Nadaraya–Watson estimator or "local constant" estimator. The kernel $K_h(x - x_i)$ is large when x_i is near x, so the weight $w_i^0(x)$ of each observation y_i is large only when x_i is near x.

Figure 12.2 of the book shows, with colors at the positions x_i, how the weights $w_i^0(x)$ for the dashed x vary (redder = larger weight); the blue curve is the graph of $\hat{m}(x)$ (Gaussian kernel, $h = 300$).

In kernel regression too, the choice of the width h affects the estimator more than the choice of kernel K. With the data of Example 12.1, compute the kernel regression estimate and see how it changes with h. The R code below uses data `x`, `y`, divides the domain of "X" into 100 points `xgrid`, and defines `kernreg()` computing the kernel regression estimate at each; `Kh()` computes the kernel $K_h(x)$.[73]

```
x = as.numeric(poll$start_date %>% as_date(), "days")
y = poll$Favorable
xgrid = round(seq(min(x),max(x), length.out = 100))
Kh = function(x,h){ exp( - x^2 /h^2)/ sqrt(2*pi)/h}
kernreg = function(xgrid, x, y, h){
  kx = sapply(x,function(xi){ Kh(xgrid-xi,h)})
  W <- kx / rowSums(kx)
  drop(W %*% y)
}
```

(The Gaussian kernel is $\exp(-x^2/(2h^2))/(\sqrt{2\pi}h)$; the book's `Kh()` omits the factor 2 in the exponent, which amounts to using width $h/\sqrt{2}$, and the normalization cancels in the weights.) With width $h = 100$ (days), `mhat = kernreg(xgrid,x,y,100)` computes $\hat{m}(x)$. Figure 12.3 of the book shows the estimates for $h = 10, 100, 300$: a small width gives a very rough curve, a large width a smoother curve. What happens to $\hat{m}(\cdot)$ if h is made infinitely large ($h \to \infty$)?[74]

12.2 Local polynomial regression

The kernel regression estimator is a special case of the local polynomial regression estimator. First see why it is called a "local constant" estimator. As least squares is the basic estimator in linear regression, the kernel regression estimator (12.1) can be written

$$\hat{m}(x) = \arg\min_{\alpha} \sum_{i=1}^{n} (y_i - \alpha)^2 w_i^0(x) = \arg\min_{\alpha} \sum_{i=1}^{n} (y_i - \alpha)^2 K_h(x - x_i). \tag{12.2}$$

(Supplied: setting the derivative in α to zero gives $\sum_i (y_i - \alpha) K_h(x - x_i) = 0$, i.e. $\alpha = \sum_i w_i^0(x) y_i$.) The kernel regression estimate $\hat{m}(x)$ at x estimates α in the constant model $y_i = \alpha + \epsilon_i$ using only the observations around it; hence it is an estimator using a locally constant model. Local polynomial regression replaces the constant model of (12.2) by a polynomial model such as $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i$. In principle any pth-degree polynomial ($p = 0, 1, 2, \dots$) can be used; for simplicity take a first-degree polynomial. Replacing the constant α of (12.2) by $\alpha + \beta x$,

$$(\hat{\alpha}_x, \hat{\beta}_x) = \arg\min_{\alpha,\beta} \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2 K_h(x - x_i). \tag{12.3}$$

That is, the linear regression line fitted locally "around" x is $y = \hat{\alpha}_x + \hat{\beta}_x x$, and the prediction at x of this linear model is the local polynomial regression estimate

$$\hat{m}_1(x) = \hat{\alpha}_x + \hat{\beta}_x x. \tag{12.4}$$

(The subscript "1" indicates a first-degree polynomial.) As the position x changes, $(\hat{\alpha}_x, \hat{\beta}_x)$ changes, so $\hat{m}_1(\cdot)$ is a curve of free form like the kernel regression curve. The local polynomial estimator

[73] `np_supp.R` defines `lcreg()`, similar to `kernreg()`; for the name `lcreg()` see the next section.

[74] If h is infinitely large, the kernel regression estimate is constant for all x: $\hat{m}(x) \equiv \bar{y}$. (Supplied, Exercise 12.1: as $h \to \infty$, $K_h(x - x_i)/K_h(x - x_j) = K((x - x_i)/h)/K((x - x_j)/h) \to K(0)/K(0) = 1$ for continuous K with $K(0) > 0$, so all weights tend to $1/n$. So the simple regression line is not a special case of the Nadaraya–Watson estimator.)

$\hat{m}_1(\cdot)$ (12.4) has the form of local linear regression (12.3) and is called the **local linear regression estimator**.

Figure 12.4 of the book shows, at two different positions x, the graph of the locally fitted regression line and the value of $\hat{m}_1(x)$. The local linear estimator $\hat{m}_1$ is the collection of the results of locally fitted regression lines.

The computation of local polynomial estimators, including local linear regression, is weighted least squares. The general form of (12.3) at a given x,

$$\arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i - \cdots - \beta_p x_i^p)^2 K_h(x - x_i), \tag{12.5}$$

can be written in matrix form. With

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \qquad X = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^p \\ 1 & x_2 & x_2^2 & \cdots & x_2^p \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^p \end{pmatrix}, \qquad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{pmatrix},$$

and $W_x = \operatorname{diag}\{K_h(x - x_1), \ldots, K_h(x - x_n)\}$, the $n \times n$ diagonal matrix with ith diagonal element $K_h(x - x_i)$, (12.5) becomes

$$\hat{\boldsymbol{\beta}}_x = \arg\min_{\boldsymbol{\beta}} (\mathbf{y} - X\boldsymbol{\beta})' W_x (\mathbf{y} - X\boldsymbol{\beta}) = (X' W_x X)^{-1} X' W_x \mathbf{y}. \tag{12.6}$$

Proof of (12.6) (supplied; Exercise 12.2). $Q(\boldsymbol{\beta}) = (\mathbf{y} - X\boldsymbol{\beta})' W_x (\mathbf{y} - X\boldsymbol{\beta})$ is a convex quadratic (W_x is positive semidefinite). Its gradient is $-2X' W_x (\mathbf{y} - X\boldsymbol{\beta})$, and setting it to 0 gives the normal equations $X' W_x X\boldsymbol{\beta} = X' W_x \mathbf{y}$. When $X' W_x X$ is invertible (at least $p + 1$ distinct x_i with positive weight), the unique minimizer is $(X' W_x X)^{-1} X' W_x \mathbf{y}$. ■

So the local polynomial estimate $\hat{m}_p(x)$ at x, with $\mathbf{x}_x = (1, x, x^2, \ldots, x^p)'$, is

$$\hat{m}_p(x) = \mathbf{x}_x' \hat{\boldsymbol{\beta}}_x = \begin{pmatrix} 1 & x & \cdots & x^p \end{pmatrix} \hat{\boldsymbol{\beta}}_x = \mathbf{x}_x' (X' W_x X)^{-1} X' W_x \mathbf{y} = \sum_{i=1}^{n} w_i^p(x)\, y_i, \tag{12.7}$$

where the weight $w_i^p(x)$ of pth-degree local polynomial regression is the ith element of the $1 \times n$ row vector $\mathbf{x}_x' (X' W_x X)^{-1} X' W_x$.[75]

The local polynomial estimate $\hat{m}_p(x)$ with $p = 0$ is the Nadaraya–Watson estimator; with $p = 1$ it is the local linear estimator; for $p \geq 2$ generally the pth-degree local polynomial estimator. The weights of the local linear estimator are

$$w_i^1(x) = \frac{b_i(x)}{\sum_{j=1}^{n} b_j(x)}, \qquad b_i(x) = \{S_2(x) + (x - x_i) S_1(x)\} K_h(x - x_i), \tag{12.8}$$

where $S_q(x) = \sum_{j=1}^{n} K_h(x - x_j)(x_j - x)^q$ for $q = 0, 1, 2, \ldots$.

Weights for $p = 0, 1$ (supplied; Exercise 12.3). For $p = 0$, $X = \mathbf{1}$ and $\mathbf{x}_x = 1$, so $w_i^0(x) = (\mathbf{1}' W_x \mathbf{1})^{-1} K_h(x - x_i) = K_h(x - x_i) / \sum_j K_h(x - x_j)$, as in (12.1). For $p = 1$, the fitted value at x is unchanged if we center the design at x, i.e. use columns $(1, x_i - x)$; then $\mathbf{x}_x = (1, 0)'$ and $\hat{m}_1(x)$ is the intercept. With $k_i = K_h(x - x_i)$ and $d_i = x_i - x$, $X' W_x X = \begin{pmatrix} S_0 & S_1 \\ S_1 & S_2 \end{pmatrix}$, and the first row of its inverse is $\frac{1}{S_0 S_2 - S_1^2}(S_2, -S_1)$. So

$$\hat{m}_1(x) = \frac{\sum_i (S_2 - S_1 d_i) k_i y_i}{S_0 S_2 - S_1^2},$$

[75] `np_supp.R` defines `lcreg()` (local constant regression) and `llreg()` (local linear regression) for $p = 0, 1$; with them the weights $w_i^p(x)$ can be examined.

and $\sum_i (S_2 - S_1 d_i)k_i = S_2 S_0 - S_1^2$. With $b_i = (S_2 - S_1 d_i)k_i = \{S_2 + (x - x_i)S_1\}K_h(x - x_i)$ this is (12.8). ■

Example 12.2

The local polynomial estimate $\hat{m}_p(x)$ can be computed with `locpoly()` of the R package `KernSmooth`. For Biden's favorability data, set the degree p with `degree` and the width h with `bandwidth`:

```
> p = 1
> h = 50
> ll = KernSmooth::locpoly(x,y,degree = p, bandwidth = h)
> head(as.data.frame(ll),3)
         x        y
1 15276.00 41.00002
2 15280.94 40.97017
3 15285.88 40.93969
```

The output of `locpoly()` is the pairs $(x, \hat{m}_p(x))$ on a grid of x. `llreg()` defined in `np_supp.R` also computes local linear estimates:

```
> xgrid = ll$x
> llregout = llreg(xgrid, x, y, h)
> cbind(xgrid, llregout$mhat) %>% head(3)
        xgrid
[1,] 15276.00 41.00000
[2,] 15280.94 41.01326
[3,] 15285.88 41.02651
```

The estimates at each x differ slightly because `llreg()` implements the definition (12.7) of the local linear estimate exactly but computes slowly, while `KernSmooth::locpoly()` gives an approximation of $\hat{m}_1(x)$ for fast computation.

The local polynomial estimate is determined by the degree p of the local polynomial and the kernel width h. Which degree and width should be chosen? The next section answers.

12.2.1 Choosing the degree of local polynomial regression

Figure 12.5 of the book shows the local polynomial estimates computed as in Example 12.2 for each of $p = 0, 1, 2$ and $h = 50, 300$. At the left and right ends of the horizontal axis the differences between the three estimates are clear. With the kernel width fixed, a higher degree gives a more complex estimated curve. At the right end of the horizontal axis, a higher degree follows the increasing pattern of the scatter plot more faithfully. That is, the higher the degree of the polynomial, the smaller the bias of the estimator:

$$\text{bias}\{\hat{m}_0(x)\} > \text{bias}\{\hat{m}_1(x)\} > \text{bias}\{\hat{m}_2(x)\} > \cdots . \tag{12.9}$$

By the bias–variance trade-off one would predict the variance grows with the degree. This is not exactly true, however. At the left end of the horizontal axis in Figure 12.5, the variability of the estimated curves for $p = 0, 1$ is similar, while that for $p = 2$ is much larger. The hypothesis suggested is

$$\text{Var}\{\hat{m}_0(x)\} = \text{Var}\{\hat{m}_1(x)\} < \text{Var}\{\hat{m}_2(x)\} = \text{Var}\{\hat{m}_3(x)\} < \cdots : \tag{12.10}$$

the variance increases with the degree, but the variance of a local polynomial estimator of odd degree equals that at the even degree one lower.

For details, consider only $p = 0, 1$. The Nadaraya–Watson estimator for $p = 0$,

$$\hat{m}_0(x) = \sum_{i=1}^{n} w_i^0(x)\, y_i, \tag{12.11}$$

and the local linear estimator for $p = 1$,

$$\hat{m}_1(x) = \sum_{i=1}^{n} w_i^1(x)\, y_i, \tag{12.12}$$

are both linear combinations of the y_i, and the bias and variance of the estimate $\hat{m}_p(x)$ at x are closely tied to the weights $w_i^p(x)$.

Theorem 12.1

In the model $Y_i = m(x_i) + \sigma(x_i)\epsilon_i$, $\sigma(x_i) > 0$, with the ϵ_i independent with mean 0 and variance 1, the bias and variance of the Nadaraya–Watson estimator (12.11) and the local linear estimator (12.12) are, for $p = 0, 1$,

$$\text{bias}[\hat{m}_p(x)] = \sum_{i=1}^{n} w_i^p(x)\{m(x_i) - m(x)\}, \tag{12.13}$$

$$\text{Var}[\hat{m}_p(x)] = \sum_{i=1}^{n} \{w_i^p(x)\, \sigma(x_i)\}^2. \tag{12.14}$$

Proof (supplied). The weights satisfy $\sum_i w_i^p(x) = 1$ (for $p = 0$ by definition; for $p = 1$ by (12.8)). Since the weights depend only on the x's, $E\hat{m}_p(x) = \sum_i w_i^p(x) m(x_i)$, and subtracting $m(x) = \sum_i w_i^p(x) m(x)$ gives (12.13). The Y_i are independent with $\text{Var}\, Y_i = \sigma^2(x_i)$, giving (12.14). ■

Theorem 12.1 assumed the error variances may differ. To express the bias and variance more simply asymptotically, we need: first, for the term $m(x_i) - m(x)$ of the bias to be simply expressed by Taylor expansion, the regression curve $m(\cdot)$ must be smooth. Second, not only the response Y_i but also the explanatory variable X_i must be random, so that the weighted means containing X_i in (12.13) and (12.14) are approximated by expectations.

To express the asymptotic bias and variance of the Nadaraya–Watson and local linear estimators, assume the following. We estimate the regression curve $m(x)$ at $x \in \mathcal{X}$ in some interval $\mathcal{X} = (a, b)$ of the explanatory variable.

Assumption A. $m(\cdot)$ is smooth on $\mathcal{X}$.[76]
Assumption B. The density $f(\cdot)$ of X_i is smooth and $f(x) > 0$ for $x \in \mathcal{X}$.

In the model

$$Y_i = m(X_i) + \sigma(X_i)\epsilon_i, \qquad i = 1, \ldots, n,$$

the bias and variance of $\hat{m}_0(x)$ and $\hat{m}_1(x)$ conditional on the observed explanatory values $\mathbb{X}_n = (X_1, \ldots, X_n)$ are:

Theorem 12.2 (García-Portugués (2021), Theorem 4.1)

Let $\hat{m}_0(x)$, $\hat{m}_1(x)$ be the Nadaraya–Watson and local linear estimators with the Gaussian kernel. For $p = 0, 1$, if Assumptions A and B hold and $n \to \infty$, $h \to 0$, $nh \to \infty$,

$$\text{bias}[\hat{m}_p(x) \mid \mathbb{X}_n] = B_p(x)h^2 + o_P(h^2), \tag{12.15}$$

[76] More precisely, the second derivative $m''(\cdot)$ exists and is continuous.

$$\operatorname{Var}[\hat{m}_p(x) \mid \mathbb{X}_n] = \frac{\sigma^2(x)}{2\sqrt{\pi} f(x)} \frac{1}{nh} + o_P((nh)^{-1}), \tag{12.16}$$

where

$$B_p(x) := \begin{cases} \dfrac{m''(x)}{2} + \dfrac{m'(x) f'(x)}{f(x)}, & p = 0, \\ \dfrac{m''(x)}{2}, & p = 1. \end{cases}$$

Sketch (supplied). For $p = 0$, $\hat{m}_0(x) - m(x) = \frac{\frac{1}{n}\sum_i K_h(x - X_i)\{m(X_i) - m(x) + \sigma(X_i)\epsilon_i\}}{\frac{1}{n}\sum_i K_h(x - X_i)}$. The denominator converges in probability to $f(x)$. For the bias part of the numerator, with $X_i = x - hu$, $E[K_h(x - X)\{m(X) - m(x)\}] = \int K(u)\{m(x - hu) - m(x)\} f(x - hu)\, du$; expanding $m(x - hu) - m(x) = -hum' + \frac{h^2u^2}{2} m'' + o(h^2)$ and $f(x - hu) = f - huf' + o(h)$ and using $\int uK = 0$, $\int u^2 K = 1$ gives $h^2\{\frac{m''f}{2} + m'f'\} + o(h^2)$; dividing by f gives $B_0 h^2$. For the variance, $\frac{1}{n^2}\sum_i K_h^2(x - X_i)\sigma^2(X_i) \approx \frac{\sigma^2(x) f(x)}{nh} \int K^2$, and $\int K^2 = \frac{1}{2\sqrt{\pi}}$ for the Gaussian kernel; dividing by $f(x)^2$ gives (12.16). For $p = 1$ the local linear fit removes the linear term exactly: the weights satisfy $\sum_i w_i^1(x)(X_i - x) = 0$ (the fit reproduces lines). So the m' contribution, including the cross term $m'f'/f$ produced by the asymmetry of the design density, cancels and only $\frac{m''}{2} h^2$ remains, while the leading variance is unchanged. ■

From Theorem 12.2, for the local polynomial estimators with $p = 0, 1$:

1. The bias (12.15) is proportional to the square of the kernel width h; the variance (12.16) is inversely proportional to nh. Both hold for $p = 0, 1$ and are the same as the asymptotic properties of the kernel density estimator. With the sample size n fixed, a larger width increases the bias but reduces the variance, and a smaller width increases the variance but reduces the bias: the bias–variance trade-off.
2. The local linear estimator has smaller bias than the Nadaraya–Watson estimator. More specifically, the term $\frac{m'(x) f'(x)}{f(x)}$ present in the bias for $p = 0$ disappears for $p = 1$. The term $\frac{m'(x) f'(x)}{f(x)}$ present only in the bias of the Nadaraya–Watson estimator is related to the distribution $f(x)$ of the explanatory variable and is also called the "design bias."[77] (This confirms roughly the hypothesis (12.9) for $p = 0, 1$.)
3. The first term of the bias, $m''(x)$ (the second derivative of $m(x)$), is larger the more sharply the regression curve $m(x)$ changes.
   - If $m''(x) = 0$, the slope of the curve $m(\cdot)$ does not change at x (it is locally linear), and the bias of the local linear estimator $\hat{m}_1(x)$ is 0.
   - If $m''(x) > 0$, the curve $m(\cdot)$ forms a "valley" near x; then $E[\hat{m}_1(x)] > m(x)$: the estimate tends to lie above the valley floor.
   - If $m''(x) < 0$, the curve forms a "peak" near x; then $E[\hat{m}_1(x)] < m(x)$: the estimate tends to lie below the peak.
4. The first term of the variance (12.16) is the same for $p = 0$ and $p = 1$. (This confirms roughly the hypothesis (12.10) for $p = 0, 1$.)

Compared with the Nadaraya–Watson estimator ($p = 0$), the local linear estimator ($p = 1$) has (asymptotically) the same variance and smaller bias. So local linear regression is generally preferred to Nadaraya–Watson. Likewise local cubic is preferred to local quadratic. For simplicity, from now on only the local linear estimator is used.

[77] Recall that in multiple regression the matrix X of the explanatory variables was called the design matrix.

12.2.2 Choosing the bandwidth

For local linear estimation of the regression curve $m(\cdot)$, a good estimator is one whose overall estimated curve $\hat{m}_1$ has small MISE, and the MISE is closely tied to the kernel width h. Integrating only the first two terms of Theorem 12.2 gives the approximation

$$\mathrm{MISE}(\hat{m}_1(X) \mid \mathbb{X}_n) \approx h^4 \int \frac{(m''(x))^2}{4} f(x)\,dx + \frac{\int \sigma^2(x)\,dx}{2\sqrt{\pi}} \frac{1}{nh}. \tag{12.17}$$

Regarding the approximated MISE (12.17) as a function of h, the value of h minimizing it is the asymptotically optimal width h^*:

$$h^* = \Big(\frac{\int \sigma^2(x)\,dx}{2\sqrt{\pi} \int (m''(x))^2 f(x)\,dx} \cdot \frac{1}{n} \Big)^{\frac{1}{5}}. \tag{12.18}$$

(Supplied: here the MISE weights the squared bias by the design density, $\int \{\cdot\}^2 f(x)\,dx$, while the variance term $\frac{\sigma^2(x)}{2\sqrt{\pi} f(x) nh}$ integrated against f gives $\frac{\int \sigma^2}{2\sqrt{\pi} nh}$; minimizing $Ah^4 + \frac{B}{nh}$ with $A = \frac{1}{4}\int (m'')^2 f$ and $B = \frac{\int \sigma^2}{2\sqrt{\pi}}$ gives $h^5 = \frac{B}{4An}$, i.e. (12.18).) The form of the denominator of (12.18) is very similar to (11.10) of Section 11.2.3. As with the optimal width in kernel density estimation, (12.18) cannot actually be used. Of the methods estimating the optimal width h^* directly, the following plug-in method is representative.

Plug-in method. The expression (12.18) of the optimal width h^* contains the second derivative $m''(x)$ of the regression curve, the marginal distribution $f(x)$ of the explanatory variable, and the error variance $\sigma^2(x)$ that can vary with position. Ruppert, Sheather and Wand (Ruppert et al., 1995) proposed estimating $\int (m''(x))^2 f(x)\,dx$ and $\int \sigma^2(x)\,dx$ directly. Roughly: divide the range $\mathcal{X}$ of the explanatory variable into N blocks $\mathcal{X}_1, \ldots, \mathcal{X}_N$; in each block estimate the regression curve as a quartic polynomial using only that block's data, call it $\tilde{m}(\cdot)$, and write the second derivative of that quartic as $\tilde{m}''(\cdot)$. Then

$$\int (m''(x))^2 f(x)\,dx = E[(m''(X))^2] \approx \frac{1}{n} \sum_{i=1}^{n} (\tilde{m}''(X_i))^2 =: \tilde{m}^2.$$

Also, from the residual sums of squares of the quartic regressions in the blocks, obtain error variance estimates $\tilde{\sigma}_1^2, \ldots, \tilde{\sigma}_N^2$ and approximate $\int \sigma^2(x)\,dx$ by their weighted average:

$$\int \sigma^2(x)\,dx \approx \sum_{j=1}^{N} \tilde{\sigma}_j^2 |\mathcal{X}_j| =: \tilde{\sigma}^2.$$

Choosing the number of blocks N by a model selection criterion such as AIC or BIC, the plug-in kernel width is

$$h^{\mathrm{DPI}} = \Big(\frac{\tilde{\sigma}^2}{2\sqrt{\pi} \tilde{m}^2} \cdot \frac{1}{n} \Big)^{\frac{1}{5}}.$$

The Ruppert–Sheather–Wand "direct plug-in" method is implemented in R as `KernSmooth::dpill` (direct plug-in for local linear regression).

Example 12.3

For Biden's favorability data, setting the width with the Ruppert–Sheather–Wand direct plug-in method gives $h^{\mathrm{DPI}} = 57.79$:

```
> (h.dpi = KernSmooth::dpill(x,y))
[1] 57.78665
```

Cross-validation. The second way of choosing the kernel width is cross-validation, often used to examine prediction performance in regression. Given $\hat{m}_1(\cdot)$ estimated from the data $\mathbb{D}$, if a new observation (X', Y') arrived, how close would $\hat{m}_1(X')$ be to Y'? To measure this we need an observation (X', Y') independent of the data $\mathbb{D}$ (or of $\hat{m}_1$ estimated from it). Leave-one-out cross-validation generates this situation from the data: for each $i = 1, \ldots, n$, from the data without the ith observation,

$$\mathbb{D}_{(-i)} = \{(x_1, y_1), \ldots, (x_n, y_n)\} \setminus \{(x_i, y_i)\},$$

compute the local linear estimate $\hat{m}_1^{(-i)}(\cdot)$, and obtain the prediction error $y_i - \hat{m}_1^{(-i)}(x_i)$ of the held-out (x_i, y_i). For a given kernel width h the mean prediction error is

$$\mathrm{CV}(h) = \frac{1}{n} \sum_{i=1}^{n} \{y_i - \hat{m}_1^{(-i)}(x_i)\}^2, \tag{12.19}$$

and the width minimizing it, $h^{\mathrm{cv}} = \arg\min_{h>0} \mathrm{CV}(h)$, is the cross-validated kernel width. As in leave-one-out validation in linear regression, computing (12.19) does not require n local linear fits: writing $\hat{m}_1(x) = \sum_{i=1}^{n} w_i^1(x) y_i$ as in (12.12), it is known that

$$\mathrm{CV}(h) = \frac{1}{n} \sum_{i=1}^{n} \left\{ \frac{y_i - \hat{m}_1(x_i)}{1 - w_i^1(x_i)} \right\}^2.$$

The leave-one-out shortcut (supplied). Removing observation i from the weighted least squares at $x = x_i$: the leave-one-out fit equals the full fit with y_i replaced by its own leave-one-out prediction $\tilde{y}_i = \hat{m}_1^{(-i)}(x_i)$, because adding a point that lies exactly on the fitted local line leaves that fit unchanged. Hence $\tilde{y}_i = \sum_{j \neq i} w_j^1(x_i) y_j + w_i^1(x_i) \tilde{y}_i$, i.e. $\tilde{y}_i = \frac{\hat{m}_1(x_i) - w_i^1(x_i) y_i}{1 - w_i^1(x_i)}$. Therefore $y_i - \tilde{y}_i = \frac{y_i - \hat{m}_1(x_i)}{1 - w_i^1(x_i)}$. ■

Example 12.4

For Biden's favorability data, compute $\mathrm{CV}(h)$ for candidate widths $h \in \{40, 42, 44, \ldots, 120\}$:

```
> h.range = seq(40, 120, by = 2)
> K = length(h.range)
> cv = vector(length = K)
> for (k in 1:K){
+   h = h.range[k]
+   llreg.out = llreg(x,x,y,h)
+   cv[k] = mean( ( (y - llreg.out$mhat) /
+                   (1- diag(llreg.out$weight)) )^2 )
+ }
> (h.cv = h.range[which.min(cv)])
[1] 70
```

$\mathrm{CV}(\cdot)$ is smallest at $h = 70$ (plot in the book). With the cross-validated width $h^{\mathrm{cv}} = 70$ and the plug-in width $h^{\mathrm{DPI}} = 57.79$ of Example 12.3, the local linear estimates are compared in Figure 12.6 of the book; the two widths are almost the same, so the estimates nearly coincide.

12.3 Confidence intervals for the regression curve

In a nonparametric model of very free form, an exact confidence interval for the regression curve is very hard to obtain. As in density estimation, with the optimal width h^* the standard error and bias of the local linear estimator are of the same size,[78] so the bias in the estimator is hard to

[78] When $h^* \asymp n^{-1/5}$, both standard error and bias are of size $n^{-2/5}$. (By comparison, in simple linear regression the standard error of the least squares estimator is $n^{-1/2}$ and the bias 0.)

remove. As in density estimation, intervals are given not for the actual regression function $m(x)$ but for the mollified regression curve

$$m_h(x) = E(\hat{m}_1(x)),$$

where h is the kernel width of $\hat{m}_1(x)$. For x in an interval $\mathcal{X} = (a, b)$ of the explanatory variable, pointwise intervals for $m_h(x)$ are obtained asymptotically and by the bootstrap. Simultaneous bands are rarely used in practice (Hall and Horowitz, 2013) and are not explained.

12.3.1 Theoretical pointwise intervals

The local linear estimator $\hat{m}_1(x)$ is a weighted mean of the responses y_i as in (12.12), so by the CLT it is asymptotically normal. Using the expression of Theorem 12.1,

$$\frac{\hat{m}_1(x) - m_h(x)}{\mathrm{se}(\hat{m}_1(x))} \dot{\sim} N(0, 1), \tag{12.20}$$

where $\mathrm{se}^2(\hat{m}_1(x)) = \mathrm{Var}(\hat{m}_1(x)) = \sum_{i=1}^n \{w_i^1(x)\sigma(x)\}^2$, and the error variance $\sigma^2(x)$ is unknown and must be estimated.

- **(Homoscedasticity)** If $\sigma^2(x) = \sigma^2$, as in linear models σ^2 is estimated by dividing the residual sum of squares by its "degrees of freedom." With the response vector $\mathbf{y} = (y_1, \ldots, y_n)'$, the "hat matrix" that computes its fitted values is the $n \times n$ matrix $W = (w_{ij})_{i,j=1,\ldots,n}$, $w_{ij} = w_j^1(x_i)$, so that $\hat{\mathbf{y}} = W\mathbf{y}$. (As in (12.12), check that $\hat{y}_i := \hat{m}_1(x_i) = \sum_{j=1}^n w_j^1(x_i) y_j$.) The residual vector of local linear regression is

  $$\mathbf{r} = \mathbf{y} - \hat{\mathbf{y}} = (I - W)\mathbf{y},$$

  and the consistent estimator of the error variance σ^2 is[79]

  $$\hat{\sigma}^2 = \frac{\mathbf{y}'(I - W)'(I - W)\mathbf{y}}{\mathrm{trace}((I - W)'(I - W))} = \frac{\sum_{i=1}^n (y_i - \hat{m}_1(x_i))^2}{\mathrm{trace}((I - W)'(I - W))} \tag{12.21}$$

  (Wasserman, 2006, Theorem 5.85).
- **(Heteroscedasticity)** Suppose $\sigma^2(x)$ varies with x. Assuming the variance function $\sigma^2(\cdot)$ is smooth, estimate $\sigma^2(\cdot)$ by smoothing the bivariate data (x_i, r_i^2) containing the squared residuals $r_i^2 = (y_i - \hat{m}_1(x_i))^2$. With the method of Yu and Jones (2004), the estimated variance function can be forced to be always positive: with $Z_i = \log r_i^2$ as response and x_i as explanatory variable, nonparametric regression gives an estimate $\hat{q}(x)$ of the regression curve, and

  $$\hat{\sigma}^2(x) = \exp(\hat{q}(x)).$$

Once the error variance is estimated, substituting

$$\widehat{\mathrm{se}}(\hat{m}_1(x)) = \Big(\sum_{i=1}^n \{w_i^1(x)\hat{\sigma}(x)\}^2 \Big)^{1/2} \tag{12.22}$$

into (12.20) gives the (asymptotic) $1 - \alpha$ interval for $m_h(x)$:

$$m_h(x) \in \hat{m}_1(x) \pm z_{1-\alpha/2} \widehat{\mathrm{se}}(\hat{m}_1(x)).$$

This interval corresponds to the plug-in large-sample interval of density estimation, but differs in estimating the standard error directly rather than through a "plug-in" of an asymptotic expression.

[79] For an $n \times n$ matrix $A = (a_{ij})$, $\mathrm{trace}(A) = \sum_{i=1}^n a_{ii}$ is the sum of the diagonal elements of A.

Example 12.5

Intervals for the local linear estimate with kernel width $h^{\text{cv}} = 70$ as in Example 12.4. Assuming homoscedastic errors, $\hat{\sigma}^2$ is computed by (12.21):

```
> n = length(x)
> llreg.x = llreg(x, x, y, h = h.cv)
> residuals = y - llreg.x$mhat
> tr = sum( diag( t(diag(n) - llreg.x$weight) %*% (diag(n) -
    llreg.x$weight)))
> (hat.sigma2 = sum(residuals^2) / tr)
[1] 12.09957
```

At each point x of the grid of the "X" variable where intervals are wanted, compute the regression estimate $\hat{m}_1(x)$ and its standard error (12.22), then the theoretical interval:

```
> xgrid = seq(min(x), max(x), length.out = 100)
> llreg.out = llreg(xgrid, x, y, h = h.cv)
> mhat = llreg.out$mhat
> se = sqrt( rowSums(llreg.out$weight^2) * hat.sigma2 )
> ci.theory = data.frame(x = xgrid,
+                        ci.low = mhat - qnorm(1-0.025)*se,
+                        mhat = mhat,
+                        ci.up = mhat + qnorm(1-0.025)*se)
```

These intervals are shown in the first panel of Figure 12.7 of the book.

12.3.2 Bootstrap pointwise intervals

Nonparametric regression is also regression, so two kinds of bootstrap resampling are possible: the case bootstrap and the residual bootstrap. With the kernel width fixed at h, the bootstrap replicate of the local linear estimate $\hat{m}_1(\cdot)$ is $\hat{m}_1^*(\cdot)$, computed from a bootstrap sample with the same width, and is easily obtained by case-bootstrap resampling. In nonparametric regression the estimator always has bias, which is also reflected in the residuals. Residual-bootstrap methods removing the effect of this bias have been developed, but exceed the level of this book; we use only the case bootstrap.

The nonparametric bootstrap resampling of the local linear estimate:

1. From the original bivariate data $\mathbb{D} = \{(x_i, y_i) : i = 1, \ldots, n\}$ estimate the kernel width h^* and with it compute the local linear estimate $\hat{m}_1(\cdot)$.
2. Draw n observations (x_i, y_i) with replacement from $\mathbb{D}$ to get $\mathbb{D}^* = \{(x_i^*, y_i^*) : i = 1, \ldots, n\}$.
3. From the bootstrap sample $\mathbb{D}^*$ compute the local linear estimate $\hat{m}_1^*(\cdot)$ with width h^*.
4. Repeat steps 2–3 B times to get the bootstrap replicates $\hat{m}_1^{*b}(\cdot)$, $b = 1, \ldots, B$.

With these replicates, estimate the standard error directly and substitute into (12.20) for the bootstrap plug-in interval, or use the percentile bootstrap interval.

Bootstrap plug-in interval. At each $x \in \mathcal{X}$ the bootstrap-estimated standard error is

$$\widehat{\text{se}}^{\text{Boot}}(\hat{m}_1(x)) = \Big(\frac{1}{B-1}\sum_{b=1}^{B}(\hat{m}_1^{*b}(x) - \overline{\hat{m}_1^*(x)})^2\Big)^{1/2},$$

and the $1-\alpha$ bootstrap plug-in interval is

$$m_h(x) \in \hat{m}_1(x) \pm z_{1-\alpha/2}\widehat{\text{se}}^{\text{Boot}}(\hat{m}_1(x)).$$

Percentile interval. With $m^*(x;\alpha/2)$ and $m^*(x;1-\alpha/2)$ the $\alpha/2$- and $(1-\alpha/2)$-quantiles of the bootstrap replicates $\hat{m}_1^{*1}(x),\ldots,\hat{m}_1^{*B}(x)$ at each $x\in\mathcal{X}$, the $1-\alpha$ bootstrap percentile interval for $m_h(x)$ is

$$m_h(x)\in\big(m^*(x;\alpha/2),\ m^*(x;1-\alpha/2)\big).$$

Example 12.6

Both the bootstrap plug-in and percentile intervals need only the bootstrap replicates of $\hat{m}_1(x)$. The R code below generates $B=1999$ of them:

```
xgrid = seq(min(x), max(x), length.out = 100)
mhat = llreg(xgrid, x, y, h = h.cv)$mhat
set.seed(1)
B = 1999
m.boot = replicate(B,{
boot.id = sample(1:n,n,replace = TRUE)
x.boot = x[boot.id]
y.boot = y[boot.id]
llreg(xgrid, x.boot, y.boot, h = h.cv)$mhat
}) %>% t()
```

With a grid $(x_{(1)},\ldots,x_{(N)})$ of length N (here $N=100$), the stored `m.boot` is a $B\times N$ matrix whose (b,j) element is $\hat{m}_1^{*b}(x_{(j)})$. For the 95% bootstrap plug-in interval compute the bootstrap standard error estimate of $\hat{m}_1(x)$:

```
se.boot = apply(m.boot,2,sd)
ci.pi = data.frame(x = xgrid,
                   ci.low = mhat - qnorm(1-0.025)*se.boot,
                   mhat = mhat,
                   ci.up = mhat + qnorm(1-0.025)*se.boot)
```

The 95% percentile interval is given by the 2.5% and 97.5% quantiles of each column of `m.boot`:

```
q1 = apply(m.boot,2,function (x) {quantile(x,0.025)})
q2 = apply(m.boot,2,function (x) {quantile(x,1-0.025)})
ci.perc = data.frame(x = xgrid,
                     ci.low =q1,
                     mhat = mhat,
                     ci.up = q2)
```

The bootstrap intervals are shown in the second and third panels of Figure 12.7 of the book.

Appendix A. Basic Usage of R

Overview

The book assumes a reader familiar with basic R. Readers new to R should consult the standard introduction, Venables et al. (2021).[80] This appendix explains the R usage needed for the statistics, resampling and visualization in the book.

A.1 Data types and operations

The simplest way to enter data in R is a vector. E.g. the recovery times $19, 22, 24, 26$ of Example 3.1 in Chapter 3 can be entered as `c(19,22,24,26)`; `c()` combines its inputs into a vector, here stored as an object named `x`:

```
x = c(19,22,24,26)
x
## [1] 19 22 24 26
```

To create a matrix directly use `matrix()`. From a vector `x` of 24 values, a 4×6 matrix:

```
x = c(128, 128, 160, 142, 157, 179,
      122, 137, 177, 177, 160, 138,
      207, 188, 181, 164, 155, 175,
      120, 208, 199, 194, 177, 195)
matrix(x, byrow = TRUE, ncol=6)
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]  128  128  160  142  157  179
## [2,]  122  137  177  177  160  138
## [3,]  207  188  181  164  155  175
## [4,]  120  208  199  194  177  195
```

The argument `byrow = TRUE` fills the matrix by rows (horizontally first). The default is `byrow = FALSE`, so omitting it fills by columns (vertically first):

```
X = matrix(x, nrow = 4)
X
##      [,1] [,2] [,3] [,4] [,5] [,6]
## [1,]  128  157  177  207  155  199
## [2,]  128  179  177  188  175  194
## [3,]  160  122  160  181  120  177
## [4,]  142  137  138  164  208  195
```

Data for statistical analysis are stored as data frames; e.g. reading the file `worldbank2020.csv` gives a data frame:

```
wbdata = read.csv("data/worldbank2020.csv")
str(wbdata)
## 'data.frame':   153 obs. of  9 variables:
##  $ Overall.rank          : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ Country.or.region     : chr  "Finland" "Denmark" ...
##  $ Score                 : num  7.81 7.65 7.56 7.5 7.49 ...
##  $ GDP.per.capita        : num  1.28 1.33 1.39 1.33 1.42 ...
##  $ Social.support        : num  1.5 1.5 1.47 1.55 1.5 ...
##  ...
```

[80] Venables et al. (2021) is freely available at https://cran.r-project.org/doc/manuals/r-release/R-intro.pdf.

`wbdata` stores 9 vectors of equal length as columns, each vector holding the observations of one variable. To extract the third variable `Score` of `wbdata` use `wbdata$Score` or `wbdata[,3]`. With `with()` the variable names inside a data frame can be called directly:

```
with(wbdata, lm(Score ~ GDP.per.capita))
##
## Call:
## lm(formula = Score ~ GDP.per.capita)
##
## Coefficients:
##    (Intercept)  GDP.per.capita
##          3.462           2.316
```

R operations are elementwise by default; e.g. the sum and product of two vectors are elementwise:

```
x = c(1,2,3,4)
y = c(5,6,7,8)
x + y
## [1]  6  8 10 12
x * y
## [1]  5 12 21 32
```

The product (`*`) of two matrices is also elementwise:

```
(X = matrix(1:6,nrow =3, ncol = 2))
##      [,1] [,2]
## [1,]    1    4
## [2,]    2    5
## [3,]    3    6
X * X
##      [,1] [,2]
## [1,]    1   16
## [2,]    4   25
## [3,]    9   36
```

Sums and differences of matrices are likewise elementwise; the results of `X + X` and `X - X` are easy to guess. For the matrix product use `%*%`:

```
(Y = matrix(1:6, nrow = 2, ncol = 3))
##      [,1] [,2] [,3]
## [1,]    1    3    5
## [2,]    2    4    6
X %*% Y
##      [,1] [,2] [,3]
## [1,]    9   19   29
## [2,]   12   26   40
## [3,]   15   33   51
```

The transpose of a matrix uses `t()`: X' is `t(X)`. The inverse uses `solve()`: if X^{-1} exists it is `solve(X)`. With these, e.g., the regression hat matrix $X(X'X)^{-1}X'$ is computed:

```
(H = X %*% solve(t(X) %*% X) %*% t(X))
##            [,1]      [,2]       [,3]
## [1,]  0.8333333 0.3333333 -0.1666667
## [2,]  0.3333333 0.3333333  0.3333333
## [3,] -0.1666667 0.3333333  0.8333333
```

What if a vector y of length 3 is multiplied by the 3×3 matrix H? Viewing y as a 1×3 matrix (row vector) the product yH is defined; viewing it as 3×1 (column vector), Hy is defined. R turns the vector y into a row or column vector automatically so that the product is defined:

```
y = c(0,1,2)
H %*% y # H times y
##              [,1]
## [1,] 4.440892e-16
## [2,] 1.000000e+00
## [3,] 2.000000e+00
y %*% H # y times H
##               [,1] [,2] [,3]
## [1,] -4.440892e-16    1    2
```

(Supplied remark: $y = (0, 1, 2)'$ lies in the column space of X, here $y = X(-\frac{2}{3}, \frac{1}{3})'$; so $Hy = y$ up to rounding error 4×10^{-16}.)

A.2 Packages and functions

An R package is a bundle of objects such as functions and data sets. The functions `matrix()`, `t()`, `solve()` above are in R's `base` package; `read.csv()` is in R's `utils` package. These two, together with packages such as `graphics` and `datasets`, are loaded automatically in an R session. To use functions and data sets of a package not included in R, run `library(package name)`. E.g. to use functions of the `tidyverse` package, run the following to "load" `tidyverse` into the R session:

```
library(tidyverse)
```

When using the package for the first time, install it with `install.packages("tidyverse")`. (Installation is needed only once, but loading must be done every time an R session starts.) `tidyverse` is in fact a collection of several packages; e.g. `ggplot2` and `dplyr` are included in it.

With many packages, the same function or data set name may be defined in two packages. Then `package::function()` or `package::dataset` specifies exactly which object of which package is meant. (If the package is installed, this calls the object directly without loading via `library()`.)

Below, the first function `select` belongs to the package `dplyr` (already loaded in the R environment by `library(tidyverse)`):

```
head( select(wbdata, Score) )
##   Score
## 1 7.809
## 2 7.646
## 3 7.560
## 4 7.504
## 5 7.488
## 6 7.449
```

`select(wbdata, Score)` selects only the variable `Score` of the data set `wbdata`. To use the `select` function of the `MASS` package, use `MASS::select()`. We have not loaded the `MASS` package, but `::` makes its functions usable:

```
MASS::select(MASS::lm.ridge(Score ~ Generosity + Social.support,
                            data = wbdata, lambda = seq(0,1,0.001)))
## modified HKB estimator is 0
## modified L-W estimator is 0
## smallest value of GCV  at 1
```

A.2.1 The pipe operator

Loading `dplyr` makes the special operator `%>%` available. The pipe operator `%>%` makes R code easier to read. E.g. for the mean (`mean()`) of a vector `x` one usually writes

```
mean(x)
## [1] 2.5
```

but in reading order one can write

```
x %>% mean
## [1] 2.5
```

The value before the pipe (vector, matrix, data frame) becomes the first input of the function `mean()` after the pipe. With the pipe, `head( select(wbdata, Score) )` becomes:

```
wbdata %>% select(Score) %>% head()
```

If the last function has no other inputs, the parentheses may be omitted.

A.2.2 Writing your own functions

To repeat the same task, it is often useful to define functions. E.g. to rank-transform observations `x` within each block `block`:

```
x = (1:6)^2;block = rep(c("a","b"), each = 3)
(temp.data = data.frame(x,block))
##    x block
## 1  1     a
## 2  4     a
## 3  9     a
## 4 16     b
## 5 25     b
## 6 36     b
out = temp.data %>% dplyr::group_by(block) %>% dplyr::mutate(ranks = rank(x))
out$ranks
## [1] 1 2 3 1 2 3
```

For blockwise ranking we used `group_by()` and `mutate()` of `dplyr`: the observations of `temp.data` are grouped by `block` (`group_by()`), and a new variable `ranks` is created by transforming the existing variable `x` within each group (`mutate()`). To repeat this for several data sets, define a function named `rankinblock`:[81]

```
rankinblock = function(x,block){
  require(dplyr)
  out = data.frame(x,block) %>% group_by(block) %>% mutate(ranks = rank(x))
  return(out$ranks)
}
```

`require(package)` plays the same role as `library(package)` but is designed for use inside functions; see `?require`. The defined function is used as:

```
rankinblock(x,block)
## [1] 1 2 3 1 2 3
```

[81] This function is in `np_supp.R`.

A.3 Expressions and functions frequent in nonparametric statistics

A.3.1 Permutations and combinations

Reordering all objects of a sequence `1:n` or vector `v` of length n is a *permutation*; in this book "shuffle" is used instead of "permutation." The number of all permutations of n objects is

$$n! = n(n-1)\cdots 2\cdot 1,$$

computed in R by

```
n = 10
factorial(n)
## [1] 3628800
```

Choosing r of the n elements of a vector `v` and arranging them is also a permutation; the number of such shuffles is $P(n,r) = n!/(n-r)! = n(n-1)\cdots(n-r+1)$. `permutations()` of the `gtools` package generates all of them (install with `install.packages("gtools")` if needed). The code below generates all shuffles choosing and ordering $r = 2$ objects from `v = 1:4` of length $n = 4$:

```
gtools::permutations(n = 4, r = 2, v = 1:4)
##       [,1] [,2]
##  [1,]    1    2
##  [2,]    1    3
##  [3,]    1    4
##  [4,]    2    1
##  [5,]    2    3
##  [6,]    2    4
##  [7,]    3    1
##  [8,]    3    2
##  [9,]    3    4
## [10,]    4    1
## [11,]    4    2
## [12,]    4    3
```

The exact permutation tests of Part I mainly use combinations rather than permutations. Choosing r objects from a vector `v` of length n without regard to order is a combination; their number is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

($\binom{n}{r}$ is read "n choose r"), computed in R by

```
choose(4,2)
## [1] 6
```

The code below generates all shuffles choosing $r = 2$ objects from `v = 1:4` of length $n = 4$ without regard to order:

```
gtools::combinations(n = 4, r = 2, v = 1:4)
##      [,1] [,2]
## [1,]    1    2
## [2,]    1    3
## [3,]    1    4
## [4,]    2    3
## [5,]    2    4
## [6,]    3    4
```

Comparing two populations, suppose we have n_1 observations `x1` from the first and n_2 observations `x2` from the second. If the two populations were in fact one, the n_1 observations `x1*` chosen at random from the pooled sample `c(x1,x2)` could have been the sample from the first population (see Chapter 3). To count all such cases, the number of combinations (not permutations) is needed; combinations too can be enumerated by shuffling the pooled sample, hence the term "shuffle."

```
set.seed(1)
x1 = rpois(n = 3, lambda = 5) # n1 = 3
x2 = rpois(n = 2, lambda = 5) # n2 = 2
(x.combined = c(x1,x2))       # n = 5
## [1] 4 4 5 8 3
```

The code below generates all shuffles choosing $r = 3$ objects from `x.combined` of length $n = 5$ without regard to order:

```
gtools::combinations(n = 5, r = 3, v = x.combined, set = F)
##       [,1] [,2] [,3]
##  [1,]    4    4    5
##  [2,]    4    4    8
##  [3,]    4    4    3
##  [4,]    4    5    8
##  [5,]    4    5    3
##  [6,]    4    8    3
##  [7,]    4    5    8
##  [8,]    4    5    3
##  [9,]    4    8    3
## [10,]    5    8    3
```

The vector `x.combined` contains the value 4 twice. In `gtools::combinations()`, the argument `set = F` forces the duplicate values of the input not to be removed. Also, `set.seed(1)` was used so that random sampling from the Poisson distribution (`rpois()`) always gives the same result. For reproducible analyses, always call `set.seed()` with a fixed number before random sampling.

A.3.2 Random shuffles and bootstrap resampling

Considering all shuffles is possible only for small sample sizes n. Even a not-large $n = 30$, choosing $n_1 = 15$, gives $\binom{30}{15} = 155117520$ (about 155 million) shuffles. So it is usual to sample at random only M of the $\binom{n}{n_1}$ possible shuffles, for a predetermined $M = 1999$ or $M = 9999$, to "approximate" all possible shuffles. Random sampling uses `sample`. The code below draws 3 elements of `x.combined` by simple sampling without replacement:

```
set.seed(1)
sample(x.combined, size = 3)
## [1] 4 8 5
sample(x.combined, size = 3)
## [1] 4 4 3
```

Being random sampling, the result differs every run. The next code adds the argument `replace = FALSE` to emphasize sampling without replacement; the default of `replace` in `sample()` is `FALSE`, so the result is the same as without it:

```
set.seed(1)
sample(x.combined, size = 3, replace = FALSE)
## [1] 4 8 5
```

The next code draws all 5 elements of `x.combined` of length 5 without replacement, the same as shuffling the 5 observations at random:

```
sample(x.combined, size = 5)
## [1] 5 4 8 4 3
```

Bootstrap resampling in Part II uses simple sampling with replacement of n objects from a sample `x.combined` of length n; set `replace = TRUE`:

```
sample(x.combined, size = 5, replace = TRUE)
## [1] 4 4 3 5 4
```

A.3.3 Repetition

Random permutation tests and bootstrap resampling repeat simple sampling with or without replacement many times. The most basic way to repeat is the control statement `for`. The code below repeats $M = 1999$ times the computation of the mean of 50 objects drawn without replacement from the variable `Score` of the data frame `wbdata`:

```
M = 1999
set.seed(2)
rand.mean = vector(length = M)
for(i in 1:M){
  rand.mean[i] = sample(wbdata$Score, size = 50) %>% mean
}
head(rand.mean)
## [1] 5.45034 5.42934 5.59312 5.60280 5.62102 5.53988
```

With `replicate()` this can be written shorter and more intuitively:

```
set.seed(2)
rand.mean = replicate(M, sample(wbdata$Score, size = 50) %>% mean)
head(rand.mean)
## [1] 5.45034 5.42934 5.59312 5.60280 5.62102 5.53988
```

`replicate()` is mostly used for repeated random sampling.

Repeated computation over each row (or each column) of a matrix is also needed. As an example take the 10×3 matrix `allperms`:

```
allperms = gtools::combinations(n = 5, r = 3, v = x.combined, set = F)
```

To compute the standard deviation of the values of each row of `allperms`, the most basic way is the control statement `for`:

```
sd.allperms = vector(length = choose(5,3))
for(i in 1:choose(5,3)){
  sd.allperms[i] = allperms[i,] %>% sd()
}
round(sd.allperms,2)
## [1] 0.58 2.31 0.58 2.08 1.00 2.65 2.08 1.00 2.65 2.52
```

Less intuitive but faster than a `for` statement is `apply()`. Reading the code below: for each row (the 1 means rows, 2 would mean columns) of the matrix `allperms`, compute the standard deviation `sd()` and round to two decimals:

```
apply(allperms, 1, sd) %>% round(2)
## [1] 0.58 2.31 0.58 2.08 1.00 2.65 2.08 1.00 2.65 2.52
```

Now compute the coefficient of variation (standard deviation / mean) of each column of `allperms`. The function computing the coefficient of variation and rounding to two decimals is given directly as the third argument of `apply()`:

```
apply(allperms, 2,
      function(x){
        cv = sd(x)/mean(x)
        round(cv,2)
      })
## [1] 0.08 0.31 0.50
```

A.4 Data visualization

The best way to understand data and their distribution is suitable visualization. Most figures of the book were produced with `ggplot()` of the `ggplot2` package included in `tidyverse`. A few examples follow; to learn more see Wickham and Grolemund (2016).[82]

Univariate data. The most representative way to visualize the distribution of a variable from its observations is the histogram:

```
x = rnorm(100, mean = 1, sd = 1)
data.frame(x = x) %>%
  ggplot(aes(x = x)) +
  geom_histogram(aes(y = ..density..), binwidth = .5,
                 fill = "grey", color = "white") +
  geom_rug()
```

To see whether the distribution of the data `x` follows the parametric $N(1,1)$, overlay the density of $N(1,1)$ on the histogram, or produce a quantile plot (qqplot). Producing a QQ envelope plot, a quantile plot with confidence bands, with `car::qqPlot()` is recommended.

```
data.frame(x = x) %>%
  ggplot(aes(x = x)) +
  geom_histogram(aes(y = ..density..), binwidth = .5,
                 fill = "grey", color = "white") +
  geom_function(fun = dnorm, args = list(mean = 1, sd = 1))
car::qqPlot(x, dist = "norm", mean = 1, sd = 1)
```

To compare the distributions of grouped univariate data by group, use box plots:

```
farm = read.csv("data/smartfarm.csv")
farm %>% ggplot(aes(x = block, y = x, color = treatment)) +
  geom_boxplot()
```

Bivariate data. For visualizing bivariate data use a scatter plot:

```
wbdata %>% ggplot(aes(x = GDP.per.capita, y = Score)) +
  geom_point()
```

(The resulting histogram, density overlay, QQ envelope plot, box plots and scatter plot are Figures D.1–D.5 of the book.)

A.5 R code of this book

All computations and visualizations of the book can be found at https://github.com/sungkyujung/npbook. The functions defined to simplify the discussion are collected in `np_supp.R`; to use them,

[82] https://r4ds.had.co.nz/

run

```
source("np_supp.R")
```